

t_c	Caloric temperature of cold fluid, °F
t_1, t_2	Inlet and outlet temperature of cold fluid, °F
Δt	True temperature difference in $Q = U_D A \Delta t$, °F
U_c, U_D	Clean and design overall coefficient of heat transfer, Btu/(hr)(ft ²)(°F)
V	Velocity, fps
v	Specific volume, ft ³ /lb
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
w'	Weight, lb
γ	Ratio of specific heats of a gas, dimensionless
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at the tube wall, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³
ϕ	$(\mu/\mu_w)^{0.14}$

Subscripts (except as noted above)

s	Shell side
t	Tube side

CHAPTER 10

STREAMLINE FLOW AND FREE CONVECTION

Streamline Flow in the Tubes of Exchangers. From the Fourier equation alone for a single tube $Q = wc(t_2 - t_1) = h_i(\pi DL) \Delta t$. When the inside tube-wall temperature t_p is constant, the temperature difference Δt in streamline flow may be replaced by the arithmetic mean of the hot and cold terminal temperature differences $\Delta t_a = [(t_p - t_1) + (t_p - t_2)]/2$. Solving for $h_i D/k$,

$$\frac{h_i D}{k} = \left(\frac{2 wc}{\pi k L} \right) \frac{(t_2 - t_1)}{(t_p - t_1) + (t_p - t_2)} \quad (10.1)$$

It is interesting to note that the highest outlet temperature attainable in a tube is the constant temperature of the hot tube wall t_p . For this case Eq. (10.1) reduces to

$$\frac{h_i D}{k} = \left(\frac{2 wc}{\pi k L} \right) \quad (10.2)$$

No observed average value of h_i can exceed that given by Eq. (10.2), and it is a useful tool by which erroneous observations may be rejected.

Graetz obtained Eq. (3.27) from purely theoretical considerations on the assumption of a parabolic distribution of velocities as the fluid flows in a tube in laminar flow. He did not include any corrections for modifications of the parabolic distribution during heating and cooling. Sieder and Tate evaluated the equivalent empirical equation [Eq. (3.32)] and obtained Eq. (6.1), which may be credited with correcting for the modifications of the velocity distribution during heating and cooling.

Streamline flow in tubes may be construed as a conduction effect, and it is also subject to the simultaneous occurrence of free convection as well. Free convection is only significant in nonviscous fluids. Fluids flow in streamline flow due to three conditions: (1) The fluid is viscous; (2) the fluid is not viscous, but the quantity is small for the flow area provided; and (3) the flow quantity and viscosity are intermediate but combine to give streamline flow. Only when free convection is suppressed owing to the high average viscosity of the liquid, say several centipoises and over, or when the temperature difference is small does Eq. (6.1) give h_i within the stated deviation as pure conduction. For cases falling under (2) or (3) above, the value of h_i from Eq. (6.1) may be

conservative, the true value of h_i being as much as 300 per cent greater due to the influence of free convection.

In the Graetz derivation the value of L in the Graetz number wc/kL or in the ratio D/L is assumed to be the length of the path over which the fluid moves with a conduction temperature gradient at right angles to the long axis of the tube. Naturally, if mixing occurs at any point in the heat-transfer tube, the distance from the inlet to the point of mixing must be regarded as the length of the path over which the temperature gradient is effective whether or not it corresponds to the total tube length of the exchanger, nL . Boussinesq¹ has advanced the theory that under ideal conditions streamline flow is not established until the liquids has traveled a length of approximately 15 tube diameters. In multipass heat-transfer equipment it is sometimes possible to consider the fluid in the tubes mixed at the end of each pass. Internal mixing is desirable, just as convection is desirable and also because it restricts the length of streamline path to the length of each pass, L .

The shorter the unmixed length the greater the value of h_i , although it is not always safe to assume that mixing occurs at the end of each tube pass of an exchanger. In modern multipass exchangers, the flow areas in the floating head and channel are usually designed to be identical with or slightly greater than the flow area of the tubes in each pass. In this way it is possible to eliminate excessive return pressure drops. If no turbulence or mixing is induced at the ends of each pass, nL is the total length of path instead of L , which leads to the calculation of safe values of h_i should mixing actually occur.

Frequently it will be found that the Reynolds number based on the viscosity at $t_a = (t_1 + t_2)/2$ is less than 2100 but near the outlet the Reynolds number based on the viscosity at t_2 is greater than 2100. Equation (6.1) is not applicable in the transitional or turbulent flow range. In calculations on multipass exchangers, the point at which $Re = 2100$ must then be obtained by trial and error, and the path beyond it excluded from calculation as streamline flow. If a portion of the tube is in the transitional range, it can best be computed by means of Fig. 24.

For a two-tube pass exchanger the maximum error in h_i calculated from Eq. (6.1) between a mixing or nonmixing assumption between passes is $(2/1)^{1/3} = 1.26$ or 26 per cent, since $h_i \propto 1/L^{1/3}$. For an eight-tube pass exchanger the error is $(8/1)^{1/3} = 2.0$ or 100 per cent. Ordinarily a decision should not be arrived at without consulting the design of the exchanger and noting whether or not provision for mixing between passes has been included.

¹ Boussinesq, J., *Compt. rend.*, **113**, 9 (1891).

Example 10.1. Crude Oil Heater: Streamline Flow. A line carries 16,000 lb/hr of 34°API crude oil. It enters the tubes at 95°F and is heated to 145°F using steam at 250°F. Consider the fluid *mixed* between passes.

The viscosities of the crude oil are

°F	μ cp
250	1.15
200	1.7
150	2.8
125	3.8
100	5.2

Available for temporary service is a horizontal 1-2 exchanger having a 15¼ in. ID shell with 86 1 in. OD, 16 BWG tubes 12'0" long laid out on 1¼-in. triangular pitch. The bundle is arranged for two tube passes, and baffles are spaced 15 in. apart. Since the heater is for temporary use, no dirt factor will be included.

Will the exchanger fulfill the requirement?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 15¼ in.	Number and length = 86, 12'0"
Baffle space = 15 in.	OD, BWG, pitch = 1 in., 16 BWG, 1¼-in. tri.
Passes = 1	Passes = 2

(1) Heat balance:

Crude, $Q = 16,000 \times 0.485(145 - 95) = 388,000$ Btu/hr

Steam, $Q = 410 \times 945.5 = 388,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid		Diff.
250	Higher Temp	145		105
250	Lower Temp	95		155
0	Differences	50		50

$$\Delta t = \text{LMTD} = 129^\circ\text{F (true counterflow)} \quad (5.14)$$

(3) T_c and t_c : This fluid is in streamline flow throughout the heater [see (6), p. 204]. For streamline flow the arithmetic mean should be used. For the first pass

$$t_a = \frac{t_1 + t_i}{2} = \frac{95 + 125}{2} = 110^\circ\text{F (approx.)}$$

On the assumption that the fluids are mixed between passes, each pass must be solved independently. Since only two passes are present in this exchanger, it is simply a matter of assuming the temperature at the end of the first pass. More than half the heat load must be transferred in the first pass; therefore assume t_i at the end of the first pass is 125°F and $t_i - t_1$ is 30°F.

Hot fluid: shell side steam

Cold fluid: tube side, crude oil

$$(4) a'_t = 0.594 \text{ in.}^2 \quad [\text{Table 10}]$$

$$a_t = N a'_t / 144n \quad [\text{Eq. (7.48)}]$$

$$= 86 \times 0.594 / 144 \times 2 = 0.177 \text{ ft}^2$$

$$(5) G_t = w/a_t$$

$$= 16,000 / 0.177$$

$$= 90,400 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$(6) D = 0.87/12 = 0.0725 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = DG_t/\mu$$

$$\text{At } t_2 = 145^\circ\text{F (the outlet temperature)}$$

$$\mu = 2.95 \times 2.42 = 7.15 \text{ lb}/(\text{ft})(\text{hr})$$

$$Re_t = 0.0725 \times 90,400 / 7.15 = 915$$

$$\text{At } t_a = 110^\circ\text{F,}$$

$$\mu = 4.8 \times 2.42 = 11.6 \text{ lb}/(\text{ft})(\text{hr})$$

$$Re_t = 0.0725 \times 90,400 / 11.6 = 565$$

$$(7) h_i = 1.86 \frac{k}{D} \left(\frac{DG_t c_{\mu} D}{\mu k L} \right)^{1/3} \phi_t$$

$$[\text{Eq. (6.1)}]$$

$$(8) \frac{c_{\mu}}{k} = \frac{0.485 \times 11.6}{0.0775} = 72.5$$

$$\frac{D}{L} = \frac{0.0725}{12} = 0.0060$$

$$(9) \frac{h_i}{\phi_t} = 1.86 \times \frac{0.0775}{0.0725}$$

$$(565 \times 72.5 \times 0.0060)^{1/3} = 12.4$$

$$(10) \text{ At } t_p = 249^\circ\text{F,}$$

$$\mu = 1.20 \times 2.42 = 2.9 \text{ lb}/(\text{ft})(\text{hr})$$

$$\phi_t = \left(\frac{\mu}{\mu_w} \right)^{0.14}$$

$$= \left(\frac{11.6}{2.9} \right)^{0.14} = 1.20$$

$$(11) h_i = \frac{h_i}{\phi_t} \phi_t = 12.4 \times 1.20$$

$$= 14.9 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$(9') h_o = 1500$$

$$(10') t_p = t_a + \frac{h_o}{h_i + h_o} (T_c - t_a)$$

$$= 110 + \frac{1500}{12.4 + 1500} (250 - 110)$$

$$= 249^\circ\text{F}$$

$$\Delta t = t_p - t_a = 249 - 110 = 139^\circ\text{F}$$

$$t_i - t_1 = \frac{h_i A_i \Delta t}{wc}$$

Internal surface per foot of length = 0.228 ft

$$A_i = \frac{86}{2} \times 12'0'' \times 0.228 = 117.5 \text{ ft}^2$$

$$t_i - t_1 = \frac{14.9 \times 117.5 \times 139}{16,000 \times 0.485} = 31.4^\circ\text{F}$$

Assumed value of $t_i - t_1 = 30.0^\circ\text{F}$ (close enough for check)

The oil now enters the second pass at 126.4°F and leaves at 152 instead of 145°F , indicating that the heater is oversurfaced if no dirt factor is required.

If four or more passes are present, the calculation is carried out in the same manner as shown above with a new assumed temperature at the end of each pass. If the calculated outlet from the last pass equals or exceeds t_2 , the heater will operate satisfactorily.

When a dirt factor is to be provided, obtain the initial outlet temperature t_2 delivered by the oversized heater, when freshly placed in service, from the heat balance and Eq. (5.18), using U_c for U . U_c is obtained from U_D and R_d by Eq. (6.10) instead of from h_{io} and h_o , since in this case it is desired to calculate the value of h_{io} which will produce the initial value of the outlet temperature. Having calculated t_2 , solve for each pass until the outlet temperature corresponds to the initial value of t_2 before dirt accumulates.

Free Convection in Tubes. Streamline flow is calculated by equations employing weight flow or mass velocity as one of the variables. However, if a horizontal tube surrounded by condensing steam carries cold

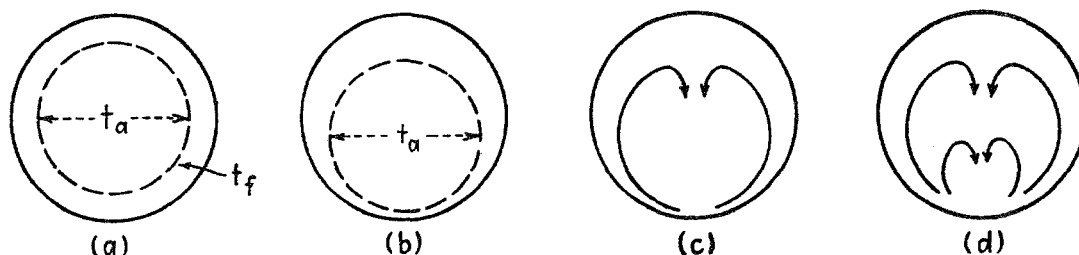


FIG. 10.1. Free convection in tubes.

liquid and its flow is suddenly halted, the liquid in the tube continues to heat. According to Eq. (6.1) the film coefficient should be zero if the mass velocity is zero. Where the heat is transferred through movements within the liquid itself without forced circulation, it occurs by free or natural convection. Some of the factors influencing free convection in liquids can be observed very readily in the laboratory owing to changes in the index of refraction which accompany changes in density. When a fluid is heated in a glass vessel on a hot plate, the convection currents are visible. Liquid at the bottom of the vessel and adjacent to the heat source is heated by conduction. The heat absorbed reduces the density of the bottom layer of liquid so that it rises and the colder liquid tends to settle.

In a *horizontal* tube the process is somewhat more orderly. Starting with a stationary liquid, heat is applied from the outside, raising the temperature of an outer layer of liquid as shown in Fig. 10.1a. The cold central core is heavier than the liquid adjacent to the wall and settles toward the bottom of the tube somewhat as shown in Fig. 10.1b. The rate of settling is retarded by the temperature-viscosity relationship between the hot fluid at the wall and the cooled core of liquid. As the

free-convection currents are established, they develop and mix with the bulk of liquid presumably as in Fig. 10.1c, and if the tube is large, mixing may be accelerated as in Fig. 10.1d. The film coefficient for free convection is a function of the inside diameter of the tube D , density of the liquid ρ , coefficient of expansion β , gravitational constant g , thermal conductivity k , viscosity μ , and lastly the temperature difference Δt_a between the hot tube wall and the bulk of the fluid.

$$h_i = f(D, \rho, \beta, c, g, k, \mu, \Delta t_a)$$

Solving by dimensional analysis

$$\frac{h_i D}{k} = C_1 \left(\frac{D^3 \rho^2 g \beta \Delta t_a}{\mu^2} \right)^a \left(\frac{c \mu}{k} \right)^b \quad (10.3)$$

where $D^3 \rho^2 g \beta \Delta t_a / \mu^2$ is the Grashof number.

Combined Free Convection and Streamline Flow in Horizontal Tubes.

Just as there is a transition region and not a single point separating streamline and turbulent flow, there must also be some transition region between free convection to a fluid standing still and streamline flow. At low linear velocities both are undoubtedly operative. Equation (6.1) was correlated from data obtained on small tubes with moderately viscous fluids and under moderate temperature differences such that the Grashof numbers were relatively small. Kern and Othmer¹ investigated this region in *horizontal* tubes under large temperature differences and tube diameters and evaluated free convection as a correction to the Sieder-Tate equation. Their final equation is

$$\frac{h_i D}{k} \left(\frac{\mu}{\mu_w} \right)^{-0.14} = 1.86 \left[\left(\frac{DG}{\mu} \right) \left(\frac{c \mu}{k} \right) \left(\frac{D}{L} \right) \right]^{1/3} \frac{2.25(1 + 0.010 Gr_a^{1/3})}{\log Re} \quad (10.4)$$

where Gr_a is the Grashof number evaluated from properties taken at the average fluid temperature $t_a = (t_1 + t_2)/2$. h_i as ordinarily calculated by Eq. (6.1) can be corrected for free convection by multiplying by

$$\psi = \frac{2.25(1 + 0.010 Gr_a^{1/3})}{\log Re} \quad (10.5)$$

Inspection of Eq. (10.4) indicates that the influence of the free-convection currents are dissipated in the transition and turbulent region. In view of the delicate nature of free-convection currents this is quite plausible. The two factors which ordinarily influence free convection most are a low viscosity and large temperature difference. Martinelli *et al.*² studied

¹ Kern, D. Q., and D. F. Othmer, *Trans. AIChE*, **39**, 517-555 (1943).

² Martinelli, R. C., C. J. Southwell, G. Alves, H. L. Craig, E. B. Weinberg, N. E. Lansing, and L. M. K. Boelter, *Trans. AIChE*, **38**, 943 (1942).

the influence of free convection in upward and downward flow in vertical tubes. They found a slight increase in the coefficient when heating water in upward flow compared with downward flow. Their final correlation is rather complicated, although they also obtained a correlation of free convection involving the Reynolds number.

Example 10.2. Kerosene Heater: Streamline Flow and Free Convection. A line carries 16,000 lb/hr of 40°API light distillate or heavy kerosene with corrosive contaminants. It enters the tubes at 95°F and is heated to 145°F using steam at 250°F. Consider the liquid *unmixed* between passes.

The viscosities of the kerosene are

°F	μ , cp
250	0.60
200	0.85
150	1.30
125	1.70
100	2.10

Available for the service is a horizontal 1-2 exchanger, having a 15¼ in. ID shell with 86 1 in. OD, 16 BWG tubes 12'0" long laid out on 1¼-in. triangular pitch. The bundle is arranged for two tube passes, and baffles are spaced 15 in. apart (same as Example 10.1).

What is the true dirt factor?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 15¼ in.	Number and length = 86, 12'0"
Baffle space = 15 in.	OD, BWG, pitch = 1 in., 16 BWG, 1¼-in. tri.
Passes = 1	Passes = 2

(1) Heat balance:

Kerosene, $Q = 16,000 \times 0.50(145 - 95) = 400,000$ Btu/hr

Steam, $Q = 4230 \times 945.5 = 400,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
250	Higher Temp	145	105
250	Lower Temp	95	155
0	Differences	50	50

$$\Delta t = \text{LMTD} = 129^\circ\text{F (true counterflow)} \quad (5.14)$$

(3) T_c and t_c : This fluid is in streamline flow throughout the heater [see (6), p. 208]. For streamline flow the arithmetic mean is

$$t_a = \frac{t_1 + t_2}{2} = \frac{95 + 145}{2} = 120^\circ\text{F}$$

Hot fluid: shell side, steam

(9') Condensation of steam:
 $h_o = 1500$

(10') t_w :

$$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_a - t_a) \quad [\text{Eq. (5.31)}]$$

$$= 120 + \frac{1500}{8.9 + 1500} (250 - 120)$$

$$= 249^\circ\text{F}$$

Cold fluid: tube side, 40° API kerosene

(4) $a'_t = 0.594 \text{ in.}^2$ [Table 10]

$a_t = N_t a'_t / 144n$ [Eq. (7.48)]

$$= 86 \times 0.594 / 144 \times 2 = 0.177 \text{ ft}^2$$

(5) $G_t = w/a_t$

$$= 16,000 / 0.177$$

$$= 90,400 \text{ lb}/(\text{hr})(\text{ft}^2)$$

(6) $D = 0.87/12 = 0.0725 \text{ ft}$ [Table 10]

$$Re_t = DG_t/\mu$$

At $t_2 = 145^\circ\text{F}$,

$$\mu = 1.36 \times 2.42 = 3.29 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$Re_t = 0.0725 \times 90,400 / 3.29 = 1990$$

At $t_a = 120^\circ\text{F}$,

$$\mu = 1.75 \times 2.42 = 4.23 \text{ lb}/(\text{ft})(\text{hr})$$

$$Re_t = 0.0725 \times 90,400 / 4.23 = 1,550$$

Streamline flow, fluid unmixed between passes

$$Ln/D = 12 \times 2 / 0.0725 = 331$$

(7) $j_H = 3.10$ [Fig. 24]

(8) At $\mu = 1.75 \text{ cp}$ and 40° API,

$$k(c\mu/k)^{1/2} = 0.24 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$$

[Fig. 16]

(9) $\frac{h_i}{\phi_t} = j_H \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/2}$ [Eq. (6.15a)]

$$h_i/\phi_t = 3.10 \times 0.24 / 0.0725 = 10.25$$

(10) $\frac{h_{io}}{\phi_t} = \frac{h_i}{\phi_t} \times \text{ID}/\text{OD}$ [Eq. (6.5)]

$$= 10.25 \times 0.87 / 1.0 = 8.91$$

(11) At $t_w = 249^\circ\text{F}$,

$$\mu_w = 0.60 \times 2.42 = 1.45 \text{ lb}/(\text{ft})(\text{hr})$$

$$\phi_t = (\mu/\mu_w)^{0.14}$$

$$= (4.23/1.45)^{0.14} = 1.16$$

(12) $h_{io} = \frac{h_{io}}{\phi_t} \phi_t$ [Eq. (6.37)]

$$= 8.91 \times 1.16$$

$$= 10.3 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$\Delta t_a = t_w - t_a = 249 - 120 = 129^\circ\text{F}$$

Since the kerosene has a viscosity of only 1.75 cp at the caloric temperature and $\Delta t_a = 129^\circ\text{F}$, free convection should be investigated.

$$\psi = \frac{2.25(1 + 0.01Gr_a^{1/3})}{\log Re} \quad [\text{Eq. (10.5)}]$$

$$\text{Grashof number, } Gr_a = D^3 \rho^2 g \beta \Delta t_a / \mu^2$$

$$s = 0.80,$$

$$\rho = 0.8 \times 62.5 = 50.0 \text{ lb}/\text{ft}^3 \quad [\text{Fig. 6}]$$

$$\beta = \frac{1}{^\circ\text{F}} = \frac{1/\rho_2 - 1/\rho_1}{(t_2 - t_1)(1/\rho_{av})}$$

$$= \frac{s_1^2 - s_2^2}{2(t_2 - t_1)s_1s_2}$$

Hot fluid: shell side, steam

Cold fluid: tube side, 40° API kerosene
 At 95°F, $s_1 = 0.810$ [Fig. 6]
 At 145°F, $s_2 = 0.792$
 $\beta = 0.000451/^\circ\text{F}$
 $Gr_o = 0.0725^3 \times 50.0^2 \times 0.00045 \times 4.18$
 $\times 10^8 \times 129/4.23^2$
 $= 1,300,000$
 $\psi = \frac{2.25(1 + 0.01 \times 1,300,000^{1/2})}{\log 1550} = 1.47$
 Corrected $h_{io} = 10.3 \times 1.47 = 15.1$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{15.1 \times 1500}{15.1 + 1500} = 14.9 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Design overall coefficient U_D :

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft} \quad (\text{Table 10})$$

$$\text{Total surface, } A = 86 \times 12'0'' \times 0.2618 = 270 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{400,000}{270 \times 129} = 11.5 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(15) Dirt factor:

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{14.9 - 11.5}{14.9 \times 11.5} = 0.0198 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary: Figures in parentheses are uncorrected for free convection

Summary

1500	h outside	(10.3) 15.1
U_C	(10.2) 14.9	
U_D	11.5	
R_d Calculated	0.0198	

If the correction for free convection were not included, it would have given the impression that the unit would not work, since U_C would be smaller than U_D . Corrected, however, it is very safe. If the viscosity were 3 or 4 centipoises and the Reynolds number remained the same because of increased weight flow, the exchanger would not be suitable.

The pressure drop may be computed as heretofore except that ϕ_t for pressure drop in streamline flow is $(\mu/\mu_w)^{0.25}$.

The Use of Core Tubes. Employing the horizontal exchanger of the preceding example, suppose that 50,000 lb/hr of a 28°API gas oil was to be heated from 105 to 130°F using steam at 250°F. Would the exchanger be satisfactory?

Solving as before, the value of the viscosity correction ϕ_t is 1.18 but the free-convection correction ψ is negligible, and U_C would be less than U_D .

Cores, dummy tubes with one end sealed (by clamping in a vise), may

be placed within the tubes of an exchanger as shown in Fig. 10.2. They constrict the cross-sectional area by forming an annulus which replaces the inside diameter of the tube with a smaller equivalent diameter and which increases the mass velocity of the tube fluid. Although the use of a core decreases the effective tube diameter and increases the mass

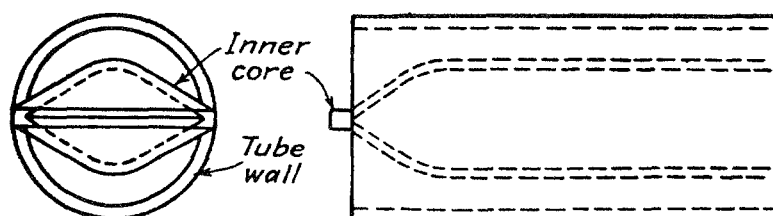


FIG. 10.2. Core detail.

velocity, it does not alter the Reynolds number from its value before introduction of the core. Using a core,

$$a_s = \frac{\pi}{4} (D_2^2 - D_1^2)$$

where D_1 is the outside diameter of the core and D_2 is the inside diameter of the tube. Wetted perimeter for heat transfer¹ = πD_2

$$D_e = 4 \frac{\pi}{4} \frac{(D_2^2 - D_1^2)}{\pi D_2} = \frac{D_2^2 - D_1^2}{D_2}$$

$$G = \frac{w}{(\pi/4)(D_2^2 - D_1^2)}$$

$$Re_t = \frac{D_e G}{\mu} = \frac{D_2^2 - D_1^2}{D_2 \mu} \frac{w}{(\pi/4)(D_2^2 - D_1^2)} = \frac{4w}{\pi D_2 \mu}$$

Without a core the result is the same.

$$Re_t = \frac{DG}{\mu} = \frac{D_2}{\mu} \frac{w}{(\pi/4)D_2^2} = \frac{4w}{\pi D_2 \mu}$$

Since the Reynolds number remains the same, the advantage of the cores is manifested by the smaller value of D_e to D in Eq. (6.1). However, the use of cores eliminates the possibility of free convection as correlated by Eq. (10.4), since the cores disrupt the natural-convection currents. Cores are most advantageously employed only when the fluid is viscous and free convection is precluded.

Any size of core may be employed, and it is customary to use 18 BWG or lighter heat-exchanger tubes for the purpose. If the tube is sealed by clamping in a vise, the core must be of such size that one-half its perimeter is greater than the inside diameter of the tube into which it is to be inserted. If the clamped width is less, the core is secured by welding

¹ Unlike the annulus of a double pipe exchanger, the wetted perimeter is the circumference at the inside diameter of the outer tube rather than the outer diameter of the inner tube.

small rods at the clamped ends, which are larger than one-half the perimeter of the tube. The cores are inserted with the sealed end at the inlet of each pass, so that in multipass exchangers half are inserted at the channel and half at the opposite end of the bundle.

Since cores produce considerable inlet and exit turbulence, it is fair to assume that substantial mixing occurs between passes. In such cases the unmixed length of path may be taken as the tube length.

The equivalent diameter for pressure drop is smaller than that for heat transfer because the wetted perimeter is the sum of the circumferences of the inside of the tube plus the outside of the core. Ordinarily, if there is a large quantity of liquid flowing but the viscosity is high, the pressure drop will be quite large. If the flow is scanty and the viscosity is small, the pressure drop will usually be negligibly small despite the use of a core.

Example 10.3. Gas Oil Heater Using Cores. In an effort to make the exchanger in Example 10.2 suitable for the flow of 50,000 lb/hr of 28°API gas oil, the use of cores will be investigated. The oil will be heated from 105 to 130°F using steam at 250°F. Comparisons will be made with Example 10.2.

The correct core cannot always be selected for the first trial, and it is usually necessary to assume several core sizes and carry out the calculation.

Viscosities of the oil in the low temperature range are

°F	μ , cp
250	2.0
200	3.1
150	5.0
125	6.3
100	8.2

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 15¼ in.	Number and length = 86, 12'0"
Baffle space = 15 in.	OD, BWG, pitch = 1 in., 16 BWG, 1¼-in. tri.
Passes = 1	Passes = 2

Assume that ½-in., 18 BWG cores are used.

(1) Heat balance:

Gas oil, $Q = 50,000 \times 0.47(130 - 105) = 587,000$ Btu/hr

Steam, $Q = 6220 \times 945.5 = 587,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
250	Higher Temp	130	120
250	Lower Temp	105	145
0	Differences	25	25

$$\Delta t = \text{LMTD} = 132.5^\circ\text{F (true counterflow)}$$

(5.14)

(3) T_c and t_c :

$$t_a = \frac{(t_1 + t_2)}{2} = \frac{130 + 105}{2} = 117.5^\circ\text{F}$$

Hot fluid: shell side, steam

$$\begin{aligned} (4') \quad a_s &= ID \times C'B/144P_T \quad [\text{Eq. (7.1)}] \\ &= 15.25 \times 0.25 \times 15/144 \times 1.25 \\ &= 0.318 \text{ ft}^2 \end{aligned}$$

$$\begin{aligned} (5') \quad G_s &= W/a_s \quad [\text{Eq. (7.2)}] \\ &= 6220/0.318 \\ &= 19,600 \text{ lb}/(\text{hr})(\text{ft}^2) \end{aligned}$$

$$\begin{aligned} (6') \quad \text{At } T_a &= 250^\circ\text{F}, \\ \mu &= 0.013 \times 2.42 = 0.0314 \text{ lb}/(\text{ft})(\text{hr}) \end{aligned}$$

$$D_s = 0.72/12 = 0.060 \text{ ft} \quad [\text{Fig. 15}]$$

$$\begin{aligned} Re_s &= D_s G_s / \mu \quad (\text{for pressure drop}) \quad [\text{Fig. 29}] \\ &= 0.06 \times 19,600 / 0.0314 = 37,400 \end{aligned}$$

$$\begin{aligned} (9') \quad \text{Condensation of steam:} \\ h_o &= 1500 \end{aligned}$$

$$\begin{aligned} (10') \quad t_w: \\ t_w &= t_a + \frac{h_o}{h_{io} + h_o} (T_a - t_a) \quad [\text{Eq. (5.31)}] \\ &= 117.5 + \frac{1500}{19.5 + 1500} (250 - 117.5) \\ &= 249^\circ\text{F} \end{aligned}$$

Cold fluid: tube side, crude oil

$$\begin{aligned} (4) \quad a'_t &= \frac{\pi}{4} (d_2^2 - d_1^2) \\ &= \frac{\pi}{4} (0.87^2 - 0.50^2) = 0.40 \text{ in.}^2 \end{aligned}$$

 $(d_2 \text{ and } d_1 \text{ are the annulus diameters, in.})$

$$\begin{aligned} a_t &= N_t a'_t / 144n \quad [\text{Eq. (7.48)}] \\ &= 86 \times 0.40 / 144 \times 2 = 0.119 \text{ ft}^2 \end{aligned}$$

$$\begin{aligned} (5) \quad G_t &= w/a_t \\ &= 50,000 / 0.119 \\ &= 420,000 \text{ lb}/(\text{hr})(\text{ft}^2) \end{aligned}$$

$$\begin{aligned} (6) \quad d_e &= \frac{d_2^2 - d_1^2}{d_2} = \frac{0.87^2 - 0.50^2}{0.87} \\ &= 0.582 \text{ in.} \end{aligned}$$

$$D_e = 0.582/12 = 0.0485 \text{ ft}$$

$$\begin{aligned} \text{At } t_a &= 117.5^\circ\text{F}, \\ \mu &= 6.9 \times 2.42 = 16.7 \text{ lb}/(\text{ft})(\text{hr}) \end{aligned}$$

$$\begin{aligned} Re_t &= D_e G_t / \mu \\ &= 0.0485 \times 420,000 / 16.7 = 1,220 \end{aligned}$$

$$\begin{aligned} \text{Assume mixing between passes } L/D_e \\ &\approx 12/0.0485 = 247 \end{aligned}$$

$$(7) \quad j_H = 3.10$$

$$\begin{aligned} (8) \quad \text{At } \mu &= 6.9 \text{ cp and } 28^\circ\text{API} \\ k(c\mu/k)^{1/3} &= 0.35 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft}) \end{aligned}$$

[Fig. 16]

$$\begin{aligned} (9) \quad \frac{h_i}{\phi_t} &= j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \quad [\text{Eq. (6.15a)}] \\ &= 3.10 \times 0.35 / 0.0485 = 22.4 \end{aligned}$$

$$\begin{aligned} (10) \quad \frac{h_{io}}{\phi_t} &= \frac{h_i}{\phi_t} \times \frac{ID}{OD} \quad [\text{Eq. (6.5)}] \\ &= 22.4 \times 0.87 / 1.0 = 19.5 \end{aligned}$$

$$\begin{aligned} (11) \quad \text{At } t_w &= 249^\circ\text{F}, \\ \mu_w &= 2.0 \times 2.42 = 4.84 \text{ lb}/(\text{ft})(\text{hr}) \end{aligned}$$

[Fig. 14]

$$\begin{aligned} \phi_t &= (\mu/\mu_w)^{0.14} \\ &= (16.7/4.84)^{0.14} = 1.18 \end{aligned}$$

$$\begin{aligned} (12) \quad h_{io} &= \frac{h_{io}}{\phi_t} \phi_t \quad [\text{Eq. (6.37)}] \\ &= 19.5 \times 1.18 \\ &= 23.0 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \end{aligned}$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{23.0 \times 1500}{23.0 + 1500} = 22.6 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Design overall coefficient U_D :

$$A = 270 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{587,000}{270 \times 132.5} = 16.4 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_c - U_D}{U_c U_D} = \frac{22.6 - 16.4}{22.6 \times 16.4} = 0.0172 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

1500	<i>h</i> outside	19.4
U_C	22.6	
U_D	16.4	
R_d Calculated 0.0172		

Note that the cores make the heater operable.

Pressure Drop

(1') For $Re_s = 37,400$, $f = 0.0016 \text{ ft}^2/\text{in.}$ (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 12 \times 12/15 = 10$ $v = 13.82 \text{ ft}^3/\text{lb}$ (Table 7) $s = \frac{1}{13.82 \times 62.5} = 0.00116$ $D_s = 15.25/12 = 1.27 \text{ ft}$ (3') $\Delta P_s = \frac{1}{2} \frac{f G_s^2 (D_s) (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ [Eq. (7.52)] $= \frac{1}{2} \times \frac{0.0016 \times 19,600^2 \times 1.27}{5.22 \times 10^{10} \times 0.060 \times 0.00116 \times 1.0}$ $= 1.4 \text{ psi}$	(1) $d'_e = (d_2 - d_1)$ [Eq. (6.4)] $= (0.87 - 0.50) = 0.37 \text{ in.}$ $D'_e = 0.37/12 = 0.0309 \text{ ft}$ $Re'_e = D'_e G_t / \mu$ $= 0.0309 \times 420,000/16.7 = 777$ $f = 0.00066 \text{ ft}^2/\text{in.}^2$ [Fig. 26] (2) For streamline-flow pressure drop: $\phi_t = (16.7/4.84)^{0.25} = 1.35$ $s = 0.85$ [Fig. 6] $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D'_e s \phi_t}$ [Eq. (7.45)] $= \frac{0.00066 \times 420,000^2 \times 12 \times 2}{5.22 \times 10^{10} \times 0.0309 \times 0.85 \times 1.35}$ $= 1.5 \text{ psi}$ [Eq. (7.46)] $\Delta P_r = \text{negligible}$
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The choice of a core was satisfactory. A large dirt factor should be used because of the inaccuracies of the calculation. It would also be advisable to investigate the use of a $\frac{1}{8}$ or $\frac{3}{4}$ in. OD core, although either of these cores will cause a considerably greater pressure drop.

Streamline Flow in Shell. When an exchanger is selected to fulfill process conditions from among existing exchangers, streamline flow in tubes should be avoided if possible. By comparison with turbulent flow, streamline flow requires the use of a larger surface for the delivery of equal heat. Sometimes streamline flow in tubes is unavoidable as when there is a great unbalance between both the sizes of the two streams and the shell-side and tube-side flow areas. If the flow area of the shell is greater than that of the tubes and one of the fluid quantities is much larger than the other, it may be imperative in meeting the allowable pressure drop to place the smaller stream

in the tubes. Still other instances of streamline flow arise when a liquid is placed in the tubes for corrosion reasons whereas from the standpoint of fluid flow it logically belongs in the shell.

Streamline flow in shells, on the other hand, is fortunately a rare problem. It occurs at a Reynolds number well below 10, as plotted in Fig. 28, due to the constantly changing flow area across the long axis of the bundle. The shell is consequently an excellent place in which to locate a scanty stream. The fouling factors of Table 12 have been predicated upon turbulent flow, and for low tube velocities they should be increased perhaps 50 to 100 per cent to provide additional protection.

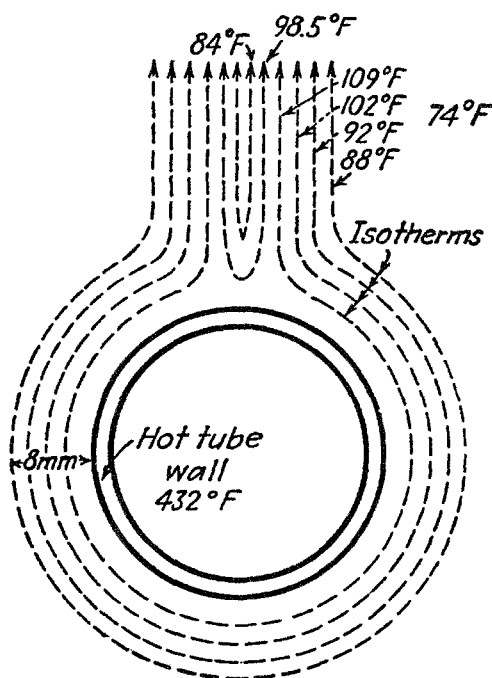


FIG. 10.3. Free convection outside tubes. (After Ray.)

Free Convection outside Tubes and Pipes. The mechanism of free convection outside a horizontal cylindrical shape differs greatly from that within it. On the outside of a pipe the convection currents are not restrained as they are within it, and heated fluid is usually free to rise through greater heights of cold fluid thereby increasing the convection. The atmosphere about a pipe has been explored by Ray,¹ and the isotherms are shown schematically in Fig. 10.3. Cold air from the relatively ambient atmosphere travels in toward the hot tube whence it is heated and rises. Numerous investigators have established the influence of the Grashof and Prandtl numbers on the correlation of free convection. Unfortunately most of the experimental information has been obtained on apparatuses such as single tubes and wires rather than industrial equipment. Accordingly, however, the film coefficient for free convection to

¹ Ray, B. B., *Proc. Indian Assoc. Cultivation Sci.*, 5, 95 (1920).

gases from horizontal cylinders can be represented by

$$\frac{h_c D}{k_f} = \alpha \left[\left(\frac{D_o^3 \rho_f^2 g \beta \Delta t}{\mu_f^2} \right) \left(\frac{c_f \mu_f}{k_f} \right) \right]^{0.25}$$

where h_c is the free convection coefficient and all the properties are evaluated at the fictitious film temperature t_f taken as the mean of the temperature of the heating surface and the bulk temperature of the fluid being heated. Thus

$$t_f = \frac{t_w + t_a}{2} \quad (10.6)$$

For the most part it is difficult to obtain good data for the various types and sizes of equipment used in industry. This is due in part to the interference and complexities of free-convection heating elements such as banks of tubes and the inability to control an atmosphere of fluid to the extent necessary for good experimental results.

Free-convection correlations from the outside surfaces of different shapes which are of direct engineering value fall largely into two classes: free convection about single tubes or pipes and free convection about vessels and walls. McAdams¹ gives an excellent review of the work done heretofore in this field. It is apparent that free-convection currents are not only influenced by the position of the surface but also by its proximity to other surfaces. A hot horizontal surface sets up currents which differ greatly from those set up by a vertical surface. McAdams has summarized these in a simplified dimensional form² for free convection to air.

$$\text{Horizontal pipes:} \quad h_c = 0.50 \left(\frac{\Delta t}{d_o} \right)^{0.25} \quad (10.7)$$

$$\text{Long vertical pipes:} \quad h_c = 0.4 \left(\frac{\Delta t}{d_o} \right)^{0.25} \quad (10.8)$$

$$\text{Vertical plates less than 2 ft high:} \quad h_c = 0.28 \left(\frac{\Delta t}{z} \right)^{0.25} \quad (10.9)$$

$$\text{Vertical plates more than 2 ft high:} \quad h_c = 0.3 \Delta t^{0.25} \quad (10.10)$$

$$\text{Horizontal plates:} \quad \text{Facing upward:} \quad h_c = 0.38 \Delta t^{0.25} \quad (10.11)$$

$$\text{Facing downward:} \quad h_c = 0.2 \Delta t^{0.25} \quad (10.12)$$

where Δt is the temperature difference between hot surface and cold fluid in °F, d_o is the outside diameter in inches, and z is the height in feet.

¹ McAdams, W. H., "Heat Transmission," 2d ed., pp. 237-246, McGraw-Hill Book Company, Inc., New York, 1942.

² Perry, J. H., "Chemical Engineers' Handbook," 3rd Ed., p. 474, McGraw-Hill Book Company, Inc., New York, 1950.

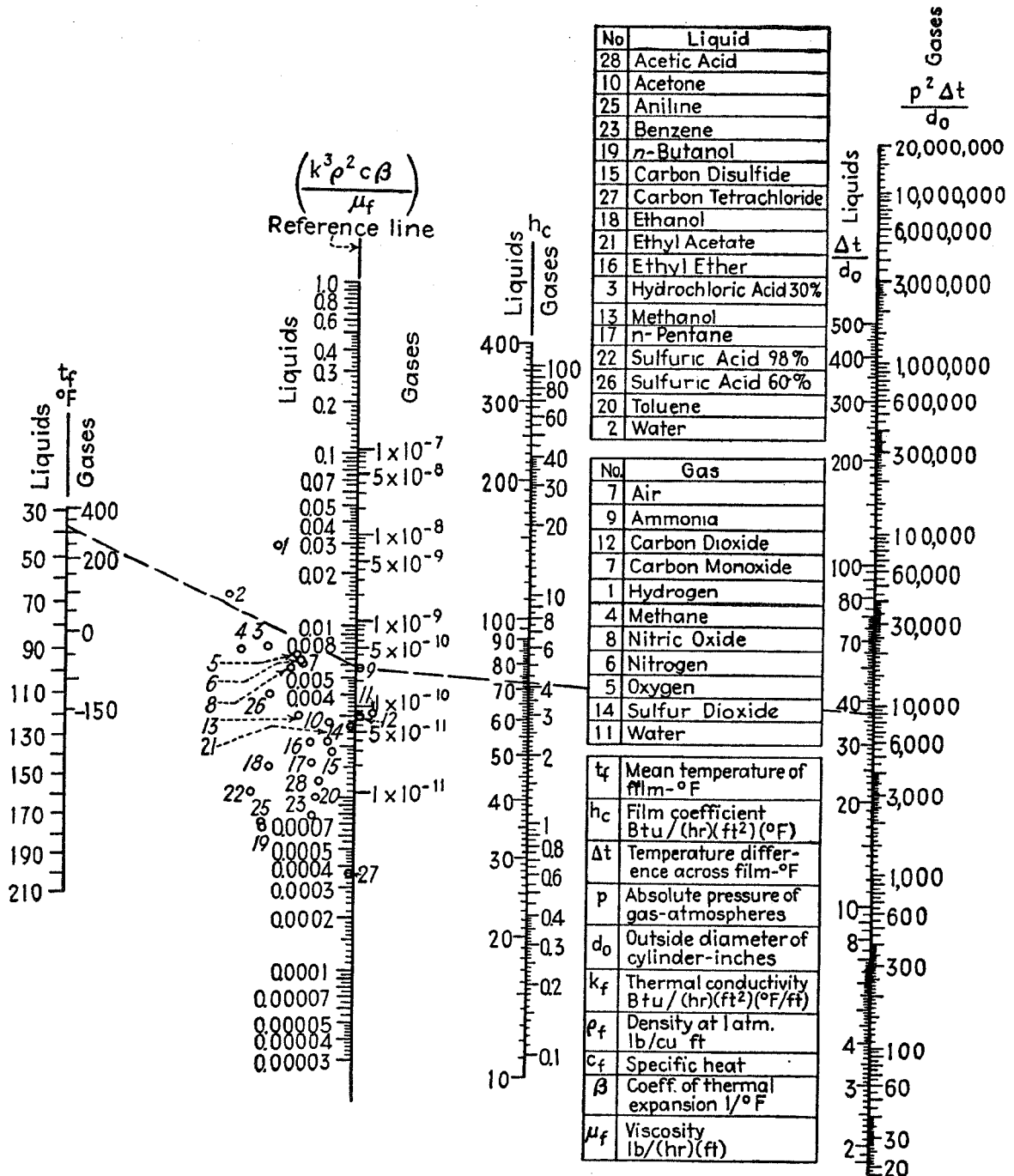


FIG. 10.4. Free convection outside horizontal pipes and tubes. [T. H. Chilton, A. P. Colburn, R. P. Generaux, and H. C. Vernon, *Trans. ASME Petroleum Mech. Eng.*, **55**, 5 (1933).]

For single horizontal pipes the dimensionless expression will apply except that α will vary between 0.47 and 0.53 between small and large pipes. Thus

$$\frac{h_c D_o}{k_f} = 0.47 \left[\left(\frac{D_o^3 \rho_f^2 g \beta \Delta t}{\mu_f^2} \right) \left(\frac{c \mu_f}{k_f} \right) \right]^{0.25} \quad (10.13)$$

Chilton, Colburn, Generaux, and Vernon¹ have developed an alignment chart which gives conservative coefficients for a single pipe but which has been used by the author and others without noticeable error for the calculations of free convection outside *banks* of tubes. The dimensional equation, plotted in Fig. 10.4 for gases and liquids, is

$$h_c = 116 \left[\left(\frac{k_f^3 \rho_f^2 c_f \beta}{\mu_f'} \right) \left(\frac{\Delta t}{d_o} \right) \right]^{0.25} \quad (10.14)$$

where μ_f' is in centipoises. Of the four axes in the alignment chart one is the reference line of values of $k_f^3 \rho_f^2 c_f \beta / \mu_f'$ which allows its use for fluids other than those whose key numbers are given in Fig. 10.4. The use of the chart for tube bundles requires that the pipes or tubes not be located too closely to the bottom of the vessel or too closely among themselves so that they interfere with the natural-convection currents. The tubes should not be located closer than several diameters from the bottom of the vessel, nor should the clearance between tubes be less than a diameter. Notwithstanding the data available, free-convection design is not very accurate, and reasonable factors of safety such as large fouling factors are recommended.

The use of the several types of free-convection correlations for fluids outside tubes is demonstrated by the typical problem which follows.

Example 10.4. Calculation of a Heating Bundle for an Aniline Storage Tank. A horizontal outdoor storage tank 5'0" ID by 12'0" long and presumed cylindrical is to

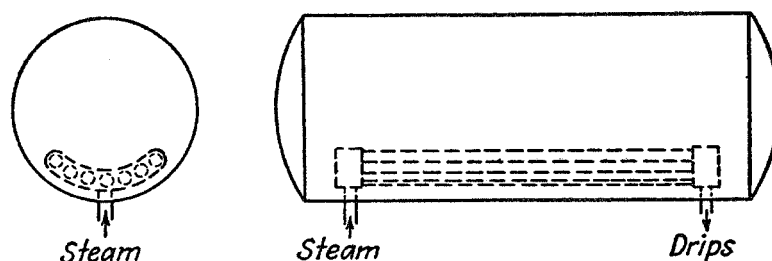


FIG. 10.5. Heating a fluid in a tank by free convection.

be used for aniline storage at a constant temperature of 100°F even though the prevailing atmospheric temperature should fall to 0°F. The tank is not insulated but is shielded from the wind. Heat will be supplied by means of exhaust steam through a bundle connected through the bottom of the tank as shown in Fig. 10.5 and consisting of 6'0" lengths of 1-in. IPS pipe. The use of a temperature controller (Chap. 21) to throttle the steam would make the entire operation automatic.

How many lengths of pipe should be provided?

Solution. The first part of the problem is to determine how much heat will be lost from the tank to the atmosphere. This gives the heat load for which the coil must be designed.

¹ Chilton, T. H., A. P. Colburn, R. P. Generaux, and H. C. Vernon, *Trans. A.S.M.E., Petroleum Mech. Eng.*, **55**, 5 (1933).

From Eq. (10.7) to (10.12) the convection coefficient to air will be somewhere between $h_c = 0.2 \Delta t^{0.25}$ and $h_c = 0.3 \Delta t^{0.25}$ for the entire tank surface. Since the equation giving the higher value of h_c is safer.

Convection loss:

$$h_c = 0.3 \Delta t^{0.25}$$

Neglecting the temperature drop through the tank metal

$$\begin{aligned} \Delta t &= 100 - 0 = 100^\circ\text{F} \\ h_c &= 0.3 \times 100^{0.25} = 0.95 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \\ \text{Radiation rate, } h_r &= \frac{0.173 \epsilon [(T_{1,\text{abs}}/100)^4 - (T_{2,\text{abs}}/100)^4]}{T_{1,\text{abs}} - T_{2,\text{abs}}} \end{aligned} \quad (4.32)$$

Assuming an emissivity of about 0.8,

$$h_r = \frac{0.173 \times 0.8(5.6^4 - 4.6^4)}{100} = 0.75 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

Combined loss:

$$\begin{aligned} h_c + h_r &= 0.95 + 0.75 = 1.70 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \\ \text{Total tank area} &= \frac{2\pi \times 5^2}{4} + \pi \times 5 \times 12 = 227.8 \text{ ft}^2 \end{aligned}$$

$$\text{Total heat loss, } Q = (h_c + h_r)A \Delta t = 1.70 \times 227.8 \times (100 - 0) = 38,800 \text{ Btu/hr}$$

This heat must be supplied by the pipe bundle.

Assuming exhaust steam to be at 212°F ,

$$\begin{aligned} \frac{\Delta t}{d_o} &= \frac{212 - 100}{1.32} = 85 \\ t_f &= \frac{212 + 100}{2} = 156^\circ\text{F (nearly)} \end{aligned}$$

From Figure 10.4, $h_c = 48 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$.

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1500 \times 48}{1500 + 48} = 46.5 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

Assume a dirt factor of $0.02 (\text{hr})(\text{ft}^2)(^\circ\text{F})/\text{Btu}$

$$U_D = \frac{U_c \times 1/R_d}{U_c + 1/R_d} = \frac{46.5 \times 50}{46.5 + 50} = 24.1 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$\begin{aligned} \text{Total surface, } A &= \frac{Q}{U_D \Delta t} \\ &= 38,800/24.1 \times (212 - 100) = 14.4 \text{ ft}^2 \\ \text{Area/pipe} &= 0.344 \times 6 = 2.06 \text{ ft}^2 \\ \text{Number of pipes} &= 14.4/2.06 = 7 \end{aligned} \quad (\text{Table 11})$$

These may be arranged in series or parallel.

PROBLEMS

10.1. 9000 lb/hr of 34°API crude oil at 100°F (see Example 10.1 for viscosities) enter the tubes of a $17\frac{1}{4}$ in. ID 1-2 exchanger containing 118 tubes 1 in. OD, 16 BWG, 12'0" long arranged for two passes in $1\frac{1}{4}$ -in. triangular pitch. It is desired to attain an outlet temperature of 150°F using steam at 250°F . What is the actual outlet temperature?

10.2. 37,000 lb/hr of an organic compound whose properties are closely approximated by the 28°API gas oil in the Appendix is to be heated by steam in the tubes of a clean exchanger because the shell baffles are half-circle support plates. The oil is to be heated from 100 to 200°F by steam at 325°F.

Available for the service is a 27 in. ID 1-2 exchanger containing 334 1 in. OD, 16 BWG, 16'0" long tubes arranged for two passes on 1¼-in. triangular pitch.

(a) Determine the dirt factor if the oil is not mixed between passes? (b) What is the actual outlet temperature of the oil if the fluid is mixed between passes?

10.3. 22,000 lb/hr of a flushing oil closely resembling 28°API gas oil is to be heated from 125 to 175°F to improve filtration. Available as a heating medium is another line corresponding to a 35°API distillate at 280°F. The temperature range of the distillate will be only 5°F so that the exchanger is essentially in counterflow.

Available for the service is a 21¼ in. ID 1-2 exchanger containing 240 ¾ in., 16 BWG tubes 8'0" long. Tubes are arranged for six passes on 1-in. square pitch. Baffles are spaced 8 in. apart. Because of contaminants the flushing oil will flow in the tubes.

Investigate all the possibilities, including the use of cores, to make the exchanger operable. Pressure drops are 10 psi, and a combined dirt factor is 0.015 (not to be confused with 0.0015).

10.4. A vertical cylindrical storage tank 6'0" ID and 8'0" tall is filled to 80 per cent of its height with ethylene glycol at 150°F. The surrounding atmosphere may fall to 0°F in the winter, but it is desired to maintain the temperature within the tank by using a rectangular bank of 1 in. IPS pipes heated with exhaust steam at 225°F. How much surface is required? How should the pipes be arranged?

NOMENCLATURE FOR CHAPTER 10

A	Heat transfer surface, ft ²
a''	External surface per linear foot, ft
a	Flow area, ft ²
B	Baffle spacing, in.
C_1, C_2	Constants, dimensionless
D	Inside diameter of tubes or pipes, ft
D_o	Outside diameter of tubes or pipes, ft
D_e, D'_e	Equivalent diameter for heat transfer and pressure drop, ft
d	Inside diameter of tubes, in.
d_o	Outside diameter of tubes, in.
f	Friction factor, ft ² /in. ²
G	Mass velocity, lb/(hr)(ft ²)
Gr	Grashof number, dimensionless
Gr_a	Grashof number at the average or bulk temperature, dimensionless
g	Acceleration of gravity, ft/hr ²
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, Btu/(hr)(ft ²)(°F)
h_{io}	Value of h_i when referred to the tube outside diameter
h_c	Heat-transfer coefficient for free convection, Btu/(hr)(ft ²)(°F)
h_r	Heat-transfer coefficient for radiation, Btu/(hr)(ft ²)(°F)
j_H	Factor for heat transfer, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Tube length or length of unmixed path, ft
n	Number of tube passes

ΔP	Pressure drop in general, psi
$\Delta P_T, \Delta P_r, \Delta P_i$	Total, tube, and return pressure drop, respectively, psi
P_T	Tube pitch, in.
R_d	Combined dirt factor, (hr)(ft ²)(°F)/Btu
Re, Re'	Reynolds number for heat transfer and pressure drop, dimensionless
s	Specific gravity, dimensionless
T_a, T_c	Average and caloric temperature of hot fluid, °F
t_1, t_2	Inlet and outlet temperatures of cold fluid, °F
t_a, t_c	Average and caloric temperatures of cold fluid, °F
t_f	Film temperature $\frac{1}{2}(t_w + t_a)$, °F
t_i	Temperature at end of first pass, °F
t_p, t_w	Inside and outside tube wall temperatures, °F
Δt	Temperature difference for heat transfer, °F
Δt_a	Temperature difference between tube wall and average fluid temperature, °F
U_c, U_D	Clean and design overall coefficients, Btu/(hr)(ft ²)(°F)
v	Specific volume, ft ³ /lb
W, w	Weight flow of hot and cold fluid, lb/hr
z	Height, ft
α	Constant
β	Coefficient of thermal expansion, 1/°F
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at tube wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_f	Viscosity at film temperature, centipoises
ρ	Density, lb/ft ³
ϕ	$(\mu/\mu_w)^{0.14}$. For pressure drop in streamline flow, $(\mu/\mu_w)^{0.25}$
ψ	Free convection correction, Eq. (10.5), dimensionless

Subscripts (except as noted above)

f	Evaluated at the film temperature
s	Shell side
t	Tube side

CHAPTER 11

CALCULATIONS FOR PROCESS CONDITIONS

Optimum Process Conditions. The experience obtained from the calculation of existing tubular exchangers will presently be applied to cases in which only the process conditions are given. Before undertaking these calculations an investigation will be made to determine whether or not several related pieces of equipment can avail themselves of the overall process temperatures in an optimum manner. This is an economic affair similar to that for the optimum outlet-water temperature

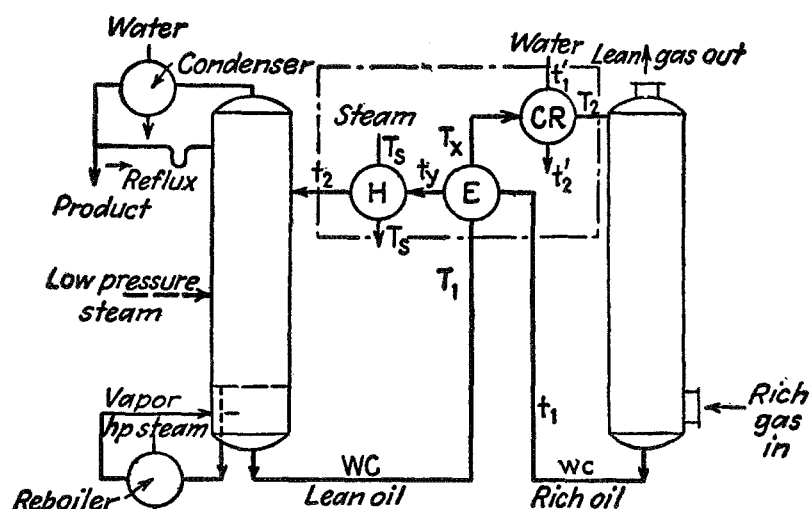


FIG. 11.1. Typical process employing heat recovery.

and the optimum use of exhaust steam discussed in Chap. 7. Often an exchanger operates in series with a cooler and a heater as shown in Fig. 11.1. The cooler regulates the final temperature of the hot fluid, and the heater adjusts the final temperature of the cold fluid¹ to the requirements of the next step in the process. The quantity of heat recovered in the exchanger alone greatly influences its size and cost, since the true temperature difference in the exchanger approaches zero as the heat recovery approaches 100 per cent. On the other hand, whatever

¹ This is actually the way to compensate and take advantage of the overperformance of an exchanger when it is clean if it has been designed for a dirty coefficient U_D . Unless equipped with a by-pass to reduce the flow through the exchanger, the temperature ranges of both fluids will exceed the process conditions. This is offset by throttling the steam and water in the heater and cooler and accordingly reducing the utility costs.

heat is not recovered in the exchanger must be removed or added through the use of additional steam in the heater and additional water in the cooler, which raises the initial costs of the two as well as the operating cost of the process. This arrangement suggests the presence of an optimum distribution of temperatures so that the fixed and the operating charges will combine to give a minimum.

Among a number of examples of heat recovery available in the chemical and power industries, that of Fig. 11.1 is typical. It shows the flow sheet of a simple vapor-recovery system such as is employed in the stripping of gasoline from natural gas, an absorption and distillation process, although the absorber and distilling column are not a part of the present analysis. Only the pertinent heat-flow lines are dash enclosed. The natural gas coming from the earth is laden with gasoline vapors which command a higher price when separated from the natural gas and condensed. The *rich gas* enters the absorber where it contacts an absorbent, usually a nonviscous oil, in which the gasoline vapors selectively dissolve. The outlet gas of reduced gasoline-vapor concentration is *lean gas*. The absorbent leaves the bottom of the absorber with the dissolved vapors as *rich oil*. It must next be fed to a distilling column where the gasoline and the oil are separated by steam distillation. The oil leaving the bottom of the distilling column is substantially free of the solute and therefore *lean oil*. Absorption is favored by low temperatures, while distillation requires higher temperatures so that the gasoline can be vaporized from the oil. The exchanger, heater, and cooler are represented in Fig. 11.1 by E , H , and CR , respectively. The temperatures to and from the absorber and distilling column will be considered fixed by equilibrium conditions whose definitions are beyond the scope of this study. The temperatures of the steam and water will also be fixed. The nucleus of the problem lies in determining the exchanger outlet temperatures T_x or t_y , either of which fixes the other, so that the total annual cost of the three heat-transfer items will be a minimum.

If C_E , C_H , and C_{CR} are the annual fixed charges per square foot for the exchanger, heater, and cooler, and if A_E , A_H , and A_{CR} are their surfaces, the five elements of cost are

1. Fixed charges of exchanger, $C_E A_E$
2. Fixed charges of heater, $C_H A_H$
3. Cost of steam, C_s , dollars/Btu
4. Fixed charges of cooler, $C_{CR} A_{CR}$
5. Cost of water, C_w , dollars/Btu

The surface of each unit, $A = Q/U \Delta t$, is dependent upon its true temperature difference. In each piece of equipment, however, the true temperature difference depends upon either T_x or t_y . To obtain an

expression for the minimum cost it will be necessary to differentiate the total annual cost with respect to either T_x or t_y and solve for T_x or t_y after setting the derivative equal to zero. For simplicity assume that only true counterflow exchangers are employed so that the true temperature difference and LMTD of all units are identical and that all the *unit* fixed charges C_E , C_H , C_{CR} are the same and independent of the total number of square feet of each unit. The equation for the total annual cost C_T is obtained after eliminating t_y through the use of the dimensionless group R .

$$C_T = \frac{C_E WC(T_1 - T_x)}{U_E[(1 - R)T_1 + (R - 1)T_x]} \ln \frac{(1 - R)T_1 - t_1 + RT_x}{T_x - t_1} \\ + \frac{C_H wc}{U_H} \ln \frac{T_s - t_1 - RT_1 + RT_x}{T_s - t_2} + C_s \theta wc(t_2 - t_1 - RT_1 + RT_x) \\ + \frac{C_{CR} WC(T_x - T_2)}{U_{CR}[(T_x - t'_2) - (T_2 - t'_1)]} \ln \frac{T_x - t'_2}{T_2 - t'_1} + C_w \theta WC(T_x - T_2) \quad (11.1)$$

where U_E , U_H , and U_{CR} are the overall coefficients, θ is the total number of annual operating hours, and the other temperatures are indicated in Fig. 11.1. When differentiated with respect to T_x and set equal to zero, the equation for the optimum value of T_x is

$$0 = C_s \theta + C_w \theta + \frac{C_H}{U_H} \frac{1}{[R(T_x - T_1) + (T_s - t_1)]} \\ + \frac{C_{CR}}{U_{CR}} \frac{(t'_1 - t'_2)}{[(T_x - T_2) + (t'_1 - t'_2)]} \ln \frac{(T_x - t'_2)}{(T_2 - t'_1)} \\ + \frac{C_{CR}}{U_{CR}} \frac{(T_x - T_2)}{[(T_x - T_2) + (t'_1 - t'_2)]} \frac{1}{(T_x - t'_2)} \\ - \frac{C_E}{U_E} \frac{(T_1 - t_1)}{[(T_1 - t_1) + R(T_x - T_1)]} \frac{1}{(T_x - T_1)} \quad (11.2)$$

T_x can be obtained by a trial-and-error calculation.

Actually where the disposition of the temperatures justifies obtaining the optimum, the operation will be too large to permit the use of true counterflow in all of the equipment. When the flow pattern in the exchanger alone deviates from true counterflow and Δt is given by Eq. (7.37), the problem is somewhat more difficult to solve and less direct. However, if F_T is placed in the denominator of the last term in Eq. (11.2), a simplified trial-and-error solution can be obtained for a system using a 1-2, 2-4, etc., exchanger. Equation (11.2) is not particularly useful, however, unless extensive data are available on the installed costs and fixed charges of exchangers, since the cost per square foot of surface also varies with the size of the exchanger. Sieder¹ has shown in Fig. 11.2 how the

¹ Sieder, E. N., *Chem. Met. Eng.*, **46**, 322-325 (1939)

relative cost (per square foot) of surface decreases from small to large exchangers. The fixed charges are usually arrived at as a percentage (30 per cent) of the initial cost per square foot, and this varies so much

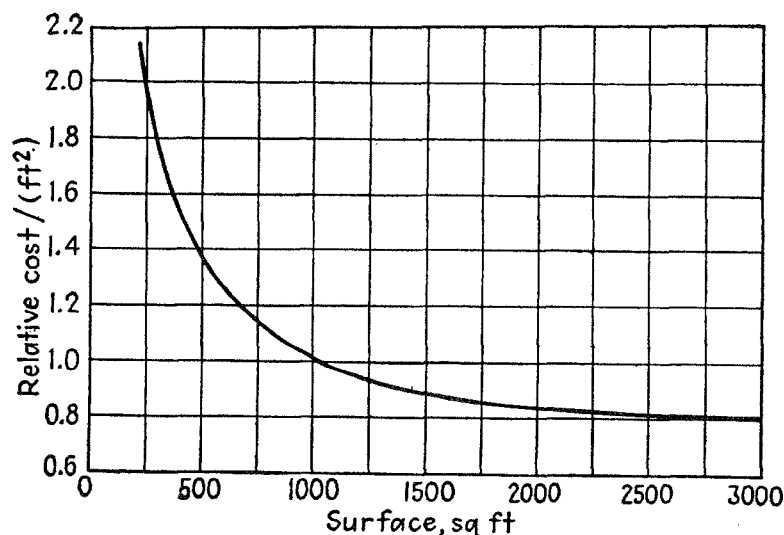


FIG. 11.2. Cost of tubular surface vs. size of exchanger. (Sieder, *Chemical Engineering*.)

with the size of the final exchanger that several successive trials are required to establish the proper range of individual costs.

Other obstacles are also encountered in the solution of a general equation for the optimum. If a typical problem is solved for a system with a 1-2 exchanger and the calculated value of T_x comes out to be less than the

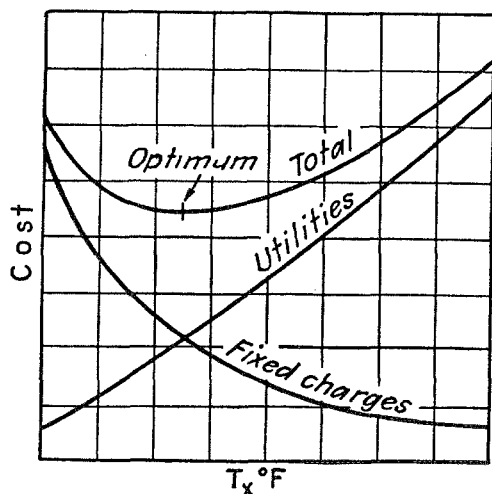


FIG. 11.3. Optimum recovery temperature.

practical limit of $F_r = 0.75 - 0.80$ in the 1-2 exchanger, the entire calculation, though valid, must be repeated using a different exchanger. The task of obtaining the optimum process temperatures can be achieved more readily by graphical means. As the temperature T_x is varied, there are two opposed costs as follows: As T_x is increased, the cost of the utilities increases but the initial cost and fixed charges on the exchanger decrease. By assuming several values of T_x the required sizes of all the equipment can be computed for each assumption. From the overall heat

balances the operating costs of the utilities can also be obtained. The total annual costs are then the sum of the two costs as plotted in Fig. 11.3 with the optimum corresponding to the point indicated.

The Optimum Exchanger. The factors which are favorable for the attainment of higher film coefficients for fluids in exchangers also increase

their pressure drops. For turbulent flow in tubes as given by Eq. (6.2) the film coefficient varies with $G_i^{0.8}$, whereas the pressure drop in Eq. (7.45) varies with G_i^2 . This means that if the tube-side coefficient is the controlling coefficient and if the design of the exchanger can be altered so as to increase the tube-side mass velocity, the overall coefficients U_c , U_D also increase and the size of the exchanger can be reduced accordingly. Offsetting this advantage is an increase in the pressure drop and pumping power costs for the greater mass velocity. As in the calculation of the optimum process conditions, there is an optimum exchanger capable of fulfilling process conditions with a minimum annual cost. The achievement of this design, however, requires an exchanger capable of providing the optimum fluid-flow velocities on the shell as well as the tube sides. This would frequently entail the use of an odd number of tube passes or an odd tube length, which is inconsistent with industrial practices. McAdams¹ has given an excellent résumé of the equations and approximations required for establishing the optimum exchanger to which the reader is referred.

Rating an Exchanger. In Chaps. 7 through 10 the calculations were carried out on existing exchangers. In the application of the Fourier equation to an existing exchanger, Q was determined from the heat balance; A from the number, outside diameter, and length of the tubes; and Δt from $F_T \times \text{LMTD}$, permitting the solution for U_D . From the fluid-flow conditions, h_o , h_{io} , U_c , and the pressure drops were calculated. The criterion of performance R_d was then obtained from U_D and U_c . When there is no available exchanger and only the process conditions are known, calculation may proceed in an orderly way by assuming the existence of an exchanger and testing it as in previous examples for a suitable dirt factor and pressure drops.

To prevent the loss of considerable time, rational methods of assuming an exchanger should be developed. By returning to the component parts of the Fourier equation $Q = U_D A \Delta t$, the heat load Q is seen to be fixed by the process conditions, while Δt is obtained by assuming a fluid-flow pattern. Thus if it is desired to set up a 1-2 exchanger, and if the process temperatures give $F_T > 0.75 - 0.80$, the remaining unknowns are U_D and A . The design or dirty coefficient U_D is in turn related by a reasonable dirt factor to U_c , which reflects the heat-transfer characteristics of the two fluids. Previous examples in the text suggest that different numerical film coefficients may be expected within definite ranges in well-designed exchangers for different classes of fluids. It is also apparent that, except where both coefficients are approximately

¹ McAdams, W. H., "Heat Transmission," 2d ed., pp. 363-367, McGraw-Hill Book Company, Inc., New York, 1942.

equal, the lower film coefficient determines the range of U_c and U_D . In the light of this experience if a trial value of U_D is assumed and then substituted into the Fourier equation to supplement calculated values of Q and Δt , it permits the trial calculation of A . To facilitate the use of reasonable trial values of U_D approximate overall coefficients have been presented in Appendix Table 8 for common pairs of fluids.

When the value of A is combined with tube length and pitch preferences, the tube counts in Appendix Table 9 become a catalogue of all the possible exchanger shells, from which only one will usually best fulfill the process conditions. Having decided which fluid will tentatively flow in the tubes, the trial number of tube passes can be approximated by a consideration of the quantity of fluid flowing in the tubes and the number of tubes corresponding to the trial value of A . The trial mass velocity should fall somewhere between 750,000 and 1,000,000 lb/(hr)(ft²) for fluids with allowable tube-side pressure drops of 10 psi.

If the trial number of tube passes has been assumed incorrectly, a change in the total number of passes changes the total surface which is contained in a given shell, since the number of tubes for a given shell diameter varies with the number of tube-pass partitions. The assumed number of tube passes for the trial surface is satisfactory if it gives a value of h_i above U_D and a pressure drop not exceeding the allowable pressure drop for the fluid. One may next proceed to the shell side by assuming a trial baffle spacing which can be varied, if in error, over a wide range without altering h_i , A , or Δt as computed previously for the tube side. It is always advantageous therefore to compute the tube side first to validate the use of a particular shell.

In calculating an exchanger the best exchanger is the smallest one with a standard layout which just fulfills the dirt-factor and pressure-drop requirements. There are only a few limitations to be considered. It is still assumed that there is no advantage in using less than the allowable pressure drop and that, in accordance with Fig. 28, 25 per cent cut segmental baffles will be employed within the minimum and maximum spacing. The extremes of the spacing range are

$$\text{Maximum spacing, } B = \text{ID of shell, in.} \quad (11.3)$$

$$\text{Minimum spacing, } B = \frac{\text{ID of shell}}{5}, \text{ or 2 in., whichever is larger} \quad (11.4)$$

These limitations stem from the fact that at wider spacings the flow tends to be axial rather than across the bundle and at closer spacings there is excessive leakage between the baffles and the shell. Owing to the convention of placing the inlet and outlet nozzles on opposite sides of the shell, the end baffles may not exactly conform to the chosen spacing

for an even number of baffles and an odd number of crosses. When using a close convenient spacing, if an odd number of baffles is indicated, the pressure-drop and heat-transfer coefficients can be calculated for the chosen spacing, although one baffle will be omitted by respacing the extreme end baffles.

The different combinations of the number of tube passes and the baffle pitch which can be employed in a given shell permit the variation of the mass velocities and film coefficients over broad limits. The number of tube passes can be varied from two to eight and in larger shells to sixteen. As indicated above, the shell-side mass velocity can be altered fivefold between the minimum and maximum baffle spacings. It is desirable to bear in mind the latitude this permits in the event that the first trial baffle spacing and tube passes have been wide of fulfilling the process conditions. In 1-2 exchangers the least performance is obtained with two tube passes and the maximum baffle space. For the tube side in turbulent flow

$$h_i \propto G_t^{0.8}$$

$$\Delta P_t \propto G_t^2 nL$$

where nL is the total length of path. Going to eight tube passes in the same shell inside diameter the changes incurred are

$$\frac{h_{i(8 \text{ passes})}}{h_{i(2 \text{ passes})}} = \left(\frac{8}{2}\right)^{0.8} = \frac{3}{1}$$

but

$$\frac{\Delta P_{t(8 \text{ passes})}}{\Delta P_{t(2 \text{ passes})}} = \frac{8^2 \times 8 \times 1}{2^2 \times 2 \times 1} = \frac{64}{1}$$

or although the heat-transfer coefficient can be increased threefold, the pressure drop must be increased 64 times to accomplish it. For streamline flow the expenditure of the larger amount of pumping energy will increase the tube-side coefficient only by

$$\frac{h_{i(8 \text{ passes})}}{h_{i(2 \text{ passes})}} = \left(\frac{8}{2}\right)^{1/3} = \frac{1.58}{1}$$

provided the fluid is in streamline flow in both cases. The shell side may be represented approximately by

$$h_o \propto G_s^{0.5}$$

$$\Delta P_s \propto G_s^2 (N + 1)$$

where N is the number of baffles and $N + 1$ the number of bundle crosses. The changes for the shell side between minimum and maximum baffle spacings are

$$\frac{h_{o,\min}}{h_{o,\max}} = \left(\frac{5}{1}\right)^{0.5} = \frac{2.23}{1}$$

but

$$\frac{\Delta P_{s,\min}}{\Delta P_{s,\max}} = \frac{5^2 \times 5}{1^2 \times 1} = \frac{125}{1}$$

Offsetting this, however, is the fact that the shell side yields a higher order of film coefficients for the smaller of the two streams if there is a great difference in their weight flow rates.

Through the use of the overall coefficients suggested in Table 8 and a judicious analysis of the summary of the first trial assumptions, it should be possible to obtain the most suitable exchanger on the second trial.

Tube Standards. There are numerous advantages in standardizing the outside diameter, gage, and length of the tubes used in a plant.

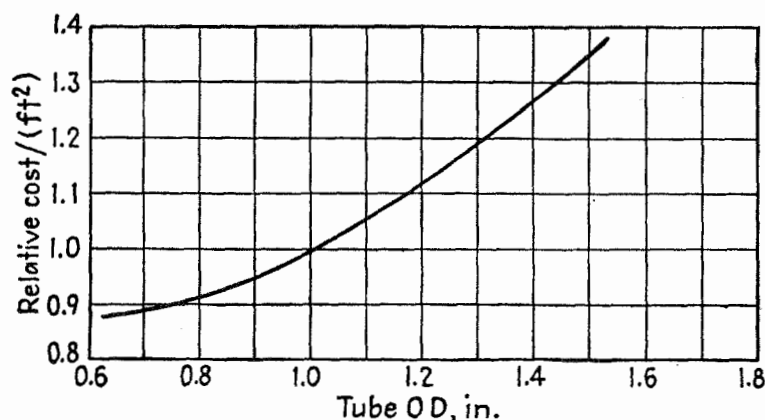


FIG. 11.4. Cost of tubular surface vs. tube outside diameter. (Sieder, *Chemical Engineering*.)

Standardization reduces the number of sizes and lengths which must be carried in storage for the replacement of tubes which develop leaks. It also reduces the number of installing and cleaning tools required for maintenance. Pitch standards have already been discussed in Chap. 7, but the selection of the tube diameter has an economic aspect. Obviously the smaller the diameter of the tube the smaller the shell required for a given surface, the greater the value of h_i , and the smaller the first cost. The nature of the variation in cost, taken from Sieder, is shown in Fig. 11.4. The difference is small between the use of $\frac{3}{4}$ and 1 in. OD tubes. The cost per square foot rises greatly, however, as the diameters increase above 1 in. OD. Similarly the longer the tubes the smaller the shell diameter for a given surface, and from Fig. 11.5 the cost variation between the use of 12-, 16-, and 20-ft tubes is not very great although 8-ft tubes sharply increase the first cost.

The lower cost of surface obtained from the smaller outside diameter and greater length tubes is offset by the fact that maintenance and particularly cleaning are costlier with long tubes of small outside diameter. If the tubes are too small, under $\frac{3}{4}$ in. OD, there are too many to be

cleaned, and there is less facility in handling and cleaning the smaller tubes. If the tubes are too long, it is difficult to remove the bundle and plot space must be allocated not only for the exchanger itself but also for the withdrawal of the bundle. Long tubes are also very difficult to replace, especially where the baffles are closely spaced. It is difficult to obtain comparative maintenance data per square foot as a function of tube diameter or length, since few industrial users appear to employ an assortment of tubes or have kept cost data. It may be significant

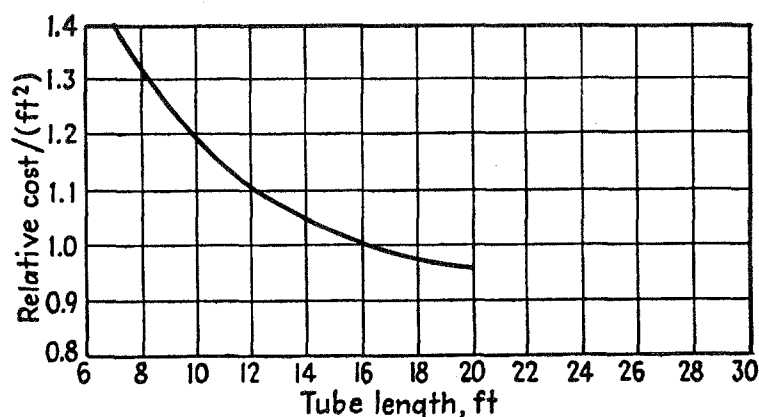


FIG. 11.5. Cost of tubular surface vs. tube length. (Sieder, *Chemical Engineering*.)

that for exchanger work with fluids of ordinary fouling characteristics the $\frac{3}{4}$ and 1 in. OD tubes are employed most frequently. For chemical evaporators, boilers, and fouling services, larger tubes are customary.

The Calculation and Design of an Exchanger. The outline for design follows:

Given:

Process conditions:

Hot fluid: $T_1, T_2, W, C, s, \mu, k, R_d, \Delta P$

Cold fluid: $t_1, t_2, w, c, s, \mu, k, R_d, \Delta P$

The tube length, outside diameter, and pitch will be specified by plant practice or may be determined from the suggestions in Chap. 7.

(1) Heat balance, $Q = WC(T_1 - T_2) = wc(t_2 - t_1)$

(2) True temperature difference, Δt :

LMTD:

$$R = \frac{T_1 - T_2}{t_2 - t_1} \quad S = \frac{t_2 - t_1}{T_1 - t_1} \quad (5.14)$$

F_T from Figs. 18 through 23

$$\Delta t = \text{LMTD} \times F_T \quad (7.42)$$

(3) Caloric temperatures, T_c and t_c (5.28) and (5.29)

Trial 1:

For the exchanger:

(a) Assume a tentative value of U_D with the aid of Table 8, and compute the surface from $A = Q/U_D \Delta t$. It is always better to assume U_D too high than too low, as this practice ensures arriving at the minimum surface. Determine the corresponding number of tubes using Table 10.

(b) Assume a plausible number of tube passes for the pressure drop allowed, and select an exchanger for the nearest number of tubes from the tube counts of Table 9.

(c) Correct the tentative U_D to the surface corresponding to the actual number of tubes which may be contained in the shell.

The performance calculation for the film coefficients should start with the tube side. If the tube-side film coefficient is relatively greater than U_D and the pressure-drop allowance is reasonably fulfilled and not exceeded, the calculation can proceed to the shell side. Whenever the number of tube passes is altered, the surface in the shell is also altered, changing the value of A and U_D . For the remainder of the calculation shown here it is assumed that the cold fluid flows in the tubes as it does in a majority but not necessarily all cases.

<i>Hot fluid: shell side</i>	<i>Cold fluid: tube side</i>
(4') Assume a plausible baffle spacing for the pressure drop allowed. Flow area, $a_s = ID \times C'B/144P_T \text{ ft}^2$ [Eq. (7.1)]	(4) Flow area, a_t : Flow area per tube a_t From Table 10 $a_t = N a'_t/144n, \text{ ft}^2$ [Eq. (7.48)] ID will be obtained from Table 10.
(5') Mass vel, $G_s = W/a_s \text{ lb}/(\text{hr})(\text{ft}^2)$ [Eq. (7.2)]	(5) Mass vel, $G_t = w/a_t \text{ lb}/(\text{hr})(\text{ft}^2)$
(6') $Re_s = D_e G_s/\mu$ [Eq. (7.3)] Obtain D_e from Fig. 28 or compute from Eq (7.4). Obtain μ at T_c	(6) $Re_t = D G_t/\mu$ Obtain D from Table 10. Obtain μ at t_c .
(7') j_H from Fig. 28	(7) j_H from Fig. 24
(8') At T_c obtain $c \text{ Btu}/(\text{lb})(^\circ\text{F})$ and $k \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ $(c\mu/k)^{1/3}$	(8) At t_c obtain $c \text{ Btu}/(\text{lb})(^\circ\text{F})$ and $k \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ $(c\mu/k)^{1/3}$
(9') $h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/3} \phi_s$ [Eq. (6.15)]	(9) $h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/3} \phi_t$ [Eq. (6.15a)]
(10') Tube-wall temp, t_w $t_w = t_c + \frac{h_o}{h_{io} + h_o} (T_c - t_c)$ Eq. [5.31]	(10) $\frac{h_{io}}{\phi_t} = \frac{h_i}{\phi_t} \frac{ID}{OD}$ [Eq. (6.5)]
(11') Obtain μ_w and $\phi_s = (\mu/\mu_w)^{0.14}$ [Fig. 24]	(11) Obtain μ_w and $\phi_t = (\mu/\mu_w)^{0.14}$ [Fig. 24]
(12') Corrected coefficient, $h_o = \frac{h_o \phi_s}{\phi_s}$ [Eq. (6.36)]	(12) Corrected coefficient, $h_{io} = \frac{h_{io} \phi_t}{\phi_t}$ [Eq. (6.37)]
Check pressure drop. If unsatisfactory, assume a new baffle spacing.	Check pressure drop. If unsatisfactory, assume a new pass arrangement.

Pressure Drop

(1') For Re_i in (6') obtain f [Fig. 29] (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $\Delta P_s = \frac{fG_i^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ [Eq. (7.44)]	(1) For Re_i in (6) obtain f [Fig. 26] (2) $\Delta P_t = \frac{fG_i^2 L n}{5.22 \times 10^{10} D_s \phi_i}$ [Eq. (7.45)] (3) $\Delta P_r = \frac{4n}{s} \frac{V^2}{2g'}$ [Eq. (7.46)] (4) $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)]
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If both sides are satisfactory for film coefficients and pressure drop, the trial may be concluded.

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} \quad (6.38)$$

(14) Dirt factor R_d : U_D has been obtained in (c) above.

$$R_d = \frac{U_C - U_D}{U_C U_D} \quad (6.13)$$

The calculation of a number of exchangers for typical sensible-heat-transfer conditions are given in this chapter. Each presents a different aspect of design. Together they should provide the perspective necessary for meeting a variety of applications encountered in modern industry. Since the method of approach involves trial-and-error calculations, the analyses and comments included in each solution should reduce the time required for subsequent calculations.

Example 11.1. Calculation of a Straw Oil-Naphtha Exchanger. 29,800 lb/hr of a 35°API light oil at 340°F is used to preheat 103,000 lb/hr of 48°API naphtha from 200 to 230°F. The viscosity of the oil is 5.0 centipoises at 100°F and 2.3 centipoises at 210°F. The viscosity of the naphtha is 1.3 centipoises at 100°F and 0.54 centipoises at 210°F. Pressure drops of 10 psi are allowable.

Because the oil may tend to deposit residues, allow a combined dirt factor of 0.005 and use square pitch. Plant practice employs $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long wherever possible.

Solution:

(1) Heat balance:

Straw oil, $Q = 29,800 \times 0.58(340 - 240) = 1,730,000$ Btu/hr

Naphtha, $Q = 103,000 \times 0.56(230 - 200) = 1,730,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
340	Higher Temp	230	110
240	Lower Temp	200	40
100	Differences	30	70

$$\text{LMTD} = 69.3^\circ\text{F} \quad (5.14)$$

$$R = \frac{100}{30} = 3.3 \quad S = \frac{30}{340 - 200} = 0.214 \quad F_T = 0.885 \quad (\text{Fig. 18})$$

A 1-2 exchanger will be satisfactory.

$$\Delta t = 0.885 \times 69.3 = 61.4^\circ\text{F} \quad (7.42)$$

(3) T_c and t_c :

$$\frac{\Delta t_c}{\Delta t_h} = 0.364 \quad K_c = 0.23 \text{ (straw oil controlling)} \quad (\text{Fig. 17})$$

$$F_c = 0.405$$

$$T_c = 240 + 0.405 \times 100 = 280.5^\circ\text{F} \quad (5.28)$$

$$t_c = 200 + 0.405 \times 30 = 212^\circ\text{F} \quad (5.29)$$

Trial:

(a) Assume $U_D = 70$: From Table 8 a value of U_D between 60 and 75 should be the maximum expected. It is always better to assume U_D too high than too low so that the final exchanger will just fulfill the requirements. Place the small stream in the shell.

$$A = \frac{Q}{U_D \Delta t} = \frac{1,730,000}{70 \times 61.4} = 403 \text{ ft}^2$$

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft} \quad (\text{Table 10})$$

$$\text{Number of tubes} = \frac{403}{16'0'' \times 0.1963} = 129 \quad (\text{Table 10})$$

(b) Assume two tube passes: The quantity of tube-side fluid is very large for the small heat load and moderately large Δt and will otherwise cause difficulty in meeting the allowable ΔP if too many tube passes are employed.

From the tube counts (Table 9): 129 tubes, two passes, $\frac{3}{4}$ in. OD on 1-in. square pitch

Nearest count: 124 tubes in a $15\frac{1}{4}$ in. ID shell

(c) Corrected coefficient U_D :

$$A = 124 \times 16'0'' \times 0.1963 = 390 \text{ ft}^2 \quad (\text{Table 10})$$

$$U_D = \frac{Q}{A \Delta t} = \frac{1,730,000}{390 \times 61.4} = 72.3$$

Hot fluid: shell side, straw oil

(4') Flow area, a_s : Since the minimum baffle space will provide the greatest value of h_o , assume

$$B = \text{ID}/5 = 15.25/5, \text{ say } 3.5 \text{ in.}$$

[Eq. (11.4)]

$$a_s = \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= 15.25 \times 0.25 \times 3.5/144 \times 1$$

$$= 0.0927 \text{ ft}^2$$

(5') Mass vel, $G_s = W/a_s$ [Eq. (7.2)]

$$= 29,800/0.0927 = 321,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

(6') At $T_c = 280.5^\circ\text{F}$,

$$\mu = 1.5 \times 2.42 = 3.63 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$D_e = 0.95/12 = 0.0792 \text{ ft}$$

[Fig. 28]

$$Re_s = D_e G_s / \mu$$

$$= 0.0792 \times 321,000/3.63 = 7000$$

Cold fluid: tube side, naphtha

(4) Flow area, $a'_t = 0.302 \text{ in.}^2$

[Table 10]

$$a_t = N_t a'_t / 144n$$

[Eq. (7.48)]

$$= 124 \times 0.302/144 \times 2 = 0.130 \text{ ft}^2$$

(5) Mass vel, $G_t = w/a_t$

$$= 103,000/0.130 = 793,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

(6) At $t_c = 212^\circ\text{F}$,

$$\mu = 0.54 \times 2.42 = 1.31 \text{ lb}/(\text{ft})(\text{hr})$$

$$D = 0.62/12 = 0.0517 \text{ ft}$$

[Table 10]

$$Re_t = D G_t / \mu$$

$$= 0.0517 \times 793,000/1.31 = 31,300$$

Hot fluid: shell side, straw oil

(7') $j_H = 46$

(8') For $\mu = 1.5$ cp and 35°API

[Fig. 16]

$k(c\mu/k)^{1/4} = 0.224 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$

(9') $h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k}\right)^{1/4} \phi_s$ [Eq. (6.15b)]

$\frac{h_o}{\phi_s} = \frac{46 \times 0.224}{0.0792} = 130$

(10'), (11'), (12') Omit the viscosity correction for the trial or $\phi_s = 1.0$.

$h_o = \frac{h_o}{\phi_s} = 130 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Cold fluid: tube side, naphtha

(7) $j_H = 102$ [Fig. 24]

(8) For $\mu = 0.54$ cp and 48°API

[Fig. 16]

$k(c\mu/k)^{1/4} = 0.167 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$

(9) $h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k}\right)^{1/4} \phi_t$ [Eq. (6.15a)]

$\frac{h_i}{\phi_t} = 102 \times 0.167/0.0517 = 329$

(10) $\frac{h_{io}}{\phi_t} = \frac{h_i}{\phi_t} \times \frac{\text{ID}}{\text{OD}} = 329 \times \frac{0.62}{0.75} = 272$ [Eq. (6.5)]

(11), (12) Omit the viscosity correction for the trial or $\phi_t = 1.0$.

$h_{io} = \frac{h_{io}}{\phi_t} = 272 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Proceed with the pressure drop calculation.

Pressure Drop

(1') For $Re_s = 7000$,

$f = 0.00225 \text{ ft}^2/\text{in.}^2$

[Fig. 29]

$s = 0.76$

[Fig. 6]

(2') No. of crosses, $N + 1 = 12L/B$

[Eq. (7.43)]

$= 12 \times 16/3.5 = 55$

$s = 0.76$

[Fig. 6]

$D_s = 15.25/12 = 1.27 \text{ ft}$

(3') $\Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ [Eq. (7.44)]

$0.00225 \times 321,000^2 \times 1.27$

$\times 55$

$= \frac{5.22 \times 10^{10} \times 0.0792 \times 0.76}{\times 1.0}$

$\times 1.0$

$= 5.2 \text{ psi}$

(1) For $Re_t = 31,300$,

$f = 0.0002 \text{ ft}^2/\text{in.}^2$

[Fig. 26]

$s = 0.72$

[Fig. 6]

(2) $\Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s s \phi_t}$ [Eq. (7.45)]

$= \frac{0.00020 \times 792,000^2 \times 16 \times 2}{5.22 \times 10^{10} \times 0.0517 \times 0.72 \times 1.0}$

$= 2.1 \text{ psi}$

(3) and (4) may be omitted for the trial. Now proceed to the shell side.

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{272 \times 130}{272 + 130} = 88.2 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) is 72.3.

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{88.2 - 72.3}{88.2 \times 72.3} = 0.0025 \text{ (hr)(ft}^2)(^\circ\text{F)}/\text{Btu} \quad (6.13)$$

Summary

130	h outside	272
U_C	88.2	
U_D	72.3	
R_d Calculated 0.0025		
R_d Required 0.0050		
5.2	Calculated ΔP	2.1
10.0	Allowable ΔP	10.0

Discussion. The first trial is disqualified because of failure to meet the required dirt factor. What conclusions may be drawn so that the next trial will produce the satisfactory exchanger? Could any advantage be gained by reversing the streams? Obviously the film coefficient for the distillate, which is controlling, would drop considerably if the streams were reversed. Could four passes be used for the tubes? Doubling the number of tube passes would approximately double the mass velocity and give *eight* times the tube-side pressure drop thereby exceeding the allowable ΔP . All the assumptions above have been reasonable. The exchanger is simply a little too small, or in other words, the value assumed for U_D must be reduced. It will be necessary to proceed anew.

Trial 2: Assume $U_D = 60$, two tube passes and the minimum shell baffle space. Proceeding as above and carrying the viscosity correction and pressure drops to completion, the new summary is given using a $17\frac{1}{4}$ in. ID shell with 166 tubes on two passes and a 3.5-in. baffle space.

Summary

115.5	h outside	213
U_C	74.8	
U_D	54.2	
R_d Calculated 0.005		
R_d Required 0.005		
4.7	Calculated ΔP	2.1
10.0	Allowable ΔP	10.0

The final exchanger will be

<i>Shell side</i>	<i>Tube side</i>
ID = $17\frac{1}{4}$ in.	Number and length = 166, 16'0"
Baffle space = 3.5 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1-in. square
Passes = 1	Passes = 2

Example 11.2. Calculation of a Lean-oil-Rich-oil Exchanger. 84,348 lb/hr of a 35°API lean absorption oil in a process identical with Fig. 11.1 leaves a stripping column to transfer its heat to 86,357 lb/hr of rich oil leaving the absorber at 100°F with a gravity of closely 36°API at 60°F. The range for the lean oil will be from 350 to 160°F, and the outlet temperature of the rich oil will be 295°F. The viscosity of the oil is 2.6 centipoises at 100°F and 1.15 centipoises at 210°F. Pressure drops of 10 psi are available, and in accordance with Table 12, a combined dirt factor of 0.004 should be allowed.

Plant practice again employs $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long and laid out on square pitch.

Solution:

(1) Heat balance:

Lean oil, $Q = 84,438 \times 0.56(350 - 160) = 8,950,000$ Btu/hr

Rich oil, $Q = 86,357 \times 0.53(295 - 100) = 8,950,000$ Btu/hr

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
350	Higher Temp	295	55
160	Lower Temp	100	60
190	Differences	195	5

$$\text{LMTD} = 57.5^\circ\text{F} \quad (5.14)$$

$$R = \frac{190}{195} = 0.975 \quad S = \frac{195}{350 - 100} = 0.78 \quad (7.18)$$

1-2 exchanger, $F_T = \text{inoperable}$ (Fig. 18)

2-4 exchanger, $F_T = \text{inoperable}$ (Fig. 19)

3-6 exchanger, $F_T = 0.725$ (Fig. 20)

4-8 exchanger, $F_T = 0.875$ (Fig. 21)

A 4-8 exchanger arrangement will be required. This may be met by four 1-2 exchangers in series or two 2-4 exchangers in series. The latter will be used.

$$\Delta t = 0.875 \times 57.5 = 50.3^\circ\text{F} \quad (7.42)$$

(3) T_c and t_c :

$$\frac{\Delta t_c}{\Delta t_h} = 1.09$$

$$K_c = 0.32 \quad (Fig. 17)$$

$$F_c = 0.48$$

$$T_c = 160 + 0.48 \times 190 = 251^\circ\text{F} \quad (5.28)$$

$$t_c = 100 + 0.48 \times 195 = 193.5^\circ\text{F} \quad (5.29)$$

Trial:

(a) Assume $U_D = 50$: Although both oils and quantities are almost identical, the temperature range of the cold rich oil and correspondingly greater viscosity will make the rich oil controlling. For this reason, allowable pressure drops being equal, the cold fluid should be placed in the shell. The coefficients will be lower than those of Table 8, since the pressure drops are more difficult to meet in a 4-8 exchanger and mass velocities must accordingly be kept down. In Example 11.1 with a similar controlling oil the value of U_C was approximately 75 with a minimum baffle spacing which did not completely take advantage of the allowable pressure drop. The

assumption of $U_D = 50$ is a compromise between medium and heavy organics and will probably be high but will help to establish the correct unit on the following trial.

$$A = \frac{Q}{U_D \Delta t} = \frac{8,950,000}{50 \times 50.3} = 3560 \text{ ft}^2$$

Use two 2-4 exchangers in series with removable longitudinal baffles.

Number of tubes per shell, $N_t = 3560/2 \times 16'0'' \times 0.1963 = 567$ (Table 10)

(b) Assume six tube passes: From previous problems a mass velocity of 700,000 gave satisfactory tube-side pressure drops. Since the number of tube passes in two units will be greater, a maximum of about 450,000 should be employed.

Six passes (flow area/tube, $a'_t = 0.302 \text{ in.}^2$)

$$G = w/a_t = w/144n/N_t a'_t = 84,438 \times 144 \times 6/567 \times 0.302 = 426,000 \text{ lb/(hr)(ft}^2\text{)}$$

From the tube counts (Table 9): 567 tubes, six passes, $\frac{3}{4}$ in. OD on 1-in. square pitch

Nearest count: 580 tubes in a 31 in. ID shell

(c) Corrected coefficient U_D :

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft}$$

(Table 10)

$$A = 2 \times 580 \times 16'0'' \times 0.1963 = 3640 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{8,950,000}{3640 \times 50.3} = 49.0$$

Hot fluid: tube side, lean oil

(4) Flow area, a_t :

$$a'_t = 0.302 \text{ in.}^2 \quad [\text{Table 10}]$$

$$a_t = N_t a'_t / 144n \quad [\text{Eq. (7.48)}]$$

$$= 580 \times 0.302 / 144 \times 6 = 0.203 \text{ ft}^2$$

(5) Mass vel, $G_t = W/a_t$

$$= 84,438 / 0.203 = 416,000 \text{ lb/(hr)(ft}^2\text{)}$$

(6) At $T_c = 251^\circ\text{F}$, $\mu = 0.88 \times 2.42$

$$= 2.13 \text{ lb/(ft)(hr)} \quad [\text{Fig. 14}]$$

$$D = 0.62/12 = 0.0517 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = DG_t/\mu$$

$$= 0.0517 \times 416,000 / 2.13 = 10,100$$

(7) $j_H = 36.5$ [Fig. 24]

(8) For $\mu = 0.88 \text{ cp}$ and 35°API

$$k(c\mu/k)^{1/3} = 0.185 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft)}$$

[Fig. 16]

$$(9) h_i = j_H \frac{k}{D} \left(\frac{c\mu}{k} \right)^{1/3} \phi_t \quad [\text{Eq. (6.15a)}]$$

$$\frac{h_i}{\phi_t} = 36.5 \times \frac{0.185}{0.0517} = 130$$

$$(10) h_{io} = h_i \times \frac{\text{ID}}{\text{OD}} \quad [\text{Eq. (6.5)}]$$

$$= 130 \times 0.62/0.75 = 107$$

(11) (12) Omit the viscosity correction for the trial, $\phi_t = 1.0$

$$h_{io} = \frac{h_{io}}{\phi_t} = 107 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)}$$

Cold fluid: shell side, rich oil

(4') Flow area, a_s : Since the quantity of fluid is large, any baffle spacing may be arbitrarily assumed.

Assume $B = 12 \text{ in.}$

$$a_s = \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= \frac{1}{2}(31 \times 0.25 \times 12/144 \times 1.0)$$

$$= 0.323 \text{ ft}^2 \text{ (2-4 exchanger)}$$

(5') Mass vel, $G_s = w/a_s$ [Eq. (7.2)]

$$= 86,357/0.323 = 267,000 \text{ lb/(hr)(ft}^2\text{)}$$

(6') At $t_c = 193.5^\circ\text{F}$,

$$\mu = 1.30 \times 2.42 = 3.15 \text{ lb/(ft)(hr)}$$

[Fig. 14]

$$D_e = 0.95/12 = 0.0792 \text{ ft} \quad [\text{Fig. 28}]$$

$$Re_s = D_e G_s / \mu$$

[Eq. (7.3)]

$$= 0.0792 \times 267,000 / 3.15 = 6,720$$

(7') $j_H = 45$

[Fig. 28]

(8') For $\mu = 1.30 \text{ cp}$ and 35°API

$$k(c\mu/k)^{1/3} = 0.213 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F/ft)}$$

[Fig. 16]

$$(9') h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \phi_s \quad [\text{Eq. (6.15)}]$$

$$\frac{h_o}{\phi_s} = 45 \times \frac{0.213}{0.0792} = 121$$

(10') Omit the viscosity correction for the trial, $\phi_s = 1.0$.

$$h_o = \frac{h_o}{\phi_s} = 121 \text{ Btu/(hr)(ft}^2\text{)}(^{\circ}\text{F)}$$

Pressure Drop

(1) For $Re_t = 10,100$,

$f = 0.00027 \text{ ft}^2/\text{in.}^2$

[Fig. 26]

$s = 0.77$

[Fig. 6]

$$(2) \Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00027 \times 416,000^2 \times 16 \times 6 \times 2}{5.22 \times 10^{10} \times 0.0517 \times 0.77 \times 1.0} = 4.3 \text{ psi}$$

(3) $G_t = 416,000$, $V^2/2g' = 0.024$

[Fig. 27]

$$\Delta P_r = \frac{4n}{s} \frac{V^2}{2g'} = \frac{4 \times 2 \times 6}{0.77} \times 0.024$$

$$= 1.5 \text{ psi} \quad [\text{Eq. (7.46)}]$$

$\Delta P_T = \Delta P_t + \Delta P_R = 4.3 + 1.5 = 5.8 \text{ psi}$

[Eq. (7.47)]

The pressure drop suggests the possibility of using eight passes but a rapid check shows the pressure drop for eight passes would exceed 10 psi.

Now proceed to the shell side.

(1') For $Re_s = 6720$,

$f = 0.0023 \text{ ft}^2/\text{in.}^2$

[Fig. 29]

$s = 0.79$

[Fig. 6]

(2') No. of crosses, $N + 1 = 12L/B$

[Eq. (7.42)]

$= 2 \times 2 \times 12 \times 16/12 = 64$

$D_s = 31/12 = 2.58 \text{ ft}$

$$(3') \Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s \phi_s} \quad [\text{Eq. (7.44)}]$$

$$= \frac{0.0023 \times 267,000^2 \times 2.58 \times 64}{5.22 \times 10^{10} \times 0.0792 \times 0.79 \times 1.0} = 8.3 \text{ psi}$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{107 \times 121}{107 + 121} = 56.8 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) is 49.0

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{56.8 - 49.0}{56.8 \times 49.0} = 0.0028 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

107	h outside	121
U_C 56.8		
U_D 49.0		
R_d Calculated 0.0028		
R_d Required 0.0040		
5.8	Calculated ΔP	8.3
10.0	Allowable ΔP	10.0

Discussion. The initial assumptions have provided an exchanger which very nearly meets all the requirements. Eight-pass units would meet the heat-transfer requirement but would give a tube-side pressure drop of 14 psi. The trial exchanger will

be somewhat less suitable when the value of ϕ_t is also taken into account. If the minimum dirt factor of 0.0040 is to be taken literally, it will be necessary to try the next size shell.

Trial 2:

Assume a 33 in. ID shell with six¹ tube passes and baffles spaced 12-in. apart, since the pressure drop increases with the diameter of the shell for a given mass velocity. The summary for the conditions are

Summary

94	h outside	118
U_C	52.3	
U_D	42.0	
R_d Calculated 0.0047		
R_d Required 0.004		
4.4	Calculated ΔP	7.9
10.0	Allowable ΔP	10.0

The two final exchangers in series will be

<i>Shell side</i>	<i>Tube side</i>
ID = 33 in.	Number and length = 676, 16'0"
Baffle space = 12 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1-in. square
Passes = 2	Passes = 6

Example 11.3. Calculation of a Caustic Solution Cooler. 100,000 lb/hr of 15°Bé caustic solution (11 per cent sodium hydroxide, $s = 1.115$) leaves a dissolver at 190°F and is to be cooled to 120°F using water at 80°F. Use a combined dirt factor of 0.002 and pressure drops of 10 psi.

The viscosity of the 11 per cent sodium hydroxide may be approximated by the methods of Chap. 7, but is sirupy, and actual data should be used if possible. The viscosity at 100°F is 1.4 centipoises and at 210°F is 0.43 centipoises. For the specific heat assume the dry salt to have a value of 0.25 Btu/lb, giving a specific heat for the solution at the mean of 0.88.

Plant practice permits the use of triangular pitch with 1 in. OD tubes for solutions in which the scale may be boiled out.

Solution:

(1) Heat balance:

$$\text{Caustic, } Q = 100,000 \times 0.88(190 - 120) = 6,160,000 \text{ Btu/hr}$$

$$\text{Water, } Q = 154,000 \times 1(120 - 80) = 6,160,000 \text{ Btu/hr}$$

¹ An eight-pass exchanger would give a pressure drop of 10.8 psi.

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
190	Higher Temp	120	70
120	Lower Temp	80	40
70	Differences	40	30

$$\text{LMTD} = 53.3^\circ\text{F}$$

Eq. (5.14)

$$R = \frac{70}{40} = 1.75 \quad S = \frac{40}{190 - 80} = 0.364$$

$$F_T = 0.815$$

(Fig. 18)

A 1-2 exchanger will be satisfactory.

$$\Delta t = 0.815 \times 53.3 = 43.5^\circ\text{F}$$

Eq. (7.42)

(3) T_c and t_c : The average temperatures T_a and t_a will be satisfactory because of the closeness of the ranges and the low viscosities.

Trial:

(a) Assume $U_D = 250$. From Table 8 this value is about the minimum for a 0.001 dirt factor and should be suitable for a trial when the required dirt factor is 0.0020.

$$A = \frac{6,160,000}{250 \times 43.5} = 567 \text{ ft}^2$$

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft}$$

(Table 10)

$$\text{Number of tubes, } N_t = \frac{567}{16'0'' \times 0.2618} = 136$$

(b) Assume four tube passes: For two tube passes $a_t = 0.258$ and $G_t = 598,000$ corresponding to a water velocity of only 2.65 fps.

From the tube counts (Table 9): 136 tubes, 4 passes, 1 in. OD on $1\frac{1}{4}$ -in. triangular pitch

Nearest count: 140 tubes in a $19\frac{1}{4}$ " ID shell

(c) Corrected coefficient U_D :

$$A = 140 \times 16'0'' \times 0.2618 = 586 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{6,160,000}{586 \times 43.5} = 242$$

Hot fluid: shell side, caustic

(4') From the previous problems a mass velocity of about 500,000 gave a reasonable pressure drop. By trial this corresponds to about a 7-in. baffle spacing.

$$\begin{aligned} a_s &= \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}] \\ &= 19.25 \times 0.25 \times 7/144 \times 1.25 \\ &= 0.1875 \text{ ft}^2 \end{aligned}$$

Cold fluid: tube side, water

$$\begin{aligned} (4) \text{ Flow area, } a'_t &= 0.546 \text{ in.}^2 \quad [\text{Table 10}] \\ a_t &= N_t a'_t / 144n \quad [\text{Eq. (7.48)}] \\ &= 140 \times 0.546 / 144 \times 4 = 0.133 \text{ ft}^2 \end{aligned}$$

Hot fluid: shell side, caustic
 (5') $G_s = W/a_s$ [Eq. (7.2)]
 $= 100,000/0.1875$
 $= 533,000 \text{ lb}/(\text{hr})(\text{ft}^2)$

(6') At $T_a = 155^\circ\text{F}$,
 $\mu = 0.76 \times 2.42 = 1.84 \text{ lb}/(\text{ft})(\text{hr})$
 $D_s = 0.72/12 = 0.06 \text{ ft}$ [Fig. 28]
 $Re_s = D_s G_s / \mu$ [Eq. (7.3)]
 $= 0.06 \times 533,000 / 1.84 = 17,400$

(7') $j_H = 75$ [Fig. 28]
 (8') At 155°F , $k = 0.9$ (k_{water}) [Table 4]
 $= 0.9 \times 0.38$

$= 0.342 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$
 $(c\mu/k)^{1/3} = (0.88 \times 1.84 / 0.342)^{1/3} = 1.68$

(9') $h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k}\right)^{1/3} \phi_s$ [Eq. (6.15b)]

$\frac{h_o}{\phi_s} = 75 \times 0.342 \times 1.68 / 0.06 = 717$

(10') (11') (12') $\phi_s = 1$ (low viscosity)

$h_o = \frac{h_o}{\phi_s} = 717 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Cold fluid: tube side, water
 (5) $G_t = w/a_t$
 $= 154,000/0.133$
 $= 1,160,000 \text{ lb}/(\text{hr})(\text{ft}^2)$

Vel, $V = G_t / 3600\rho$
 $= 1,160,000 / 3600 \times 62.5 = 5.16 \text{ fps}$

(6) At $t_a = 100^\circ\text{F}$,
 $\mu = 0.72 \times 2.42 = 1.74 \text{ lb}/(\text{ft})(\text{hr})$ [Fig. 14]

$D = 0.834/12 = 0.0695 \text{ ft}$ [Table 10]
 (Re_t is for pressure drop only)

$Re_t = DG_t / \mu$
 $= 0.0695 \times 1,160,000 / 1.74 = 46,300$

(9) $h_i = 1240 \times 0.94 = 1165$ [Fig. 25]

(10) $h_{io} = h_i \times \text{ID}/\text{OD}$ [Eq. (6.5)]
 $= 1165 \times 0.834/1.0$
 $= 972 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Pressure Drop

(1') For $Re_s = 17,400$,
 $f = 0.0019 \text{ ft}^2/\text{in.}^2$ [Fig. 29]

(2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)]

$= 12 \times 16/7 = 28$
 $D_s = 19.25/12 = 1.60 \text{ ft}$

(3') $\Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s \phi_s}$ [Eq. (7.44)]
 $= \frac{0.0019 \times 533,000^2 \times 1.60 \times 28}{5.22 \times 10^{10} \times 0.06 \times 1.115 \times 1.0}$
 $= 7.0 \text{ psi}$

(1) For $Re_t = 46,300$,
 $f = 0.00018 \text{ ft}^2/\text{in.}^2$ [Fig. 26]

(2) $\Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)]
 $= \frac{0.00018 \times 1,160,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0695 \times 1.0 \times 1.0}$
 $= 4.3 \text{ psi}$ [Eq. (7.46)]

(3) $\Delta P_r = \frac{4n}{s} \frac{V^2}{2g}$ [Fig. 27]
 $= \frac{4 \times 4}{1} \times 0.18 = 2.9 \text{ psi}$

(4) $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)]
 $= 4.3 + 2.9 = 7.2 \text{ psi}$

Now proceed to the shell side.

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{972 \times 717}{972 + 717} = 413 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) is 242.

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{413 - 242}{413 \times 242} = 0.0017 \text{ (hr)}(\text{ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Summary

717	h outside	972
U_C	413	
U_D	242	
R_d Calculated 0.0017		
R_d Required 0.0020		
7.0	Calculated ΔP	7.2
10.0	Allowable ΔP	10.0

Discussion. Adjustment of the baffle space to use the full 10 psi will still not permit the exchanger to make the 0.002 dirt factor. The value of U_D has been assumed too high. Try the next size shell.

Trial 2:

Try a 21¼ in. ID shell with four tube passes and a 6-in. baffle space. This corresponds to 170 tubes.

Summary

720	h outside	840
U_C	390	
U_D	200	
R_d Calculated 0.0024		
R_d Required 0.002		
9.8	Calculated ΔP	4.9
10.0	Allowable ΔP	10.0

The use of six tube passes exceeds the allowable tube side pressure drop. The final exchanger will be

<i>Shell side</i>	<i>Tube side</i>
ID = 21¼ in.	Number and length = 170, 16'0"
Baffle space = 6 in.	OD, BWG, pitch = 1 in., 14 BWG, 1¼-in. tri.
Passes = 1	Passes = 4

Example 11.4. Calculation of an Alcohol Heater. 115,000 lb/hr of absolute alcohol (100 per cent ethyl alcohol, $s = 0.78$) is to be heated under pressure from a stor-

age temperature of 80 to 200°F using steam at 225°F. A dirt factor of 0.002 is required along with an allowable alcohol pressure drop of 10 psi.

Plant practice is established using 1-in. OD tubes, 14 BWG, 12'0" long. Triangular pitch is satisfactory for clean services.

Solution:

(1) Heat balance:

$$\text{Alcohol, } Q = 115,000 \times 0.72(200 - 80) = 9,950,000 \text{ Btu/hr}$$

$$\text{Steam, } Q = 10,350 \times 962 = 9,950,000 \text{ Btu/hr}$$

(2) Δt : LMTD (true counterflow):

Hot Fluid		Cold Fluid	Diff.
225	Higher Temp	200	25
225	Lower Temp	80	145
0	Differences	120	120

$$\text{LMTD} = 68.3^\circ\text{F}$$

(5.14)

(3) T_c and t_c : Use T_a and t_a because of the low alcohol viscosity.

Trial.

(a) Assume $U_D = 200$: From Table 8 values of U_D from 200 to 700 may be expected when a dirt factor of 0.001 is employed. Since the dirt factor required is 0.002, the very maximum value of U_D would be 500 corresponding to the dirt alone.

$$A = \frac{Q}{U_D \Delta t} = \frac{9,950,000}{200 \times 68.3} = 728 \text{ ft}^2$$

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft}$$

(Table 10)

$$\text{Number of tubes, } N_t = \frac{728}{12'0'' \times 0.2618} = 232$$

(b) Assume two tube passes: Only one or two passes are required for steam heaters. From the tube counts (Table 9): 232 tubes, two passes, 1 in. OD on 1¼-in. triangular pitch

Nearest count: 232 tubes in a 23¼ in. ID shell

(c) Corrected coefficient U_D :

$$A = 232 \times 12'0'' \times 0.2618 = 728 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{9,950,000}{728 \times 68.3} = 200$$

Hot fluid: tube side, steam

$$\begin{aligned} (4) \text{ Flow area, } a'_t &= 0.546 \text{ in.}^2 \text{ [Table 10]} \\ a_t &= N_t a'_t / 144n \quad \text{[Eq. (7.48)]} \\ &= 232 \times 0.546 / 144 \times 2 = 0.44 \text{ ft}^2 \end{aligned}$$

Cold fluid: shell side, alcohol

(4') To obtain a mass velocity between 400,000 and 500,000 use a 7-in. baffle space.

$$\begin{aligned} a_s &= ID \times C'B / 144P_T \quad \text{[Eq. (7.1)]} \\ &= 23.25 \times 0.25 \times 7 / 144 \times 1.25 \text{ ft}^2 \\ &= 0.226 \end{aligned}$$

Hot fluid: tube side, steam	Cold fluid: shell side, alcohol
(5) $G_t = W/a_t$ $= 10,350/0.44$ $= 23,500 \text{ lb}/(\text{hr})(\text{ft}^2)$	(5') $G_s = w/a_s$ [Eq. (7.2)] $= 115,000/0.226$ $= 508,000 \text{ lb}/(\text{hr})(\text{ft}^2)$
(6) At 225°F , $\mu = 0.013 \times 2.42 = 0.0314 \text{ lb}/(\text{ft})(\text{hr})$ [Fig. 15]	(6') At $t_a = 140^\circ\text{F}$, $\mu = 0.60 \times 2.42 = 1.45 \text{ lb}/(\text{ft})(\text{hr})$ [Fig. 14]
$D = 0.834/12 = 0.0695 \text{ ft}$	$D_e = 0.72/12 = 0.06 \text{ ft}$ [Fig. 28]
$Re_t = DG_t/\mu$ (for pressure drop only) $= 0.0695 \times 23,500/0.0314$ $= 52,000$	$Re_s = D_e G_s/\mu$ [Eq. (7.3)] $= 0.06 \times 508,000/1.45 = 21,000$
	(7') $j_H = 83$
	(8') At 140°F , $k = 0.085 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ [Table 4] $(c\mu/k)^{1/3} = (0.72 \times 1.45/0.085)^{1/3} = 2.31$
(9) $h_{io} = 1500 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$	(9') $h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/3} \phi_s$ [Eq. (6.15b)] $\frac{h_o}{\phi_s} = 83 \times 0.085 \times 2.31/0.06 = 270$
	(10') $\phi_s = 1.0$, $h_o = \frac{h_o}{\phi_s} = 270 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Pressure Drop

(1) For $Re_t = 52,000$, $f = 0.000175 \text{ ft}^2/\text{in.}^2$ [Fig. 26] From Table 7 the specific volume is approximately $21 \text{ ft}^3/\text{lb}$. $\rho = 1/21 = 0.0477 \text{ lb}/\text{ft}^3$ $s = 0.0477/62.5 = 0.00076$	(1') For $Re_s = 21,000$, $f = 0.0018 \text{ ft}^2/\text{in.}^2$ [Fig. 29] $s = 0.78$
(2) $\Delta P_t = \frac{1}{2} \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. 7.53] $= \frac{1}{2} \times \frac{0.000175 \times 23,500^2 \times 12 \times 2}{5.22 \times 10^{10} \times 0.0695 \times 0.00076 \times 1.0}$ $= 0.42 \text{ psi}$	(2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 12 \times 12/7 = 21$ $D_s = 23.25/12 = 1.94 \text{ ft}$
(3) ΔP_r : Negligible because of partial condensation at end of first pass.	(3') $\Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_e s \phi_s}$ [Eq. (7.44)] $= \frac{0.0018 \times 508,000^2 \times 1.94 \times 21}{5.22 \times 10^{10} \times 0.06 \times 0.78 \times 1.0}$ $= 7.8 \text{ psi}$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{1500 \times 270}{1500 + 270} = 229 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) is 200.

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{229 - 200}{229 \times 200} = 0.000633 \text{ (hr)(ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Summary

1500	h outside	270
U_c	229	
U_D	200	
R_d Calculated 0.000633		
R_d Required 0.002		
0.42	Calculated P	7.8
Neg.	Allowable P	10.0

Discussion. This is clearly an instance in which U_D was assumed too high. It is now a question of how much too high. With the aid of the summary it is apparent that in a larger shell a clean overall coefficient of about 200 may be expected. To permit a dirt factor of 0.002 the new U_D should be

$$\frac{1}{U_D} = \frac{1}{U_c} + R_d = \frac{1}{200} + 0.002$$

$$U_D = 143$$

Trial 2:

$$A = \frac{Q}{U_D \Delta t} = \frac{9,950,000}{143 \times 68.3} = 1020 \text{ ft}^2$$

$$\text{No. of tubes} = \frac{1020}{12'0'' \times 0.2618} = 325$$

Nearest count: 334 tubes in a 27 in. ID shell (Table 9).

The same baffle pitch should be retained, since the pressure drop increases with the inside diameter.

Summary

1500	h outside	250
U_C	214	
U_D	138.5	
R_a Calculated 0.0025		
R_a Required 0.002		
0.23	Calculated ΔP	7.1
Neg.	Allowable ΔP	10.0

If a 25-in. exchanger had been used, the dirt factor would be less than 0.002 and a 6-in. baffle space would give a pressure drop exceeding 10 psi. The final exchanger is

Shell side	Tube side
ID = 27 in.	Number and length = 334, 12'0"
Baffle space = 7 in.	OD, BWG, pitch = 1 in., 14 BWG, 1¼-in. tri.
Passes = 1	Passes = 2

Split Flow. Sometimes it is not possible to meet the pressure-drop requirements in a 1-2 or 2-4 exchanger. Instances will occur when (1) the true temperature difference or U_D is very great and a small exchanger is indicated for the quantity of heat to be transferred, (2) one fluid stream has a very small temperature range compared with the other, or (3) the allowable pressure drop is small. In gases and vapors the last is the more critical because of the low density of the gas or vapor. In liquids an excellent example of (2) is found in the quenching of steel, where it is customary to cool a large volume of circulated quench oil over a small range. It is also characteristic of a number of near-constant temperature operations such as the removal of heat from exothermic reactions by continuous recirculation of the reacting fluids through an external 1-2 cooler.

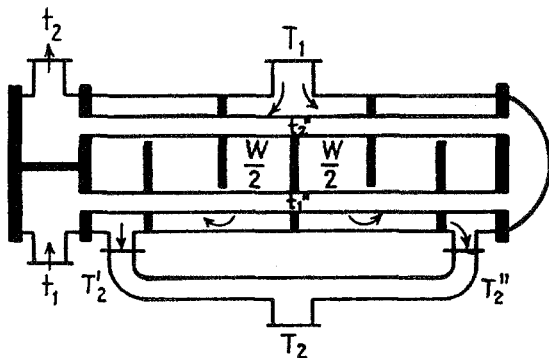


FIG. 11.6. Split-flow exchanger.

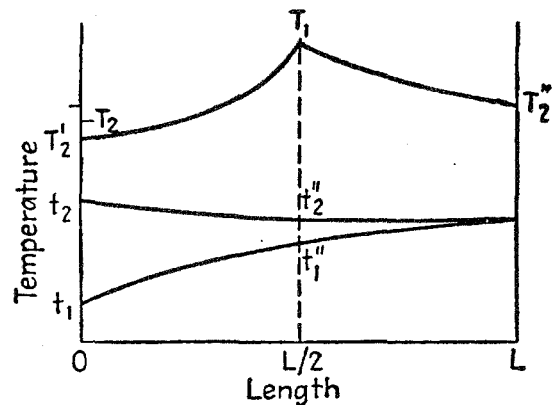


FIG. 11.7. Temperature relations in split flow.

The failure to meet the allowable pressure drop by the conventional methods in a 1-2 exchanger should be construed as an indication that fluid flow and not heat transfer is the controlling factor. Reducing the length of the tubes and increasing the diameter of the shell provides one means of reducing the pressure drop, but other means are available. By locating the shell inlet nozzle at the center instead of at the end of the shell and using two outlet nozzles as shown in Fig. 11.6, the shell-side pressure drop will be but one-eighth as great as in a conventional 1-2 exchanger of the same shell diameter. The reduction is due to halving the mass velocity and halving the length of the shell-side path. This type of flow is known as *split flow*.

As seen in Fig. 11.7 the temperature relations in a split-flow exchanger are not in true counterflow or identical with a 1-2 exchanger, being dis-

continuous at the shell mid-point. A direct solution of the equation¹ for the true temperature difference is somewhat tedious, since the values of t_1'' and t_2'' are related to the actual temperature differences and heat transferred in both parts of the exchanger on either side of the shell inlet. If $T_2 > t_2$, it is satisfactory to obtain Δt by multiplying the LMTD by the value of F_T obtained for a 1-2 exchanger. For services in which there is a temperature cross the actual split-flow equation should be used.

Another type of flow giving even lower pressure drops is *divided flow*, which is usually reserved for low-pressure gases, condensers, and reboilers. It will be discussed in Chap. 12.

Example 11.5. Calculation of a Flue Gas Cooler. 10,500 cfm of flue gas (mol. wt. = 30) at 2 psig and 250°F is to be cooled to 125°F with an allowable pressure drop of 1.0 psi. Cooling will be effected by water from 80 to 100°F and with an allowable pressure drop of 10 psi. An overall dirt factor of 0.005 should be provided with a reasonable minimum water velocity.

Plant practice uses 1 in. OD, 14 BWG tubes on square pitch for all services, and because it is sometimes difficult to meet the pressure drop in gas coolers, no tube length is specified.

Solution:

(1) Heat balance:

Gas: 10,500 cfm of flue gas at 250°F

$$\text{Total gas} = \frac{10,500 \times 60 \times 30}{359 \times (711/492) \times (14.7/16.7)} = 41,300 \text{ lb/hr}$$

$$\text{Gas, } Q = 41,300 \times 0.25(250 - 125) = 1,290,000 \text{ Btu/hr}$$

$$\text{Water, } Q = 64,500 \times 1(100 - 80) = 1,290,000 \text{ Btu/hr}$$

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
250	Higher Temp	100	150
125	Lower Temp	80	45
125	Differences	20	105

¹ Unpublished notes, D. Q. Kern and C. L. Carpenter. The equation in terms of terminal temperatures for a split-flow exchanger with two tube passes is

$$\Delta t = \frac{(T_1 - T_2)\lambda/2}{2.3 \log \left\{ \frac{\left(\frac{(T_1 - T_2)2\lambda n}{n+1} e^{(T_1 - T_2)\left(\frac{\lambda}{2} - 1\right)/2\Delta t} - (T_1 - T_2)(\lambda + 2) + t_1 + t_2 \right) - \frac{4T_2 + 2(n-1)T_1}{n+1}}{(T_1 - T_2)(\lambda - 2) + t_1 + t_2 - \frac{4T_2 + 2(n-1)T_1}{n+1}} \right\}}$$

where $n = Q_B/Q_A$

$$\lambda = \frac{\sqrt{4R^2 + 1}}{R}$$

and

Q_B/Q_A = ratio of heat transfer in each half

$$\text{LMTD} = 87.4^{\circ}\text{F} \quad (5.14)$$

$$R = \frac{125}{20} = 6.25 \quad S = \frac{20}{250 - 80} = 0.118 \quad F_T = 0.935 \quad (\text{Fig. 18})$$

A conventional 1-2 exchanger will be satisfactory.

$$\Delta t = 0.935 \times 87.4 = 81.6^{\circ}\text{F} \quad (7.42)$$

- (3) T_c and t_c : The average temperatures T_a and t_a will be satisfactory because of the small variations in the individual viscosities.

Trial:

(a) Assume $U_D = 15$: From the examples of Chap. 9, at atmospheric pressure and 2 psi allowable pressure drop a coefficient of about 20 might be anticipated. Since the allowable pressure drop in this example is only 1.0 psi, the trial value of U_D must be reduced accordingly. Assume 12'0" tubes to increase the shell cross section.

$$A = \frac{Q}{U_D \Delta t} = \frac{1,290,000}{15 \times 81.6} = 1055 \text{ ft}^2$$

$$a'' = 0.2618 \text{ ft}^2/\text{lin ft}$$

$$\text{Number of tubes} = \frac{1055}{12'0'' \times 0.2618} = 336$$

(b) Assume eight tube passes: Because of the low design coefficient gas exchangers are large for the amount of cooling medium required.

From the tube counts (Table 9): 336 tubes, eight passes, 1 in. OD on 1¼-in. square pitch

Nearest count: 358 tubes in a 31 in. ID shell

(c) Corrected coefficient U_D :

$$A = 358 \times 12'0'' \times 0.2618 = 1125 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{1,290,000}{1125 \times 81.6} = 14.0$$

When solved in a manner identical with the preceding examples and using the smallest integral number of bundle crosses (five) corresponding to a 28.8-in. spacing the summary is

Summary

24.0	h outside	392
U_C	22.7	
U_D	14.0	
R_D Calculated	0.027	
R_d Required	0.005	
5.2	Calculated ΔP	1.0
10	Allowable ΔP	10.0

Discussion. The exchanger selected as a solution to the requirements combines two conditions which have not been met previously: The dirt factor is considerably greater than necessary, and the pressure drop is five times greater than the allowable. Had 8-ft tubes been used in place of the 12-ft tubes for $U_D = 15$, the shell inside diameter would have been 37 in. The baffles could have been spaced 32 in. apart to provide three bundle crosses, but the resulting pressure drop would be 1.7 psi. This would be unsatisfactory, since gases require large inlet connections and the flow distribution on

the first and third bundle crosses would be poor and the conditions of allowable pressure drop would still not be met. The solution is found in a split-flow arrangement.

Trial 2. Split flow:

(a) Assume $U_D = 15$. Referring to the summary of the first trial it is evident that, if the pressure drop is to be met, the mass velocity must be reduced so that the new gas film coefficient will be considerably below the value of 24.0 obtained for ordinary flow.

(b) Assume 12 tube passes. The low water coefficient of 392 corresponds to a velocity of only 1.7 fps which is extremely low for corrosion and dirt even where good water may be employed. Since the size of the shell will not be altered appreciably, having a large inside diameter, it is justifiable that 12 passes be employed. Fewer tube passes would be needed if tube cores were inserted in the tubes. These may be calculated in the manner of Example 10.3. When using more than 8 passes in large shells, the tube count for 8 passes should be reduced by 5 per cent for 12 passes and 10 per cent for 16 passes. For smaller shells it is advisable to avoid the use of 12 and 16 passes. Using the same shell as in Trial 1 for 12 passes, the new tube count will be $358 \times 0.95 = 340$ tubes.

(c) Corrected coefficient U_D :

$$A = 340 \times 12'0'' \times 0.2618 = 1070 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{1,240,000}{1070 \times 81.6} = 14.8$$

Hot fluid: shell side, flue gas

(4') Flow area,

$$a_s = ID \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

There must be an odd number of crosses in each half of the shell and the largest spacing is 31 in.

72 in./31 in. = 2 crosses; say 3 crosses (odd)

$$\text{Actual spacing} = \frac{12 \times 12}{2 \times 3} = 24 \text{ in.}$$

$$a_s = 31 \times 0.25 \times 24/144 \times 1.25 = 1.03 \text{ ft}^2$$

(5') Mass vel, split flow:

$$\begin{aligned} G_s &= \frac{1}{2}W/a_s \quad [\text{Eq. (7.2)}] \\ &= \frac{1}{2} \times 41,300/1.03 \\ &= 20,000 \text{ lb}/(\text{hr})(\text{ft}^2) \end{aligned}$$

(6') At $T_a = 187.5^\circ\text{F}$,

$$\mu = 0.0206 \times 2.42 = 0.050 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Fig. 15}]$$

$$D_s = 0.99/12 = 0.0825 \quad [\text{Fig. 28}]$$

$$\begin{aligned} Re_s &= D_s G_s / \mu \quad [\text{Eq. (7.3)}] \\ &= 0.0825 \times 20,000 / 0.05 = 33,000 \end{aligned}$$

$$(7') j_H = 105 \quad [\text{Fig. 28}]$$

(8') At 187.5°F ,

$$\begin{aligned} k &= 0.015 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft}) \quad (\text{Table 5}) \\ (c\mu/k)^{1/4} &= (0.25 \times 0.050/0.015)^{1/4} = 0.94 \end{aligned}$$

Cold fluid: tube side, water

(4) Flow area,

$$a'_t = 0.546 \text{ in.}^2 \quad [\text{Table 10}]$$

$$\begin{aligned} a_t &= N_t a'_t / 144n \\ &= 340 \times 0.546 / 144 \times 12 = 0.107 \text{ ft}^2 \end{aligned}$$

(5) $G_t = w/a_t$

$$\begin{aligned} &= 64,500/0.107 \\ &= 602,000 \text{ lb}/(\text{hr})(\text{ft}^2) \end{aligned}$$

$$\begin{aligned} \text{Vel, } V &= G_t / 3600\rho \\ &= 602,000/3600 \times 62.5 = 2.68 \text{ fps} \end{aligned}$$

(6) At $t_a = 90^\circ\text{F}$,

$$\mu = 0.81 \times 2.42 = 1.96 \text{ lb}/(\text{ft})(\text{hr}) \quad [\text{Fig. 14}]$$

$$D = 0.834/12 = 0.0695 \text{ ft}$$

$$\begin{aligned} Re_t &= D G_t / \mu \\ &= 0.0695 \times 602,000 / 1.96 = 21,300 \end{aligned}$$

(Re_t is for pressure drop only)

Hot fluid: shell side, flue gas

$$(9') \quad h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/3} \phi_s \quad [\text{Eq. (6.15)}]$$

$$\frac{h_o}{\phi_s} = 105 \times 0.015 \times 0.94 / 0.0825$$

$$= 17.9$$

(10') (11') (12') $\phi_s = 1.0$

$$h_o = \frac{h_o}{\phi_s} = 17.9 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

Cold fluid: tube side, water

(9) $h_i = 710 \times 0.94 = 667$ [Fig. 25]
 (10) $h_{io} = h_i \times \text{ID}/\text{OD}$
 $= 667 \times 0.83/1.0$
 $= 557 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

Pressure Drop

(1') For $Re_s = 33,000$,
 $f = 0.0017 \text{ ft}^2/\text{in.}^2$ [Fig. 29]
 (2') No. of crosses, $N + 1 = 3$ [Eq. (7.43)]
 $D_s = 31/12 = 2.58 \text{ ft}$
 $s = 0.0012$

(3') $\Delta P_s = \frac{fG_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s}$ [Eq. (7.44)]
 $= \frac{0.00167 \times 20,000^2 \times 2.58 \times 3}{5.22 \times 10^{10} \times 0.0825 \times 0.0012 \times 1.0}$
 $= 1.0 \text{ psi}$

(1) For $Re_t = 21,300$,
 $f = 0.00012 \text{ ft}^2/\text{in.}^2$ [Fig. 26]

(2) $\Delta P_t = \frac{fG_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$ [Eq. (7.45)]

$$= \frac{0.00022 \times 602,000^2 \times 12 \times 12}{5.22 \times 10^{10} \times 0.0695 \times 1.0 \times 1.0}$$

$$= 3.1 \text{ psi}$$

(3) $\Delta P_r = 4n/s(V^2/2g')$ [Fig. 27]
 $= (4 \times 12/1)0.052 = 2.5 \text{ psi}$

(4) $\Delta P_T = \Delta P_t + \Delta P_r$ [Eq. (7.47)]
 $= 3.1 + 2.5 = 5.6 \text{ psi}$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{557 \times 17.9}{557 + 17.9} = 17.3 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) = 14.8

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{17.3 - 14.8}{17.3 \times 14.8} = 0.0098 \text{ (hr)}(\text{ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Summary

17.9	h outside	557
U_C	17.3	
U_D	14.8	
R_d Calculated 0.0098		
R_d Required 0.005		
1.0	Calculated ΔP	5.6
1.0	Allowable ΔP	10.0

A 16-pass unit would also be suitable but is not warranted. The final exchanger is

<i>Shell side</i>	<i>Tube side</i>
ID = 31 in.	Number and length = 340, 12'0"
Baffle space = 24 in.	OD, BWG, pitch = 1 in., 14 BWG, 1¼-in. square
Passes = split flow	Passes = 12

PROBLEMS

For the following process conditions determine the size and arrangement of exchanger to fulfill the conditions allowing pressure drops of 10 psi each stream and a combined dirt factor of 0.004. Employ 1-2 exchangers wherever possible.

11.1. 60,000 lb/hr of 42°API kerosene is cooled from 400 to 225°F by heating 35°API distillate from 100 to 200°F.

Use ¾ in. OD, 16 BWG tubes, 16'0" long on 1½-in. triangular pitch.

11.2. 120,000 lb/hr of aniline is cooled from 275 to 200°F by heating 100,000 lb/hr of benzene from 100 to 200°F.

Use ¾ in. OD, 14 BWG tubes, 16'0" long on 1-in. square pitch.

11.3. 84,000 lb/hr of 42°API kerosene is cooled from 300 to 100°F using water from 85 to 120°F. Calculate the requirement using the 1-2 exchangers in series.

Use ¾ in. OD, 16 BWG tubes, 16'0" long on 1-in. triangular pitch.

11.4. 22,000 lb/hr of 35°API distillate is heated from 200 to 300°F by 28°API gas oil from an inlet temperature of 500°F.

Use 1 in. OD, 14 BWG tubes, 12'0" long on 1¼-in. triangular pitch.

11.5. 68,000 lb/hr of 56°API gasoline is cooled from 200 to 100°F using water at 85°F.

Use 1 in. OD, 14 BWG tubes, 12'0" long on 1¼-in. square pitch.

11.6. 32,000 lb/hr of oxygen at 5 psig is cooled from 300 to 150°F using water at 85°F. (Allowable pressure drop for oxygen 2.0 psi.)

Use ¾ in. OD, 16 BWG tubes, 12'0" long on 1-in. square pitch.

NOMENCLATURE FOR CHAPTER 11

A	Heat-transfer surface, ft ²
A_{CR}, A_E, A_H	Heat-transfer surface of cooler, exchanger, and heater, ft ²
a	Flow area, ft ²
a''	External surface per linear foot, ft
B	Baffle spacing, in.
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
C_S	Cost of steam, dollars/Btu
C_T	Total annual cost, dollars
C_W	Cost of water, dollars/Btu
C_{CR}, C_E, C_H	Fixed charges of cooler, exchanger, and heater, respectively, dollars/(ft ²)(year)
C'	Clearance between tubes, in.
c	Specific heat of fluid, Btu/(lb)(°F)
D	Inside diameter of tubes, ft
D_s	Inside diameter of shell, ft
D_e, D'_e	Equivalent diameter for heat transfer and pressure drop, ft
d	Inside diameter of tubes, in.
d_s, d'_s	Equivalent diameter for heat transfer and pressure drop, in.
F_c	Caloric fraction, dimensionless
F_T	Temperature-difference factor, $\Delta t = F_T \times \text{LMTD}$, dimensionless

f	Friction factor, $\text{ft}^2/\text{in.}^2$
G	Mass velocity, $\text{lb}/(\text{hr})(\text{ft}^2)$
g	Acceleration of gravity, ft/hr^2
g'	Acceleration of gravity, ft/sec^2
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
h_{io}	Value of h_i when referred to the tube outside diameter, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
j_H	Factor for heat transfer, dimensionless
K_c	Caloric constant, dimensionless
k	Thermal conductivity, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$
L	Tube length, ft
LMTD	Log mean temperature difference, $^\circ\text{F}$
N	Number of shell-side baffles
N_t	Number of tubes
n	Number of tube passes
P_T	Tube pitch, in.
$\Delta P_T, \Delta P_t, \Delta P_r$	Total, tube and return pressure drop, psi
Q	Heat flow, Btu/hr
R	Temperature group $(T_1 - T_2)/(t_2 - t_1)$, dimensionless
R_d	Dirt factor, $(\text{hr})(\text{ft}^2)(^\circ\text{F})/\text{Btu}$
Re, Re'	Reynolds number for heat transfer and pressure drop, dimensionless
S	Temperature group $(t_2 - t_1)/(T_1 - t_1)$, dimensionless
s	Specific gravity, dimensionless
T_1, T_2	Inlet and outlet temperature of hot fluid, $^\circ\text{F}$
T_a	Average temperature of hot fluid, $^\circ\text{F}$
T_c	Caloric temperature of hot fluid, $^\circ\text{F}$
T_s	Temperature of steam, $^\circ\text{F}$
T_x	Optimum exchanger hot-fluid outlet temperature, $^\circ\text{F}$
t_y	Optimum exchanger cold-fluid outlet temperature, $^\circ\text{F}$
t_a	Average temperature of cold fluid, $^\circ\text{F}$
t_c	Caloric temperature of cold fluid, $^\circ\text{F}$
t_w	Tube-wall temperature, $^\circ\text{F}$
t'_1, t'_2	Inlet and outlet water temperatures, $^\circ\text{F}$
t''_1, t''_2	Temperatures in a split-flow exchanger, $^\circ\text{F}$
Δt	True temperature difference in $Q = U_D A \Delta t$, $^\circ\text{F}$
$\Delta t_c, \Delta t_h$	Cold and hot terminal temperature difference, $^\circ\text{F}$
U_C, U_D	Clean and design overall coefficient of heat transfer, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
U_{CR}, U_E, U_H	Overall coefficient of heat transfer for cooler, exchanger, and heater, respectively, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$
V	Velocity, fps
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
ϕ	Viscosity ratio, $(\mu/\mu_w)^{0.14}$
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at tube-wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
θ	Annual operating hours

Subscripts (except as noted above)

s	Shell
t	Tube

CHAPTER 12

CONDENSATION OF SINGLE VAPORS

Introduction. A fluid may exist as a gas, vapor, or liquid. The change from liquid to vapor is *vaporization*, and the change from vapor to liquid is *condensation*. The quantities of heat involved in the vaporization or condensation of a pound of fluid are identical. For a pure fluid compound at a given pressure the change from liquid to vapor or vapor to liquid occurs at but one temperature which is the *saturation* or *equilibrium* temperature. Since vapor-liquid heat-transfer changes usually occur at constant or nearly constant pressure in industry, the vaporization or condensation of a single compound normally occurs isothermally. When a vapor is removed upon formation from further contact with a liquid, the addition of heat to the vapor causes *superheat*, during which it behaves like a gas. If a mixture of vapors instead of a pure vapor is condensed at constant pressure, the change does not take place isothermally in most instances. The general treatment of vapor mixtures differs in certain respects from single compounds and will be studied in the next chapter with the aid of the phase rule of J. Willard Gibbs.

Condensation occurs at very different rates of heat transfer by either of the two distinct physical mechanisms, which will be discussed presently, *dropwise* or *filmwise* condensation. The condensing film coefficient is influenced by the texture of the surface on which condensation occurs and also by whether the condensing surface is mounted vertically or horizontally. In spite of these apparent complications condensation, like streamline flow, lends itself to direct mathematical study.

Dropwise and Filmwise Condensation. When a saturated pure vapor comes into contact with a cold surface such as a tube, it condenses and may form liquid droplets on the surface of the tube. These droplets may not exhibit an affinity for the surface and instead of coating the tube fall from it, leaving bare metal on which successive droplets of condensate may form. When condensation occurs by this mechanism, it is called *dropwise* condensation. Usually, however, a distinct film may appear as the vapor condenses and coats the tube. Additional vapor is then required to condense into the liquid film rather than form directly on the bare surface. This is *film* or *filmwise* condensation. The two mechanisms are distinct and independent of the quantity of vapor condensing per square foot of surface. *Filmwise* condensation is therefore

not a transition from *dropwise* condensation because of the rapidity at which condensate forms on the tube. Due to the resistance of the condensate film to the heat passing through it the heat-transfer coefficients for dropwise condensation are four to eight times those for filmwise condensation. Steam is the only pure vapor known to condense in a dropwise manner, and special conditions are required for its occurrence. These are described by Drew, Nagle, and Smith¹ and principally result from the presence of dirt on the surface or the use of a contaminant which adheres to the surface. Materials have been identified by Nagle² which promote the dropwise condensation of steam although these also introduce an impurity into the steam. Dropwise condensation also occurs when several materials condense simultaneously as a mixture and where the condensate mixture is not miscible, as in the case of a hydrocarbon and steam. However, during various periods in the normal operation of a steam condenser the mechanism may initially be filmwise condensation, shift to dropwise condensation, and at some later time revert to film condensation. Because of the lack of control it is not customary in calculations to take advantage of the high coefficients which have been obtained in dropwise-condensation experiments. This chapter consequently deals with the calculation of condensers for various conditions and is based solely upon film-condensation heat-transfer coefficients.

It is fortunate that the phenomenon of film condensation lends itself to mathematical analysis, and the nature of condensation on a cold surface may be considered one of self-diffusion. The saturation pressure of vapor in the vapor body is greater than the saturation pressure of the cold condensate in contact with the cold surface. This pressure difference provides the potential for driving vapor out of the vapor body at a great rate. Compared with the small resistance to heat transfer by diffusion from the vapor into the condensate, the film of condensate on the cold tube wall contributes the controlling resistance. It is the slowness with which the heat of condensation passes through this film that determines the condensing coefficient. The ultimate form of an equation for the condensing coefficient may be obtained from dimensional analysis where the average condensing coefficient $\bar{h}$ is a function of the properties of the condensate film, k , ρ , g , μ , and L , Δt , and λ , the last being the latent heat of vaporization. Nusselt theoretically derived the relationships for the mechanism of film condensation, and the results he obtained are in excellent agreement with experiments.

Process Applications. In chemical industry it is a common practice to separate a liquid mixture by distilling off the compounds which have

¹ Drew, T. B., W. M. Nagle, and W. Q. Smith, *Trans. AIChE*, **31**, 605-621 (1935)

² Nagle, W. M., U.S. Patent 1,995,361.

lower boiling points in the pure condition from those having higher boiling points. In a solution of several compounds each exerts a partial pressure and the most volatile cannot be boiled off from the rest without carrying some of the heavier or higher boiling compounds along with it. The proportion of heavier compounds carried off when a solution starts to boil is less than existed in the original solution before boiling commenced. If the vapor coming off initially is condensed, it has a lower boiling point than the original solution, indicating the increase in the proportion of the more volatile compounds. By successively boiling off

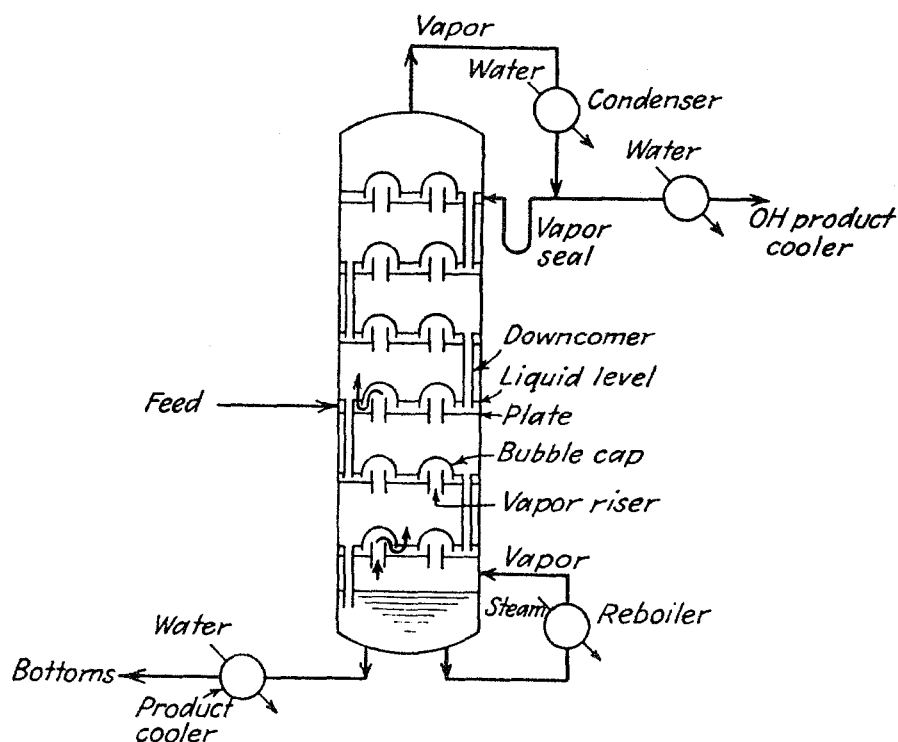


FIG. 12.1. Distilling column with auxiliaries.

only part of a liquid mixture, condensing the vapor formed, and boiling off only a part of the condensate, it is possible to obtain a nearly pure quantity of the most volatile compound by numerous repetitions of the procedure. Thus the separation by distillation is accomplished by partial vaporization and subsequent condensation.

In distillation it is customary to obtain a number of partial vaporizations and condensations by directly contacting a vapor and a liquid cooling medium in a continuous distilling column. The bubble cap distilling column shown in Fig. 12.1 is representative of modern practice and derives its name from a series of inverted slotted caps which are placed over vapor risers on each plate of the column. Vapor from below a plate enters the risers and is broken into bubbles as it passes through the slotted bottoms of the bubble caps and thence through the layer of liquid

maintained by the *downcomer* on each plate. The *feed*, which is usually a liquid, is a mixture of more and less volatile compounds and enters the distilling column at the feed plate where the volatile compounds are partially vaporized by the rising vapors as the feed travels across the plate. The remainder of the liquid on the plate is less volatile than the feed and overflows to the plate below through the downcomer. The boiling points of the liquids on each of the lower plates is consequently higher.

To vaporize a portion of the feed, vapor from below the feed plate must exchange heat with the liquid on the feed plate, thereby driving the more volatile compounds to the plate above the feed. By supplying heat at the bottom of the column where the increased concentration of the least volatile compounds represents the highest boiling temperature in the system, a thermal gradient is established plate by plate between the bottom of the column and the top. Heat supplied at the bottom by vaporization in a *reboiler* is transmitted to the top of the column plate-by-plate due to the temperature differences corresponding to the differences in boiling points between plates. Continuous distillation requires the presence of liquid at all times on the plates, so that vapors of the less volatile compounds in the feed may be condensed and carried downward. To accomplish this, some volatile liquid from the condenser, which represents one plate above the top plate and which is therefore colder, is introduced onto the top plate and flows downward in the column. The volatile liquid which is poured back into the column from the condenser is the *reflux*. The quantity of volatile components removed from the system at the top and having the same composition as the reflux is called the *distillate* or *overhead* product. The heavier compounds removed at the bottom are variously called *waste* or *residue* or, if they are of value, *bottoms product*. The quantitative aspects of the heat balance are treated in Chap. 14.

It is the condensing temperature in the condenser which determines the operating pressure of the distilling column, since the saturation temperature of a vapor varies with its pressure. The overhead product must condense in the condenser at a temperature sufficiently high so that its latent heat can be removed by cooling water. The size of the condenser is dependent upon the difference between the condensing temperature and the range of the cooling water. If the condensing temperature is very close to the cooling water range at atmospheric pressure, the distillation pressure must be elevated to permit the attainment of a larger Δt .

In the power industry the term *surface condenser* is reserved for tubular equipment which condenses steam from the exhaust of turbines and engines. Since a turbine is primarily designed to obtain mechanical work

from heat, the maximum conversion is obtained in the turbine by maintaining a low turbine-discharge temperature. If the turbine were to discharge to the atmosphere, the lowest attainable steam temperature would be 212°F, but if the steam were to discharge into a condenser under vacuum, it would be possible to operate at discharge temperatures of 75°F and lower and to convert the enthalpy difference from 212 to 75°F into useful work.

Condensation on Surfaces—Nusselt's Theory. In condensation on a vertical surface a film of condensate is formed as shown in Fig. 12.2 and further condensation and heat transfer to the surface occurs by conduction through the film which is assumed to be in laminar flow downward.

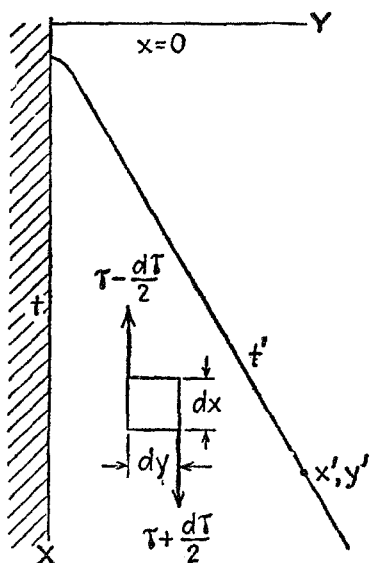


FIG. 12.2. The vertical condensate film.

The thickness of this film greatly influences the rate of condensation, since the heat accompanying the removal of vapors from the vapor phase encounters the condensate film as a resistance which may be quite large. The thickness of the film is a function of the velocity of drainage which varies with the deviation of the surface from a vertical position. For a vertical surface the thickness of the film cumulatively increases from top to bottom. For this reason the condensing coefficient for a vapor condensing on a vertical surface decreases from top to bottom, and for the attainment of a large condensing coefficient the height of the surface should not be very great. The velocity of drainage for equal quantities of condensate is also a function of the viscosity of the condensate: The

lower the viscosity the thinner the film. For all liquids the viscosity decreases as the temperature increases, and the condensing coefficient consequently increases with the condensate temperature. The derivations given in this chapter through Eq. (12.34) are those of Nusselt.¹ The following assumptions are involved:

1. The heat delivered by the vapor is latent heat only.
2. The drainage of the condensate film from the surface is by laminar flow only, and the heat is transferred through the film by conduction.
3. The thickness of the film at any point is a function of the mean velocity of flow and of the amount of condensate passing at that point.
4. The velocity of the individual layers of the film is a function of the relation between frictional shearing force and the weight of the film (see Chap. 3).

¹ Nusselt, W., *Z. Ver. deut. Ing.*, **60**, 541 (1916).

5. The quantity of condensate is proportional to the quantity of heat transferred, which is in turn related to the thickness of the film and of the temperature difference between the vapor and the surface.

6. The condensate film is so thin that the temperature gradient through it is linear.

7. The physical properties of the condensate are taken at the mean film temperature.

8. The surface is assumed to be relatively smooth and clean.

9. The temperature of the surface of the solid is constant.

10. The curvature of the film is neglected.

Condensation. Vertical Surfaces. In Fig. 12.2 the rate at which heat passes from the vapor through the liquid condensate film and into the cooling surface per unit area is given by

$$\frac{Q}{A} = \frac{k(t' - t)}{y'} = \lambda W' = h(t' - t) \quad (12.1)$$

where λ is the latent heat of vaporization, W' the pounds of condensate formed per hour per square foot, and y' is the thickness of the condensate film at the generalized point in the figure whose coordinates are x' , y' . The other symbols have their conventional meaning.

The rate at which the vapor condenses is then given by

$$W' = \frac{k(t' - t)}{\lambda y'} \quad (12.2)$$

The liquid flows downward over the vertical surface with a velocity u varying from zero at the tube-film interface and increasing outward to the condensate-vapor interface. The velocity also increases vertically as the condensate flows downward.

Consider a small cube of unit depth $dz = 1$, defined by $dx dy$ 1 in the moving condensate film of Fig. 12.2. On the side near the cold vertical surface there is a tangential force acting upward and tending to support the cube. On the side away from the cold vertical surface there is a tangential force acting downward due to the more rapid movement of the liquid downward as the distance from the surface is increased. If the resultant force upward through the cube is designated by τ , then the respective forces are $\tau - d\tau/2$ and $\tau + d\tau/2$. The differential tangential force must be offset by gravity acting down.

$$\rho dx dy 1 = \left(\tau - \frac{d\tau}{dy} \frac{dy}{2} \right) - \left(\tau + \frac{d\tau}{dy} \frac{dy}{2} \right) = -d\tau \quad (12.3)$$

On unit area, $dx dz = 1$

$$\rho = - \frac{d\tau}{dy}$$

From the basic definition of viscosity in Chap. 3 the tangential stress is defined by Eq. (3.3), using (lb-force)(hr)/ft² as the dimensions for the viscosity. Since it is customary to use the dimensions (lb-mass)/(ft)(hr) for the dimensions, Eq. (3.3) becomes

$$\tau = \frac{\mu}{g} \frac{du}{dy} \quad (12.4)$$

$$\frac{d\tau}{dy} = \frac{\mu}{g} \frac{d^2u}{dy^2} = -\rho \quad (12.5)$$

Take ρ/μ as constant.

$$\frac{d^2u}{dy^2} = -\frac{\rho g}{\mu} \quad (12.6)$$

$$u = -\frac{\rho g y^2}{2\mu} + C_1 y + C_2 \quad (12.7)$$

The constants C_1 and C_2 must now be evaluated. Since the liquid adheres to the wall, u must equal zero at $y = 0$, making C_2 equal zero.

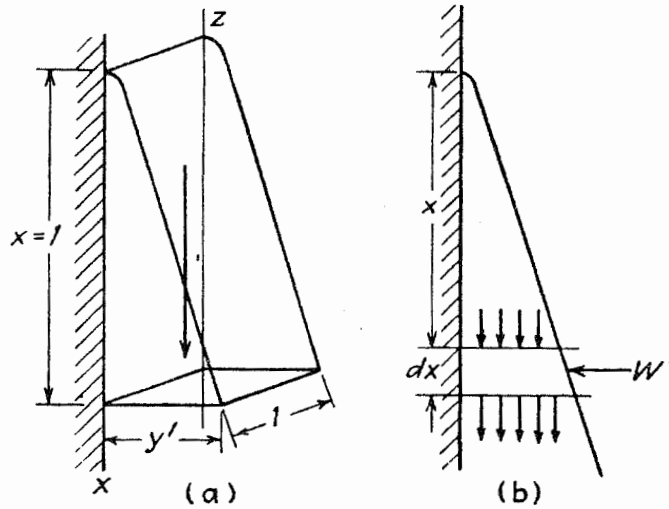


FIG. 12.3. Vertical condensate film flow.

At the outer boundary of the film (condensate-vapor interface) there is no tangential stress and from $\tau = \mu du/dy$ when $y = y'$,

$$\left(\frac{du}{dy}\right)_{y=y'} = 0 = -\frac{\rho g y'}{\mu} + C_1 \quad C_1 = \frac{\rho g y'}{\mu} \quad (12.8)$$

$$u = \frac{\rho g}{\mu} \left(y y' - \frac{y^2}{2} \right)$$

At a distance x from the top of the condensing surface the average velocity downward $\bar{u}$ is given by

$$\bar{u} = \frac{1}{y'} \int_0^{y'} u dy = \frac{\rho g}{3\mu} y'^2 \quad (12.9)$$

When the value of x from the top of a vertical wall is taken to be unity as shown in Fig. 12.3a, the quantity of downward flow across a horizontal plane of area $1y'$ of the condensate is

$$1y'\bar{u}\rho$$

At $x + dx$ there is a gain in the amount of downward flow of condensate as shown in Fig. 12.3b. Using the value for $\bar{u}$ from Eq. (12.9), multiplying by $\rho y'$, and differentiating with respect to x to obtain the increase from x to $x + dx$,

$$d(\rho\bar{u}y') = d\left(\frac{\rho^2 g}{3\mu} y'^3\right) = \frac{\rho^2 g y'^2}{\mu} dy' \quad (12.10)$$

And this increase must come from condensation out of the vapor and into the condensate film

$$d(\rho\bar{u}y') = W'1 dx$$

where W' is the condensate flow out of the vapor and "normal" to the falling condensate layer per unit area as in Fig. 12.3b. From Eq. (12.2), however, W' was defined in terms of the heat transfer as

$$W' = \frac{k(t' - t)}{\lambda y'}$$

Substituting for W' in Eq. (12.10) the value from Eq. (12.2),

$$\frac{k}{\lambda y'} (t' - t) dx = \frac{\rho^2 g y'^2}{\mu} dy' \quad (12.11)$$

$$(t' - t) dx = \frac{\rho^2 g \lambda y'^3}{k\mu} dy' \quad (12.12)$$

For a limited range set $t' - t$, ρ , λ , μ , and k constant and integrate.

When $y' = 0$, $x = 0$

$$y'^4 \frac{\rho^2 g \lambda}{4\mu k} = (t' - t)x \quad (12.13)$$

$$y' = \left[\frac{4\mu k}{\rho^2 \lambda g} (t' - t)x \right]^{1/4} \quad (12.14)$$

The heat-transfer coefficient across the condensate layer at the distance x from the origin per unit of interfacial area is given from Eq. (12.1) by

$$h_x = \frac{Q_x/A_x}{t' - t} = \frac{k}{y'} \quad (12.15)$$

Substitute y' from Eq. (12.14):

$$h_x = \left[\frac{k^3 \rho^2 \lambda g}{4\mu (t' - t)} \right]^{1/4} \frac{1}{x^{1/4}} \quad (12.16)$$

The total heat through the condensate layer from 0 to x is Q_x

$$\begin{aligned} Q_x &= \int_0^x h_x(t' - t) dx = \int_0^x \left[\frac{k^3 \rho^2 \lambda g}{4\mu(t' - t)} \right]^{1/4} (t' - t) \frac{dx}{x^{1/4}} \\ &= \frac{4^{3/4}}{3} \left(\frac{k^3 \rho^2 \lambda g}{\mu} \right)^{1/4} [(t' - t)x]^{3/4} \end{aligned} \quad (12.17)$$

If the average coefficient between the two points is $\bar{h}$,

$$\begin{aligned} \bar{h} &= \frac{(Q_x)_{x=L}}{(t' - t)L} = \frac{4^{3/4}}{3} \frac{\left(\frac{k_f^3 \rho_f^2 \lambda g}{\mu_f} \right)^{1/4} [(t' - t)L]^{3/4}}{(t' - t)L} \\ \bar{h} &= 0.943 \left(\frac{k_f^3 \rho_f^2 \lambda g}{\mu_f L \Delta t_f} \right)^{1/4} \end{aligned} \quad (12.18)$$

where k_f , ρ_f , and μ_f are evaluated at the film temperature t_f and where the film temperature is

$$t_f = \frac{1}{2}(t' + t) = \frac{1}{2}(T_v + t_w) \quad (12.19)$$

and

$$\Delta t_f = t_f - t_w$$

In the above, as in the derivation which follows, the stress caused by the passage of the saturated vapor over the condensate-vapor interface has been neglected. It can be included, although it is not of practical consequence. The variation of the film thickness and local heat-transfer coefficient are shown in Fig. 12.4. The shapes of the curves follow the thickness and consequently the resistance of the condensate film.

Inclined Surfaces. Consider a cube making the angle α as shown in Fig. 12.5. The gravity component acting in a plane parallel to the surface is $\rho \sin \alpha$, and Eq. (12.3) becomes

$$\rho \sin \alpha dy dx \frac{1}{2} = \left(\tau - \frac{d\tau}{dy} \frac{dy}{2} \right) - \left(\tau + \frac{d\tau}{dy} \frac{dy}{2} \right) = -d\tau \quad (12.20)$$

On unit area $dx dz = 1$

$$\rho \sin \alpha = - \frac{d\tau}{dy} \quad (12.21)$$

Equation (12.6) becomes

$$\frac{d^2 u}{dy^2} = - \frac{g\rho}{\mu} \sin \alpha \quad (12.22)$$

Equation (12.7) becomes

$$u = - \frac{g\rho}{2\mu} y^2 \sin \alpha + C_1 y + C_2 \quad (12.23)$$

At the start of condensation on the tube where $y = 0$ and there is no velocity along the tube, $u = 0$ and $C_2 = 0$

$$C_1 = \frac{g\rho y'}{\mu} \sin \alpha$$

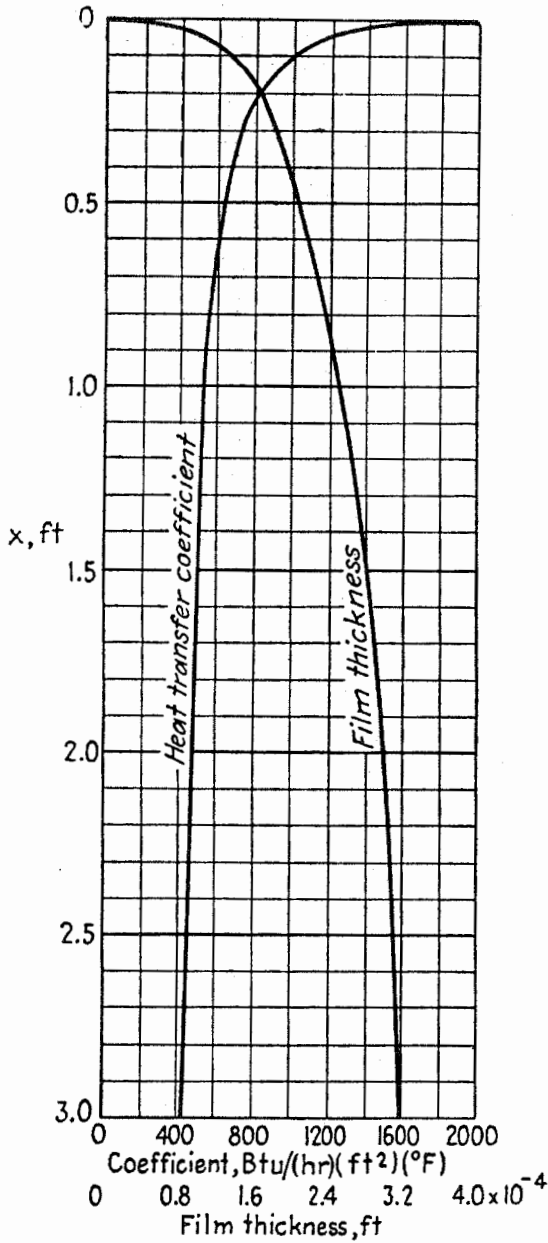


FIG. 12.4. Vertical film thickness and condensing coefficients for a descending film. (After Nusselt.)

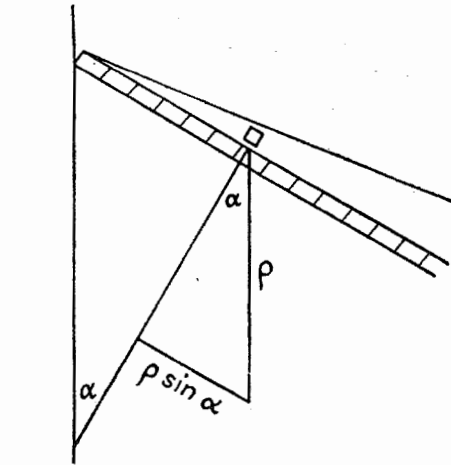


FIG. 12.5. Film on an inclined surface

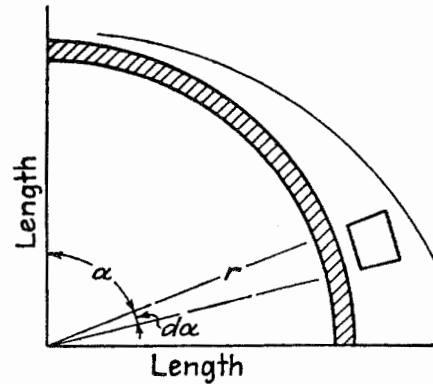


FIG. 12.6. Condensate film on a horizontal tube.

and Eq. (12.9) becomes

$$\bar{u} = \frac{g\rho y'^2}{3\mu} \sin \alpha \quad (12.24)$$

and Eq. (12.18) becomes

$$\bar{h} = 0.943 \left(\frac{k_f^3 \rho_f^2 \lambda g \sin \alpha}{\mu_f L \Delta t_f} \right)^{1/4} \quad (12.25)$$

Horizontal Tubular Surfaces. Consider a cube of unit length at radius r making an angle α with the vertical as seen in Fig. 12.6. The mass flow of vapor into the condensate film through the area $r d\alpha$ and with a film

thickness y' is given by the conduction equation.

$$W' = \frac{k(t' - t)r d\alpha}{\lambda y'}$$

The condensation must give rise to an increase in the inclined falling film. For a differential amount of condensation the increase through the condensing area $r d\alpha$ is $d(\rho \bar{u} y')$ and for Eq. (12.10)

$$d(\rho \bar{u} y') = \frac{\rho^2 g}{3\mu} d(y'^3 \sin \alpha) = W' dx$$

Substituting for W' , Eq. (12.11) becomes

$$\begin{aligned} \frac{k(t' - t)r d\alpha}{\lambda y'} &= \frac{\rho^2 g}{3\mu} d(y'^3 \sin \alpha) \\ \frac{3\mu k(t' - t)r d\alpha}{\rho^2 g \lambda} &= y' d(y'^3 \sin \alpha) \end{aligned}$$

Let

$$\begin{aligned} m &= \frac{3\mu k(t' - t)r}{\rho^2 g \lambda} \\ m d\alpha &= y' d(y'^3 \sin \alpha) \end{aligned}$$

Differentiating

$$\begin{aligned} m d\alpha &= y'(3y'^2 \sin \alpha dy' + y'^3 \cos \alpha d\alpha) \\ &= 3y'^3 \sin \alpha dy' + y'^4 \cos \alpha d\alpha \end{aligned} \quad (12.26)$$

In Eq. (12.26) the term $3y'^3 dy'$ appears, but $d(y'^4) = 4y'^3 dy'$ and

$$3y'^3 dy' = \frac{3}{4} dy'^4.$$

Rearranging Eq. (12.26) and substituting,

$$d\alpha = \frac{3}{4m} \sin \alpha dy'^4 + \frac{y'^4}{m} \cos \alpha d\alpha$$

Let $y'^4/m = \psi^4 = z$

$$d\alpha = \frac{3}{4} \sin \alpha dz + z \cos \alpha d\alpha \quad (12.27)$$

$$\frac{3}{4} \sin \alpha \frac{dz}{d\alpha} + z \cos \alpha - 1 = 0 \quad (12.28)$$

Equation (12.28) is a linear differential equation whose solution is

$$z = \frac{1}{\sin^{3/2} \alpha} \left(\frac{4}{3} \int \sin^{1/2} \alpha d\alpha + C_3 \right) \quad (12.29)$$

When $\alpha = 0$, $C_3 = 0$,

$$z = \frac{4}{3} \frac{1}{\sin^{3/2} \alpha} \int \sin^{1/2} \alpha d\alpha \quad (12.30)$$

The value of this integral for different values of α may be determined by graphical methods. From the substitution in Eq. (12.27)

$$y' = \psi m^{1/4} = \psi \left[\frac{3\mu k(t' - t)r}{\rho^2 g \lambda} \right]^{1/4} \quad (12.31)$$

As shown in Eq. (12.15), $h_x = k/y'$.

The thickness of the film actually decreases slightly as α increases from 0 to 5° , and then it increases steadily and breaks into drops. The local-heat transfer coefficient at any point is then

$$h_\alpha = \frac{k}{y'} = \frac{1}{\psi} \left[\frac{k^3 \rho^2 \lambda g}{3\mu(t' - t)r} \right]^{1/4} \quad (12.32)$$

The average heat-transfer coefficient $\bar{h}_\alpha$ of the segment between angles α_1 and α_2 is

$$\bar{h}_\alpha \Big|_{\alpha_1}^{\alpha_2} = \frac{k}{m^{1/4}(\alpha_2 - \alpha_1)} \int_{\alpha_1}^{\alpha_2} \frac{d\alpha}{\psi} \quad (12.33)$$

Employing graphical methods as before, where D_o is the outside diameter of the tube, the average heat-transfer coefficients are found to be

$$\begin{aligned} \bar{h}_\alpha \Big|_{0^\circ}^{90^\circ} &= 0.860 \left(\frac{k^3 \rho^2 \lambda g}{\mu D_o \Delta t_f} \right)^{1/4} \\ \bar{h}_\alpha \Big|_{90^\circ}^{180^\circ} &= 0.589 \left(\frac{k^3 \rho^2 \lambda g}{\mu D_o \Delta t_f} \right)^{1/4} \end{aligned}$$

From 0 to 180° which is one-half the tube, the other half being symmetrical,

$$\bar{h} = 0.725 \left(\frac{k_f^3 \rho_f^2 \lambda g}{\mu_f D_o \Delta t_f} \right)^{1/4} \quad (12.34)$$

The variation of the film thickness and heat-transfer coefficient for steam on a horizontal tube is shown in Fig. 12.7. As in the preceding case it is governed by the resistance of the condensate film to conduction.

Development of Equations for Calculations. McAdams¹ found from the correlation of the data of several investigations that observed condensing coefficients for steam on vertical tubes were 75 per cent greater than the theoretical coefficients calculated by Eq. (12.18). The values calculated from Eq. (12.18) agree, however, for a condensate in streamline flow with the values calculated from Eq. (6.1) for ordinary streamline flow.

When a liquid descends vertically on the outside of a tube, it is certainly

¹ McAdams, *op. cit.*, p. 264.

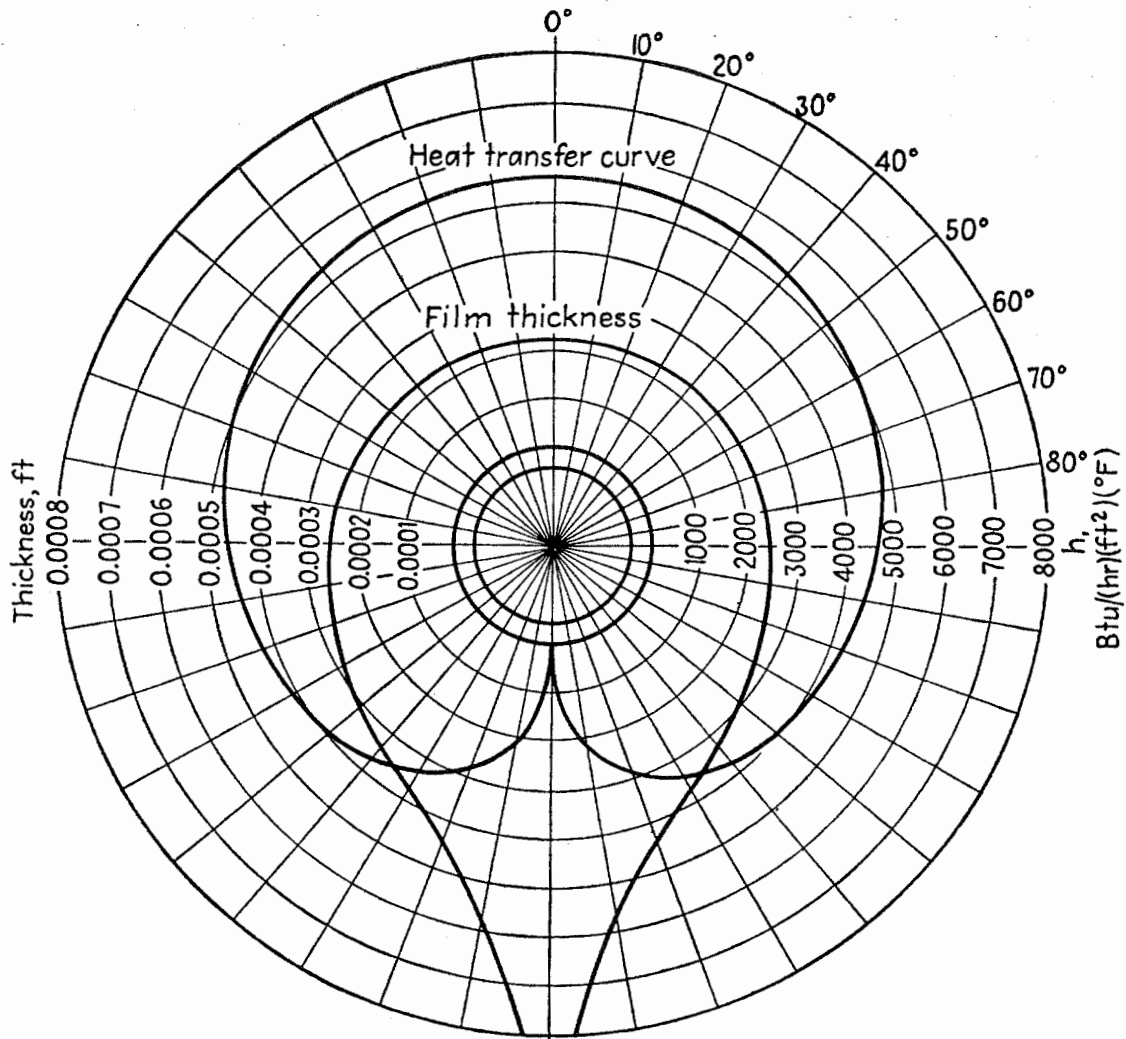


FIG. 12.7. Plot of the heat-transfer coefficient and film thickness of water on a horizontal tube. (After Nusselt.)

in streamline flow at the top of the tube, where the accumulation of condensate is small. If a relatively large amount of vapor is condensed on the tube, it is possible that at some point below the top the film will

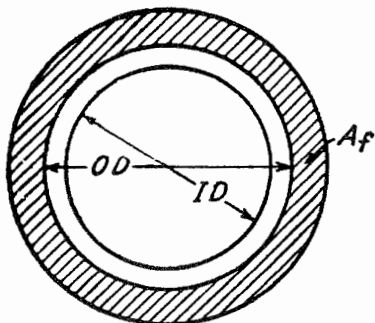


FIG. 12.8. Vertical descending film.

change to turbulent flow. This may be estimated from the diameter and length of the tube, the viscosity of the condensate, and the quantity being condensed. Referring to the tube as shown in Fig. 12.8, the crosshatched area outside the tube represents condensate film as seen at any point looking down. This is similar to the flow in the annulus of a double pipe exchanger except that the outer surface of the film is not formed by a concentric pipe. In

the case of the double pipe exchanger the equivalent diameter was taken as four times the hydraulic radius.

Then

$$D_e = 4r_h = 4 \times \frac{\text{free flow area}}{\text{wetted perimeter}}$$

and

$$Re = \frac{D_e G}{\mu}$$

For *vertical* tubes let

A_f = cross-sectional area (shaded)

P = wetted perimeter per tube

$$D_e = 4 \times A_f / P$$

Letting the loading per tube be $w' = W/N_t$, where N_t is the number of tubes,

$$\begin{aligned} G &= w' / A_f \quad \text{lb}/(\text{hr})(\text{ft}^2) \\ Re &= D_e G / \mu = (4A_f / P)(w' / A_f) / \mu = 4w' / \mu P \end{aligned} \quad (12.35)$$

Calling the condensate loading per linear foot G' ,

$$G' = \frac{w'}{P}, \quad \text{lb}/(\text{hr})(\text{lin ft}) \quad (12.36)$$

Eq. (12.35) becomes

$$Re = \frac{4G'}{\mu}$$

The total heat load is given by $Q = \lambda w'$.

$$\bar{h} = \frac{Q}{A \Delta t_f} = \frac{\lambda w'}{PL \Delta t_f} = \frac{\lambda}{L} \frac{G'}{\Delta t_f} \quad (12.37)$$

Substituting in Eq. (12.18),

$$\bar{h} = 0.943 \left(\frac{k_f^3 \rho_f^2 g}{\mu_f} \frac{\bar{h}}{G'} \right)^{1/4} \quad (12.38)$$

Multiplying the right term by $(4\mu/4\mu)^{1/4}$,

$$\begin{aligned} \bar{h}^{3/4} &= 0.943 \left(\frac{4k_f^3 \rho_f^2 g}{\mu_f^2} \frac{\mu_f}{4G'} \right)^{1/4} \\ \bar{h} \left(\frac{\mu_f^2}{k_f^3 \rho_f^2 g} \right)^{1/3} &= 1.47 \left(\frac{4G'}{\mu_f} \right)^{-1/3} \end{aligned} \quad (12.39)$$

For *horizontal* tubes Eq. (12.39) becomes

$$\bar{h} \left(\frac{\mu_f^2}{k_f^3 \rho_f^2 g} \right)^{1/3} = 1.51 \left(\frac{4G''}{\mu_f} \right)^{-1/3} \quad (12.40)$$

where the loading for a single horizontal tube is

$$G'' = \frac{W}{LN_t} \quad (12.41)$$

Using the corresponding loading as given by either Eq. (12.36) or (12.41) as the case may be, Eqs. (12.39) and (12.40) may both be represented by

$$\bar{h} \left(\frac{\mu_f^2}{k_f^3 \rho_f^2 g} \right)^{1/3} = 1.5 \left(\frac{4G'}{\mu_f} \right)^{-1/3} = 1.5 \left(\frac{4G''}{\mu_f} \right)^{-1/3} \quad (12.42)$$

Equations (12.39) and (12.40) were obtained for condensation on single tubes. In a vertical-tube bundle the presence of one or more tubes does not alter the assumptions on which the derivation was predicated. However, on horizontal tubes in tube bundles it has been found that the splashing of the condensate as it drips over successive rows of tubes causes G'' to be more nearly inversely proportional to $N_t^{2/3}$ rather than N_t , so that it is preferable to use a fictitious value for horizontal tubes

$$G'' = \frac{W}{LN_t^{2/3}} \quad \text{lb/(hr)(lin ft)} \quad (12.43)$$

Figure 12.9 is a line chart of solutions of Eq. (12.42) prepared for convenience. Its use requires that the film be in streamline flow corresponding to an average Reynolds number of about 1800 to 2100 for the flow gradient assumed by the condensate. For steam at atmospheric pressure Eq. (12.42) reduces to the equations given by McAdams:¹ For horizontal tubes

$$\bar{h} = \frac{3100}{D_o^{1/4} \Delta t_f^{1/4}} \quad (12.44a)$$

and for vertical tubes

$$\bar{h} = \frac{4000}{L^{1/4} \Delta t_f^{1/4}} \quad (12.44b)$$

where Δt_f ranges from 10 to 150°F.

It is frequently desirable to apply Eqs. (12.39), (12.40), and (12.42) to the calculation of condensers which are modifications of the 1-2 exchanger with condensation in the shell. Such condensers have baffled tube bundles. The baffles do not affect the condensing film coefficients in horizontal condensers, since the coefficients are independent of the vapor mass velocity, but they do influence the accumulation of condensate on the tubes of vertical condensers. Moreover, in condensers with multi-pass tubes the tube-wall temperature is different at every point in each pass, whereas the surface temperature was assumed constant in the derivations. A correction for the latter cannot be accounted for in the calculations except by the treatment of small surface increments of each pass individually. The error introduced by using the mean tube-wall temperature as being effective over all the surface is apparently too small to justify the lengthier calculation. Since the baffle holes are ordinarily

¹ *Ibid.*, p. 270.

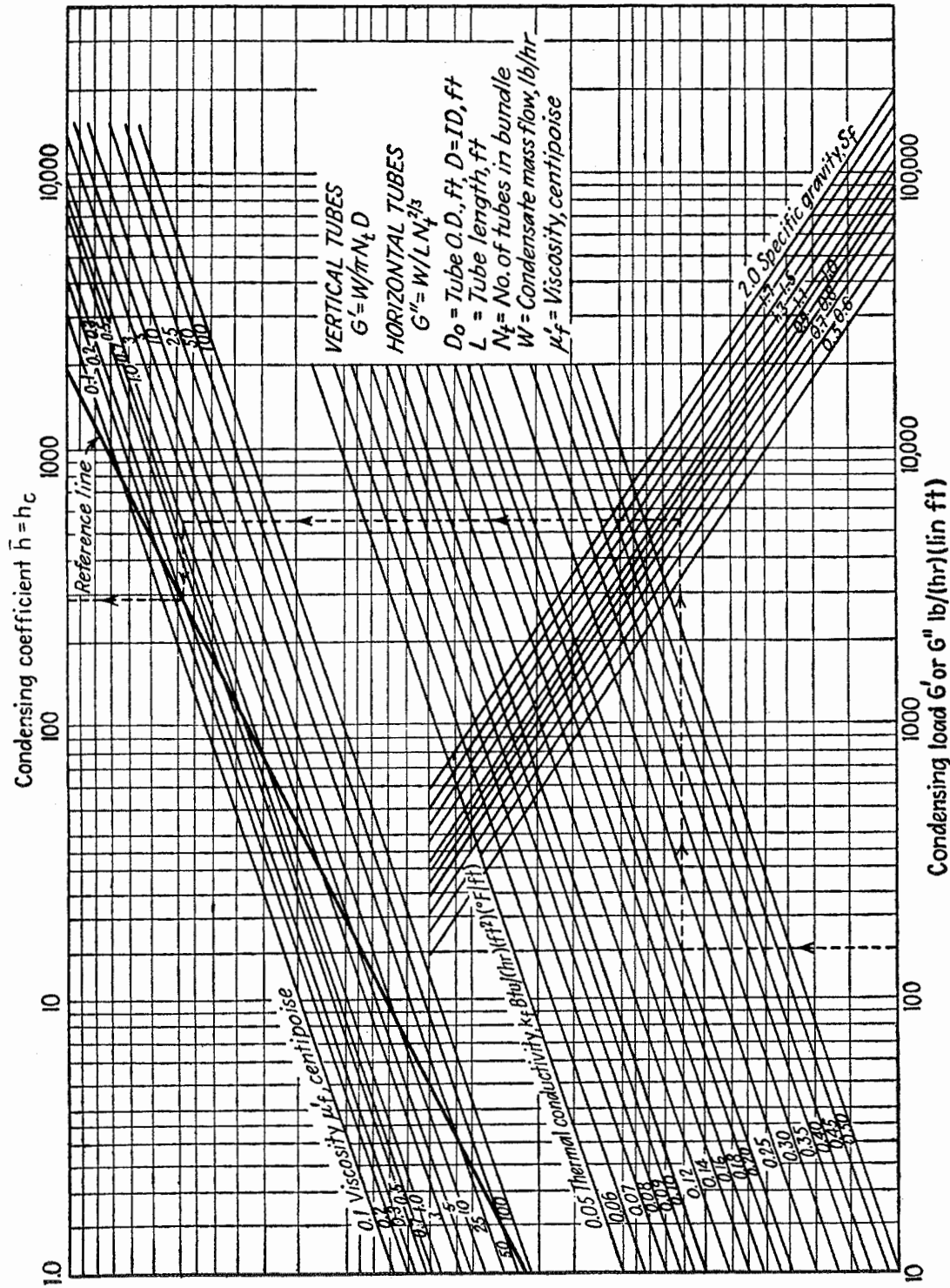


Fig. 12.9. Condensing coefficients. Graphical solution of Eq. (12.42).

about $\frac{1}{32}$ in. greater in diameter than the tube outside diameter, the baffles in vertical condensers prevent the condensate film from reaching a thickness greater than $\frac{1}{64}$ in. before impinging on the baffle. This, however, is a favorable limitation except at high tube loadings where the condensate film might otherwise grow sufficiently to change to turbulent flow.

Comparison between Horizontal and Vertical Condensation. The value of the condensing film coefficient for a given quantity of vapor on a given surface is significantly affected by the position of the condenser. In a vertical tube about 60 per cent of the vapor condenses in the upper half of the tube. By combining Eqs. (12.18) and (12.34) the ratio of the theoretical horizontal and vertical condensing coefficients is given by

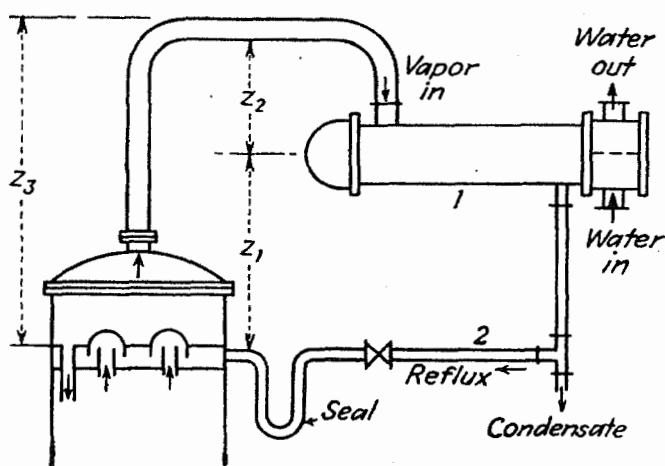


FIG. 12.10. Condenser with gravity return of reflux.

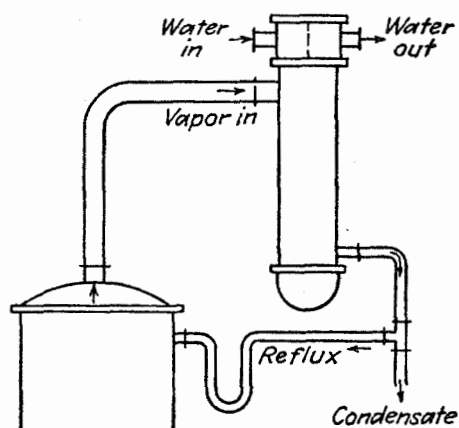


FIG. 12.11. Vertical condenser with condensation in shell and gravity return of reflux.

$(0.725/0.943)(L/D)^{1/4}$. For a $\frac{3}{4}$ in. OD tube 16'0" long, the horizontal coefficient should be 3.07 times as great as the vertical coefficient, provided that the condensate film is in streamline flow throughout. Ordinarily, however, the advantage is not quite so great due to the other modifications which prevail such as a transition of the vertical film into turbulent flow.

For the condensation of exhaust steam from turbines with vacuum discharge the condenser surface is usually very great, from 10,000 to 60,000 ft² per shell, and the economics are such that tubes up to 26 ft long are employed. These large condensers are designed with overall transfer coefficients as high as 800 Btu/(hr)(ft²)(°F) as treated later. Condensers for such services are universally installed in a horizontal position to facilitate the distribution of the vapor and the removal of the condensate.

When a condenser is employed on a distillation column, several specific factors must be taken into consideration. A typical arrangement of such

a condenser is shown in Fig. 12.10, in which the reflux is returned to the column by gravity. The condensate leg 1-2 of height z_1 , must supply sufficient hydrostatic head to return the condensate to the column through the seal. In Fig. 12.11 a vertical condenser is employed for the same service, but it is not well suited for gravity return of the condensate, since it must be elevated considerably above the column, which in many cases is very tall by itself. Maintenance and structural support for the vertical condenser may be costly and considerably more difficult. On the other hand if it is desired not only to condense the overhead vapor but also to subcool the condensate, the vertical condenser is admirably suited. *Subcooling* is the operation of cooling the condensate below its saturation temperature, and this is done very frequently when the overhead product is a volatile liquid to be sent to storage. By subcooling it is possible to avoid large evaporation losses during initial storage. The combination of condensation and subcooling in a single unit eliminates the need for a separate overhead product cooler, as shown in Fig. 12.1.

Condensation inside Tubes. *Horizontal Condensers.* The equations developed so far give excellent results when applied to condensation outside tubes, although the deviations in commercial condensers have not been reported except in isolated instances. Often, however, the condensate is corrosive, or it is desired to recover the latent heat from the vapor by using it to preheat the feed to a column. In such cases it may be preferable to condense the vapor in the tubes rather than on the tubes wherein the original derivation is no longer applicable. Within the tubes of a single-pass horizontal condenser each tube condenses an equal amount of vapor and there is no change in the coefficient due to the splashing of condensate from one tube row to another. As the condensate flows along the inside bottoms of the tubes, however, it builds up a thicker condensate resistance film than that anticipated in the derivation. Little is available of a theoretical nature to permit a rational analysis, but it has been found that the film coefficient may be safely computed by Eq. (12.40) when G'' , which is theoretically W/LN_t , is replaced by the fictitious loading

$$G'' = \frac{W}{0.5LN_t} \quad \text{lb/(hr)(lin ft)} \quad (12.45)$$

Equation (12.45) is especially useful when condensation occurs in the inner pipe of a double pipe exchanger. For condensation in the tubes of a condenser with multipass tubes it is preferable to compute the average film coefficient for each pass. The condensate formed in the first pass is carried through the second pass by one or more of the lower tubes in the pass, which may flow full of condensate and therefore expose no surface

for condensation. The calculation is then carried out by trial and error to determine the true tube loading in each pass.

Vertical Condensers. Condensation inside vertical tubes follows essentially the same mechanism as condensation outside vertical tubes if the interference of the shell baffles is neglected. Since the condensate film has the ability to grow continuously in its descent down the inside or outside of tubes, it may change from streamline to turbulent flow at some height between the top and the bottom. The local condensing coefficient decreases continuously from the top downward until at some point the film changes to turbulent flow. After the transition to turbulent flow the coefficient increases in accordance with the usual behavior of forced convection. By semiempirical means Colburn¹ has combined the effect

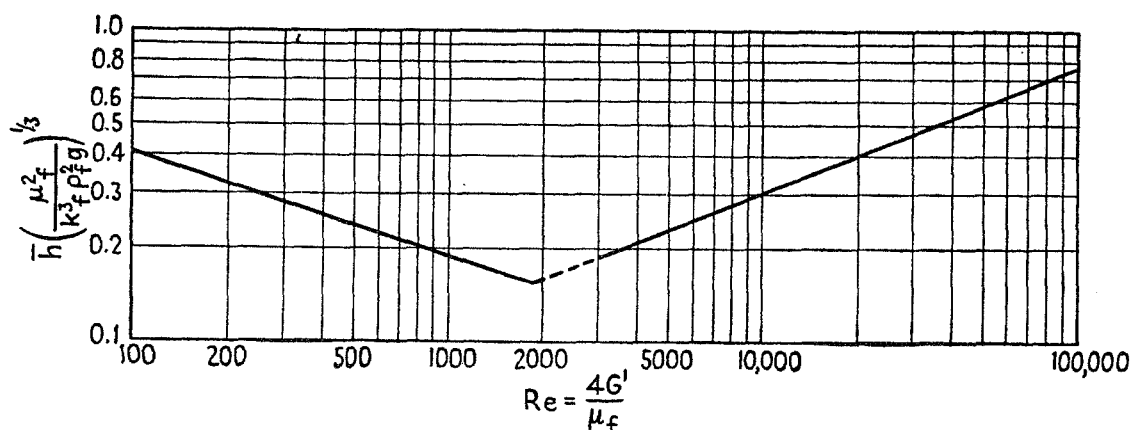


FIG. 12.12. Semiempirical condensation curve for vertical tubes. (Colburn, *Transactions of American Institute of Chemical Engineers*.)

of streamline flow in the upper portion of the tube with that of turbulent flow below the point where $4G'/\mu_f = 2100$. This required the selection of a heat-transfer factor for forced convection such that h at the transition point was roughly the same for both Nusselt condensation and turbulent flow. He then obtained the mean coefficient for the *entire height* of the tube by weighting the average coefficient for the upper portion of the tube and h for forced convection in the lower portion of the tube. The mean coefficients of condensation for the entire tube when $4G'/\mu_f > 2100$ are given in Fig. 12.12. This plot contains the values recommended by McAdams. The distance from the top of the tube at which the change from laminar to turbulent flow occurs may be obtained from Nusselt's semiempirical derivation based on the transition occurring at $4G'/\mu_f = 1400$ for steam and employing Eqs. (12.8) and (12.14). If x_c is the distance in feet from the top,

$$x_c = \frac{2668\lambda\mu_f^{3/2}}{\rho^{2/3}kg^{1/3}(T_v - t_w)} \quad \text{ft} \quad (12.46)$$

¹ Colburn, A. P., *Trans. AIChE*, **30**, 187-193 (1934).

Where there is evidence that the transition occurs at a higher value than $4G'/\mu_f = 1400$, as for organic vapors, the value of x_c should be multiplied by the ratio of the corrected value of $4G'/\mu_f$ divided by 1400.

Naturally if a vertical condenser is to operate with condensation inside the tubes, it will have but one tube pass as shown in Fig. 12.13.

Condenser Calculations. Condensers are better classified by what goes on inside them than by their process locations or services. Often in addition to condensation they may also perform vapor desuperheating or condensate subcooling so that a separate shell need not be employed for the sensible-heat transfer. A convenient classification in which each class is indicative of a different modification of the calculation is given below:

1. Single vapors (the vapor of a pure compound or a constant boiling mixture)
 - a. Saturated vapor: Total or partial condensation outside tubes
 - b. Superheated vapor: Desuperheating and condensation outside tubes
 - c. Saturated vapor: Condensation and subcooling outside tubes
 - d. Condensation inside tubes: Desuperheating, condensing, subcooling
 - e. Condensation of steam
2. Vapor mixtures (Chap. 13). The application of the phase rule.
 - a. Binary mixture
 - b. Vapor mixture with long condensing range
 - c. Vapor mixtures forming immiscible condensates
 - d. Single vapor or vapors with noncondensable gas
 - e. Vapor mixtures and noncondensable gases forming immiscible condensates

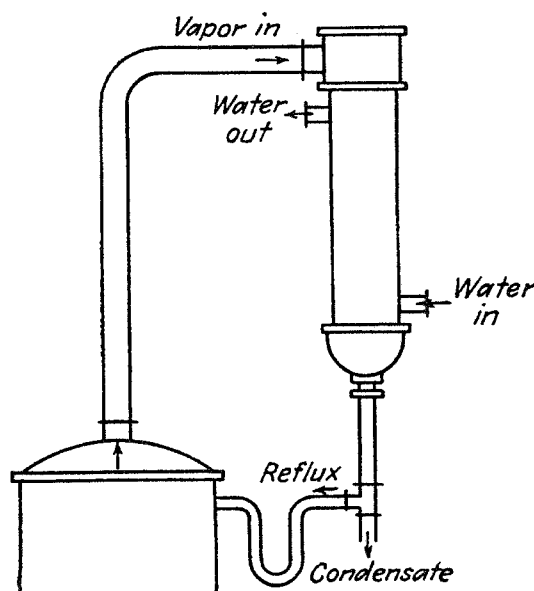


FIG. 12.13. Vertical condenser with condensation in the tubes and gravity return of reflux.

For cases 1a through 1d the majority of condensers are modifications of the 1-2 exchanger and may be referred to as 1-2 condensers. The use of a 1-2 exchanger as a condenser usually requires provision for a larger entrance space so that the vapor is not subjected to too great a pressure drop on entering the shell. This may be accomplished in either of three ways: The vapor may be introduced through a vapor belt as shown in Fig. 12.14 or by means of a flared shell inlet nozzle as shown in Fig. 12.15. A third method is the elimination of some of the tubes from the bundle which lie close to the inlet nozzle.

Although condensation reduces the volume of the vapor, it occurs at constant pressure except for the frictional pressure drop between the inlet and the outlet. In a horizontal condenser using conventional seg

mental baffles it is imperative that they be arranged for side-to-side flow and not up-and-down flow as shown in Fig. 7.6. This involves merely rotating the bundle 90° before bolting the channel to the shell flange. Failure to provide side-to-side flow causes pools of condensate to form between each pair of baffles whose cutout areas are on the top of the shell, thereby impeding the passage of vapor.

The Allowable Pressure Drop for a Condensing Vapor. In the original Nusselt assumptions the condensing coefficient was considered to be independent of the vapor velocity across the bundle of the condenser and to depend only upon G' or G'' , the loading as pounds of condensate per hour per linear foot. It is customary for the sake of good vapor distribution to have the vapor travel across the bundle as rapidly as the pressure-drop considerations will allow and to space the baffles accordingly. To account for the shrinkage in the total pounds of vapor as it travels the

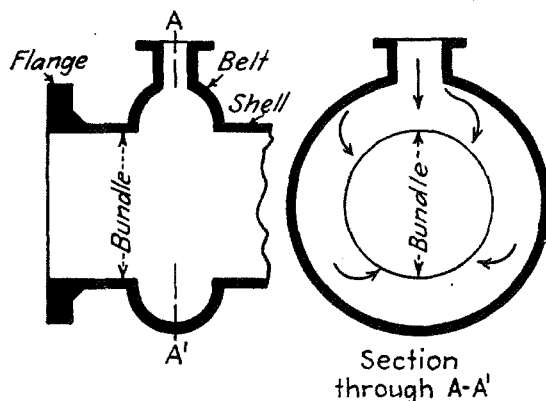


FIG. 12.14. Vapor-belt detail.

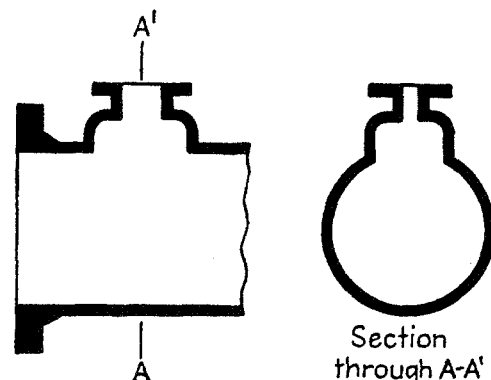


FIG. 12.15. Flared-nozzle detail.

length of the bundle, the baffle spacing may be *staggered* to give a nearly constant vapor mass velocity. At the inlet the baffles are spaced far apart but are spaced more closely toward the outlet. The use of staggered pitch is not universally accepted, since it reduces the adaptability of the equipment to other services in the event that the original process is discontinued. With uniform baffle pitch the condenser may be readily adapted to gas-liquid and liquid-liquid heat exchange.

In distillation the allowable pressure drop for a vapor coming off a column is usually very small if the condenser is installed for the gravity return of the reflux to the column. Referring to Fig. 12.10 the hydraulic circuit consists of the weight of the overhead vapor line $z_3 - z_2$, condenser pressure drop ΔP_c , the weight of the condensate leg z_1 , and the pressure drop in the condensate return line, which is usually neglected. The equation is given closely in pounds per square inch by

$$\frac{\rho_v z_1}{144} + \Delta P_c = \frac{\rho_l z_1}{144}$$

where ρ_v = density of vapor, lb/ft³

ρ_l = density of liquid, lb/ft³

ΔP_c = pressure drop in condenser, psi

Expansion and contraction losses from and to the tower have also been neglected. For all practical purposes the minimum available hydrostatic head must exceed $\rho_l z_1 = 144 \Delta P_c$. To this must usually be added an additional factor to permit the use of a flow-control valve in the reflux line. For a liquid with a specific gravity of 1.0 the minimum elevation of the condensate line above the level of the top plate for a 2.0 psi allowable pressure drop in the condenser is $(2.0/14.7) \times 34 = 4.62$ ft, and for liquids of lower specific gravity the elevation of the condenser will be proportionately greater. For this reason the condenser is frequently installed at the ground level and a condensate return pump is substituted for gravity return of reflux. This is especially true at higher pressures, where hydraulic control is more difficult and serious surging may occur at the top of the column. To prevent surging the condensate leaving the condenser first enters a condensate accumulator and then goes to the pump. With pumped return a vapor pressure drop of 5 psi is often allowed in the condenser. For gravity systems the allowable vapor pressure drop in the condenser will be usually 1 to 2 psi.

In the condensation of a pure saturated vapor, the vapor enters the condenser at its saturation temperature and leaves as a liquid. The pressure drop is obviously less than that which would be calculated for a gas at the inlet specific gravity of the vapor and greater than that which would be computed using the outlet specific gravity of the condensate. The mass velocity of inlet vapor and outlet liquid are, however, the same. In the absence of extensive correlations reasonably good results are obtained when the pressure drop is calculated for a mass velocity using the total weight flow and the average specific gravity between inlet and outlet. This method can be further simplified, as in the condensation of steam, by taking one-half the conventional pressure drop computed entirely on inlet conditions. Thus for condensation in the shell,

$$\Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_e s} \quad (12.47)$$

where s is the specific gravity of vapor. For condensation in tubes

$$\Delta P_t = \frac{1}{2} \frac{f G_t^2 L n}{5.22 \times 10^{10} D_e s} \quad (12.48)$$

where s is the specific gravity of the vapor. No contraction or expansion losses need be considered. Both Eqs. (12.47) and (12.48) are on the safe side, since the mass velocity of the vapor decreases nearly linearly

in the presence of a large Δt from inlet to outlet whereas the pressure drop decreases as the square of the velocity.

When circumstances make it difficult to meet the allowable drop through a 1-2 condenser, a lower pressure drop can be obtained by resorting to split flow as shown in Fig. 12.16. Still another arrangement

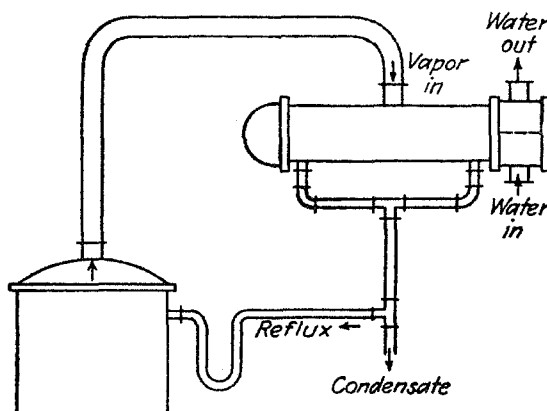


FIG. 12.16. Condenser with split flow and gravity return of reflux.

whose characteristics will be discussed later is the double-flow or *divided-flow* condenser shown in Fig. 12.17. This condenser consists of a conventional bundle with a removable longitudinal baffle and transverse support plate as well as smaller baffles to induce side-to-side flow of the vapor and condensate. The support plate, in addition to supporting all

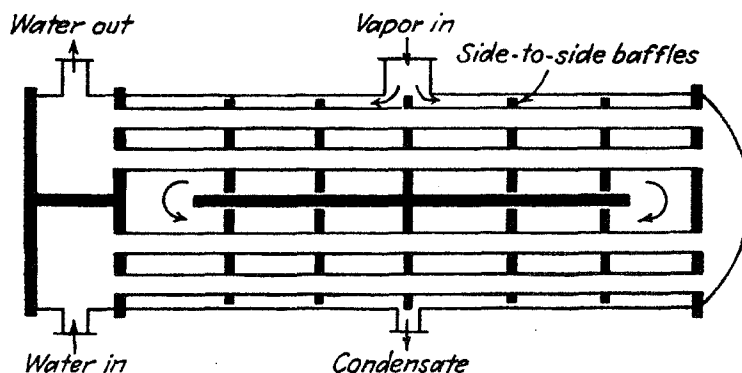


FIG. 12.17. Divided-flow 1-2 condenser.

the tubes, also serves to divide the flow. To prevent the condensate and vapor in the upper half of the shell from passing between the longitudinal baffle and the inside diameter of the shell, it is customary to provide sealing strips, which force the vapor and condensate to travel the half lengths of the divided flow condenser.

Example 12.1. Calculation of a Horizontal *n*-Propanol Condenser. A horizontal 1-2 condenser is required for the condensation of 60,000 lb/hr of substantially pure *n*-propanol (propyl alcohol) coming from the top of a distilling column operating at

15 psig, at which pressure it boils at 244°F. Water at 85°F will be used as the cooling medium. A dirt factor of 0.003 is required with allowable pressure drops of 2.0 psi for the vapor and 10.0 psi for the water.

Because of the location of the condenser, assume that 8'0" tubes are used. Tubes are to be 3/4 in. OD, 16 BWG on 1 5/16-in. triangular pitch.

Solution:

(1) Heat balance:

$$n\text{-Propanol, } Q = 60,000 \times 285 = 17,100,000 \text{ Btu/hr}$$

$$\text{Water, } Q = 488,000 \times 1(120 - 85) = 17,100,000 \text{ Btu/hr}$$

(2) Δt :

Hot Fluid		Cold Fluid	Diff.
244	Higher Temp	120	124
244	Lower Temp	85	159
0	Differences	35	35

$$\Delta t = \text{LMTD} = 141^\circ\text{F} \quad (5.14)$$

The exchanger is in true counterflow, since the shell-side fluid is isothermal.

(3) T_c and t_c : The influence of the tube-wall temperature is included in the condensing film coefficient. The mean $t_a = 102.5^\circ\text{F}$ can be used for t_c .

Trial:

(a) Assume $U_D = 100$: Condensing film coefficients will generally range from 150 to 300. Assuming a film coefficient of 1000 for water U_C will range from 130 to 230.

$$A = \frac{Q}{U_D \Delta t} = \frac{17,100,000}{100 \times 141} = 1213 \text{ ft}^2$$

$$\text{Number of tubes} = \frac{1213}{8'0'' \times 0.1963} = 773$$

(b) Assume four tube passes: The quantity of water is large, but the condenser will have a large number of tubes, making a two-pass assumption inadvisable.

From the tube counts (Table 9): 773 tubes, four passes, 3/4 in. OD on 1 5/16-in. triangular pitch

Nearest count: 766 tubes in a 31 in. ID shell

(c) Corrected coefficient U_D :

$$A = 766 \times 8'0'' \times 0.1963 = 1205 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{17,100,000}{1205 \times 141} = 101$$

Hot fluid: shell side, n-propanol

(4') Assume maximum baffle space. This will be 32 1/2, 31, and 32 1/2 in. equal to 96 in. or 2 baffles and 3 crosses for side-to-side flow.

$$\begin{aligned} a_s &= \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}] \\ &= 31 \times 0.1875 \times 31/144 \times 0.937 \\ &= 1.34 \text{ ft}^2 \end{aligned}$$

Cold fluid: tube side, water

$$\begin{aligned} (4) \text{ Flow area, } a_t' &= 0.302 \text{ in.}^2 \quad [\text{Table 10;} \\ a_t &= N_t a_t' / 144n \quad [\text{Eq. (7.48)}] \\ &= 766 \times 0.302 / 144 \times 4 = 0.402 \text{ ft}^2 \end{aligned}$$

Hot fluid: shell side, *n*-propanol

$$(5') G_s = W/a_s \text{ (for pressure drop only)} \quad [\text{Eq. (7.2)}]$$

$$= 60,000/1.34 = 44,700 \text{ lb/(hr)(ft}^2\text{)}$$

$$\text{Loading, } G'' = W/LN_i^{3/4} \quad [\text{Eq. (12.43)}]$$

$$= 60,000/8 \times 766^{3/4} = 89.3 \text{ (lb)/(hr)(lin ft)}$$

$$\text{Assume } \bar{h} = h_o = 200$$

$$\text{From (10) } h_{io} = 1075$$

$$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_v - t_a) \quad [\text{Eq. (5.31)}]$$

$$= 102.5 + 200/1275(244 - 102.5) = 125^\circ\text{F}$$

$$t_f = (T_v + t_w)/2 \quad [\text{Eq. (12.19)}]$$

$$= (244 + 125)/2 = 184.5^\circ\text{F}$$

$$k_f = 0.094 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)} \quad [\text{Table 4}]$$

$$s_f = 0.80 \quad [\text{Table 6}]$$

$$\mu_f = 0.62 \text{ cp} \quad [\text{Fig. 14}]$$

$$\text{From Fig. 12.9 or Eq. (12.42)}$$

$$\bar{h} = h_o = 172 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Cold fluid: tube side, water

$$(5) G_t = w/a_t = 488,000/0.402 = 1,210,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$\text{Vel, } V = G_t/3600\rho = 1,210,000/3600 \times 62.5 = 5.42 \text{ fps}$$

$$(6) \text{ At } t_a = 102.5^\circ\text{F,}$$

$$\mu = 0.72 \times 2.42 = 1.74 \text{ lb/(ft)(hr)}$$

$$D = 0.62/12 = 0.0517 \text{ ft} \quad [\text{Table 10}]$$

$$Re_t = DG_t/\mu \text{ (for pressure drop only)}$$

$$= 0.0517 \times 1,210,000/1.74 = 36,200$$

$$(9) h_i = 1300 \quad [\text{Fig. 25}]$$

$$(10) h_{io} = h_i \times \text{ID/OD} \quad [\text{Eq. (6.5)}]$$

$$= 1300 \times 0.62/0.75 = 1075 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Based on $\bar{h} = 172$ instead of the assumed 200 a new value of t_w and t_f could be obtained to give a more exact value of $\bar{h}$ based on fluid properties at a value of t_f more nearly correct. It is not necessary in this example because the condensate properties will not change materially.

Pressure Drop

$$(1') \text{ At } T_v = 244^\circ\text{F}$$

$$\mu_{\text{vapor}} = 0.010 \times 2.42 = 0.0242 \text{ lb/(ft)(hr)} \quad [\text{Fig. 15}]$$

$$D_s = 0.55/12 = 0.0458 \text{ ft} \quad [\text{Fig. 28}]$$

$$Re_s = D_s G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.0458 \times 44,700/0.0242 = 84,600$$

$$f = 0.00141 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 29}]$$

$$(2') \text{ No. of crosses, } N + 1 = 3$$

$$\text{Mol. wt.} = 60.1$$

$$\text{Density, } \rho = \frac{60.1}{359(704/492)(14.7/29.7)} = 0.238 \text{ lb/ft}^3$$

$$s = 0.238/62.5 = 0.00381$$

$$D_s = 3\frac{1}{2} = 2.58 \text{ ft}$$

$$(3') \Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s} \quad [\text{Eq. (12.47)}]$$

$$= \frac{1}{2} \frac{0.0014 \times 44,700^2 \times 2.58 \times 3}{5.22 \times 10^{10} \times 0.0458 \times 0.00381} = 1.2 \text{ psi}$$

$$(1) \text{ For } Re_t = 36,200,$$

$$f = 0.00019 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 26}]$$

$$(2) \Delta P_t = \frac{f G_t^2 L_n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00019 \times 1,210,000^2 \times 8 \times 4}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0} = 3.3 \text{ psi}$$

$$(3) \Delta P_r = (4n/s)(V^2/2g') \quad [\text{Eq. (7.46)}]$$

$$= (4 \times 4/1)0.20 = 3.2 \text{ psi}$$

$$(4) \Delta P_r = \Delta P_t + \Delta P_r \quad [\text{Eq. (7.47)}]$$

$$= 3.3 + 3.2 = 6.5 \text{ psi}$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1075 \times 172}{1075 + 172} = 148.5 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

(14) Dirt factor R_d :¹ U_D from (c) = 101

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{148.5 - 101}{148.5 \times 101} = 0.0032 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Summary

172	h outside	1075
U_C	148.5	
U_D	101	
R_d Calculated 0.0032		
R_d Required 0.003		
1.2	Calculated ΔP	6.5
2.0	Allowable ΔP	10.0

The first trial exchanger is satisfactory and will be

<i>Shell side</i>	<i>Tube side</i>
ID = 31 in.	Number and length = 766, 8'0"
Baffle space = 31 in. (approx.)	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, $1\frac{5}{16}$ -in. tri.
Passes = 1	Passes = 4

It is interesting at this point to compare a vertical condenser with the horizontal condenser which fulfilled the process conditions of Example 12.1. The horizontal and vertical condensing film coefficients are both affected by W and N_t , and the best basis for comparison is obtained when the number of tubes in both condensers is the same. To this end a vertical condenser will be assumed which employs the same tube count as that of the preceding example except that the tube length may be 12 or 16 ft as needed to account for the lower coefficients obtained in vertical condensation.

Example 12.2. Design of a Vertical *n*-Propanol Condenser. Process conditions will be taken from Example 12.1. To prevent corrosion of the shell the water will flow in the tubes.

Solution:

- (1) Heat balance: same as Example 12.1, $Q = 17,100,000$ Btu/hr
- (2) Δt : same as Example 12.1, $\Delta t = 141^\circ\text{F}$
- (3) T_c and t_c : same as Example 12.1

¹ In condensation calculations the omission of the resistance of the tube metal may introduce a significant error and should be checked.

Trial:

(a) Assume $U_D = 70$. The equation for the condensing film coefficient gives greater values for horizontal tubes than for vertical tubes. It will consequently be necessary to reduce the value of U_D .

$$A = \frac{Q}{U_D \Delta t} = \frac{17,100,000}{70 \times 141} = 1730 \text{ ft}^2$$

Nearest common tube length:

$$\frac{1730}{766 \times 0.1963} = 11.5 \text{ ft} \quad (\text{use } 12'0'')$$

(b) The layout of Example 12.1, using $\frac{3}{4}$ in. OD tubes on $\frac{15}{16}$ -in. triangular pitch and four passes will be retained for comparison.

(c) Corrected coefficient U_D :

$$A = 766 \times 12'0'' \times 0.1963 = 1805 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{17,100,000}{1805 \times 141} = 67.2$$

Hot fluid: shell side, n-propanol

(4') See (1') under Pressure Drop

$$D_o = 0.75/12 = 0.0625 \text{ ft}$$

(5') Loading, $G' = W/3.14N_t D_o$

[Eq. (12.36)]

$$= 60,000/3.14 \times 766 \times 0.0625$$

$$= 399 \text{ lb/(hr)(lin ft)}$$

Assume $\bar{h} = h_o = 100$

$$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_v - t_a) \quad [\text{Eq. (5.31)}]$$

$$= 102.5 + (100/1175)(244 - 102.5)$$

$$= 114.5^\circ\text{F}$$

$$t_f = \frac{1}{2}(T_v + t_w) \quad [\text{Eq. (12.19)}]$$

$$= (244 + 114.5)/2 = 179^\circ\text{F}$$

$$k_f = 0.0945 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$$

[Table 4]

$$s_f = 0.76$$

$$\mu'_f = 0.65 \text{ cp } (4G'/\mu = 1025) \quad [\text{Fig. 14}]$$

From Fig. 12.9 or Eq. (12.41),

$$\bar{h} = h_o = 102 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Cold fluid: tube side, water

(4)-(10) Same as Example 12.1

$$h_{io} = 1075$$

Pressure Drop

(1') It will be necessary to arrange the 12-ft bundle into a minimum number of bundle crosses or $N + 1 = 5$. The spacing will be

$$B = 14\frac{4}{5} = 29 \text{ in.}$$

$$a_s = ID \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= 31 \times 0.1875 \times 29/144 \times 0.937$$

$$= 1.25 \text{ sq ft}$$

(1) Same as Example 12.1 except for the tube length.

Pressure Drop

$$G_s = W/a_s \quad [\text{Eq. (7.2)}]$$

$$= 60,000/1.25 = 48,000 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$\text{At } T_v = 244^\circ\text{F}, \mu_{\text{vapor}} \quad [\text{Fig. 15}]$$

$$= 0.010 \times 2.42 = 0.0242 \text{ lb}/(\text{ft})(\text{hr})$$

$$D_e = 0.55/12 = 0.0458 \text{ ft} \quad [\text{Fig. 28}]$$

$$Re_s = D_e G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.0458 \times 48,000 / 0.0242 = 91,000$$

$$f = 0.0014 \text{ ft}^2/\text{in.}^2$$

$$(2') \text{ No. of crosses,}$$

$$N + 1 = 5 \quad [\text{Eq. (7.43)}]$$

$$s = 0.00381 \quad [\text{Example 12.1}]$$

$$D_s = 3/12 = 2.58 \text{ ft}$$

$$(3') \Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_{es}} \quad [\text{Eq. (12.47)}]$$

$$= \frac{1}{2} \times \frac{0.00142 \times 48,000^2 \times 2.58 \times 5}{5.22 \times 10^{10} \times 0.0458 \times 0.00381}$$

$$= 2.3 \text{ psi}$$

The pressure drop is high, and if it cannot be compensated for by elevating the condenser, it will be necessary to use the half-circle support baffles as in Example 7.8.

$$(2) \Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t}$$

$$= \frac{0.00019 \times 1,210,000^2 \times 12 \times 4}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0}$$

$$= 5.0 \text{ psi}$$

$$(3) \Delta P_r \text{ as in Example 12.1} = 3.2 \text{ psi}$$

$$(4) \Delta P_T = \Delta P_t + \Delta P_r$$

$$= 5.0 + 3.2 = 8.2 \text{ psi}$$

(13) Clean overall coefficient U_C :

$$U_C = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{1075 \times 102}{1075 + 102} = 93.2 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

(14) Dirt factor R_d : U_D from (c) = 67.2

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{93.2 - 67.2}{93.2 \times 67.2} = 0.00415 \quad (6.13)$$

Summary

102	h outside	1075
U_C	93.2	
U_D	67.2	
R_d Calculated 0.00415		
R_d Required 0.0030		
2.3	Calculated ΔP	8.2
2.0	Allowable ΔP	10.0

Discussion. The condenser is somewhat safe from a heat-transfer standpoint but exceeds the allowable pressure drop, although not seriously. The advantage of horizontal condensation may be observed from the U_C of 148.5 in the horizontal condenser as compared with 93.2 in a vertical condenser in an identical service. The vertical condenser has advantages, however, when the condensate is to be subcooled.

The final vertical condenser is

<i>Shell side</i>	<i>Tube side</i>
ID = 31 in.	Number and length = 766, 12'0"
Baffle space = 29 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16, $1\frac{5}{16}$ -in. tri.
Passes = 1	Passes = 4

Partial Condensers and the Balanced Pressure Drop. Sometimes it is desirable to condense only a portion of the vapor in a condenser such as may be needed only for reflux. Such a condenser is a *partial condenser* although the term *dephlegmator* was formerly used. The calculation of a partial condenser does not alter the method of computing the condensing film coefficient. The calculation of the pressure drop for a partial condenser is obtained with sufficient accuracy from the average of the pressure drops based on inlet and outlet conditions. The pressure drop for the inlet conditions has already been treated and for the outlet conditions it is obtained for practical purposes by computing the mass velocity and Reynolds number from the weight flow of vapor still in the vapor phase at the outlet.

A more precise calculation of the pressure drop involves an additional factor. During partial condensation in a horizontal condenser the vapor stream travels in the upper portion of the shell and the layer of condensate travels in parallel with it on the bottom of the shell. Both must traverse the length of the exchanger with the *same* pressure drop, since the terminal pressures for both are identical. This is called the condition of *balanced pressure drops*, and it applies whenever two fluids flow in parallel in the same channel. For condensation inside horizontal tubes, particularly in tube bundles, the pressure drop can be computed by trial and error by assuming the *average* segments of the pipe in which the condensate and vapor flow. The equivalent diameter can then be calculated for each portion from the free-flow area and the wetted perimeter. The mass velocity, Reynolds number, and pressure drops can similarly be calculated. If the pressure drops for the assumed division of the segments do not check, a new assumption must be made and the calculation repeated. The principle of balanced pressure drops is particularly important when it is desired to subcool the condensate to a very specific temperature and the area for sensible-heat transfer must be determined closely. When dealing with the shell side of 1-2 horizontal condensers, it is extremely difficult to predict accurately the effective flow area at the

bottom of the shell unless it can be determined by planimeter from the actual layout, although the principle of balanced pressure drop must also apply.

Influence of Impurities upon Condensation. In distillation operations the volatile component is always only partially separated from the less volatile components and the overhead product is never 100 per cent pure. Instead it may contain from a trace to a substantial concentration of the heavier components, and it does not condense isothermally except when the overhead is a constant boiling mixture or a mixture forming two immiscible liquids. When the temperature range over which the condensation of a mixture occurs is small, perhaps not exceeding 10 to 20°F, it may be treated as a pure compound with the true temperature difference being the LMTD for 1-1 condensers or $F_T \times \text{LMTD}$ for 1-2 condensers. The use of the conventional LMTD in either case assumes that the heat load removed from the vapor per degree of temperature decrease is uniform. Particularly where close approaches to the temperature of the cooling medium are involved, this may lead to serious error as demonstrated in Chap. 13. For the majority of services the assumption does not cause a serious error.

Another type of impurity causing a deviation from isothermal condensation is the presence of a trace of noncondensable gas, such as air, mixed with the vapor. A noncondensable gas is actually a superheated gas which is not cooled to its saturation temperature while the vapor itself is condensed. A common example is the presence of air in the condensation of steam. The presence of only 1 per cent of air by volume may reduce the condensing coefficient of steam by 50 per cent. The mechanism of the condensation becomes one of diffusion of the vapor through the air, the latter serving as a resistance to the transfer of heat. Under conditions of superatmospheric pressure there is little hazard that air will leak into a system except for the small quantities which may have been dissolved in the feed water before it was vaporized. In vacuum operation the possibility of air leakage into the system requires that provision be made for its continuous removal.

Condensation of a Superheated Vapor. The condensation of a superheated vapor differs from that of a saturated vapor in so far as there is sensible heat to be removed. The superheat may be the result of an additional heat absorption by a dry saturated vapor after removal from contact with the liquid from which it was formed or from passing a saturated vapor through a pressure-reducing valve. For the condensation of superheated steam McAdams¹ cites a number of investigators who have found that the entire superheated and condensing heat loads may be

¹ McAdams, *op. cit.*, p. 279.

considered transferred by the temperature difference between the steam saturation temperature and the tube wall temperature, or

$$Q = \bar{h}A(T_s - t_w) \quad (12.49)$$

where $\bar{h}$ is the condensing film coefficient and T_s is the saturation temperature of the superheated steam corresponding to its inlet pressure. This is actually a compromise rule. It simplifies the calculation by balancing the greater temperature difference that should be used for the desuperheating alone against a film coefficient which may be somewhat lower than the condensing coefficient. In the light of less conclusive evidence one may speculate on what possibility occurs in a desuperheater-condenser. Consider a horizontal true counterflow or 1-1 condenser with the vapor in the shell. When the superheated vapor enters at the hot end of the condenser, the temperature of the tube wall may be lower than the saturation temperature of the vapor. The superheated vapor contacting the tube wall condenses at its saturation temperature and may possibly even be subcooled. As the condensate drips from the tubes, it probably exchanges heat by reflashing into the hot superheated vapor at a great transfer rate, thereby effecting most of the desuperheating. Here, then, the desuperheating is probably controlled by the rate of subcooling, which should be relatively high for a film on a cold tube wall provided the condensate does not drain too rapidly to be subcooled.

It is also possible to have a reverse condition. If the cooling medium has been heated so that the tube-wall temperature near the hot terminal is greater than the saturation temperature, the tube wall will be dry toward the hot end and desuperheating will occur only as if the superheated steam were a dry gas. In this case it is necessary to divide the unit into a desuperheating and a condensing zone, and this method will be treated presently.

In the condensation of vapors other than steam it must be noted that a different relationship exists between the relative heat content of the superheat and the latent heat of evaporation. When steam at atmospheric pressure and superheated 100°F is condensed, the desuperheat represents slightly less than 5 per cent of the total heat load. If an organic vapor whose boiling point is in the same range, such as *n*-heptane, is superheated 100°F and then condensed, the desuperheat represents over 25 per cent of the heat load. Furthermore, water vapor has a very low density compared with organic vapors, and the mechanics of reflashing and diffusion in the vapor phase is probably more active for steam than for organic vapors. In condensers with multiple tube passes, condensation does not occur where the tube-wall temperature is higher than the saturation temperature of the vapor, so that it is possible to have only part of the surface wetted.

The Weighted Clean Coefficient and Temperature Difference in Desuperheater-Condensers. In establishing the true temperature difference for counterflow and 1-2 exchangers it was assumed that no partial phase change occurred in the exchanger. The desuperheater-condenser is a distinct violation of this assumption, requiring the development of a new method of computing the true temperature difference. In Fig. 12.18 the temperatures during the condensation of a pure vapor are shown. In Fig. 12.19 are shown the temperatures during the desuperheating and condensation of a superheated vapor. It is convenient to consider the

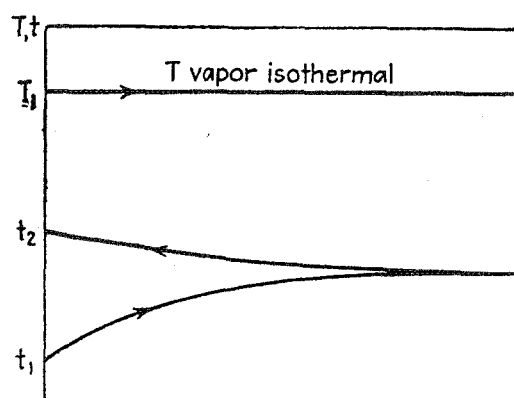


FIG. 12.18. Temperature distribution vs. tube length during isothermal condensation in a 1-2 condenser.

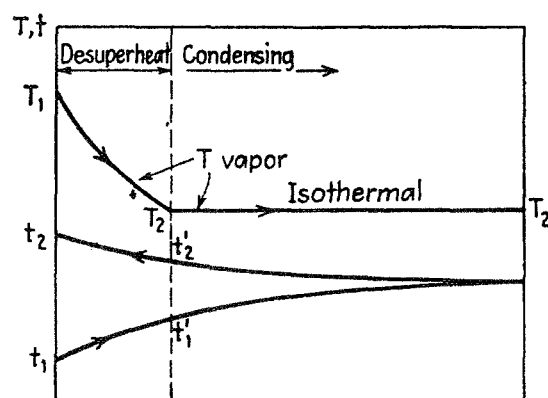


FIG. 12.19. Temperature distribution vs. tube length in a 1-2 desuperheater-condenser.

condenser as being subdivided into two zones *in series*, one for desuperheating and the other for condensation. The length L in the liquid condensate loading, $G'' = W/LN_i^{2/3}$, for a horizontal desuperheater-condenser is then the length of tube corresponding only to condensation and not the overall tube length. It will be seen that this requires an estimate of the length of tube corresponding only to condensation before $\bar{h}$ can be obtained. This difficulty in calculation does not arise in vertical units.

The temperature relations in the condensing zone of Fig. 12.19 are identical with those of a 1-2 exchanger, but those of the desuperheating zone differ from any type encountered previously. For a two-tube pass 1-2 desuperheater-condenser with desuperheating occurring at the channel end, the temperature t'_1 may be obtained from differential equations for the heat transfer over the two tube passes of the desuperheating zone. Then t'_1 is given by¹

$$t'_1 = \frac{1}{2} \left(\frac{W\lambda}{wc} + 2T_1 + t_1 - t_2 \right) \pm \frac{1}{2} \sqrt{\left(t_2 - t_1 - 2T_1 - \frac{W\lambda}{wc} \right)^2 - 4 \left[(2T_1 - t_2)t_1 + \left(\frac{WC}{wc} \right) T_1 \right]}$$

¹ Unpublished notes, D. Q. Kern and C. L. Carpenter.

By a heat balance t'_2 can also be found. For four or more tube passes the calculation of the temperature difference is considerably more involved but may be simplified semiempirically. For desuperheating and condensation in a counterflow apparatus no problem arises. In a 1-2 condenser if the condensation is isothermal or the condensing range is small, and if there is no temperature cross between the condensing range and the range of the cooling medium, it is possible, with small error, to consider the entire heat load delivered in counterflow regardless of the nozzle orientation. Temperature crosses in parallel flow-counterflow desuperheater-condensers should always be avoided if at all possible. Then the temperature rise of the water during condensation corresponds to the condensing heat load. This method is not so unsafe in the desuperheating zone as it appears if the tube-wall temperature at the hot terminal of the condenser is lower than the saturation temperature of the vapor. The calculation of the surface required for desuperheating by using a dry gas film coefficient without regard to possible desuperheating by reflash-ing in the desuperheating zone more than compensates for the error in the temperature difference encountered in real problems. If d and c indicate the desuperheating and condensing zones, respectively, and q is the heat transferred in each, the two zones may be computed by $q_d = U_d A_d (\Delta t)_d$ and $q_c = U_c A_c (\Delta t)_c$ where U_d and U_c are the clean overall coefficients and A_d and A_c the corresponding surfaces.

The use of the two zones permits the calculation of individual values of U_d and U_c , for each zone, but in the preceding examples the practice was established to judge the suitability of an exchanger by the size of the dirt factor, which was obtained from the difference between the clean and design overall coefficients. In each zone of a desuperheater-condenser there is a separate clean coefficient, and each is effective over an independent surface. The two overall coefficients may be replaced by a single value, the *weighted overall clean coefficient*, which is obtained from

$$U_c \text{ (weighted)} = \frac{\Sigma U_c A_c}{\Sigma A_c} = \frac{U_c A_c + U_d A_d}{A_c + A_d} \quad (12.50)$$

where U_c is the *weighted clean overall coefficient*.

To calculate a value of U_D from the Fourier equation $Q = U_D A \Delta t$ it is necessary first to obtain a single value for Δt . This may be weighted in terms of the heat transferred, respectively, by the temperature differences for condensation and desuperheating. Since $U_D A = Q/\Delta t$,

$$\Delta t \text{ (weighted)} = \frac{Q}{\sum \frac{q}{\Delta t}} = \frac{Q}{\frac{q_c}{(\Delta t)_c} + \frac{q_d}{(\Delta t)_d}} \quad (12.51)$$

where Δt is the weighted temperature difference and $(\Delta t)_c$ and $(\Delta t)_d$ are LMTD in counterflow. The assumptions in the weighting method should be completely understood before attempting the solution of problems. They serve as a consistent means, if not entirely precise, for comparing the performance and design of desuperheater-condensers.

In the calculation of desuperheater-condensers for the fulfillment of process conditions, it is difficult to establish a fast rule for assuming a reasonable value of U_D for the trial calculation. It is not only dependent upon the film coefficients of the superheated and saturated vapor but to an equal extent upon the distribution of the heat load between condensation and superheat which may have any value. By weighting U_D from the individual heat loads and anticipated individual design coefficients it permits a reasonably good approximation. Thus for a trial calculation

$$U_D \text{ (weighted)} = \frac{Q}{\frac{q_c}{(U_D)_c} + \frac{q_d}{(U_D)_d}} \quad (12.52)$$

Example 12.3. Calculation of a Butane Desuperheater-condenser. 27,958 lb/hr of isobutane with small amounts of *n*-butane issues from a gas reactor at 200°F and 85 psig. The gas becomes saturated at 130°F and is completely condensed at 125°F. Cooling is by well water from 65 to 100°F.

A minimum combined dirt factor of 0.003 is required along with allowable pressure drops of 2.0 psi for the butanes and 10.0 psi for the water.

Available for the service is a 23¼ in. ID horizontal 1-2 gas exchanger with 352 ¾ in. OD, 16 BWG tubes 16'0" long laid out on 1-in. triangular pitch for four passes. The baffles are spaced 12 in. apart.

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 23¼ in.	Number and length = 352, 16'0"
Baffle space = 12 in.	OD, BWG, pitch = ¾ in., 16 BWG, 1 in.-tri
Passes = 1	Passes = 4

(1) Heat balance:

$$\begin{aligned} \text{Butanes: Desuperheating range, } q_d &= 27,958 \times 0.44(200 - 130) \\ &= 860,000 \text{ Btu/hr} \end{aligned} \quad (\text{Fig. 5 or 9})$$

Condensing range (From Fig. 9):

$$\text{Enthalpy of } n\text{-butane vapor at 100 psia and 130°F} = 309 \text{ Btu/lb}$$

$$\text{Enthalpy of } n\text{-butane liquid at 100 psia and 125°F} = 170 \text{ Btu/lb}$$

$$q_c = 27,958(309 - 170) = 3,880,000 \text{ Btu/hr}$$

$$Q = 860,000 + 3,880,000 = 4,740,000 \text{ Btu/hr}$$

$$\text{Water, } Q = 135,500 \times 1(100 - 65) = 4,740,000 \text{ Btu/hr}$$

$$\Delta t_{\text{water}} = 3,880,000/135,500 = 28.7^\circ\text{F}$$

(2) Δt weighted:

Desuperheat $(\Delta t)_d$				Condensation $(\Delta t)_c$			
Hot Fluid		Cold Fluid	Diff.	Hot Fluid		Cold Fluid	Diff.
200	Higher Temp	100	100	130	Higher Temp	93.7	36.3
130	Lower Temp	93.7	36.3	125	Lower Temp	65.0	60.0
70	Differences	6.3	63.7	5	Differences	28.7	23.7

$$(\Delta t)_d = \text{LMTD} = 63.0^\circ\text{F} \quad (5.14)$$

$$\frac{q_d}{(\Delta t)_d} = \frac{860,000}{63.0} = 13,650$$

$$(\Delta t)_c = \text{LMTD} = 47.0^\circ\text{F} \quad (5.14)$$

$$\frac{q_c}{(\Delta t)_c} = \frac{3,880,000}{47} = 82,500$$

$$\text{Weighted } \Delta t = \frac{Q}{\Sigma q/\Delta t} = \frac{4,740,000}{13,650 + 82,500} = 49.3^\circ\text{F} \quad (12.51)$$

(3) T_c and t_c : Average values will be satisfactory.

<i>Hot fluid: shell side, butanes</i>		<i>Cold fluid: tube side, water</i>	
(4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)]		(4) $a'_t = 0.302 \text{ in.}^2$	[Table 10]
$= 23.25 \times 0.25 \times 12/144 \times 1.0$		$a_t = N a'_t/144n$	
$= 0.484 \text{ ft}^2$		$= 352 \times 0.302/144 \times 4 = 0.185 \text{ ft}^2$	
<i>Desuperheating (dry gas):</i>			
(5') $G_s = W/a_t$ [Eq. (7.2)]		(5) $G_t = w/a_t$	
$= 27,958/0.484$		$= 135,500/0.185$	
$= 57,800 \text{ lb}/(\text{hr})(\text{ft})$		$= 732,000 \text{ lb}/(\text{hr})(\text{ft}^2)$	
(6') At $T_a = 165^\circ\text{F}$, the mean for the superheated gas,		Vel, $V = G_t/3600\rho$	
$\mu = 0.01 \times 2.42 = 0.0242 \text{ lb}/(\text{ft})(\text{hr})$		$= 732,000/3600 \times 62.5 = 3.25 \text{ fps}$	
[Fig. 15]		(6) At $t_a = 82.5^\circ\text{F}$,	
$D_e = 0.73/12 = 0.0608 \text{ ft}$ [Fig. 28]		$\mu = 0.87 \times 2.42 = 2.11 \text{ lb}/(\text{ft})(\text{hr})$	
$Re_s = D_e G_s/\mu$		[Fig. 14]	
$= 0.0608 \times 57,800/0.0242 = 145,000$		$D = 0.62/12 = 0.0517 \text{ ft}$	
(7') $j_H = 239$ [Fig. 28]		$Re_t = D G_t/\mu$ (for pressure drop only)	
(8') At 165°F ,		$= 0.0517 \times 732,000/2.11 = 18,000$	
$k = 0.012 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$ [Table 5]			
$(c\mu/k)^{1/3} = (0.44 \times 0.0242/0.012)^{1/3}$			
$= 0.96$			
(9') $h_o = j_H \times \frac{k}{D_e} \left(\frac{c\mu}{k}\right)^{1/3}$ [Eq. (6.15b)]		(9) $h_i = 800$	[Fig. 25]
$= 239 \times 0.012 \times 0.96/0.0608$		$h_{io} = h_i \times \text{ID}/\text{OD}$	(6.5)
$= 45.2 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$		$= 800 \times 0.62/0.75$	
		$= 662 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$	

Clean overall coefficient U_d , desuperheat:

$$U_d = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{662 \times 45.2}{662 + 45.2} = 42.3 \quad (6.38)$$

Clean surface required for desuperheating:

$$A_d = \frac{q_d}{U_d(\Delta t)_d} = \frac{860,000}{42.3 \times 63.0} = 323 \text{ ft}^2$$

Condensation

(5') Assume condensation occurs over 60% of the tube length.

$$L_c = 16'0'' \times 0.60 = 9.6 \text{ ft}$$

$$G'' = W/LN_t^{2/3} = \frac{27,958}{9.6 \times 352^{2/3}} = 58.3 \text{ lb/(hr)(lin ft)} \quad [\text{Eq. (12.43)}]$$

Assume $\bar{h} = h_o = 200$;

From (9), $h_{io} = 662$

Average temp of condensation:

$$T_v = (130 + 125)/2 = 127.5^\circ\text{F}$$

Average temp during condensation

$$t_a = 82.5^\circ\text{F}$$

$$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_v - t_a) \quad [\text{Eq. (5.31)}]$$

$$= 82.5 + \frac{200}{662 + 200} (127.5 - 82.5)$$

$$= 93^\circ\text{F}$$

$$t_f = (T_v + t_w)/2 = (127.5 + 93)/2 = 110^\circ\text{F} \quad [\text{Eq. (12.19)}]$$

k_f is not given in Table 4 for butane or isobutane, but from the values for pentane and hexane it should be about 0.076 at 86°F and 0.074 at 140°F. A value of 0.075 will be used.

$$s_f = 0.55 \text{ (110}^\circ\text{API)} \quad [\text{Fig. 6}]$$

$$\mu_f' = 0.14 \text{ cp} \quad [\text{Fig. 14}]$$

$$\bar{h} = h_o = 207 \quad [\text{Fig. 12.9}]$$

The assumption of $\bar{h} = 200$ is satisfactory.

Clean overall coefficient U_c , condensation:

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{662 \times 207}{662 + 207} = 158 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

Clean surface required for condensation:

$$A_c = \frac{q_c}{U_c(\Delta t)_c} = \frac{3,880,000}{158 \times 47} = 523 \text{ ft}^2$$

Total clean surface required A_c :

$$A_c = A_d + A_c = 323 + 523 = 846 \text{ ft}^2$$

Check of assumed condensing length L_c :

$$\frac{A_c}{A_c + A_d} \times 100 = \frac{523}{846} \times 100 = 62\%$$

Assumed length = 60% (satisfactory)

(13) Weighted clean overall coefficient:

$$U_c = \frac{\Sigma UA_c}{\Sigma A_c} = \frac{42.3 \times 323 + 158 \times 523}{846} = 114 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (12.50)$$

(14) Design overall coefficient U_D :

$$\begin{aligned} a'' &= 0.1963 \text{ ft}^2/\text{lin ft} && \text{(Table 10)} \\ \text{Total surface} &= 352 \times 16'0'' \times 0.1963 = 1105 \text{ ft}^2 \\ U_D &= \frac{Q}{A \Delta t} = \frac{4,740,000}{1105 \times 49.3} = 87.2 \end{aligned}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_c - U_D}{U_c U_D} = \frac{114 - 87.2}{114 \times 87.2} = 0.0027 \quad (6.13)$$

Pressure Drop

(1') Desuperheat:

For $Re_s = 145,000$, $f = 0.0013 \text{ ft}^2/\text{in.}^2$

[Fig. 29]

(2') No. of crosses

$(L_d = 16'0'' \times 0.40 = 6.4 \text{ ft})$

$12L/B = 6.4 \times 12/12 = 6$ [Eq. (7.43)]

Mol. wt. = 58.1

$$\begin{aligned} \text{Density, } \rho &= \frac{58.1}{359(625/492)(14.7/99.7)} \\ &= 0.863 \text{ lb/ft}^3 \end{aligned}$$

$$s = 0.863/62.5 = 0.0138$$

$$D_s = 23.25/12 = 1.94 \text{ ft}$$

$$\begin{aligned} (3') \Delta P_s &= \frac{f G_s^2 D_s (N+1)}{5.22 \times 10^{10} D_s s \phi_s} \\ & \quad \text{[Eq. (7.44)]} \\ &= \frac{0.0013 \times 57,800^2 \times 1.94 \times 6}{5.22 \times 10^{10} \times 0.0608 \times 0.0138 \times 1.0} \\ &= 1.1 \text{ psi} \end{aligned}$$

(1') Condensation:

Use of the same Reynolds number will be satisfactory. $s = 0.0146$

(2') No. of crosses ($L_c = 9.6 \text{ ft}$)

$$N+1 = 12 \times 9.6/12 = 10 \quad \text{[Eq. (7.43)]}$$

$$\begin{aligned} (3') \Delta P_s &= \frac{1}{2} \frac{f G_s^2 D_s (N+1)}{5.22 \times 10^{10} D_s s} \\ & \quad \text{[Eq. (12.47)]} \\ &= \frac{1}{2} \times \frac{0.0013 \times 57,800^2 \times 1.94 \times 10}{5.22 \times 10^{10} \times 0.0608 \times 0.0146} \\ &= 0.90 \text{ psi} \end{aligned}$$

$$(4') \Delta P_s = 1.1 + 0.9 = 2.0 \text{ psi (total)}$$

(1) For $Re_t = 17,900$, $f = 0.0023 \text{ ft}^2/\text{in.}$
[Fig. 26]

$$\begin{aligned} (2) \Delta P_t &= \frac{f G_t^2 L_n}{5.22 \times 10^{10} D_s \phi_t} \quad \text{[Eq. (7.45)]} \\ &= \frac{0.0023 \times 730,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0} \\ &= 3.0 \text{ psi} \end{aligned}$$

$$\begin{aligned} (3) \Delta P_r &= (4n/s)(V^2/2g') \quad \text{[Eq. (7.46)]} \\ & \quad \text{[Fig. 27]} \\ &= (4 \times 4/1)0.075 = 1.2 \text{ psi} \end{aligned}$$

$$\begin{aligned} (4) \Delta P_T &= \Delta P_t + \Delta P_r \\ &= 3.0 + 1.2 = 4.2 \text{ psi} \end{aligned}$$

Summary

45.2/207	h outside	662
U_c	114	
U_D	87.2	
R_d Calculated 0.0027		
R_d Required 0.0030		
2.0	Calculated ΔP	4.2
2.0	Allowable ΔP	10.0

The Vertical Condenser-subcooler. It is often desirable to subcool a vapor to a temperature lower than the saturation temperature of the vapor. This occurs in distillation when the overhead product is volatile and it is desired to send it to storage at a lower temperature to prevent excessive evaporation losses. Vertical condensers are excellent for use as condenser-subcoolers whether they be of the 1-2 type as shown in Fig. 12.11 with condensation in the shell or the 1-1 type in Fig. 12.13 with condensation inside or outside the tubes.

If a saturated vapor passes into the shell of a vertical condenser, it is possible to divide it into two distinct zones operating in series, the upper for condensation and the lower for subcooling. This is accomplished by means of a *loop seal* as shown in Fig. 12.20. The object of the loop seal is to prevent the condensate from draining from the exchanger at so rapid a rate that it emerges without subcooling. Since the shell will flow full of liquid in the subcooling zone, when a loop seal is used, the film coefficients for subcooling can be computed through the use of Fig. 28 in the same manner employed heretofore for liquids. The same is true for the tube side of a 1-1 exchanger using tube-side heat transfer data, Fig. 24, for calculating coefficients.

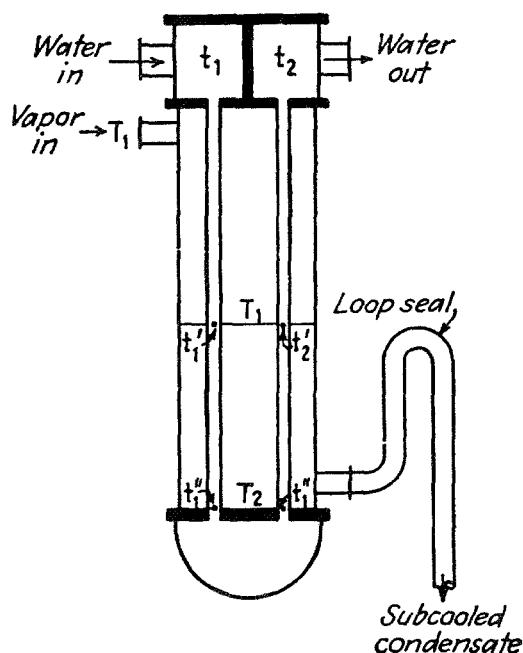


FIG. 12.20. Loop seal on vertical condenser-subcooler.

In establishing the true temperature difference for the condenser-subcooler in Fig. 12.20 there is no margin of safety such as occurs in the desuperheater-condenser. In the subcooling zone the temperatures are identical with a 1-2 exchanger having water inlet and outlet temperatures t'_1 and t'_2 , whereas in the condensing zone the pattern again differs. For a precise calculation of the true temperature difference it is necessary to estimate t'_1 and t'_2 and to solve for the two condensing passes separately. This entails a lengthy trial-and-error calculation. For four or more tube passes the solution is even lengthier.

The cooling medium is heated over a greater temperature range in the first condensing pass than in the second, so that $T_1 - t_1$ is greater than $T_1 - t_2$. The temperature t'_1 at the bottom of the subcooling zone is actually higher than the mean of t_1 and t_2 .

As in the case of the desuperheater-condenser, the calculation of the effective temperature difference may be simplified by means of a similar assumption. Avoid process conditions which involve a cross between the cooling medium outlet temperature and the subcooled condensate outlet temperatures. Then for 1-2 condensers with any even number of tube passes consider the cooling medium to be in counterflow and calculate the weighted temperature difference accordingly.

Example 12.4. Calculation of a Vertical Condenser-subcooler. 21,000 lb/hr of mixed *n*-pentane and *i*-pentane comes from a distilling column at 130°F and 25 psia and condenses completely at 125°F. The condensate is to be subcooled from 125 to 100°F for storage. Cooling will be effected by water from 80 to 100°F. A shell-side pressure drop of 2.0 psi is permissible for the vapor and 10.0 psi for water. A minimum fouling factor of 0.003 should be provided.

Available for the service is a 25 in. ID vertical 1-2 condenser containing 370 $\frac{3}{4}$ in. OD, 16 BWG, 16'0" tubes laid out on 1-in. square pitch. Shell baffles are 12 in. apart, and the bundle is arranged for four tube passes.

Will the condenser be a satisfactory vertical condenser-subcooler?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 25 in.	Number and length = 370, 16'0"
Baffle space = 12 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1 in.-square
Passes = 1	Passes = 4

(1) Heat balance:

Condensing range (130 to 125°F): (Data from Fig. 9).

Enthalpy of *n*-pentane vapor at 25 psia and 130°F = 315 Btu/lb

Enthalpy of *n*-pentane liquid at 25 psia and 125°F = 170 Btu/lb

$q_c = 21,000(315 - 170) = 3,040,000$ Btu/hr

Subcooling range (125 to 100°F):

$$q_s = 21,000 \times 0.57(125 - 100) = 300,000 \text{ Btu/hr}$$

$$Q = \Sigma q = 3,040,000 + 300,000 = 3,340,000 \text{ Btu/hr}$$

$$\text{Water, } Q = 167,000 \times 1(100 - 80) = 3,340,000 \text{ Btu/hr}$$

$$\text{During condensation } \Delta t_{\text{water}} = 3,040,000/167,000 = 18.2^\circ\text{F}$$

(2) Δt weighted:

$$\text{Condensing, } (\Delta t)_c = 36.4; q_c/(\Delta t)_c = 3,040,000/36.4 = 83,500 \text{ Btu/(hr)}(^\circ\text{F})$$

$$\text{Subcooling, } (\Delta t)_s = 30.2; q_s/(\Delta t)_s = 300,000/30.2 = 9,930$$

$$\underline{93,430 \text{ Btu/(hr)}(^\circ\text{F})}$$

$$\Delta t = Q / \sum \frac{q}{\Delta t} = \frac{3,340,000}{93,430} = 35.8^\circ\text{F} \quad (12.51)$$

(3) T_c and t_c : Average values will be satisfactory.

<i>Hot fluid: shell side, pentanes</i>	<i>Cold fluid: tube side, water</i>
(4') Condensation:	(4) $a'_t = 0.302 \text{ in.}^2$ [Table 10]
$D_o = 0.75/12 = 0.0625 \text{ ft}$	$a_t = N_t a'_t / 144n$ [Eq. (7.48)]
$G' = W/\pi N_t D_o$ [Eq. (12.42)]	$= 370 \times 0.302 / 144 \times 4 = 0.194 \text{ ft}^2$
$= 21,000 / 3.14 \times 370 \times 0.0625$	(5) $G_t = w/a_t$
$= 290 \text{ lb/(hr)}(\text{lin ft})$	$= 167,000 / 0.194$
Assume $\bar{h} = h_o = 125$	$= 860,000 \text{ lb/(hr)}(\text{ft}^2)$
Average temp of condensing vapor:	Vel, $V = G_t / 3600\rho = 860,000 / 3600$
$T_v = (130 + 125)/2 = 127.5^\circ\text{F}$	$\times 62.5$
$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_v - t_a)$ [Eq. (5.31)]	$= 3.84 \text{ fps}$
$= 90 + \frac{125}{777 + 125} (127.5 - 90) = 95^\circ\text{F}$	(9) $h_i = 940$ [Fig. 25]
$t_f = (T_v + t_w)/2 = (127.5 + 95)/2$	$h_{io} = h_i \times \text{ID}/\text{OD}$ [Eq. (6.5)]
$= 111^\circ\text{F}$ [Eq. (12.19)]	$= 940 \times 0.62/0.75$
$k_f = 0.077 \text{ Btu/(hr)}(\text{ft}^2)(^\circ\text{F}/\text{ft})$	$= 777 \text{ Btu/(hr)}(\text{ft}^2)(^\circ\text{F})$
[Table 4]	
$s_f = 0.60$ (92°API) [Fig. 6]	
$\mu_f = 0.19 \text{ cp}$ [Fig. 14]	
$\bar{h} = h_o = 120$ (vs. 125 assumed. t_f will not be changed materially by recalculation.)	

Clean overall coefficient for condensation U_c :

$$U_c = \frac{h_{io} h_o}{h_{io} + h_o} = \frac{777 \times 120}{777 + 120} = 104 \text{ Btu/(hr)}(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

Clean surface required for condensation A_c :

$$A_c = \frac{q_c}{U_c(\Delta t)_c} = \frac{3,040,000}{104 \times 36.4} = 803 \text{ ft}^2$$

(4') Subcooling:

$$a_s = ID \times C'B/144P_T \quad [\text{Eq. (7.1)}]$$

$$= 25 \times 0.25 \times 12/144 \times 1.0 = 0.521 \text{ ft}^2$$

$$(5') G_s = W/a_s \quad [\text{Eq. (7.2)}]$$

$$= 21,000/0.521$$

$$= 40,300 \text{ lb}/(\text{hr})(\text{ft}^2)$$

(6') At $T_a = 112.5^\circ\text{F}$,

$$\mu = 0.19 \times 2.42 = 0.46 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$D_e = 0.95/12 = 0.0792 \text{ ft}$$

$$Re_s = D_e G_s / \mu \quad [\text{Eq. (7.3)}]$$

$$= 0.0792 \times 40,300 / 0.46 = 6950$$

(7') $j_H = 46.5$ (8') At 112.5°F ,

$$k = 0.077 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft}) \quad [\text{Table 4}]$$

$$(c\mu/k)^{1/2} = (0.57 \times 0.46 / 0.077)^{1/2} = 1.51$$

$$(9') h_o = j_H \frac{k}{D_e} \left(\frac{c\mu}{k} \right)^{1/2} \quad [\text{Eq. (6.15b)}]$$

$$= 46.5 \times 0.077 \times 1.51 / 0.0792$$

$$= 68.0 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

$$(6) \text{ At } 90^\circ\text{F}, \mu = 0.82 \times 2.42 = 1.98$$

[Fig. 14]

$$D = 0.62/12 = 0.0517 \text{ ft}$$

$$Re_t = DG_t/\mu \text{ (for pressure drop only)}$$

$$= 0.0517 \times 860,000 / 1.98 = 22,500$$

$$(9) h_{io} = 777 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

Clean overall coefficient for subcooling U_s :

$$U_s = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{777 \times 68.0}{777 + 68.0} = 62.5 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

Clean surface required for subcooling A_s :

$$A_s = \frac{q_s}{U_s(\Delta t)_s} = \frac{300,000}{62.5 \times 30.2} = 159 \text{ ft}^2$$

Total clean surface required A_C :

$$A_C = A_e + A_s = 803 + 159 = 962 \text{ ft}^2$$

(13) Weighted overall clean coefficient U_C :

$$U_C = \frac{\Sigma U A_C}{\Sigma A_C} = \frac{104 \times 803 + 62.5 \times 159}{962} = 97.1 \quad (12.50)$$

(14) Design overall coefficient U_D :

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft} \quad (\text{Table 10})$$

$$\text{Total surface} = 370 \times 16'0'' \times 0.1963 = 1160 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{3,340,000}{1160 \times 35.8} = 80.5$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{97.1 - 80.5}{97.1 \times 80.5} = 0.0021 \text{ (hr)}(\text{ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Pressure Drop

Height of zones:

Condensing: $L_c = LA_c/A_c$

$$= 16 \times 803/962 = 13.4 \text{ ft}$$

(1') Condensation:

At $T_v = 127.5^\circ\text{F}$,

$$\mu = 0.0068 \times 2.42 = 0.0165 \text{ lb/(ft)(hr)}$$

[Fig. 15]

$$Re_s = D_s G_s / \mu$$

$$= 0.0792 \times 40,300 / 0.0165 = 193,000$$

$$f = 0.0012 \text{ ft}^2/\text{in.}^2$$

[Fig. 29]

$$\text{Mol. wt.} = 72.2$$

$$\rho_{\text{vapor}} = \frac{72.2}{359(590/492)(14.7/25)}$$

$$= 0.284 \text{ lb/ft}^3$$

$$s = 0.284/62.5 = 0.00454$$

(2') No. of crosses, $N + 1 = 12L/B$

$$= 12 \times 13.4/12 = 13.4, \text{ say } 14$$

$$D_s = 2\frac{5}{12} = 2.08 \text{ ft}$$

$$(3') \Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_{es}}$$

[Eq. (12.47)]

$$= \frac{1}{2} \times \frac{0.0012 \times 40,300^2 \times 2.08 \times 14}{5.22 \times 10^{10} \times 0.0792 \times 0.00454}$$

$$= 1.6 \text{ psi}$$

(4') ΔP of subcooling is negligible.(1) For $Re_t = 22,500$, $f = 0.00022 \text{ ft}^2/\text{in.}^2$ [Fig. 26]

$$(2) \Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00022 \times 860,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0}$$

$$= 3.9 \text{ psi}$$

$$(3) \Delta P_r = (4n/s)(V^2/2g') \quad [\text{Fig. 27}]$$

$$= (4 \times 4/1)0.10 = 1.6 \text{ psi}$$

[Eq. (7.46)]

$$(4) \Delta P_T = \Delta P_t + \Delta P_r \quad [\text{Eq. (7.47)}]$$

$$= 3.9 + 1.6 = 5.5 \text{ psi}$$

Summary

$12\frac{5}{8}$	h inside	777
U_C	97.1	
U_D	80.5	
R_d Calculated	0.0021	
R_d Required	0.003	
1.6	Calculated ΔP	5.5
2.0	Allowable ΔP	10.0

The dirt factor is too small to warrant installation of the unit.

The Horizontal Condenser-subcooler.¹ The horizontal condenser can also be equipped with a loop seal as shown in Fig. 12.21 to provide subcooling surface. This can also be accomplished by means of a *dam* baffle, as shown in Fig. 12.22. The loop seal has the advantage of external adjustability. In either case the flow of vapor is predominantly the same as in a condenser. The condensing and subcooling zones are in parallel instead of in series as in the vertical unit. This requires that the condensing vapor and the condensate both travel the length of the exchanger with a balanced pressure drop, and since the specific gravity of the condensate is so much greater than that of the vapor, the flow area required for the condensate must indeed be very small. The calculation of a horizontal condenser-subcooler brings up the problem of balancing

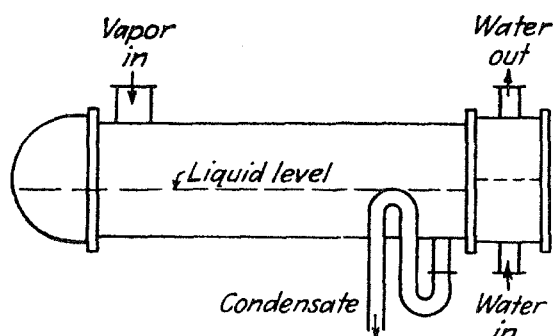


FIG. 12.21. Loop seal on horizontal condenser-subcooler.

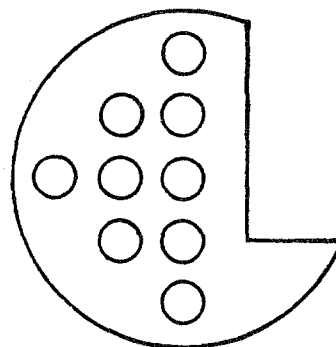


FIG. 12.22. Dam baffle.

the pressure drops and at the same time balancing the performance for the sensible and condensing heat loads to correspond to the assumed proportion of the bundle submerged for subcooling.

The calculation of the weighted clean overall coefficient to be employed here presupposes the presence of the two zones in parallel. It is further assumed that the surface for subcooling is not more than 50 per cent of the total surface. If subcooling represents more than 50 per cent, it would often be preferable to use a separate shell for subcooling alone where a higher condensate velocity can be attained. When the condensation surface requirements are more than 50 per cent of the total surface requirement, the vapor may be assumed to travel as in an ordinary condenser and its film coefficient is computed by Eq. (12.43), except that N_t in G'' is the number of tubes which are not submerged. The value of L used in calculating the vapor loading is the overall tube length. For simplicity, subcooling may be assumed to occur at a heat transfer coefficient corresponding to free convection although all the correlations for

¹ The method of computing this type of apparatus is arbitrary, although it gives overall coefficients which have proved satisfactory in a large number of cases.

free convection in Chap. 10 were for heating only. For light materials, such as organic solvents and petroleum fractions like kerosene and lighter, it is safe to assume a subcooling film coefficient of about 50, and for heavier condensates, such as aniline, straw oil, etc., it is about 25 or lower.

The calculation of the weighted temperature difference is also modified. In the orientation of the tube passes in a horizontal 1-2 condenser-subcooler the coldest pass is always placed in contact with the flooded portion of the shell, and it is a safe practice to consider the temperature rise of the cooling medium proportional to the transferred heat load. Thus if the subcooling heat load is 25 per cent of the total heat load, the cooling medium will rise the first 25 per cent of its temperature range while traveling through the subcooling passes.

The pressure drop in the shell should be computed on the basis of balancing the pressure drops in the two zones, but this requires a number of additional assumptions which cannot be entirely justified. Instead, for any given submergence, it is possible to approximate the pressure drop by calculating the average pressure drop for the vapors alone, using a segmental flow area equivalent to the vapor space and assuming that the pressure drop so calculated is identical with that of the condensate.

Example 12.5. Calculation of a Horizontal Condenser-subcooler. Using the same process conditions and condenser as in Example 12.4, which were unsatisfactory, what dirt factor would be achieved if the condenser-subcooler was operated horizontally?

It is understood that the baffle pitch will have to be somewhat greater if the unit is operated horizontally, since flooding the horizontal shell reduces the vapor flow area and therefore increases the pressure drop. But the baffle pitch is not regarded as influencing the transfer coefficients in the horizontal condenser-subcooler, and this example may be considered an illustration of the effectiveness of horizontal vs. vertical surface in condenser-subcoolers.

Solution:

Exchanger: Same as Example 12.4.

- (1) Heat balance: Same as Example 12.4.
- (2) Δt weighted:

Temperature rise of water during condensation = 18.2°F

Condensation (Δt_c)

Hot Fluid		Cold Fluid	Diff.
130	Higher Temp	100	30
125	Lower Temp	81.8	43.2

$$(\Delta t)_c = \text{LMTD} = 36.4^\circ\text{F}$$

[Eq. (5.14)]

Subcooling (Δt_s)

Hot Fluid		Cold Fluid	Diff.
125	Higher Temp	81.8	43.2
100	Lower Temp	80.	20.

$$(\Delta t)_s = \text{LMTD} = 30.2^\circ\text{F}$$

[Eq. (5.14)]

$$\text{Condensing, } (\Delta t)_c = 36.4 \quad \frac{q_c}{(\Delta t)_c} = \frac{3,040,000}{36.4} = 83,500 \text{ Btu/(hr)(}^\circ\text{F)}$$

$$\text{Subcooling, } (\Delta t)_s = 30.2 \quad \frac{q_s}{(\Delta t)_s} = \frac{300,000}{30.2} = 9,930$$

$$\quad \quad \quad 93,430 \text{ Btu/(hr)(}^\circ\text{F)}$$

$$\Delta t = Q / \sum \frac{q}{\Delta t} = \frac{3,340,000}{93,430} = 35.8^\circ\text{F} \quad (12.51)$$

(3) T_c and t_c : Average values will be satisfactory.

Trial:

Assume shell is flooded to a height of $0.3D_s$. Originally subcooling represented $15\frac{9}{62} = 16.5$ per cent of the total surface. For the horizontal condenser $\bar{h}$ will be much higher than for the vertical, hence about 25 per cent of the surface will be required for subcooling.

From mathematical tables determine the area of shell cross section which is flooded a'_s to obtain the number of submerged tubes.

For $0.3D_s$, $C_1 = 0.198$ in the formula

$$a'_s = C_1 D_s^2 = 0.198 \times 25^2 = 124 \text{ in.}^2$$

$$\text{Number of submerged tubes} = 370 \times \frac{124}{(\pi/4) \times 25^2} = 93 \text{ (approx.)}$$

$$\text{Number of tubes for condensation} = 370 - 93 = 277$$

$$\text{Flooded surface} = (93/370)100 = 25\%$$

Condensation:

$$G'' = \frac{W}{LN_t^{2/3}} = \frac{21,000}{16 \times 277^{2/3}} = 30.9 \quad (12.43)$$

Assume the same film temperature as before.

$$\bar{h} = h_o = 251$$

(Fig. 12.9)

$$h_{io} = 777 \text{ from Example 12.4}$$

Clean overall coefficient for condensation U_c :

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{777 \times 251}{777 + 251} = 190 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

Clean surface required for condensation A_c :

$$A_c = \frac{q_c}{U_c(\Delta t)_c} = \frac{3,040,000}{190 \times 36.4} = 440 \text{ ft}^2$$

Subcooling: (Free-convection rate assume $h = 50$):

Clean overall coefficient for subcooling U_s :

$$U_s = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{777 \times 50}{777 + 50} = 47.0 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.9)$$

Clean surface required for subcooling A_s :

$$A_s = \frac{q_s}{U_s(\Delta t)_s} = \frac{300,000}{47.0 \times 30.2} = 211 \text{ ft}^2$$

Total clean surface required A_C :

$$A_C = A_c + A_s = 440 + 211 = 651 \text{ ft}^2$$

(13) Weighted clean overall coefficient U_C :

$$U_C = \frac{\Sigma UA}{\Sigma A_C} = \frac{190 \times 440 + 47.0 \times 211}{651} = 144 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (12.50)$$

(14) Design overall coefficient U_D :

$$U_D = \frac{Q}{A \Delta t} = \frac{3,340,000}{1160 \times 35.8} = 80.5 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

(15) Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{144 - 80.5}{144 \times 80.5} = 0.0054 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Pressure Drop

It will be necessary to spread the baffles to a spacing of 18 in. to compensate for the reduction in crossflow area due to the flooded subcooling zone. The tube-side pressure drop will be the same as before. Assume bundle flooded to $0.3D_s$.

$$a_s = 0.7 \times \text{ID} \times C' \frac{B}{144 P_T} = 0.7 \times 25 \times 0.25 \times \frac{18}{144 \times 1.0} = 0.547 \text{ ft}^2 \quad (7.1)$$

$$G_s = \frac{W}{a_s} = \frac{21,000}{0.547} = 38,400 \text{ lb/(hr)(ft}^2\text{)} \quad (7.2)$$

$$Re_s = \frac{D_s G_s}{\mu} = 0.0792 \times \frac{38,400}{0.0165} = 185,000 \quad (7.3)$$

$$f = 0.00121 \text{ ft}^2/\text{in.}^2 \quad (\text{Fig. 29})$$

$$\text{Number of crosses, } N + 1 = \frac{12L}{B} = 12 \times \frac{16}{18} = 11 \quad (7.43)$$

$$D_s = 2.08 \text{ ft}$$

$$\Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_{es}} = \frac{1}{2} \frac{0.00121 \times 38,400^2 \times 2.08 \times 11}{5.22 \times 10^{10} \times 0.0792 \times 0.00454} = 1.0 \text{ psi} \quad (12.47)$$

Summary

25 1/50	<i>h</i> inside	777
U_C	144	
U_D	80.5	
R_d Calculated 0.0054		
R_d Required 0.003		
1.0	Calculated ΔP	5.5
2.0	Allowable ΔP	10.0

Remarks. The horizontal condenser-subcooler should be equipped with a loop seal or dam baffles approximately one-third the height of the shell.

It is possible to draw some general conclusions from the summaries of Examples 12.4 and 12.5 which shed light on the relative orders of magnitude between horizontal and vertical condenser-subcoolers. The vertical condenser has the advantage of well-defined zones, but it is restricted by the height available at the top of a column when employed with a column and the smaller size of the condensing film coefficient. On the other hand, the horizontal condenser-subcooler, though less adaptable to close calculations, gives considerably higher overall clean coefficients. The majority of the condenser-subcoolers used in industry are consequently of the horizontal type.

Vertical Refluxing inside Tubes.

It is often necessary to treat a solid material with a volatile boiling liquid or to maintain a liquid mixture at

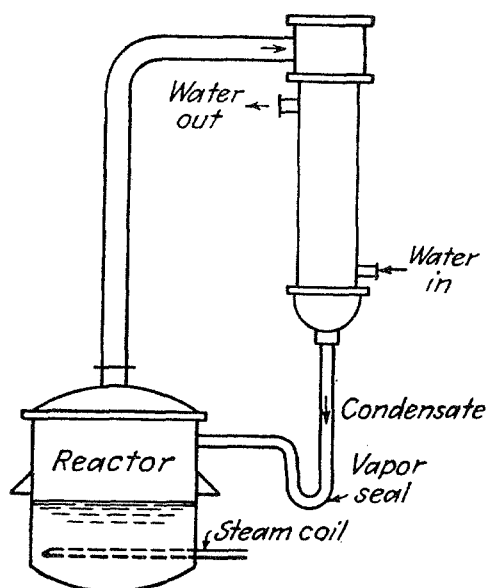


FIG. 12.23. Reflux-type condenser.

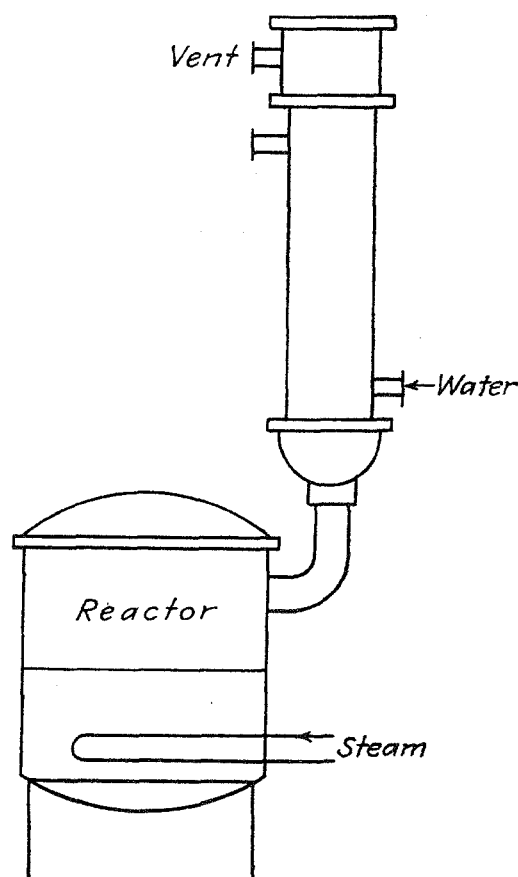


FIG. 12.24. Knock-back condenser.

its boiling point so as to complete a chemical reaction. To maintain a constant pressure the volatile liquid is boiled overhead in the reactor and returned at a continuous rate as shown in Fig. 12.23. The rate at which the volatile liquid is boiled overhead is called the *boil-up rate* and is usually expressed in pounds per hour per square foot of liquid surface. A condenser operated in a closed circuit in this manner is a *reflux-type* condenser. When the vapor enters the bundle at the bottom as in Fig. 12.24 it is a special case of the reflux type known as a *knock-back* condenser. The reflux type of condenser is not intended to produce subcooling, and the condensate drains back freely into the treating vessel. Very often the boil-up rate is so great that the condensing film may be

partially in turbulent flow, whence the film coefficient is calculated from Colburn's relationship as given in Fig. 12.12.

Greater headroom is required for a knock-back condenser, although it has some advantage over a horizontal condenser due to its excellent drainage. If the Reynolds number for a vertical condenser, $4G'/\mu_f$, is less than 1800 to 2100 corresponding to the transition point in Fig. 12.12, all the condensation will occur in streamline flow as computed by Eq. (12.42). If the Reynolds number is greater than about 1800, the upper surface where the condensate film is in streamline flow may be computed separately from that below it in turbulent flow. Knock-back condensers differ from reflux-type condensers, since the vapor enters at the bottom and may not, in fact, even reach the top.

Example 12.6. Calculation of Vertical Reflux-type CS₂ Condenser. A solid material is treated in a vessel with carbon disulfide and traces of corrosive sulfur compounds at 25 psig corresponding to a boiling point of 176°F and a total boil-up of 30,000 lb/hr. A negligible pressure drop will be required for the tube side, and 10 psi is available for the water, which enters at 85°F. A combined dirt factor of 0.003 will be necessary

Available for the purpose is 17¼ in. ID 1-1 exchanger having 177 ¾ in. OD, 16 BWG tubes 16'0" long. They are laid out for 1-in. square pitch, and the baffle spacing is 6 in.

Will the exchanger be satisfactory?

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 17¼ in.	Number and length = 177, 16'0"
Baffle space = 6 in.	OD, BWG, pitch = ¾ in., 16 BWG, 1-in. square
Passes = 1	Passes = 1

(1) Heat balance:

Carbon disulfide, $Q = 30,000 \times 140 = 4,200,000$ Btu/hr (Fig. 12)

Water, $Q = 120,000 \times 1(120 - 85) = 4,200,000$ Btu/hr

(2) Δt : $\Delta t = \text{LMTD} = 72.1^\circ\text{F}$ (5.14)

(3) T_c and t_c : The use of average temperatures will be satisfactory.

In this particular problem it will be advisable to compute the shell side first, since the water rate is needed to establish the tube-wall temperature for condensation.

<i>Hot fluid: tube side, carbon disulfide</i>	<i>Cold fluid: shell side, water</i>
Assume $h_{io} = 300$	(4') $a_s = \text{ID} \times C'B/144P_T$ [Eq. (7.1)]
$t_w = t_a + \frac{h_{io}}{h_{io} + h_o} (T_v - t_a)$	$= 17.25 \times 0.25 \times 6/144 \times 1.0$
$= 102.5 + 30\%_{1086} (176 - 102.5)$	$= 0.18 \text{ ft}^2$
$= 122.5^\circ\text{F}$	(5') $G_s = w/a_s$ [Eq. (7.2)]
$t_f = (122.5 + 176)/2 = 149^\circ\text{F}$	$= 120,000/0.18$
[Eq. (12.19)]	$= 667,000 \text{ lb}/(\text{hr})(\text{ft}^2)$

<i>Hot fluid: tube side, carbon disulfide</i> $\mu_f = 0.28 \times 2.42 = 0.68 \text{ lb/(ft)(hr)}$ [Fig. 24] $k_f = 0.09 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$ [Fig. 14] $s_f = 1.26; \rho_f = 62.5 \times 1.26 = 78.8 \text{ lb/ft}^3$ $D = 0.62/12 = 0.0517 \text{ ft}$ $G' = W/\pi N_i D$ [Eq. (12.42)] $= 30,000/3.14 \times 177 \times 0.0517$ $= 1045 \text{ lb/(hr)(lin ft)}$ $Re_i = 4G'/\mu_f$ [Eq. (12.37)] $= 4 \times 1045/0.68 = 6150$ The film will be in turbulent flow. $\bar{h}(\mu_f^2/k_f^3\rho_f^2g)^{1/4} = 0.251$ [Fig. 12.12] $h_i = \bar{h} = 0.251 \left(\frac{k_f^3\rho_f^2g}{\mu_f^2} \right)^{1/4}$ $= 0.251 \left(\frac{0.093^3 \times 78.8^2 \times 4.17 \times 10^8}{0.68^2} \right)^{1/4}$ $= 400$ $h_{io} = 400 \times 0.62/0.75$ $= 331 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$ [Eq. (6.5)]	<i>Cold fluid: shell side, water</i> (6') At $t_a = 102.5^\circ\text{F}$, $\mu = 0.70 \times 2.42 = 1.70 \text{ lb/(ft)(hr)}$ [Fig. 14] $D_s = 0.95/12 = 0.0792 \text{ ft}$ [Fig. 28] $Re_s = D_s G_s/\mu$ [Eq. (7.3)] $= 0.0792 \times 667,000/1.70 = 31,000$ (7') $j_H = 103$ [Fig. 28] (8') At $t_a = 102.5^\circ\text{F}$, $k = 0.36 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$ $(c\mu/k)^{1/2} = (1 \times 1.70/0.36)^{1/2} = 1.68$ (9') $h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k} \right)^{1/2}$ [Eq. (6.15b)] $= 103 \times 0.36 \times 1.68/0.0792$ $= 786 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$
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Clean overall coefficient U_C :

$$U_C = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{331 \times 786}{331 + 786} = 233 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)} \quad (6.38)$$

Design overall coefficient U_D :

$$\text{Total surface} = 177 \times 16'0'' \times 0.1963 = 556 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{4,200,000}{556 \times 72.1} = 105 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F)}$$

Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{233 - 105}{233 \times 105} = 0.00522 \text{ (hr)(ft}^2\text{)(}^\circ\text{F)/Btu} \quad (6.13)$$

Pressure Drop

(1) Flow area: $a'_i = 0.302 \text{ in.}^2$ [Table 10] $a_i = N_i a'_i/144n$ [Eq. (7.48)] $= 177 \times 0.302/144 \times 1 = 0.372 \text{ ft}^2$ (2) $G_i = W/a_i$ [Eq. (6.3)] $= 30,000/0.372$ $= 80,500 \text{ lb/(hr)(ft}^2\text{)}$ At inlet, $\mu = 0.012 \times 2.42 = 0.029 \text{ lb/(ft)(hr)}$ [Fig. 15] $D = 0.62/12 = 0.0517 \text{ ft}$ $Re = 0.0517 \times 80,500/0.029 = 143,000$ $f = 0.000138 \text{ ft}^2/\text{in.}^2$ [Fig. 26] Mol. wt. = 76.1 $\rho = \frac{76.1}{359(636/492)(14.7/39.7)} = 0.443 \text{ lb/ft}^3$	(1') For $Re_s = 31,000$ $f = 0.0017 \text{ ft}^2/\text{in.}^2$ [Fig. 29] (2') No. of crosses, $N + 1 = 12L/B$ [Eq. (7.43)] $= 12 \times 16/6 = 32$ $D_s = 17.25/12 = 1.44 \text{ ft}$
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Pressure Drop

$$s = 0.443/62.5 = 0.0071$$

$$(3) \Delta P_t = \frac{1}{2} \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s} \quad [\text{Eq. (12.48)}]$$

$$= \frac{0.000138 \times 80,500^2 \times 16}{5.22 \times 10^{10} \times 0.0517 \times 0.0071}$$

$$= 0.4 \text{ psi}$$

$$(3') \Delta P_s = \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s \phi_s} \quad [\text{Eq. (7.44)}]$$

$$= \frac{0.0017 \times 667,000^2 \times 1.44 \times 32}{5.22 \times 10^{10} \times 0.0792 \times 1.0 \times 1.0}$$

$$= 8.4 \text{ psi}$$

Summary

331	<i>h</i> outside	786
U_C	233	
U_D	105	
R_a Calculated 0.00522		
R_a Required 0.004		
0.4	Calculated ΔP	8.4
neg.	Allowable ΔP	10.0

The unit is satisfactory.

THE CONDENSATION OF STEAM

The Surface Condenser. Any saturated vapor can be condensed by a direct spray of cold water under appropriate conditions of temperature and pressure provided contamination of the condensate by water is not objectionable. Steam, on the other hand, as generated in power stations, is an extremely pure form of water substantially free of the impurities causing scale. The term *surface condenser* is reserved for tubular apparatus employed for the condensation of steam.

In the design and operation of a steam turbine or engine the exhaust temperature of the heat is kept as low as possible so that a maximum change in enthalpy occurs during the conversion of heat into work. This is a natural deduction of the Carnot cycle. The exhaust temperature is limited only by the coldness and abundance of the cooling medium and the allowance of a reasonable temperature difference. The economics and optimum energy distribution of power cycles is beyond consideration here, but with cooling water available at 70°F or thereabouts a turbine

will exhaust at 75°F, which corresponds to a vacuum saturation pressure. Consequently, such apparatus is usually designed for vacuum operation on the steam side.

The surface condenser, as a development of the power rather than chemical industry, is handled differently from preceding illustrations, and the purpose here is to indicate the exceptions. Equations (12.44a) and (12.44b) for the condensation of steam at atmospheric pressures have

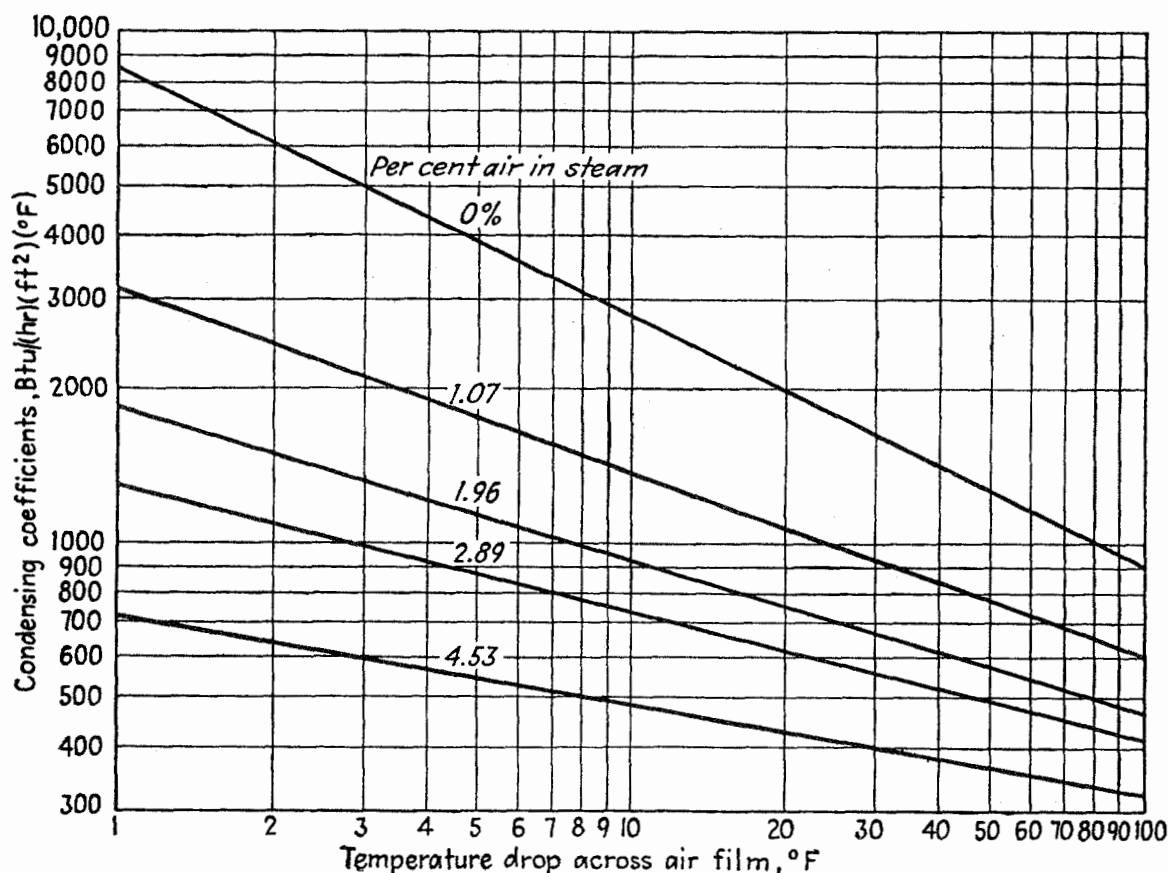


FIG. 12.25. Influence of air on the condensing coefficient of 230°F steam. (Data of Othmer.)

already been presented. These equations give steam-condensing coefficients at atmospheric pressure and are of little use in estimating the condensation rate under a vacuum of 1 to 1½ in. Hg abs. Furthermore, air dissolved in the boiler feed water, despite deaeration, tends to accumulate in the condenser, where it impedes heat transfer. Othmer¹ has shown that, when as little as 1 per cent of air by volume is mixed with the inlet steam, the condensing coefficient falls from 2000 to 1100 with a temperature difference between the steam and cooling medium of 20°F. When the air concentration is 2 per cent by volume, the condensing coefficient falls from 2000 to 750 at the same temperature difference. It is also found that during steady-state transfer the air tends to surround the

¹ Othmer, D. F., *Ind. Eng. Chem.*, **21**, 576 (1929).

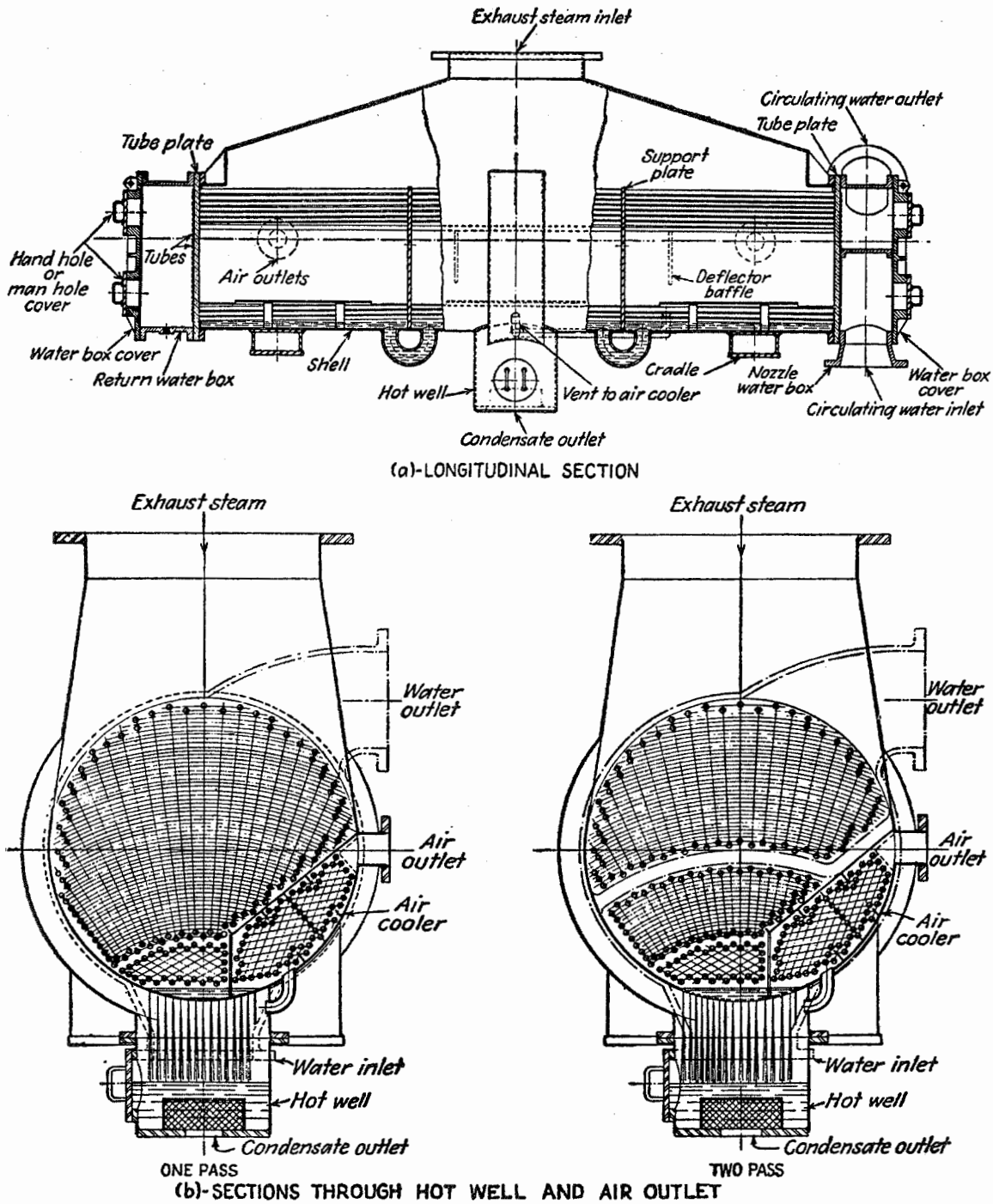


FIG. 12.26. Surface condenser. (Foster Wheeler Corporation.)

tube surface, setting up a resistance through which the steam must diffuse. Othmer's data are reproduced in Fig. 12.25.

Surface condensers are usually much larger than other types of tubular equipment, some containing over 60,000 ft² of condensing surface. The attainment of a very small steam-side pressure drop directly affects the pressure of the condenser exhaust and the cycle efficiency for a given cooling-water inlet temperature. A typical small surface condenser is shown in Fig. 12.26. To permit a low pressure drop and a deep penetra-

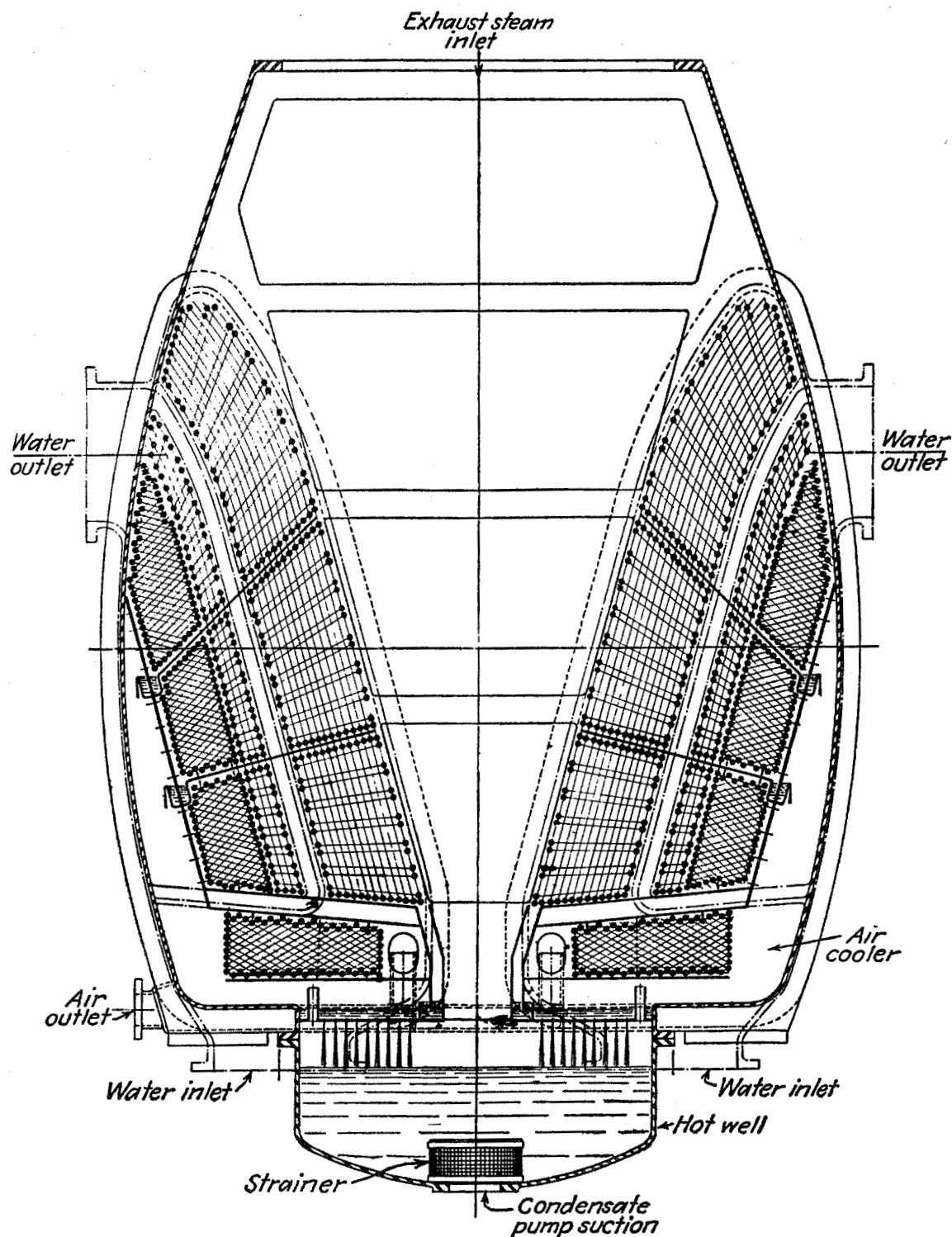


FIG. 12.27. Large dual-bank surface condenser. (Foster Wheeler Corporation.)

tion of the steam into the bundle tubes are laid out for *crossflow*, using *radial* pitch as shown in Fig. 12.26b. Another important consideration is the removal of air, since an accumulation of air increases the total pressure in the condenser and raises the condensation temperature. Only when surface condensers contain 15,000 ft² or less are they apt to have

cylindrical bodies. When they contain surface in excess of 15,000 ft², they are made in the boxlike manner shown in the elevation of Fig. 12.27. In large condensers the steam is not required to pass through the entire tube bundle but instead through a divided portion of the tube bundle.

The principal parts of surface condensers, in addition to the shell and water boxes or channels, are a large exhaust steam inlet port, shell-side air-removal outlets, and a hot well equipped with condensate and air outlets. The condenser shown in Fig. 12.26 is arranged horizontally for two passes or double flow on the water side as are the majority of such condensers. The tubes are usually 22 to 26 ft in length with support plates through which all the tubes pass, since the steam travels in crossflow. The lower portion of the bundle is separated to permit the cooling of the air-vapor mixture which leads to the vacuum apparatus. This is usually an air jet pump as discussed in Chap. 14. Since the steam condenses isothermally, the LMTD in crossflow is identical with that for counterflow. The method of calculation outlined in the remainder of this chapter is in agreement with the recommended practices of the Heat Exchange Institute, whose member companies include the larger manufacturers of surface condensers.

Definitions. There are a number of definitions which apply to surface condensers. The more important are given here: The *steam load* is expressed in pounds per hour of steam having an assumed heat rejection to the cooling water of 950 Btu/lb. When used in connection with the exhaust of a steam engine the rejection is taken as 1000 Btu/lb. The *condenser vacuum* is the difference between atmospheric pressure and the pressure measured at the steam inlet and is expressed in in. Hg at a temperature of 32°F. The *absolute pressure* in a condenser is the difference between the barometric pressure and condenser vacuum and is expressed in in. Hg abs. The *heat head* is the difference between the inlet circulating water temperature and the temperature corresponding to the absolute pressure at the steam inlet to the condenser, T_s . The *temperature rise* refers to the circulating-water temperatures $t_2 - t_1$, and the terminal difference is defined as $T_s - t_2$.

Since the condensate is also under vacuum, it collects in a hot well at the bottom of the condenser and requires pumping out. The *condensate depression* is the difference between the temperature of the condensate in the hot well and the temperature corresponding to the absolute pressure of the steam at the inlet to the condenser. It is the actual number of degrees the condensate is subcooled, and must be maintained within close limits since subcooling reduces the saturation pressure and hence the suction pressure on the intake of the condensate pump, reducing its performance. It is also customary to compute the total surface in a

surface condenser using the tube length between the facing sides of the tube sheets instead of the overall tube length. This is equivalent to the *effective surface* described in Chap. 7.

There are several conventions employed in surface condensers which are rarely violated. Surface condensers are seldom designed to operate at an absolute pressure below 0.7 in. Hg abs with a terminal difference of less than 5°F, with a dissolved oxygen content in the condensate of less than 0.03 cm³/liter, or with a steam loading [not to be confused with G'

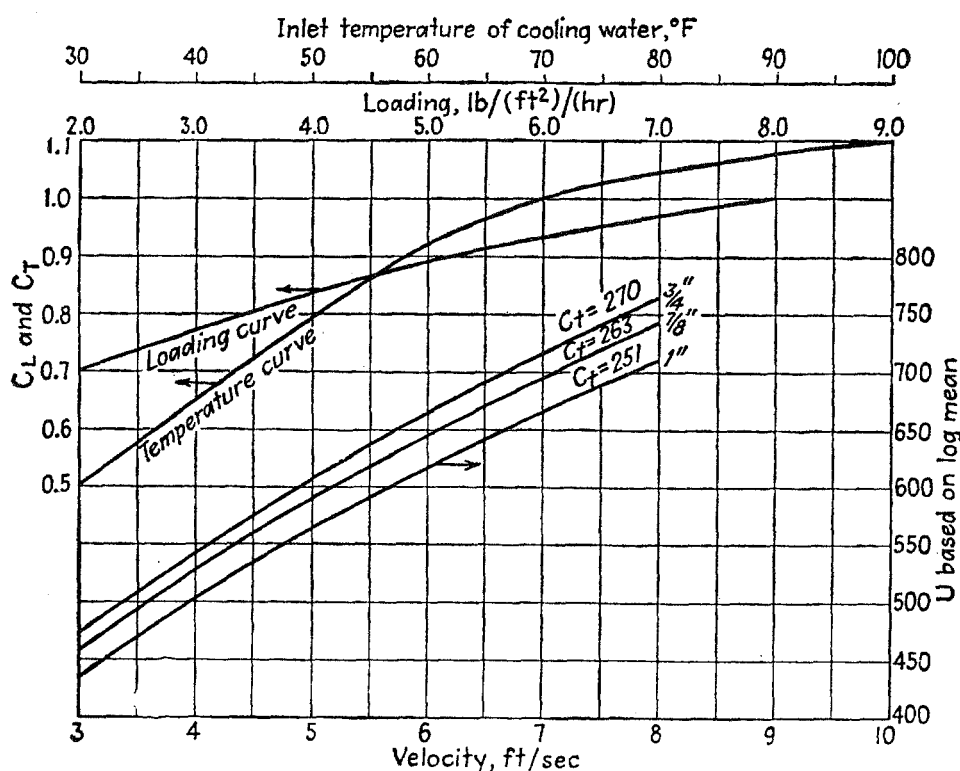


FIG. 12.28. Overall coefficients in surface condensers. (Heat Exchange Institute.)

or G'' , lb/(hr)(lin ft)] exceeding 8 lb/(hr)(ft²). Water velocities of less than 3 fps are not used. The quantity of air which leaks into the system must be estimated for the design of the air pump whether a mechanical pump or a jet-type ejector is used. The Heat Exchange Institute gives a graph of *overall* clean heat-transfer coefficients to be used as shown in Fig. 12.28 and with the maximum loading of 8 lb/(hr)(ft²) of steam condensed when the inlet water is at 70°F. The clean overall coefficient for different tube diameters and a loading of 8 lb/(hr)(ft²) can also be obtained by

$$U_c = C_t \sqrt{V} \quad (12.53)$$

where C_t is a constant as shown in Fig. 12.28 for each tube outside diameter and V is the water velocity in feet per second. Dirt factors are defined as a percentage of the clean overall coefficients called the *cleanli-*

ness factor. Thus a cleanliness factor of 85 per cent means that the design overall coefficient will be 85 per cent of the clean overall coefficient. The design overall coefficient U_D for a loading of 8 lb/(hr)(ft²) and inlet water temperature of 70° is

$$U_D = C_{cl}C_t \sqrt{V} \quad (12.54)$$

And the coefficient for any other loading and temperature is

$$U_D = C_L C_T C_{cl} C_t \sqrt{V} \quad (12.55)$$

where C_L is the loading correction equal to $\sqrt[4]{\text{loading}/8}$ and C_T is the temperature correction.

Surface Condenser Calculation. The condenser pressure-drop calculation is far more complex than for previous types of equipment and hinges very greatly upon the geometric design of the surface condenser. There is no published correlation between the pressure drop of condensing steam in crossflow and radial pitch, and the calculations performed here determine the surface required for the heat load without layout refinements. The calculations, however, permit the evaluation of a condenser for conditions other than those for which it was designed.

The pressure drop on the tube side is conventionally determined by a formula of the Williams and Hazen type using a constant of 130, so that

$$\Delta P_t = 0.0067 \frac{V^{1.84}}{d^{1.16}} \quad (12.56)$$

where V = water velocity, fps

d = tube ID, in.

It differs somewhat on the unsafe side from Eq. (7.45) at velocities of 6 fps and less but agrees very closely at velocities approaching 10 fps. The return losses suggested by the Heat Exchange Institute correspond to less than the four velocity heads given by Eq. (7.46). The relationships for the condenser are

$$Q = w_1(t_2 - t_1) = 500G_o(t_2 - t_1)$$

where G_o are the gallons per minute and multiplication by 500 gives the flow in pounds per hour.

$$\begin{aligned} \Delta t = \text{LMTD} &= \frac{(T_s - t_1) - (T_s - t_2)}{\ln (T_s - t_1)/(T_s - t_2)} = \frac{t_2 - t_1}{\ln (T_s - t_1)/(T_s - t_2)} \\ A &= \frac{Q}{U \Delta t} = \frac{500G_o(t_2 - t_1)}{U \frac{(t_2 - t_1)}{\ln (T_s - t_1)/(T_s - t_2)}} = \frac{500G_o}{U} \ln \frac{T_s - t_1}{T_s - t_2} \end{aligned} \quad (12.57)$$

$$t_2 = T_s - \frac{T_s - t_1}{\text{antilog } UA/(2.3 \times 500G_o)} \quad (12.58)$$

The outlet temperature of the water t_2 depends upon gallons per minute circulated G_o or the velocity V . But U is also dependent upon V . Equation (12.58) can be solved directly for t_2 , since

$$A = La''N_t$$

where a'' are the square feet of external surface per foot of tube length and

$$\frac{UA}{2.3 \times 500G_o} = \frac{U}{2.3 \times 500} \frac{La''N_t}{7.5 \times 60V(a'_t/144)N_t} = \frac{0.00279ULna''}{Va'_t}$$

and

$$t_2 = T_s - \frac{T_s - t_1}{\text{antilog } 0.00279ULna''/Va'_t} \quad (12.59)$$

Example 12.7. Calculation of a Surface Condenser. A steam turbine exhausts 250,000 lb/hr of steam at a pressure of 1.5 in. Hg. Cooling water is available at 70°F. A preference has been designated for the use of $\frac{3}{8}$ in. OD, 18 BWG tubes and a cleanliness factor of 85 per cent.

To permit estimation of the cost of the condenser on the basis of dollars/ft², how much surface will be required and what will be the operating quantity and range of the cooling water?

Solution. Assume the maximum loading of 8 lb/(hr)(ft²) and a water velocity of 7.5 fps.

$$\begin{aligned} C_{ci} &= 0.85 & C_T &= 1.0 & C_L &= 1.0 \\ U_D &= C_{ci}C_T C_L C_i \sqrt{V} \\ &= 0.85 \times 1.0 \times 1.0 \times 263 \sqrt{7.5} = 612 \\ A &= \frac{250,000}{8} = 31,250 \text{ ft}^2 \end{aligned} \quad (12.55)$$

The necessary outlet terminal water temperature will be

$$t_2 = T_s - \frac{T_s - t_1}{\text{antilog } 0.000279 \times ULna''/Va'_t} \quad (12.59)$$

For a $\frac{3}{8}$ in. OD, 18 BWG tube, $a'' = 0.229 \text{ ft}^2/\text{ft}$, ID = 0.777 in., and $a'_t = 0.475 \text{ in.}^2$ (computed from Table 10).

$$T_s \approx 1.5 \text{ in. Hg} = 91.72^\circ\text{F}$$

Since this will be a relatively large condenser, assume a maximum tube length of 26'0" and two tube passes.

$$t_2 = 91.72 - \frac{91.72 - 70}{\text{antilog } (0.000279 \times 612 \times 26 \times 2 \times 0.229)/(7.5 \times 0.475)} = 85.90^\circ\text{F}$$

The circulation rate is

$$G_o = \frac{250,000 \times 950}{(85.90 - 70) \times 500} = 29,800 \text{ gpm}$$

Individual Film Coefficients for Steam Condensers. The method of calculating surface condensers has been based on the use of overall coefficients instead of individual coefficients. Individual coefficients are diffi-

cult to obtain from direct experimentation but can be computed from the overall coefficients. Unlike condensers employing organic vapors, the condensing coefficient for steam is considerably greater than that for the water, and under these circumstances the tube metal resistance is a sufficiently sizable portion of the total resistance to require inclusion in the calculation. For a clean tube the overall coefficient of heat transfer may be equated thus:

$$\frac{1}{U_c} = R_c + R_m + R_w$$

and for a dirty tube:

$$\frac{1}{U_D} = R_c + R_m + R_d + R_w$$

where R_c is the resistance of the condensing vapor, R_m is the resistance of the tube metal, and R_w the resistance of the water in the tubes, all based on the tube outside diameter. Wilson¹ has shown that the sum of the first three resistances for a series of tests is essentially constant and that R_w is the controlling resistance. Furthermore, the film coefficient for water in tubes is proportional to the 0.8 power of the velocity through the tubes, R_w can then be replaced by $1/a_1 V^{0.8}$, and the equations above can be rewritten with a_0 and a_1 constants.

$$\frac{1}{U} = a_0 + \frac{1}{a_1 V^{0.8}} \quad (12.60)$$

A plot of $1/U$ vs. $1/V^{0.8}$ on rectangular coordinates should yield a straight line permitting the evaluation of both constants a_0 and a_1 . a_0 is actually the intercept for an infinite water velocity, and when the metal and dirt resistances are subtracted, the true value of the condensing coefficient can be obtained.

$$R_c = \frac{1}{h_c} = a_0 - R_m - R_d$$

PROBLEMS

12.1. 62,000 lb/hr of pure ethyl alcohol at 2.0 psig is to be condensed by water from 85 to 120°F. A dirt factor of 0.003 should be provided. A pressure drop of 2 psi is allowable for the vapor and 10.0 for the water.

Calculate the required size of a 1-2 horizontal condenser using 1 in. OD, 14 BWG tubes 16'0" long on 1¼-in. triangular pitch.

12.2. Using the data of Prob. 12.1 calculate the size of a 1-2 vertical condenser required to fulfill the conditions.

12.3. Using the data of Prob. 12.1 calculate the size of a 1-1 vertical condenser with condensation in the tubes.

¹ Wilson, R. E., *Trans. ASME*, **37**, 47 (1915).

12.4. 24,000 lb/hr of nearly pure methyl ethyl ketone vapor at 2 psig (boiling point 180°F) is to be condensed and cooled to 160°F by water from 85 to 120°F. Pressure drops of 2.0 psi for the vapor and 10.0 for the water are permissible.

Available for the service is a 25 in. ID 1-2 horizontal condenser with 468 $\frac{3}{4}$ in. OD, 16 BWG tubes, 16'0" long arranged for four passes on $1\frac{5}{16}$ -in. triangular pitch. Baffles are spaced 25" apart.

(a) What is the true temperature difference?

(b) What are the dirt factor and the pressure drops?

12.5. For the data of Prob. 12.4 calculate the true temperature difference and size of a horizontal 1-2 condenser-subcooler required to fulfill the conditions using $\frac{3}{4}$ in. OD, 16 BWG tubes 12'0" long on $1\frac{5}{16}$ -in. triangular pitch with a dirt factor of 0.003.

12.6. 50,000 lb/hr of ethyl acetate at 35 psig (boiling point 248°F) enters a horizontal 1-2 desuperheater-condenser at 300°F and leaves at 248°F. Cooling is effected by water from 85 to 120°F.

Available for the service is a 27 in. ID 1-2 condenser containing 432 $\frac{3}{4}$ in. OD, 16 BWG tubes 12'0" long arranged for four passes on 1-in. square pitch. Baffles are spaced 24 in. apart.

Calculate the true temperature difference and the dirt factor and pressure drops.

12.7. 57,000 lb/hr of nearly pure hexane enters the shell of a vertical 1-2 condenser at 5 psig and 220°F. The condensing range is from 177 to 170°F, at which temperature it goes to storage. Cooling water is used between 90 and 120°F.

Available for the service is a 31 in. ID condenser containing 650 $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long on four-pass layout using 1-in. triangular pitch. Baffles are spaced 18 in. apart.

Calculate the true temperature difference, the dirt factor, and the pressure drops.

12.8. 59,000 lb/hr of a mixture of light hydrocarbons, principally propane, enters a 1-2 horizontal condenser at the initial condensing temperature of 135°F at 275 psig. The condensing range is from 135 to 115°F, at which temperature 49,000 lb/hr is condensed by cooling water from 90 to 110. The remainder of the condensation occurs with refrigerated water.

Available for the service is a 37 in. ID 1-2 horizontal condenser with 1100 $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long arranged for four passes on $1\frac{5}{16}$ -in. triangular pitch. Baffles are spaced 36 in. apart.

What is the dirt factor, and what are the pressure drops?

NOMENCLATURE FOR CHAPTER 12

A	Heat-transfer surface, ft ²
A_c, A_d, A_s	Heat-transfer surface for condensation, desuperheating, and sub-cooling, respectively, ft ²
A_f	Cross-sectional area of a film, ft ²
A_c	Total clean heat-transfer surface, ft ²
a	Flow area, ft ²
a''	External surface per linear foot, ft
a_s'	Submerged cross section of shell, ft ²
a_t'	Flow area per tube, in. ²
a_0, a_1	Constants
B	Baffle spacing, in.
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F)
C_d, C_T, C_L	Cleanliness, temperature, and loading factors, dimensionless
C_i	Tube factor, dimensionless

C'	Clearance between tubes, in.
C_1, C_2, C_3	Constants
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of tubes, ft
D_o	Outside diameter of tubes, ft
D_e	Equivalent diameter for heat transfer and pressure drop, ft
d	Inside diameter of tubes, in.
d_o	Outside diameter of tubes, in.
f	Friction factor, ft ² /in. ²
G	Mass velocity, lb/(hr)(ft ²)
G'	Condensate loading for vertical tubes, lb/(hr)(ft)
G''	Condensate loading for horizontal tubes, lb/(hr)(ft)
G_o	Circulating water rate, gpm
g	Acceleration of gravity, ft/hr ²
g'	Acceleration of gravity, ft/sec ²
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, Btu/(hr)(ft ²)(°F)
h_{io}	Value of h_i when referred to the tube OD, Btu/(hr)(ft ²)(°F)
h_x	Condensing film coefficient at a distance x from the top of the tube Btu/(hr)(ft ²)(°F)
h_α	Condensing film coefficient at angle α° , Btu/(hr)(ft ²)(°F)
$\bar{h}$	Average value of the condensing film coefficient between two points, Btu/(hr)(ft ²)(°F)
j_H	Factor for heat transfer, dimensionless
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
L	Tube length, ft
L_c	Tube length exposed to condensation, ft
LMTD	Log mean temperature difference, °F
m	A constant
N	Number of shell-side baffles
n	Number of tube passes
N_t	Number of tubes effective for condensation
P	Perimeter, ft
P_T	Tube pitch, in.
ΔP	Pressure drop in general, psi
$\Delta P_T, \Delta P_i, \Delta P_r$	Total, tube, return pressure drop, respectively, psi
ΔP_c	Pressure head of condensate, psi
Q	Heat flow, Btu/hr
Q_x	Heat flow at distance x from top of tube, Btu/hr
q_c, q_d, q_s	Heat flow for condensation, desuperheating and subcooling respectively, Btu/hr
R_d	Combined dirt factor, (hr)(ft ²)(°F)/Btu
R_c, R_m, R_w	Resistance of condensate film, tube metal, and water, respectively, (hr)(ft ²)(°F)/Btu
r	Radius of tube, ft
r_h	Hydraulic radius, r_h = flow area/wetted perimeter, ft
Re	Reynolds number, dimensionless
s	Specific gravity, dimensionless
T, T_1, T_2	Temperature in general, inlet and outlet of hot fluid, respectively, °F

T_a	Average temperature of hot fluid, °F
T_c	Caloric temperature of hot fluid, °F
T_s	Steam temperature, °F
t, t_1, t_2	Cold-fluid temperature in general, inlet, and outlet, respectively, °F
t_a	Average temperature of cold fluid, °F
t_c	Caloric temperature of cold fluid, °F
t_f, t_w	Temperature of film and tube wall respectively, °F
t'	Temperature at outer surface of condensate film, °F
Δt	True temperature difference in $Q = U_D A \Delta t$, °F
$(\Delta t)_c, (\Delta t)_d, (\Delta t)_s$	True or fictitious temperature difference for condensation, desuperheating, and subcooling, respectively, °F
Δt_f	$(T_v - t_w)/2$, °F
U, U_c, U_D	Overall coefficient of heat transfer, clean coefficient, and design coefficient, respectively, Btu/(hr)(ft ²)(°F)
U_c, U_d, U_s	Clean overall coefficient for condensation, desuperheating, and subcooling respectively, Btu/(hr)(ft ²)(°F)
u	Film velocity along axis of tube, ft/hr
$\bar{u}$	Average film velocity, ft/hr
V	Velocity, fps
W	Weight flow in general, weight flow of hot fluid, lb/hr
W'	Rate of condensation, lb/(hr)(ft ²)
w	Weight flow of cold fluid, lb/hr
w'	Rate of condensation per tube W/N_t , lb/(hr)(tube)
x, x'	Length of film, ft
x_c	Distance from top of tube at which change from streamline to turbulent flow occurs, ft
y, y'	Width of film, ft
z	Distance, ft; height of a hydrostatic leg, ft; a synthetic function
α	Angle of tube, deg
λ	Latent heat of condensation or vaporization, Btu/lb
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ'	Viscosity, centipoises
μ_w	Viscosity at tube-wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³
τ	Tangential stress, lb/ft ²
ϕ	Viscosity ratio, $(\mu/\mu_w)^{0.14}$, dimensionless
ψ	A function

Subscripts (except otherwise noted)

c	Condensate, condensing
f	Film or film temperature
l	Liquid
s	Shell side
v	Vapor
t	Tube side

CHAPTER 13

CONDENSATION OF MIXED VAPORS

Introduction. In the preceding chapter it was assumed that the condensing vapor consisted of a pure or substantially pure compound which condensed isothermally. If the vapor was mixed with another compound having a slightly different boiling point, the mixture condensed over a small condensing range. It was further assumed that, where there was a condensing range, the latent heat of condensation was transferred to the cooling medium uniformly over the entire condensing range.

However, consider a mixture of two diverse fluids having a condensing range of 100°F. In order to effect the first 10 per cent reduction in the vapor temperature it may be necessary to remove 50 per cent of the total heat load from the vapor mixture, since the less volatile component condenses more rapidly as the vapor temperature is reduced. Previously, in using the LMTD for condensing vapors in counterflow it was assumed that during the first 10 per cent reduction in the vapor temperature only 10 per cent of the heat was removed. The substitution of the logarithmic mean for the true temperature difference based on assumed uniform condensing characteristics for the vapor mixture may lead to a conservative or an unsafe value of Δt and the selection of the wrong condenser. The problems imposed by the condensation of a vapor mixture do not end there. Depending upon the nature of the mixture, the average condensing coefficient $\bar{h}$ may not remain constant throughout the condensing range and may vary greatly with the *composition* of the vapor mixture as the less volatile components are condensed. The latter is particularly true of a mixture consisting of a vapor and a noncondensable gas, such as steam and air.

The Phase Rule.¹ The different types of vapor mixtures may be studied qualitatively through the use of the phase rule of J. Willard Gibbs. Phases are defined as homogeneous amounts of matter in the solid,² liquid, or gaseous form which are distinguishable from each other by the presence of an interface between any two phases. A vapor is the coexistence of a gas and a liquid and, like any boiling liquid and its vapor, consists of two phases.

¹ The treatment of the phase rule is simplified here for the particular applications in this book.

² In nonthermodynamic texts these are frequently referred to as states.

There are many organic chemical compounds which are not soluble in water, forming *immiscible* mixtures with it. If a cylindrical vessel is filled to the top with a mixture of pentane and water, an interface appears between the water on bottom ($s = 1.0$) and the pentane on top ($s = 0.8$) and the *system* contains two liquid phases. While agitation may be used to disperse the water into the pentane layer as droplets, it does not affect the number of phases, since only the nature of the interface is of consequence and not its shape. On the other hand, a mixture of two gases readily forms a homogeneous mixture in a vessel, and as in all gas mixtures, all the gaseous compounds form but one phase. An interesting relationship is obtained if a mixture of water and pentane is brought to a boil in an enclosed vessel: There is a mixture of pentane and water in the gas phase and two liquid phases, so that the bounded matter or *system* consists of three phases.

Suppose a mixture of two mutually soluble or miscible compounds is placed in an insulated vessel at its boiling point as shown in Fig. 13.1.

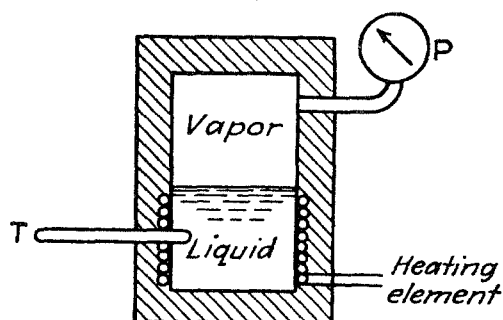


FIG. 13.1. Equilibrium vessel.

If the vapor and liquid are in *equilibrium*, no heat can escape and the system continues to vaporize and condense indefinitely. The boiling point of the system is related to the total pressure of the system as well as the ratio of the two compounds in the original liquid. Assume next that it is desired to change the pressure on the system. Does the liquid

still continue to boil at the same temperature if the enthalpy remains constant? Do the chemical compositions of the vapor and liquid phases remain the same? If these questions can be answered, it is possible without running experiments to determine whether or not a given mixture condenses isothermally. If condensation is isothermal, the true temperature difference may be taken as being identical with the LMTD. If condensation is not isothermal, then other methods must be applied for the calculation of the true temperature difference, the commonest of which will be developed later in this chapter.

Gibbs¹ formulated a rule to determine the number of independent variables pertaining to a system in equilibrium which must be specified for the number of phases to be fixed permanently. Thus if a system consists of several phases in equilibrium, what conditions in the original system can be varied while keeping the kind (compositions) and number

¹ Gibbs, J. W., *Trans. Conn. Acad. Arts Sci.*, III, 108-248 (1876).

of phases from changing? Changes in the relative sizes of the phases are of no consequence. The numbers of variables, such as the temperature, pressure, and compositions of the system, which the process engineer is *free* to fix so as to fix the number and kinds of phases permanently are the *degrees of freedom* (of selection). Sometimes it is possible to fix the number of phases in a system by simply fixing the pressure. This is true of a pure boiling fluid such as water. If it is desired to boil water by establishing a gas and a liquid phase, can it be done under any combination of pressure and temperature the engineer desires to fix? If the pressure is regarded as a degree of freedom, since it may be freely and independently chosen, and it is fixed at 14.7 psia, it suffices to fix all the conditions for the existence of the two phases. But the temperature may not be freely and independently chosen, since it is not an independent variable but in the equilibrium system it is a property *dependent* upon the saturation (pressure) curve of the liquid. It is consequently not possible to have the water present in two phases at a pressure of 14.7 psia at any other temperature than 212°F.

The boiling of a single compound is a relatively simple matter. If several compounds are mixed together, it is considerably more difficult to determine how the compositions, pressure, and temperature serve to fix permanently the number and kinds of phases in the original system. Gibbs's phase rule allows the rapid determination of the number of degrees of freedom in systems of greater complexity. The phase rule is written

$$F = C - P + 2 \quad (13.1)$$

where F is the number of degrees of freedom, C the number of individual chemical compounds or chemical substances in the system, and P is the number of phases. Nothing in the phase rule determines how many phases will result from the admixture of any particular group of chemical components. The number of phases which a group of chemical compounds may form in a boiling system must be known beforehand from a knowledge of the miscibility or immiscibility of the components as liquids plus the addition of one gas phase into or from which vaporization and condensation may occur. This information can be obtained from the solubility tables included in chemical handbooks.

Applications of the Phase Rule. There are nine common types of vapor mixtures encountered in heat transfer, and these are given in Table 13.1. To demonstrate the use of the phase rule for determining the isothermal or nonisothermal nature of the condensation of a vapor mixture in a condenser three of the types will be illustrated. It is again emphasized that, in ordinary process design, condensers operate at sub-

TABLE 13.1. COMMON CONDENSATION REQUIREMENTS

Case	Type of components	Example	Degrees of freedom	Temp† during condensation
1	Pure vapor	Water	1	Isothermal
2	Two miscible	Butane-pentane	2	Decreasing
3	<i>n</i> -Miscible	Butane-pentane-hexane	<i>n</i> *	Decreasing
4	Vapor and noncondensable	Steam-air	2	Decreasing
5	<i>n</i> -Miscible + noncondensable	Butane-pentane-air	<i>n</i> + 1	Decreasing
6	Two immiscible	Pentane-steam	1	Isothermal
7	<i>n</i> -Miscible + one immiscible	Butane-pentane-steam	<i>n</i>	Decreasing
8	<i>n</i> -Immiscible + noncondensable	Pentane-steam-air	2	Decreasing
9	<i>n</i> -Miscible + one immiscible + noncondensable	Butane-pentane-steam-air	<i>n</i> + 1	Decreasing

* Where there are more than three degrees of freedom, the additional degrees represent concentrations which must be fixed. Thus a mixture of butane, pentane, and hexane requires fixing not only the butane in relation to hexane but also the amount of pentane in relation to hexane.

† For constant-pressure processes.

stantially constant pressure and that one degree of freedom is usually fixed by the operating pressure of the process.

For the analyses below reference should be made to Fig. 13.2 which shows the inlet end of a condenser. Directly under the inlet nozzle at

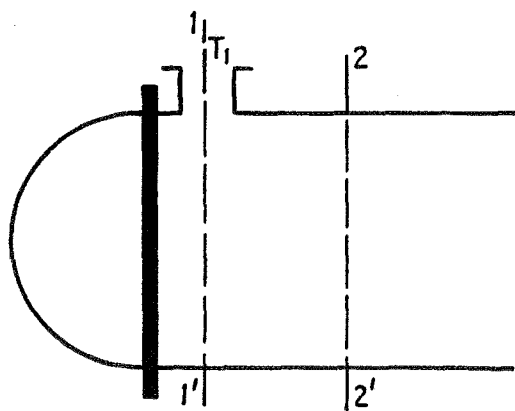


FIG. 13.2. Conditions at condenser inlet.

1-1' the inlet temperature is T_1 , and at another section such as 2-2', after partial condensation has occurred, the temperature is designated as T_2 , which may or may not be identical with T_1 . When a vapor is to be condensed, it is convenient to consider condensation as the transfer of material between a gas phase and one or more liquid phases. Since condensation occurs because the heat-transfer surface is below the dew point of the vapor, it is proper to assume that the heat-trans-

fer surface directly under the inlet nozzle is wet and supplies the liquid phases immediately after the vapor enters the condenser.

Case 2. Condensation of a Mixture of Two Miscible Components.
Example: Butane-pentane.

$$\begin{aligned} F &= C - P + 2 \\ &= 2 - 2 + 2 = 2 \quad (2 \text{ deg of freedom}) \end{aligned}$$

At the inlet to the condenser the pressure p_1 and the composition C_1 of the vapor are fixed by the process operation preceding the condenser such as distillation. The system is completely defined by the process which has already fixed the two degrees of freedom p_1 and C_1 , and T_1 at 1-1' is *fixed* therefore as a dependent variable as in the case of water boiling at 14.7 psia. Between 1-1' and 2-2' heat is removed and with it a quantity of the high-boiling compound and a lesser quantity of the low-boiling compound condense into the liquid phase. The composition C_2 of the vapor at 2-2' differs from C_1 at 1-1', the vapor being leaner in the high-boiling compound. Apply the phase rule again at 2-2'. There must still be two degrees of freedom, since it is desired to retain the same number of phases. The pressure is constant at p_1 , but the composition has changed to C_2 . According to the laws governing vapor-liquid equilibrium, the temperature T_2 at p_1 , C_2 cannot be the same as T_1 at p_1 , C_1 , since the latter is a point on a vapor-liquid equilibrium line just as 212°F and 14.7 psia is a point on the equilibrium line for water. The temperature of condensation must consequently vary from 1-1' to 2-2', and the condensation is not isothermal.

Case 4. Condensation of a Vapor from a Noncondensable Gas. Example: Steam-air-water.

$$F = 2 - 2 + 2 = 2 \quad (2 \text{ deg of freedom})$$

The operating pressure is fixed, and C_1 is fixed at the inlet to the condenser by the ratio of steam to air in the inlet gas. Since two degrees of freedom p_1 and C_1 are fixed, the temperature T_1 is again a dependent variable at 1-1'. Moving along to 2-2', where the pressure on the system is still p_1 , some of the steam is condensed while the gas is not condensed and a new composition C_2 results. The temperature T_2 for p_1 , C_2 must differ from T_1 at p_1 , C_1 for the same reason as before. The temperature at 2-2' or any other point is actually the dew point for the mixture remaining in the gas phase and varies from the inlet to outlet as the composition of the vapor phase changes. Condensation is not isothermal.

Case 6. Condensation of a Mixture of Two Immiscible Components
Example: Pentane-steam-water.

$$F = 2 - 3 + 2 = 1 \quad (1 \text{ deg of freedom})$$

The condensate consists of two liquid phases. With but one degree of freedom the system is completely fixed at 1-1' by selecting the operating pressure p_1 . The temperature T_1 then corresponds to p_1 . At 2-2' the pressure is the same, p_1 , and since this fixes the system, T_2 and T_1 must be identical, both being dependent upon p_1 . Condensation is isothermal.

1. CONDENSATION OF A BINARY MIXTURE

Vapor-liquid Relationships. In the derivation of the Nusselt equation for a single component it was assumed that the rate at which the vapor enters the condensate film is nearly infinite. A vapor entering a condenser at its saturation pressure condenses because the surface is at a temperature below its dew point or saturation temperature. The rate at which the vapor passes from the vapor phase to the liquid phase, however, is dependent upon the mechanism of diffusion. This means that, if the temperature of the condensate film is lower than the saturation temperature of the bulk of the vapor, the pressure of the component at the condensate film is less than that of the vapor and a pressure differential is established. The direction of the differential promotes the flow out of the vapor phase.

In the condensation of a binary mixture the problem of diffusion is slightly more involved, although it does not produce a significant resistance in series with the resistance of the condensate film. In a binary mixture unless it is a constant boiling mixture such as 95 per cent ethanol-water, the higher boiling component condenses in greater proportion near the condenser inlet. The rate of condensation of both components is related to their individual pressure differentials between the bulk of the vapor and the condensate film. Colburn and Drew¹ pointed out that the *ratio* of the partial pressures exerted by the condensate film for a binary mixture is dependent upon the temperature of the film, which in turn is related to the temperature of the cold surface and the cooling water range. Thus for a given binary mixture entering a condenser not only the overall rate but also the chemical composition of the condensate are influenced by the temperature of the cooling water. This is not of particular concern in the total condensation of a vapor, but it can influence the composition of the product from a partial condenser. Colburn and Drew give equations for calculating the composition and temperature at the vapor-condensate interface for binary mixtures. It is customary, however, to assume the temperature of the condensate at the vapor-liquid interface to be the same as that of the vapor.

In the condensation of the overhead vapor coming from a binary dis-

¹ Colburn, A. P., and T. B. Drew, *Trans. AICHE*, **33**, 197-215 (1937).

tillation the vapor is almost entirely composed of the more volatile component and the presence of the second component establishes a condensing range as predicted by the phase rule. The second component, being less volatile, condenses more rapidly near the inlet than the outlet. It is conceivable, then, for a binary mixture with a condensing range of 20°F that the higher boiling component comes out in greater proportion during the first 10° than it does during the second 10°. Similarly, more of the heat may be removed from the vapor in the second 10° than in the first. The use of the LMTD is slightly in error on the unsafe side in the case of binary overhead vapors, although the size of the error does not usually justify its rejection. Where a number of compounds are contained in the overhead vapor as in multicomponent distillation, the temperature range and the distribution of the heat transfer differ greatly from the straight-line relationship of Q vs. t on which the LMTD is predicated. Methods of treating multicomponent mixtures are dealt with in the next section.

Film coefficients for the transfer of heat by condensation from binary mixtures may be treated in the same manner as heretofore for a single vapor using the weighted film properties of the mixture.

2. CONDENSATION OF A MULTICOMPONENT MIXTURE

Vapor-liquid Relationships in Mixtures. The phase rule has been used only qualitatively, but it is important for the identification of the various types of mixture problems. Except in binary mixtures or mixtures of several compounds whose boiling points in the pure condition do not differ greatly, the condensation of the vapor mixture occurs over a long temperature range. The fraction of the total heat load delivered during a fractional decrease in the vapor temperature need not be uniform over the entire condensing range, and this invalidates the use of the logarithmic mean alone or $F_T \times \text{LMTD}$ in the case of a 1-2 condenser. The solution of such problems requires the determination or calculation of the *condensing curve* for the mixture.

When a single vapor is in equilibrium with its liquid, the vapor and liquid have the same composition. For a mixture some of the components are more volatile than others (except in constant-boiling mixtures) and the vapor and liquid in equilibrium have different compositions, the percentage of the more volatile components being greater in the vapor. The following discussion applies particularly to mixtures which form *ideal solutions*, although suggestions for its application to nonideal solutions are included.

An ideal solution is one in which the presence of the several components

has no effect upon the behavior of each and which is governed by Dalton's and Raoult's laws. For such a system Dalton's law states that the total pressure is the sum of the partial pressures in the phase above the liquid solution. Raoult's law states that the partial pressure of a component in the phase above a liquid solution is equal to the product of its pressure as a pure component and its mol fraction in the solution. The latter is not true of nonideal solutions in which the presence of the various components tends to reduce the partial pressures of the others so that the total pressure is not the sum of the products of the mol fractions and vapor pressures in the pure condition. Materials of an electrolytic or ionic nature deviate from the laws of ideal solutions to a great extent. For an ideal solution:

Dalton's law:

$$p_t = p_1 + p_2 + p_3 \quad p_1 = p_t y_1 \quad (13.2)$$

and Raoult's law:

$$p_1 = p_{p1} x_1 \quad (13.3)$$

where p_t is the total pressure, p_1 the partial pressure of component 1, p_{p1} the vapor pressure of the pure component 1 at the temperature of the solution, y_1 the mol fraction of component 1 in the vapor phase, and x_1 the mol fraction of component 1 in the liquid. Subscripts 2, 3, etc., refer to other components. The mol fraction is the ratio of the number of mols of a single component to the total number of mols in the mixture and is sometimes abbreviated by *mf*. The mol per cent is the mol fraction multiplied by 100.

Solving Eqs. (13.2) and (13.3) for p_1 ,

$$p_1 = p_{p1} x_1 = p_t y_1 \quad (13.4)$$

Rearranging,

$$y_1 = \frac{p_{p1} x_1}{p_t} \quad (13.5)$$

Solutions which are ideal at moderate pressures appear to deviate from ideality at high pressures, each component tending in some degree to lower the pressure of the other. The total pressure is then no longer the summation of the partial pressures, and Eq. (13.5) is not valid. *Fugacities* or corrected pressures, designated by the letter *f*, are then introduced. The fugacities are the partial pressures of the compounds such that the various criteria of ideality may be retained, and their values originate with experimental pressure-volume-temperature studies on the actual chemical compounds.

$$f_{p1} x_1 = f_t y_1 \quad (13.6)$$

where f_t replaces the total pressure p_t

$$y_1 = \frac{f_{p1}x_1}{f_t} \quad (13.7)$$

or

$$y_1 = K_1x_1 \quad (13.8)$$

where $K_1 = f_{p1}/f_t$

$$\Sigma y_1 = \Sigma (K_1x_1) \quad (13.9)$$

K is called the *equilibrium constant*.

When a mixture is in vapor-liquid phase equilibrium, the vapor possesses a greater percentage of the more volatile components than the

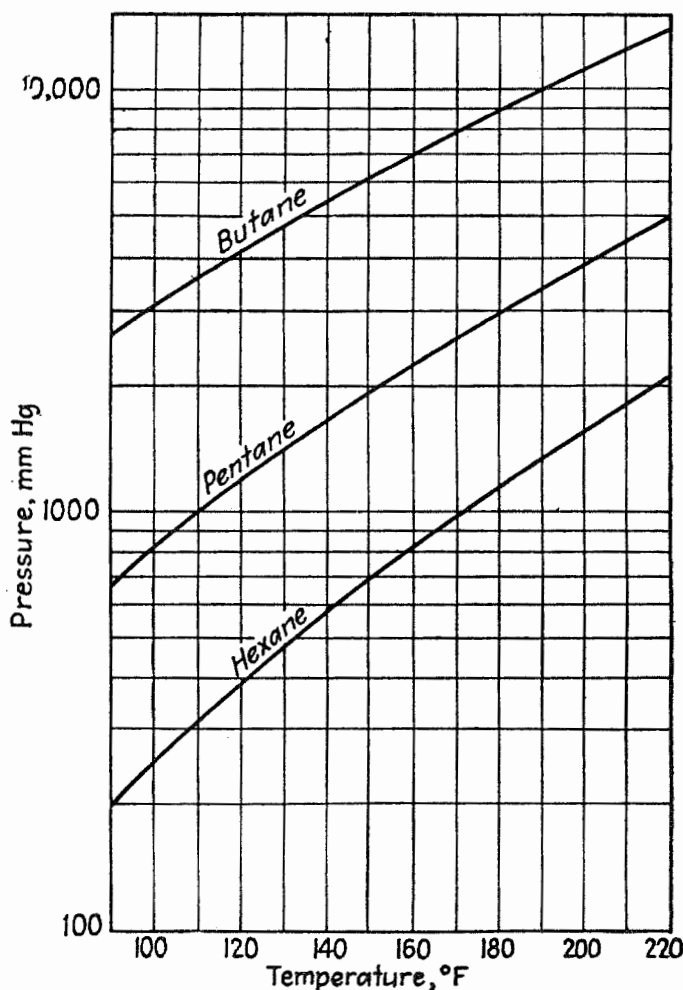


FIG. 13.3. Vapor pressure curves.

liquid. It is possible from Eqs. (13.4) to (13.9) at any given total pressure to calculate the mol fraction y_1 of the component in the vapor phase which may coexist with a mol fraction x_1 of the same component in the

liquid phase. The vapor pressures of the pure components must be known at the temperature at which the vapor-liquid equilibrium is assumed to exist. Given butane, pentane, and hexane, as seen in Fig. 13.3, the butane possesses the highest vapor pressure at any given temperature. If several mixtures of the three compounds are formulated with equal ratios of pentane to hexane, the one with the largest percentage of butane would start to boil most nearly at the boiling point of pure butane. Other mixtures having a preponderance of hexane boil nearer the boiling point of hexane. Any mixture of the three compounds starts to boil at an initial temperature higher than the boiling point of pure butane and lower than the boiling point of pure hexane, since boiling starts only when the sum of the partial pressures, $p_1 + p_2 + p_3$, is infinitesimally greater than the total pressure on the system.

Boiling is presumed to start when the first bubble is forced out of the solution. This temperature is called the *bubble point* of the mixture, the term *boiling point* being reserved for a pure compound. If the mixture is boiled at *constant pressure* with a full expansion of the volume of vapor formed, the liquid phase does not disappear until the last droplet is vaporized. For the mixtures referred to above the last droplet consists mainly of hexane and it disappears at a higher temperature than that at which the initial bubble of the mixture formed. Upon complete vaporization the composition of the total vapor is the same as the composition of the liquid before vaporization began. Conversely, if the equilibrium were to start with all the mixture in the vapor phase, the liquid phase begins with the formation of the first droplet of condensate which is identical in composition to the last droplet vaporized. The temperature of formation of the first droplet by heat extraction from the vapor phase is again called the *dew point*. The difference between the bubble point and dew point is the *boiling range*, which must exist for any miscible mixture as predicted by the phase rule.

Example 13.1. Calculation of the Bubble Point, Dew Point, and Vapor Composition of a Mixture. The following mixture is to be heated and vaporized at atmospheric pressure. What is the temperature at which boiling will start (*i.e.*, the bubble point), and what is the composition of the first vapor formed?

Compound	Lb/hr	Mol. wt.	Mol/hr	<i>mf</i>
Butane, C_4H_{10}	1,625	58.12	28.0	0.077
Pentane, C_5H_{12}	16,200	72.15	224.0	0.613
Hexane, C_6H_{14}	9,750	86.17	113.0	0.310
	27,575		365.0	1.000

Solution:

(a) Bubble point: At atmospheric pressure, $p_t = 14.7 \text{ psi} = 760 \text{ mm Hg}$.

$p_t = 760 \text{ mm}$		Assume $T = 100^\circ\text{F}$		Assume $T = 96^\circ\text{F}$		Assume $T = 97^\circ\text{F}$	
	x_1, mf	$p_{p1, 100^\circ\text{F}}$	$p_1 = p_{p1}x_1$	$p_{p1, 96^\circ\text{F}}$	$p_1 = p_{p1}x_1$	$p_{p1, 97^\circ\text{F}}$	$p_1 = p_{p1}x_1$
C_4^*	0.077	3,170	244	2,990	230	3,040	234
C_5	0.613	790	484	725	444	740	454
C_6	0.310	250	77.5	229	71	234	72.4
	1.000	$p_t = \Sigma p_1 = 805.5$ Too high			745 Too low		760.4 Check

* The use of the letter C with a subscript is the usual abbreviation for a straight-chain organic molecule where the subscript indicates the number of carbon atoms. If not straight-chain as isobutane, the subscript designates the number of carbon atoms but the C is preceded by i -. Thus butane C_4 , isobutane i - C_4 , pentane C_5 , isopentane i - C_5 , etc.

The composition of the first bubble is found from Eq. (13.5), $y_1 = \frac{p_{p1}x_1}{p_t}$

	$\frac{p_{p1}x_1}{p_t} = y_1$
C_4	$234/760.4 = 0.308$
C_5	$454/760.4 = 0.597$
C_6	$72.4/760.4 = 0.095$
	1.000

(b) Similarly at what temperature will the mixture start to boil if the system is under a pressure of 35 psia and what will the composition be? $p_t = 35 \text{ psia} = 1810 \text{ mm}$.

	x_1, mf	Assume $T = 150^\circ\text{F}$		Assume $T = 149^\circ\text{F}$		$\frac{p_{p1}x_1}{p_t} = y_1$
		$p_{p1, 150^\circ\text{F}}$	$p_1 = p_{p1}x_1$	$p_{p1, 149^\circ\text{F}}$	$p_1 = p_{p1}x_1$	
C_4	0.077	6100	469	6050	467	$467/1810 = 0.258$
C_5	0.613	1880	1153	1850	1135	$1135/1810 = 0.627$
C_6	0.310	680	211	670	208	$208/1810 = 0.115$
	1.000	$P_t = \Sigma p_1 = 1833$ Too high			1810 Check	1.000

(c) Parts (a) and (b) may have been solved by Eqs. (13.6) to (13.9) from tabulated values of f_p/f_t which were computed from experimental data and are shown in Fig. 7 in the Appendix. In Fig. 7 values of the equilibrium constant K are plotted for a number of hydrocarbons against temperature and pressure. It is, in effect, the vapor pressure curve for the 18 compounds indicated. Thus if a focal point of $K = 1$ is chosen for any compound, a line connecting pressure and temperature through $K = 1$ gives the boiling temperature of the pure compound corresponding to any pressure. Since $K = 1$, from Eq. (13.8), $f_p/f_t = 1$, and the pressure of a pure compound f_p or p_p

must be identical with the total pressure f_t or p_t when there is no other fluid present. At 760 mm or 14.7 psia the bubble point computed from the fugacity data represented by Fig. 7 will be higher than in (a) owing to the mutual reduction of partial pressures resulting from nonideality.

BUBBLE POINT

	x_1, mf	Assume $T = 95^\circ\text{F}$		Assume $T = 100^\circ\text{F}$		Assume $T = 102^\circ\text{F}$	
		$K_{95^\circ\text{F}}$	$y_1 = K_1 x_1$	$K_{100^\circ\text{F}}$	$y_1 = K_1 x_1$	$K_{102^\circ\text{F}}$	$y_1 = K_1 x_1$
C_4	0.077	3.13	0.241	3.35	0.258	3.45	0.266
C_5	0.613	0.92	0.564	1.00	0.613	1.02	0.625
C_6	0.310	0.30	0.093	0.335	0.104	0.35	0.109
	1.000	$\Sigma y_1 = 0.898$ Too low			0.975 Too low		1.000 Check

(d) The use of K values gives y_1 directly and permits use of the total mol fraction of $\Sigma y_1 = 1.00$ as the criterion for equilibrium. Similarly for 35 psia

	x_1	Assume $T = 150^\circ\text{F}$		Assume $T = 153^\circ\text{F}$	
		$K_{150^\circ\text{F}}$	$y_1 = K_1 x_1$	$K_{153^\circ\text{F}}$	$y_1 = K_1 x_1$
C_4	0.077	2.80	0.216	2.90	0.223
C_5	0.613	1.01	0.619	1.06	0.650
C_6	0.310	0.40	0.124	0.415	0.1285
	1.000	$\Sigma y_1 = 0.959$ Too low			1.0015 Check

(e) The temperature at which the liquid phase disappears if boiling occurs in an enclosed vessel is the temperature at which only the last droplet is left. But this is the same as the physical picture of the dew point when the first droplet is formed. The initial compositions are then the vapor mol fractions or y 's, and for the liquid formed $x_1 = y_1/K_1$.

Dew point: At $p_t = 14.7$ psia, 760 mm.

	y_1	Assume $T = 130^\circ\text{F}$		Assume $T = 120^\circ\text{F}$		Assume $T = 123^\circ\text{F}$	
		$K_{130^\circ\text{F}}$	$x_1 = \frac{y_1}{K_1}$	$K_{120^\circ\text{F}}$	$x_1 = \frac{y_1}{K_1}$	$K_{123^\circ\text{F}}$	$x_1 = \frac{y_1}{K_1}$
C_4	0.077	5.0	0.015	4.4	0.0175	4.60	0.0167
C_5	0.613	1.65	0.371	1.40	0.437	1.49	0.412
C_6	0.310	0.62	0.500	0.51	0.608	0.545	0.568
	1.000	$\Sigma x = 0.886$			1.062		0.9967 Check

The dew point is 123°F .

At $p_t = 35$ psia

	y_1	Assume $T = 174^\circ\text{F}$	
		$K_{174^\circ\text{F}}$	$x_1 = \frac{y_1}{K_1}$
C_4	0.077	3.70	0.0208
C_5	0.613	1.38	0.444
C_6	0.310	0.58	0.533
	1.000	$\Sigma x = 0.998$	

The dew point is 174°F .

The Operating Pressure of a Condenser. In Example 13.1 may be found the quantitative requirements for setting the operating pressure of a distilling column and condenser. If the condenser is operated at atmospheric pressure, the condensing range is from 123 to 102°F . The application of 1–2 condensers for these temperatures is not very satisfactory when cooling water is available at 85°F , since the water temperature range must be kept small to prevent a large temperature cross above the 102°F condensate outlet. It would be necessary for atmospheric pressure condensation to operate the condenser with a Δt of about 16.5°F with the use of a large amount of water. If the distilling column pressure were raised to 35 psia, the range of condensation would be from 174 to 153°F . The cooling-water range could be from 85 to 120°F , and the Δt would be about 57°F , requiring approximately one-quarter as much surface as at atmospheric pressure. It must be realized, however, that raising the pressure on the column increases the column first cost and also the temperature of the heating medium in the reboiler. The selection of the optimum process operating pressure is a matter of economic analysis. The total annual operating cost for several pressures, which includes utilities and fixed charges, is plotted against operating pressure, the optimum occurring when the total annual cost is a minimum.

Relative Volatilities. Another method of obtaining the composition during phase equilibrium is by *relative volatilities*. This method utilizes the principle that in a mixture of several components some are more and some are less volatile (have larger or smaller K 's) than an intermediate compound. Although the K 's may change greatly even over a small temperature range, the ratios of the equilibrium constants relative to each other remain nearly constant. By means of this premise it is possible to eliminate the successive trial-and-error calculation provided the first trial is reasonably close. For a three component system

$$\begin{aligned}
 y_1 &= K_1 x_1 & y_2 &= K_2 x_2 & y_3 &= K_3 x_3 \\
 \frac{y_1}{y_2} &= \frac{K_1 x_1}{K_2 x_2} & \frac{y_3}{y_2} &= \frac{K_3 x_3}{K_2 x_2}
 \end{aligned} \tag{13.10}$$

$K_1/K_2 = \alpha_{1-2}$ is the relative volatility of compound 1 to compound 2 and $K_3/K_2 = \alpha_{3-2}$ is the relative volatility of compound 3 to compound 2.

For vaporization:

$$y_1 + y_2 + y_3 = 1.00 \quad (13.11)$$

Relative to y_2

$$\frac{y_1}{y_2} + 1 + \frac{y_3}{y_2} = \frac{1.00}{y_2} \quad (13.12)$$

Substituting,

$$\alpha_{1-2} \frac{x_1}{x_2} + 1 + \alpha_{3-2} \frac{x_3}{x_2} = \frac{1.00}{y_2}$$

Rearranging,

$$y_2 = \frac{x_2}{\alpha_{1-2}x_1 + x_2 + \alpha_{3-2}x_3} = \frac{x_2}{\Sigma\alpha x} \quad (13.13)$$

and since

$$\frac{x_2}{y_2} = \alpha_{1-2} \frac{x_1}{y_1} \quad y_1 = \frac{\alpha_{1-2}x_1}{\Sigma\alpha x} \quad y_2 = y_2 \quad y_3 = \frac{\alpha_{3-2}x_3}{\Sigma\alpha x} \quad (13.14)$$

For condensation:

$$x_1 + x_2 + x_3 = 1.00 \quad (13.15)$$

$$\frac{x_1}{x_2} + 1 + \frac{x_3}{x_2} = \frac{1}{x_2}$$

$$\frac{y_1}{y_2 \alpha_{1-2}} + 1 + \frac{y_3}{y_2 \alpha_{3-2}} = \frac{1}{x_2}$$

$$\frac{y_1}{y_2 \alpha_{1-2}} + 1 + \frac{y_3}{y_2 \alpha_{3-2}} = \frac{1}{x_2}$$

$$x_2 = \frac{y_2}{y_1/\alpha_{1-2} + y_2 + y_3/\alpha_{3-2}}$$

$$x_1 = \frac{y_1/\alpha_{1-2}}{\Sigma y/\alpha} \quad x_2 = \frac{y_2}{\Sigma y/\alpha} \quad x_3 = \frac{y_3/\alpha_{3-2}}{\Sigma y/\alpha} \quad (13.16)$$

Example 13.2. Calculation of the Bubble Point and Vapor Composition by Relative Volatilities. As before, make the assumption that the bubble point is 95°F, which is considerably off.

Bubble point: $p_t = 14.7$ psia. Assume $T = 95^\circ\text{F}$.

	x_1	$K_{95^\circ\text{F}}$	$\alpha_{95^\circ\text{F}}$	$\alpha_1 x_1$	$y_1 = \frac{\alpha_1 x_1}{\Sigma \alpha x}$
C_4	0.077	3.13	3.40	0.262	0.269
C_5	0.613	0.92	1.0	0.613	0.628
C_6	0.310	0.30	0.326	0.101	0.1035
	1.000		$\Sigma \alpha x = 0.976$		1.000
		$K_2 = \frac{y_2}{x_2} = \frac{(\alpha_2 x_2 / \Sigma \alpha x)}{x_2} = \frac{0.628}{0.613} = 1.025$			

Look up the temperature of C_5 in Fig. 7 corresponding to a K_2 of 1.025 and $p_t = 14.7$ psia.

$$K_2 = 1.025 \quad T = 102^\circ\text{F} \quad (\text{Checks Example 13.1 on first trial})$$

Dew point; $p_t = 14.7$ psia. Assume $T = 130^\circ\text{F}$.

	y_1	$K_{130^\circ\text{F}}$	$\alpha_{130^\circ\text{F}}$	$\frac{y_1}{\alpha_1}$	$x_1 = \frac{y_1/\alpha_1}{\sum y/\alpha}$
C_4	0.077	5.0	3.03	0.0254	0.0175
C_5	0.613	1.65	1.00	0.613	0.419
C_6	0.310	0.62	0.376	0.824	0.5635
	1.000		$\sum y/\alpha = 1.462$		1.0000
		$K_2 = \frac{y_2}{x_2} = y_2 / \frac{y/\alpha}{\sum y/\alpha} = \frac{0.613}{0.419} = 1.46$			

Look up the temperature of C_5 corresponding to a $K_2 = 1.46$ and $p_t = 14.7$ psia.

$$K_2 = 1.46 \quad T = 122^\circ\text{F}$$

The value computed from K values was 123°F , showing a small variation in the actual relationship of the volatilities.

*** Calculation of Compositions between the Dew Point and Bubble Point.** The calculation of dew points and bubble points from mol fractions is usually an unnecessary step, and it is desirable to carry out multi-component calculations on the total number of the mols directly.

Since

$$y_1 = K_1 x_1$$

it is the same as

$$\frac{V_1}{V} = K_1 \frac{L_1}{L}$$

where V_1 is the number of mols of a component in the vapor and L_1 the number of mols of the component in the liquid. V and L are the total number of mols of vapor and liquid, respectively.

At the dew point,

$$V_1 = K_1 L_1 \quad (13.17)$$

At the bubble point,

$$L_1 = \frac{V_1}{K_1} \quad (13.18)$$

In an enclosed vessel at equilibrium at *any temperature* between the dew point and bubble point if Y is the original number of mols of vapor consisting of Y_1 , Y_2 , and Y_3 , etc., for each compound, the amount condensed is given by

$$Y = V + L \quad Y_1 = V_1 + L_1$$

where V is the total mols of vapor remaining and L is the total mols of liquid formed. Then

$$V_1 = Y_1 - L_1$$

and

$$V_1 = K_1 L_1 \frac{V}{L}$$

$$L_1 = \frac{Y_1}{1 + K_1(V/L)} \quad (13.19)$$

To determine the mols condensed at any given temperature between the dew point and bubble point, assume a ratio of V/L and calculate by Eq. (13.19) the number of mols of liquid formed for the assumed value of V/L . If the ratio of the mass of vapor remaining to the mols of liquid formed does not check the assumed value of V/L , a new assumption must be made.

Differential Condensation. The equilibrium in a condenser causes *differential condensation*. Consider a condenser as shown in Fig. 13.4 divided into a number of condensing intervals as 0-0 to 1-1, 1-1 to 2-2, etc. At 0-0 there is perhaps one droplet of condensate, but in the zone

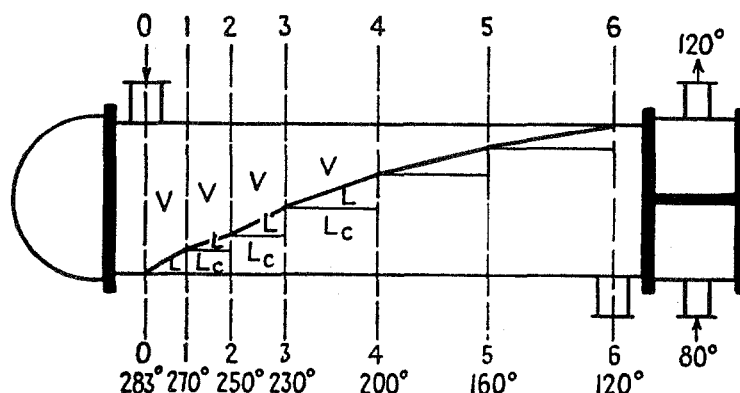


Fig. 13.4. Differential condensation in a condenser.

from 0-0 to 1-1 considerable condensate is formed. At 1-1 there is now a layer of condensate, and the total mols of vapor plus condensate is the same as at 0-0, but the mols of vapor is less and its composition differs from the original feed. The equilibrium at 1-1 differs from the dew point in so far as there is now a definite number of mols of liquid present instead of one droplet. In the interval from 1-1 to 2-2 where a new equilibrium is established, the vapor is in equilibrium not merely with the liquid which has been formed by condensation during the interval but also with a liquid whose composition consists of all the precondensed liquid from previous intervals. The similarity between this behavior and that which occurs in a constant-pressure batch vessel can be demonstrated. If L_c are the total mols of liquid condensed in the zone 0-0 to 1-1 before

the zone from 1-1 to 2-2, L_{c1} the mols of component 1 condensed from 0-0 to 1-1, and L' and L'_1 the total mols and mols of component 1 formed in the zone from 1-1 to 2-2 in Fig. 13.4, then at 2-2

$$\begin{aligned}\frac{V_1}{V} &= K_1 \frac{L'_1 + L_{c1}}{L' + L_c} \\ V_1 &= Y_1 - L'_1 \\ \frac{Y_1 - L'_1}{V} &= K_1 \frac{L'_1 + L_{c1}}{L' + L_c} \\ L'_1 &= Y_1 - K_1 \frac{V(L'_1 + L_{c1})}{L' + L_c} \\ L'_1 &= \frac{Y - K_1(V/L' + L_c)L_{c1}}{1 + K_1(V/L' + L_c)}\end{aligned}\quad (13.20)$$

At any point, however, $L' + L_c = L$, $L'_1 + L_{c1} = L_1$, and Eq. (13.20) reduces to Eq. (13.19). Figure 13.4 is obviously idealized and does not take into account the hydrodynamics of such a condensate layer.

The Calculation of the Condensing Curve for a Multicomponent Mixture. The calculation of the weighted temperature difference is dependent upon the shape of the condensing curve which is a plot of vapor heat content vs. vapor temperature for the condensing range. The weighted temperature difference is then obtained by taking increments of the condensing range and computing the average temperature difference between the vapor and water temperatures. From a practical standpoint it is necessary to choose only a reasonable number of temperatures, since the solution of Eq. (13.19) is by trial and error. The best selection of the intervals should give equal increments of $dQ/\Delta t$ or UA , but this is rather difficult to accomplish by inspection. It is usually helpful to observe whether the vapor to be condensed has fronts or tails. Tails indicate that for a high dew point the main portion of the vapor does not condense until a considerably lower temperature is reached. Thus in example 13.1 if a small number of mols of C_9 or C_{10} was added to the mixture, the dew point would be considerably higher than before although the major heat load removed by condensation would start only at the temperature which previously was the dew point or near 130°F at 14.7 psia. With small amounts of C_9 and C_{10} the temperature of the vapor would fall rapidly upon the removal of a small amount of heat. Similarly in the case of fronts, the presence of the small amount of propane C_3 reduces the bubble point, although the heat removal between the old and new bubble points would not necessarily represent a significant percentage of the total heat load removed in the condenser.

The Weighted Temperature Difference. In the desuperheater-condenser and condenser-subcooler it was assumed that the counterflow

temperature differences could be applied over the entire length of the shell to obtain the weighted Δt . In the case of a multicomponent mixture, because it requires an integration to obtain the weighted Δt , it is similarly very convenient to assume that the cooling medium is in counter-flow with the multicomponent vapor even if a 1-2 condenser is employed. The percentage rise in the temperature of the cooling medium at any cross section of the shell is then taken as proportional to the percentage of the heat load removed from any condenser cross section to the exit. The weighted Δt is then the averaged temperature difference between the condensing curve (vapor heat content vs. T_v plot) and a straight line representing the cooling medium. When a 1-2 condenser is employed, if the value of F_T based on inlet and outlet temperatures is not very nearly 1.0, the assumption above may not be admissible. If q is the heat load for an interval on the condensing curve, then the weighted Δt is obtained from the total heat load Q divided by the summation of the values of $q/\Delta t_{av}$ where Δt_{av} is the average temperature difference for the interval.

Heat-transfer Coefficients for a Multicomponent Mixture. When a multicomponent mixture is condensed, the condensing range between the dew point and the bubble point may be greater than 100° . The liquid which is formed near the inlet differs greatly in composition from that formed near the outlet and must be cooled to the bubble point at the end of the shell before it drains from the condenser. The film coefficient in the condenser differs somewhat from the Nusselt assumptions in that the first liquid to condense is the higher boiler and higher boilers in any homologous chemical series are more viscous than the lower ones. For vertical or horizontal tubes this means that the Reynolds number for the inlet portion of the tubes might well be lower than might be calculated using the mean properties of the total mixture for the condensate film. Nevertheless, to avoid the tediousness of an integration to determine the changes in $\bar{h}$ due to changing liquid properties for differential changes in area dA , it is possible to use a method of averaging the inlet and outlet film coefficients. This can be done by using the mean properties of the mixture, or, if there is a great difference in the characteristics of the condensate between inlet and outlet, to calculate h at both terminals and take the mean. The compositions of the final condensate on the tubes of vertical and horizontal condensers condensing the same mixture are not identical, since the condensate is cumulative on vertical tubes. There appears to be little need, however, in view of the other assumptions which also apply, to take this into account except with viscous condensates.

In either vertical or horizontal condensers it is helpful to consider, as in the development of Eq. (13.20), that the phase equilibrium at a cross section exists between the residual vapor and all the liquid formed up to

that point. In condensing a multicomponent mixture between its dew point and bubble point the vapor is cooled and the condensate formed at the inlet leaves at the temperature of the outlet, which may be colder by some 100°F or more. The vapor and condensate must be cooled sensibly as they travel the length of the shell, although this is not the same as desuperheating or subcooling, since it occurs concurrently with condensation rather than in distinct zones. The clean surface required for condensation is frequently calculated from the *entire* heat load, the weighted Δt , and a value of U_{clean} , using h obtained from either of the two methods described above. The main problem of sensible-heat transfer appears to be in cooling the condensate rather than the vapor, since the vapor coefficient in the presence of condensation is quite high as discussed for steam in Chap. 12. One method of treating the sensible-heat transfer is to allow additional surface equivalent to the percentage which the sensible-heat load is to the total heat load. The sensible-heat-transfer surface is then an *additional* percentage of the condensing surface. This is equivalent to the use of a sensible-heat-transfer coefficient about one-half the condensing coefficient. Another method is to compute condensate cooling surface using the free-convection coefficient of about 50 but applied only to the average liquid cooling sensible-heat load. Actually both methods give about the same answer. The effectiveness of the surface required for sensible-heat transfer, and particularly for cooling the condensate, is assured by submerging all the additional surface allowance through the use of a loop seal or dam baffle as discussed previously. The overall clean coefficient is then the weighted coefficient based on the total clean surface. If the vapor enters above the dew point or condensate leaves below the bubble point, the desuperheating and subcooling zones are weighted along with the condensing zone as in Chap. 12.

Example 13.3. Condenser Calculations for a Multicomponent Mixture. The overhead vapor from a distilling column operating at 50 psia contains only saturated hydrocarbons such as propane, butane, and hexane and has the following analysis:

	Lb/hr	Mol/hr
C_3	7,505	170.5
C_4^*	16,505	284.0
C_6	4,890	56.8
C_7	34,150	341.1
C_8	32,400	284.0
	95,450	1136.4

* A trace of C_5 has been combined half with C_4 and half with C_6 to simplify the calculation.

It is to be condensed in a 1-2 horizontal condenser using cooling water from 80 to 120°F. Pressure drops of 2.0 psi for the vapor and 10.0 psi for the water should be adequate. A minimum dirt factor of 0.004 should be provided. Available for the service is a 33 in. ID 1-2 condenser having 774 $\frac{3}{4}$ in. OD, 16 BWG tubes 16'0" long and laid out on 1-in. triangular pitch. The bundle is arranged for four passes, and baffles are spaced, except at the inlet and outlet, 30 in. apart.

- Determine the condensing range.
- Compute the condensing curve.
- Compute the weighted Δt .
- Establish the suitability of the condenser.

Solution. (a) Condensing range: This is the temperature spread between the dew point [Eq. (13.17)] and the bubble point [Eq. (13.18)].

Dew point: Assume $T = 283^\circ\text{F}$

Bubble point: Assume $T = 120^\circ\text{F}$

	V_1	$K_1 \text{ at } 283^\circ\text{F}$	$\frac{V_1}{K_1}$	L_1	$K_1 \text{ at } 120^\circ\text{F}$	$K_1 L_1$
C_3	170.5	13.75	12.40	170.5	4.1	700
C_4	284.0	6.18	46.0	284.0	1.39	395
C_5	56.8	1.60	35.5	56.8	0.17	9.66
C_7	341.1	0.825	414.	341.1	0.06	20.44
C_8	284.0	0.452	628.	284.0	0.023	6.54
	1136.4		1147.0 Check	1136.4		1132.6 Check

Assume intervals at 270, 250, 230, 200, 160°, and the dew point, and solve for V/L by Eq. (13.19) to obtain the condensation in each interval and from it the heat load for the interval.

Range: 283 to 270°F

Trial: Assume $V/L = 4.00$.

	Y_1	K_{270°	$\frac{K_1 V}{L}$	$1 + \frac{K_1 V}{L}$	$L_1 = \frac{Y_1}{1 + K_1(V/L)}$
C_3	170.5	12.75	51.0	52.0	3.28
C_4	284.0	5.61	22.4	23.4	12.13
C_5	56.8	1.40	5.60	6.60	8.60
C_7	341.1	0.705	2.82	3.82	89.3
C_8	284.0	0.375	1.50	2.50	113.7
	1136.4			$L = \Sigma L_1 = 227.0$	

$$V = 1136.4 - 227.0 = 909.4 \rightarrow V = L + V \rightarrow V = 909.4$$

$$V/L \text{ calculated} = 4.00$$

$$V/L \text{ assumed} = 4.00 \text{ Check}$$

If the assumed and calculated values of V/L had not checked, a new value would have been assumed. A check in this type of calculation generally infers a variation of 0.01 or less as the ratio of V/L decreases. For the next range, 270 to 250°F, proceed as above and obtain the actual mols of condensation for the interval by subtracting the liquid in equilibrium at 270°F from that in equilibrium at 250°F. Then L_{c1} are the mols of the individual compounds formed before the interval and L'_1 are the mols of the individual compounds formed in the interval. L'_1 is obtained by subtracting L_{c1} from L_1 , which is in turn obtained from a checked assumption of V/L where $\Sigma L_1 = L$. A summary of the point-to-point calculations is given in Table 13.2.

(b) Condensing curve: This step requires the calculation of the heat load between intervals. Except at the inlet and outlet of the condenser there is a change in both vapor and liquid quantities over the interval. The heat changes are determined from the change in enthalpies as given in Fig. 10. A representative interval is calculated following Table 13.2.

The LMTD would be 87.7°F, and the error resulting from its use would have been $(13/100.7) \times 100 = 12.9$ per cent on the safe side. In any system without serious fronts or tails the use of the LMTD is usually satisfactory, although there is no assurance that the error will always be on the safe side.

The condensing curve for true counterflow is shown as a straight line in Fig. 13.5. The actual condensing curve is shown as a curved line, and the area enclosed by the two represents the actual increase in temperature potential which is available.

TABLE 13.2. POINT-TO-POINT COMPOSITIONS

$$L_1 = L_{c1} + L'_1 \quad L = (L_1)$$

T_{vapor} (DP) 283°		270°		250°			230°			200°		
	Y_1	L_1	Y_1	L_{c1}	L'_1	Y_1	L_{c1}	L'_1	Y_1	L_{c1}	L'_1	
C_3	170.5	3.28	167.2	3.28	5.85	161.4	9.13	8.02	153.3	17.15	16.40	
C_4	284.0	12.13	271.9	12.13	20.81	251.1	32.94	26.04	225.0	58.98	46.6	
C_6	56.8	8.60	48.2	8.60	12.00	36.2	20.60	11.02	25.2	31.62	11.88	
C_7	341.1	89.3	251.8	89.3	94.5	157.3	183.8	63.6	93.7	247.4	51.6	
C_8	284.0	113.7	170.3	113.7	82.4	87.9	196.1	41.7	46.2	237.8	28.2	
	1136.4	227.0	909.4	227.0	215.6	693.9	442.5	150.3	543.4	593.0	154.7	
$V/L =$		4.00		1.567			0.916			0.520		

200°		160°			120°(BP)		
	Y_1	L_{c1}	L'_1	Y_1	L_{c1}	L'_1	
C_3	136.9	33.55	40.2	96.7	73.8	170.5	
C_4	178.4	105.6	83.7	94.7	189.3	284.0	
C_6	13.3	43.5	9.30	4.0	52.8	56.8	
C_7	42.1	299.0	31.8	10.3	330.8	341.1	
C_8	18.0	266.0	14.30	3.7	280.3	284.0	
	388.7	747.7	179.3	209.4	927.0	1136.4	
$V/L =$		0.520		0.226			

HEAT LOAD FOR THE INTERVAL 270 TO 250°

	Mol. wt.	$H_{v,270^\circ}$	Y_1	H_v	$H_{l,270^\circ}$	L_{c1}	H_1
C_3	44	324	167.2	2,384,000	210	3.28	30,300
C_4	58	334	271.9	5,260,000	212	12.13	149,200
C_6	86	352	48.2	1,460,000	226	8.60	167,000
C_7	100	359	251.8	9,030,000	236	89.3	2,105,000
C_8	114	368	170.3	7,150,000	239	113.7	3,100,000
				25,284,000			5,551,500
				5,551,500			

$$H_{270^\circ} = 30,835,500 \text{ Btu/hr}$$

	Mol. wt.	$H_{v,250^\circ}$	Y_1	H_v	$H_{l,250^\circ}$	L_{c1}	H_1
C_3	44	313	161.4	2,221,000	195	9.13	78,400
C_4	58	323	251.1	4,700,000	197.5	32.94	377,000
C_6	86	341	36.2	1,060,000	212.5	20.60	376,000
C_7	100	350	157.3	5,500,000	224	183.8	4,110,000
C_8	114	358	87.9	3,580,000	225	196.1	5,040,000
				17,061,000			9,981,400
				9,981,400			

$$H_{250^\circ} = 27,042,400 \text{ Btu/hr}$$

$$Q_{270-250^\circ} = 30,835,500 - 27,042,000 = 3,793,100 \text{ Btu/hr}$$

HEAT LOAD FOR THE ENTIRE RANGE

$T_{\text{vapor}}, ^\circ\text{F}$	H	$q = \Delta H$	Δt_w^*	$t_w, ^\circ\text{F}$	Δt_{av}	$\frac{q}{\Delta t}$	$q_{\text{cumulative}}$
283	34,312,000	3,476,500	6.55	120			0
270	30,835,500	3,793,100	7.15	113.4	159.8	21,780	3,476,500
250	27,042,000	2,839,400	5.35	106.3	150.1	25,210	7,269,600
230	24,203,000	3,359,000	6.34	100.9	136.4	20,800	10,109,000
200	20,844,000	3,931,100	7.42	94.6	117.2	28,620	13,468,000
160	16,912,900	3,803,900	7.17	87.2	88.2†	44,550	17,399,100
120	13,109,000			80.0	54.9†	69,450	21,203,000
		21,203,000	39.98			$\Sigma UA = 210,410$	

* Water requirement: t_w is the water temperature.

$$\frac{21,203,000}{120 - 80} = 530,000 \text{ lb/hr of cooling water} \approx 1060 \text{ gpm}$$

† LMTD

$$\text{Weighted } \Delta t = \frac{\Sigma q}{\Sigma UA} = \frac{21,203,000}{210,410} = 100.7^\circ\text{F}$$

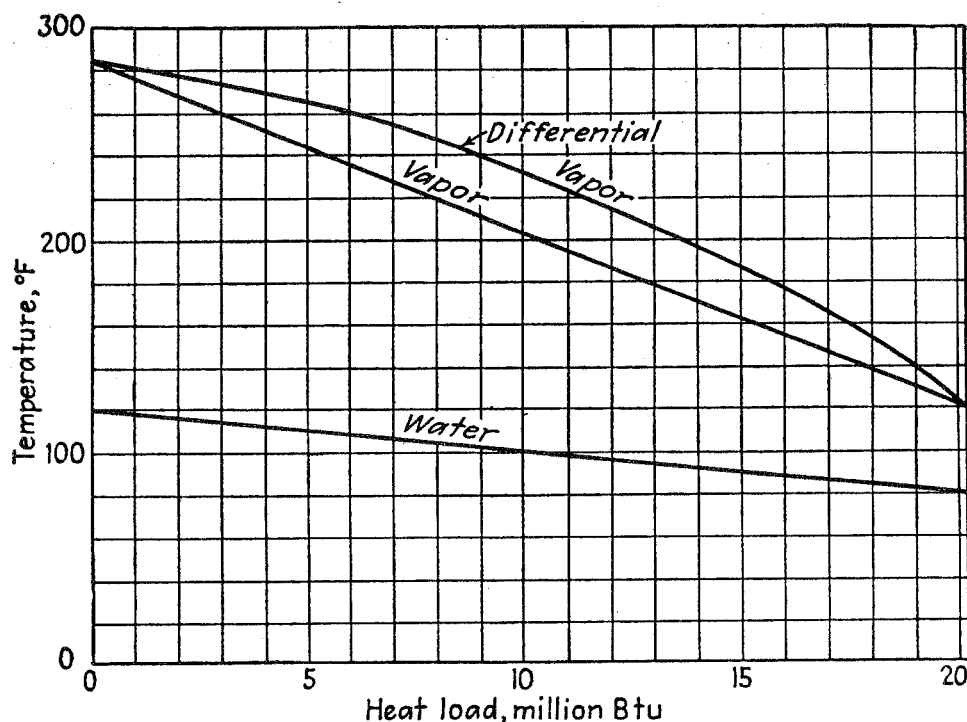


FIG. 13.5. Straight-line and differential condensing curves

The calculation of the exchanger for Example 13.3 follows:

Exchanger:

Shell side	Tube side
ID = 33 in.	Number and length = 774, 16'0"
Baffle space = 30 in.	OD, BWG, pitch = $\frac{3}{4}$ in., 16 BWG, 1-in. tri.
Passes = 1	Passes = 4

(1) Heat balance:

$$Q_{\text{shell}} = 21,203,000 \text{ Btu/hr}$$

$$Q_{\text{water}} = 530,000 \times 1(120 - 80) = 21,203,000 \text{ Btu/hr}$$

$$\text{Average molecular weight} = \frac{95,450}{1136.4} = 84$$

This corresponds very closely to hexane (mol. wt. = 86.2) whose properties will be used throughout.

$$\text{Condensate sensible heat load} = 95,450 \times \frac{0.6(283 - 120)}{2} = 4,670,000 \text{ Btu/hr}$$

$$\text{Submergence} = 4,670,000 \times \frac{100}{21,203,000} = 22\% \text{ (approx)}$$

(2) Δt : Weighted $\Delta t = 100.7^\circ\text{F}$

(3) T_c and t_c : The use of average temperatures will be satisfactory.

Hot fluid: shell side, vapor	Cold fluid: tube side, water
(4') Unsubmerged tubes	(4) Flow area, $a'_t = 0.302 \text{ in.}^2$
$= 774 \times (1 - 0.22) = 604$	[Table 10]
$G'' = W/LN_t^{2/3} = 95,450/16 \times 604^{2/3}$	$a_t = N_t a'_t / 144n$ [Eq. (7.48)]
$= 83.7$ [Eq. (12.43)]	$= 774 \times 0.302 / 144 \times 4 = 0.406 \text{ ft}^2$

Hot fluid: shell side, vapor	Cold fluid: tube side water
Assume $\bar{h} = 200$	(5) $G_t = w/a_t$
$T_v = (283 + 120)/2 = 201.5^\circ\text{F}$	$= 530,000/0.406$
$t_w = t_a + \frac{h_o}{h_{io} + h_o} (T_v - t_a)$ [Eq. (5.31)]	$= 1,300,000 \text{ lb}/(\text{hr})(\text{ft}^2)$
$= 100 + \frac{200}{1120 + 200} (201.5 - 100)$	$V = G_t/3600\rho = 1,300,000/3600 \times 62.5$
$= 115^\circ\text{F}$	$= 5.79 \text{ fps}$
$t_f = \frac{1}{2}(T_v + t_w) = \frac{1}{2}(201.5 + 115)$	(6) $h_i = 1355$ [Fig. 25]
$= 158^\circ\text{F}$	$h_{io} = h_i \times \text{ID}/\text{OD}$
$s_f = 0.60$ [Fig. 6]	$= 1355 \times 0.62/0.75$
$\mu_f = 0.21 \text{ cp}$ [Fig. 14]	$= 1120 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$ [Eq. (6.5)]
$k_f = 0.077 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}/\text{ft})$ [Table 4]	
$\bar{h} = 206$ [Fig. 12.9]	

Clean overall coefficient U_c , condensation:

$$U_c = \frac{h_{io}h_o}{h_{io} + h_o} = \frac{1120 \times 206}{1120 + 206} = 174 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F}) \quad (6.38)$$

Clean surface required for condensation:

$$A_c = \frac{q_c}{U_c \Delta t} = \frac{21,203,000}{174 \times 100.7} = 1210 \text{ ft}^2$$

Clean surface required for subcooling:

$$A_s = 1210 \times 0.22 = 267 \text{ ft}^2$$

Total surface required:

$$A_C = 1210 + 267 = 1477 \text{ ft}^2$$

Weighted overall clean coefficient U_C :

$$U_C = \frac{Q}{A \Delta t} = \frac{21,203,000}{1477 \times 100.7} = 143$$

Design overall coefficient U_D :

$$a'' = 0.1963 \text{ ft}^2/\text{lin ft}$$

(Table 10)

$$\text{Total surface} = 774 \times 16'0'' \times 0.1963 = 2430 \text{ ft}^2$$

$$U_D = \frac{Q}{A \Delta t} = \frac{21,203,000}{2430 \times 100.7} = 86.7 \text{ Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$$

Dirt factor R_d :

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{143 - 86.7}{143 \times 86.7} = 0.00455 \text{ (hr)}(\text{ft}^2)(^\circ\text{F})/\text{Btu} \quad (6.13)$$

Pressure Drop

$$\begin{aligned} (1') a_s &= \text{ID} \times C'B/144P_T \quad [\text{Eq. (7.1)}] \\ &= 33 \times 0.25 \times 30/144 \times 1.0 \\ &= 1.72 \text{ ft}^2 \end{aligned}$$

Submergence may be neglected unless the calculated pressure drop is close.

$$\begin{aligned} G_s &= W/a_s \quad [\text{Eq. (7.2)}] \\ &= 95,450/1.72 = 55,500 \text{ lb}/(\text{hr})(\text{ft}^2) \end{aligned}$$

(1) At 100°F ,

$$\mu = 0.72 \times 2.42 = 1.74 \text{ lb}/(\text{ft})(\text{hr})$$

[Fig. 14]

$$D = 0.62/12 = 0.0517 \text{ ft}$$

$$Re_t = DG_t/\mu$$

$$= 0.0517 \times 1,300,000/1.74 = 38,600$$

$$f = 0.00019 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 26}]$$

Pressure Drop

At $T_1 = 283^\circ\text{F}$,

$$\mu = 0.009 \times 2.42 = 0.0218 \text{ lb}/(\text{ft})(\text{hr})$$

$$D_s = 0.73/12 = 0.0608 \text{ ft} \quad [\text{Fig. 15}]$$

$$Re_s = D_s G_s / \mu$$

$$= 0.0608 \times 55,500 / 0.0218 = 155,000$$

$$f = 0.00125 \text{ ft}^2/\text{in.}^2 \quad [\text{Fig. 29}]$$

$$(2') \text{ No. of crosses } (N + 1) = 12L/B \quad [\text{Eq. (7.43)}]$$

$$= 12 \times 16/30 = 7$$

$$\rho = \frac{84}{359 \times 743/492 \times 14.7/50} = 0.527 \text{ lb}/\text{ft}^3$$

$$s = 0.527/62.5 = 0.00844$$

$$D_s = 33/12 = 2.75 \text{ ft}$$

$$(3') \Delta P_s = \frac{1}{2} \frac{f G_s^2 D_s (N + 1)}{5.22 \times 10^{10} D_s s} \quad [\text{Eq. (12.47)}]$$

$$= \frac{1}{2} \frac{0.00125 \times 55,500^2 \times 2.75 \times 7}{5.22 \times 10^{10} \times 0.0608 \times 0.00844} = 1.4 \text{ psi}$$

$$(2) \Delta P_t = \frac{f G_t^2 L n}{5.22 \times 10^{10} D_s \phi_t} \quad [\text{Eq. (7.45)}]$$

$$= \frac{0.00019 \times 1,300,000^2 \times 16 \times 4}{5.22 \times 10^{10} \times 0.0517 \times 1.0 \times 1.0} = 7.6 \text{ psi}$$

$$(3) \Delta P_r = (4n/s)(V^2/2g') \quad [\text{Eq. (7.46)}]$$

$$= 4 \times 4 \times 0.23 = 3.7 \text{ psi} \quad [\text{Fig. 27}]$$

$$(4) \Delta P_T = \Delta P_t + \Delta P_r = 7.6 + 3.7 = 11.3 \text{ psi} \quad [\text{Eq. (7.47)}]$$

Summary

206*	<i>h</i> outside	1120
U_C	143	
U_D	86.7	
R_d Calculated 0.00455		
R_d Required 0.004		
1.4	Calculated ΔP	11.3
2.0	Allowable ΔP	10.0

* Condensation only.

The slightly excessive tube-side pressure drop for water should not be objectionable. The bundle should be submerged about 25 to 30 per cent.

3. CONDENSATION OF A MIXTURE OF MISCIBLES AND ONE IMMISCIBLE

This case arises in the steam distillation of organics which are miscible among themselves but not with water. The stripping of the volatile compounds from absorption oil is a typical example as shown in Fig 11.1.

The introduction of the steam permits part of the total operating pressure in the distilling column to be contributed by the steam, so that the mixture of volatiles and oil need not be raised to a high temperature to accomplish distillation. In this way and without resorting to vacuum a greater separation can be made between the volatiles and the oil, which boils over a somewhat higher range than that at which the distillation is carried out. If the overhead vapor is a mixture of a single compound and steam, the condensation occurs isothermally. If the mixture contains more than one compound miscible with the first but immiscible with water, there is a condensing range. Since the total pressure in the latter consists of water plus the miscibles, the equilibrium relationships for the miscibles corresponds to the sum of their partial pressures instead of the total pressure on the system. The partial pressure of the steam depends only upon the saturation pressure corresponding to its temperature in the mixture as given by the properties of saturated steam in Table 7. The total pressure is constant, but the relative partial pressures of the miscibles and steam change from point to point. The problems involved in the calculation of the condensing curve for this system are included in the more comprehensive problem demonstrated in Sec. 5 of this chapter. Ordinarily the calculation is so lengthy that the use of the LMTD or the use of $F_T \times \text{LMTD}$ is justifiable as the case may be.

Hazelton and Baker¹ carried out experimental work on a single vertical tube condensing benzene, toluene, and chlorbenzene with steam. It was found that the presence of the organic tended to favor the dropwise condensation of the steam. Film coefficients were found to be independent of the temperature drop across the condensate film and independent of the properties of the liquid condensed with water. Hazelton and Baker were able to correlate their work with the experimental results of others. For vertical tubes they obtained

$$\bar{h} = 79 \left[\frac{(\text{wt-\%})_A \lambda_A + (\text{wt-\%})_B \lambda_B}{(\text{wt-\%})_A L} \right]^{1/4} \quad (13.21a)$$

where A and B refer, respectively, to the organic and the water in the condensate film and L is the tube length in feet. They were able to correlate the work of several investigators on horizontal tubes by

$$\bar{h} = 61 \left[\frac{(\text{wt-\%})_A \lambda_A + (\text{wt-\%})_B \lambda_B}{(\text{wt-\%})_A D_o} \right]^{1/4} \quad (13.21b)$$

For tube bundles in horizontal exchangers it is unlikely that the coefficient differs greatly from the value predicted by Eq. (13.21b).

¹ Hazelton, R., and E. M. Baker, *Trans. AIChE*, **40**, 1-29 (1944).

4. CONDENSATION OF A VAPOR FROM A NONCONDENSABLE GAS

As in the case of a gas compressor intercooler (Chap. 9) if a mixture of a vapor and a gas is cooled in a constant-pressure operation, the temperature at which the first droplet of condensate appears is the *dew point*. The dew point is the saturation temperature of the vapor corresponding to its partial pressure in the mixture. The calculation of the dew point of a mixture of a vapor and a noncondensable gas was demonstrated in Example 9.3. The calculation of the compressor intercooler is covered in this section.

When a mixture of a vapor and a noncondensable gas is fed to a condenser and the temperature of the tubes is below the dew point, a film

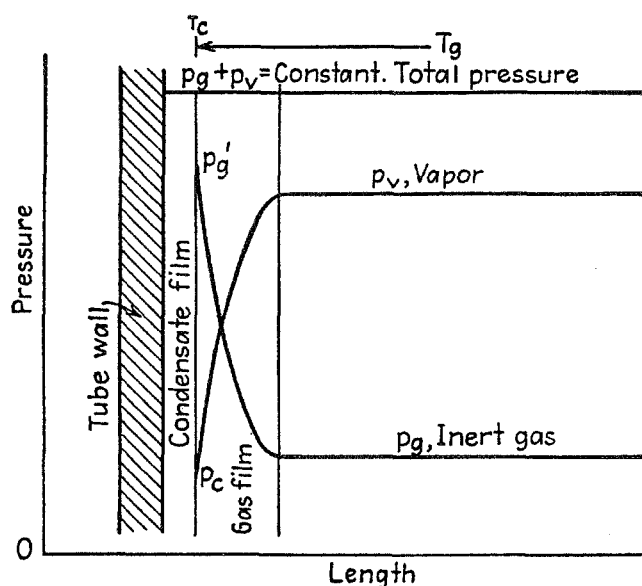


FIG. 13.6. Condensation potentials with noncondensables.

of condensate forms on the tubes. The relationship of the partial pressures is shown in Fig. 13.6. A film of noncondensable gas and vapor collects about the condensate film as suggested by the data of Othmer in Fig. 12.25. If an equilibrium is presumed to exist on the surface of the condensate film, the partial pressure of the vapor at the tube wall corresponds to the cold condensate p_c and the partial pressure of the vapor in the gas film lies between that at the condensate film p_c and that in the gas body p_v . In order for the vapor in the gas body to continue condensing into the condensate film, it must be driven across the gas film by the difference between the partial pressure of the vapor in the gas body and in the condensate. The passage of one component through another is called *diffusion* or *mass transfer*, and in a diffusion system the rate at which the steam condenses is no longer dependent entirely upon Nusselt's condensing mechanism but upon the laws governing diffusion. When

vapor diffuses through a noncondensable gas film and liquefies at the tube wall, it also carries with it its latent heat of condensation. In addition to the partial-pressure difference promoting diffusion there is also a temperature difference between the gas body T_g and the condensate film T_c by which the gas is sensibly cooled. It will be shown that the rate of diffusion and the rate of heat transfer are not independent of each other.

Relationships between Heat and Mass Transfer. The theory of diffusion will be treated in greater detail in Chap. 17, which is devoted to diffusional heat transfer.

It was shown in Chap. 3 that there is an analogous behavior between heat transfer and fluid friction when a fluid flows in a tube. Another analogy exists for a system in which a transfer of matter by diffusion is accompanied by a transfer of heat. A number of authors have contributed to the refinement and extension of this analogy, but the derivations employed here are essentially those of Colburn,¹ Colburn and Hougen,² and Chilton and Colburn.³ The paper of Colburn and Hougen forms the basis for design calculations.

It was shown in Chap. 9 that when dealing with gases it is advantageous to express the heat-transfer factor j_H for a fluid flowing inside a tube by

$$j_H = \frac{hD}{k} \left(\frac{c\mu}{k} \right)^{-1/3} = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{2/3} \left(\frac{DG}{\mu} \right) \quad (13.22)$$

A new factor j_h may then be defined as $j_h = j_H/(DG/\mu)$ or

$$j_h = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{2/3} \quad (13.23)$$

By employing j_h it is possible to eliminate the variation of k with temperature in making a sensible-heat-transfer calculation, since $(c\mu/k)$ is nearly constant over a wide temperature range. Since

$$Q = wc(t_2 - t_1) = h\pi DL \Delta t$$

and $G = 4w/\pi D^2$, on substituting for both h and G Eq. (13.23) becomes

$$j_h = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{2/3} = \frac{t_2 - t_1}{\Delta t} \frac{D}{4L} \left(\frac{c\mu}{k} \right)^{2/3} \quad (13.24)$$

Multiplying the last term by $\pi D/\pi D$,

$$j_h = \frac{h}{cG} \left(\frac{c\mu}{k} \right)^{2/3} = \frac{t_2 - t_1}{\Delta t} \left(\frac{a}{A} \right) \left(\frac{c\mu}{k} \right)^{2/3} \quad (13.25)$$

where a is the flow area $\pi D^2/4$, and A is the tube surface πDL .

¹ Colburn, A. P., *Trans. AIChE*, **29**, 174 (1933).

² Colburn, A. P., and O. A. Hougen, *Ind. Eng. Chem.*, **26**, 1178 (1934).

³ Chilton, T. H., and A. P. Colburn, *Ind. Eng. Chem.*, **26**, 1183 (1934).

When a vapor is absorbed from a gas, which is not saturated with molecules of the solvent, diffusion may occur in two directions: Molecules of the vapor may pass into the absorbent, and molecules of the absorbent may pass into the gas. In the passage of water vapor from the gas body into a condensate film consisting of liquid water alone the transfer of matter is in one direction and the mols of matter transferred from the gas to the liquid is given by

$$dN_d = d \left(\frac{G a p_v}{M_m p_t} \right) = K_G \Delta p dA \quad (13.26a)$$

where A = diffusion surface, ft^2

a = flow area of gas and vapor, ft^2

G = mass velocity, $\text{lb}/(\text{hr})(\text{ft}^2)$

K_G = mass diffusion coefficient, $\text{mol}/(\text{hr})(\text{ft}^2)(\text{atm})$

M_m = mean molecular weight of the vapor and noncondensable, lb/mol

N_d = material transferred, mol/hr

p_c = partial pressure in atmospheres of the vapor at condensate film, atm

p_v = partial pressure in atmospheres of the vapor in gas body, atm

p_t = total pressure on the system in atmospheres, atm

Δp = instantaneous driving potential in atmospheres, $p_v - p_c$, atm

The coefficient K_G is simply the dimensional rate constant which makes dN_d equal to the right side of the equation. K_G is determined experimentally, and therefore K_G and U are similar in function, K_G being to Δp in mass transfer what U is to Δt in heat transfer. If the initial concentration of the condensable vapor is small, it is convenient to use the simplification that the mass velocity of the mixture does not vary appreciably during the diffusion of the vapor out of the gas and that

$$\frac{G}{M_m} = \frac{G_i}{M_i} \left(\frac{p_t}{p_t - p_v} \right)$$

where the subscript i refers to the inert gas. Replacing $p_t - p_v$ by p_g , the pressure of the inert gas in the gas body, and keeping p_t constant in the tube with constant a , the differentiation of the second term in Eq. (13.26a) yields

$$dN_d = \frac{G a dp_v}{M_m p_t} = K_G \Delta p dA \quad (13.26b)$$

The second and third terms can be readily regrouped to give

$$\frac{K_g p_{gf}}{G/M_m} = \left(\frac{dp_v}{\Delta p} \right) \left(\frac{p_{gf}}{p_t} \right) \left(\frac{a}{dA} \right)$$

where p_{gf} is the log mean of p_g of the inert in the gas body and $p'_g = p_t - p_c$ the pressure of the inert at the condensate film. In integrated form it becomes

$$\frac{K_g p_{gf}}{G/M_m} = \left(\frac{p_1 - p_2}{\Delta p} \right) \left(\frac{p_{gf}}{p_g} \right) \left(\frac{a}{A} \right) \quad (13.27)$$

where p_1 and p_2 are the partial pressures of the diffusing component at A_1 and A_2 .

When a fluid flows along a surface, the particles within the fluid exchange momentum with the stationary film at the surface, causing a pressure drop in the fluid in the direction of flow. This assumption led to Eq. (3.51) in the Reynolds analogy. It is entirely conceivable that a similar condition occurs when a vapor traveling along a surface condenses against the condensate film which it enters by moving at right angles to the direction of flow and giving up its momentum. In Eq. (3.51) and subsequent equations it became apparent that the ratio of the loss of momentum by skin friction to the total momentum of the stream depended upon how much stationary surface there was for the total amount of fluid flowing. For a given quantity of fluid flowing in a tube and a given total diffusion or heat-transfer surface, the amount of skin friction will be greater if the path consists of a long small-bore tube than for a short tube of large diameter. The index of these possibilities is the ratio A/a or, when used in a diffusion factor, its reciprocal a/A . Just as it was found in the refinement of the Reynolds analogy that the ratio of μ/k influences the heat transfer, so it can be inferred that the properties of the fluid also affect diffusion. The properties associated with skin friction are contained in the dimensionless Schmidt number $\mu/\rho k_d$, where k_d is the diffusion coefficient (diffusivity) in square feet per hour of one gas through another and μ and ρ are the viscosity and density of the mixture. If the influence of μ/k_d on diffusion is comparable to that of μ/k in heat transfer, then it is reasonable to multiply Eq. (13.27) by $(\mu/\rho k_d)$. Assuming $p_{gf}/p_g = 1.0$, designating the diffusion factor j_d , and arbitrarily using the two-thirds power

$$j_d = \left(\frac{p_1 - p_2}{\Delta p} \right) \left(\frac{a}{A} \right) \left(\frac{\mu}{\rho k_d} \right)^{2/3} \quad (13.28)$$

Attention is now directed to the similarity between Eqs. (13.25) and (13.28). From an extension of the Reynolds analogy to distillation,

where the analogy between mass and heat transfer is very close, there is good reason to believe that j_d and j_h are the same function of the Reynolds number and equal. The relationship between diffusion and heat transfer is then obtained by equating Eqs. (13.25) and (13.28) and solving for K_G .

$$K_G = \frac{h(c\mu/k)^{2/3}}{cp_{gf}M_m(\mu/\rho k_d)^{2/3}} \quad (13.29)$$

The principal deduction from Eq. (13.29) is that the rates of diffusion and heat transfer do not occur independently. When the vapor concentration is high, as in many industrial applications, Eq. (13.28) must be calculated for incremental changes in the surface, since p_{gf}/p_g will no longer be unity.

The overall coefficient of heat transfer varies greatly during the condensation of a vapor from a noncondensable gas which is initially at its dew point because the potential for diffusion varies greatly as the vapor is removed from the gas body leaving a higher percentage of inert. At the inlet the composition of a mixture of vapor and noncondensable may be almost all vapor and the film coefficient may be very nearly that of the pure condensing coefficient for the vapor alone. But after much of the vapor has condensed, the outlet may consist of substantially pure noncondensable gas with a low accompanying film coefficient. It is often possible to have a variation of U_c for the condensation of steam from air from 1500 Btu/(hr)(ft²)(°F) at the inlet to a value of 15 at the outlet.

Inasmuch as the film coefficient varies from inlet to outlet, the heat distribution may also vary owing to a differential rate of change in the enthalpy of the vapor mixture as the temperature falls. In other words, although the temperature of the gas declines 50 per cent of the total gas-temperature range, it is probably untrue that 50 per cent of the total heat load will have been delivered. This is not merely the case of finding the true temperature difference but the heat-transfer coefficient also varies as $dQ = q$ varies. The surface is then defined by the fundamental equation

$$A = \int \frac{dQ}{(U \Delta t)} \quad (13.30)$$

Equation (13.30) cannot be integrated unless U and Δt are expressed as functions of Q . A much simpler method is to integrate numerically $dQ = q$ for small but finite intervals.

Diffusivities. For an excellent account of diffusion and the experimental methods of determining the diffusivity of one gas through another,

the reader is referred to Sherwood's excellent book.¹ Gilliland² has established an empirical equation for the determination of the diffusivity of one gas through another which is given by

$$k_d = 0.0166 \frac{T^{3/2}}{p_t(v_A^{1/3} + v_B^{1/3})^2} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2} \quad (13.31)$$

where k_d = diffusivity, ft²/hr

p_t = total pressure, atm

v_A, v_B = molecular volumes of diffusing and inert gases computed from the data on atomic volumes in Table 13.3

T = absolute temperature °K(°C abs)

M_A, M_B = molecular weights of the diffusing and inert gases, respectively

TABLE 13.3. ATOMIC VOLUMES

Bromine.....	27.0
Sulfur.....	25.6
Oxygen.....	7.4
In methyl esters.....	9.1
In higher esters and ethers.....	11.0
In acids.....	12.0
Carbon.....	14.8
Chlorine.....	24.6
Hydrogen.....	3.7
Nitrogen.....	15.6
In primary amines.....	10.5
In secondary amines.....	12.0
For benzene ring formation deduct.....	15
For naphthaline deduct.....	30
For the hydrogen molecule use.....	$v = 14.3$
For air use.....	$v = 29.9$

Example 13.4. Calculation of the Diffusivity of a Mixture. Calculate the diffusivity of a steam-CO₂ mixture at 267°F and 30 psig.

For steam, H₂O: $v_A = 2 \times 3.7 + 7.4 = 14.8$ $M_A = 18$

For CO₂: $v_B = 14.8 + 2 \times 7.4 = 29.6$ $M_B = 44$

$T = 267^\circ\text{F} \approx 273 + 130 = 403^\circ\text{K}$ $30 \text{ psig} \approx \frac{14.7 + 30}{14.7} = 3.04 \text{ atm}$

$$k_d = 0.0166 \times \frac{403^{3/2}}{3.04(14.8^{1/3} + 29.6^{1/3})^2} \left(\frac{1}{18} + \frac{1}{44} \right)^{1/2} = 0.41 \text{ ft}^2/\text{hr}$$

Development of an Equation for Heat Transfer. Chilton and Colburn have shown that the results of their analogy, culminating in Eq. (13.29),

¹ Sherwood, T. K., "Absorption and Extraction," McGraw-Hill Book Company, Inc., New York, 1937.

² Gilliland, E. R., *Ind. Eng. Chem.*, **26**, 516 (1934).

hold on the safe side for flow inside tubes, flow across a single tube, and flow along plane surfaces. In each case the appropriate value of h is substituted in Eq. (13.29). It appears likely that these equations are valid in most cases of a vapor forming a nonviscous, noncontrolling condensate film. The use of h_o calculated for the gas from Fig. 28 and substituted in Eq. (13.29) for the condensation of steam from air and CO_2 in the shells of horizontal baffled condenser has been known to hold successfully in a number of applications.

To establish an equation which may be solved from point to point for U and Δt as in Eq. (13.30) it should be necessary only to sum up all the resistances in series at an average cross section in each increment of q . In the condensation of a vapor from a noncondensable gas the quantity of heat which leaves the gas film must equal the quantity picked up by the cooling water. The total heat flow across the gas film is the sum of the latent heat carried by vapor diffusion into the condensate film plus the sensible heat removed from the gas because of the temperature difference $T_g - T_c$. The heat load expressed in terms of the shell-side, tube-side, and overall potentials per square foot of surface when the mixture of gas and vapor flows in the shell is

$$h_o(T_g - T_c) + \overset{(a)}{K_g M_v \lambda (p_v - p_c)} = \overset{(b)}{h_{io}(T_c - t_w)} = \overset{(c)}{U(T_g - t_w)} \quad (13.32)$$

where h_o = shell-side dry-gas coefficient, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

h_{io} = tube-side water coefficient, $\text{Btu}/(\text{hr})(\text{ft}^2)(^\circ\text{F})$

T_g = temperature of the gas, $^\circ\text{F}$

T_c = temperature of the condensate, $^\circ\text{F}$

t_w = water temperature, $^\circ\text{F}$

p_v = partial pressure of vapor in the gas body, atm

p_c = partial pressure of the vapor at the condensate film, atm

M_v = molecular weight of vapor, dimensionless

λ = latent heat, Btu/lb

The possibility of subcooling the vapor at the tube wall has been omitted from the heat balance, since it is not usually significant compared with the larger latent effects. If subcooling is of consequence, it means that the amount of heat delivered at the tube wall is greater than that which entered the gas film.

In the application of Eqs. (13.29), (13.30), and (13.32) to the solution of an actual condenser it is assumed that there is a single value of T_g and T_c at any cross section and hence of p_v and p_c . In a condenser having several tube passes with water in the tubes this is obviously not possible. If there is a long condensing range, a small cooling-water range, and no temperature cross so that F_T for a 1-2 condenser would be substantially

1.0, then little harm results in the assumption of a true counterflow distribution of water temperatures. At whatever point the gas gives up half of the total heat load, the water can be assumed to have received half the total heat load. This may appear rather arbitrary, but in the final analysis it is a problem of maintaining $T_g - T_c$ and $p_v - p_c$ in their proper relation, and both require the presence of the same single value of T_c at any point. Otherwise the solution must be extended to cover each of the passes as a related condenser.

The Calculation of a Vapor-Noncondensable Mixture. The method of applying Eqs. (13.29), (13.30), and (13.32) are outlined below:

1. A complete exchanger must be assumed to fix the shell-side and tube-side flow areas. The surface is obtained by integration on the assumption of true counterflow.
2. From the process conditions compute h_o and h_{io} for the gas and cooling medium, respectively. The use of an average value for h_{io} is acceptable but not for h_o , since the mass velocity of the gas changes from point to point.
3. From the value of h_o obtain K_G/p_{gf} from Eq. (13.29).
4. Fix the first interval of calculation by fixing T_g , which also fixes the heat load q for the interval.
5. Assume values of T_c , the condensate temperature, so that Eqs. (13.32a) and (13.32b) balance. For each assumed value of T_c it is necessary to compute a new value of p_{gf} , since the pressure of the vapor at the condensate film is the saturation pressure corresponding to T_c .
6. When Eqs. (13.32a) and (13.32b) balance, the total heat load transferred per square foot of each is the same as the load which must have been transferred overall, $U(T_g - t_c)$.
7. From q obtained in 4 and $U(T_g - t_c)$ obtain dA for the interval.
8. Proceed with the next interval by assuming a lower value of T_g .

Example 13.5. Calculation of a Steam-Carbon Dioxide Condenser. Steam is to be condensed from carbon dioxide in the following exchanger:

21¼ in. ID shell with 12-in. baffle spacing.

246 tubes, ¾ in. OD, 16 BWG, 12'0" on 1-in. square pitch.

The tube bundle has four passes.

The hot stream is a mixture of 4500 lb of steam and 1544 lb of CO₂ at 30 psig entering at the dew point and leaving at 120°F. Cooling water will enter at 80°F and leave at 115°F.

The diffusivity of steam-CO₂ mixtures calculated by the Gilliland formula is 0.41 ft²/hr, and $(\mu/\rho k_a)^{2/3}$ may be taken for simplicity as constant at the average value of 0.62 between inlet and outlet.

(a) Determine the weighted Δt . (b) Determine the dirt factor for the condenser.

Solution. As a simplification it may be assumed that the heat-transfer coefficient at the average water temperature is constant throughout the exchanger. This does not alter the earlier assumption that the water temperature is considered in true counterflow with the condensate.

Basis: One hour

Entering	Lb/hr	Mol/hr
CO ₂	1544	35
H ₂ O	4500	250
Total	6044	285

Total pressure = $30 + 14.7 = 44.7$ psia ≈ 3.05 atm where $\text{psi}/14.7 = \text{atm}$ Partial pressure of water = $25\frac{0}{285} \times 44.7 = 39.2$ psi ≈ 2.68 atmDew point = 267° from Table 7 ≈ 2.68 atm.Mean molecular weight, $M_m = 6044\frac{4}{285} = 21.2$ (a) Weighted temperature difference Δt :

Overall balances:

Inlet: Water vapor pressure, $p_v = 2.68$ atmInert pressure, $p_g = 3.05 - 2.68 = 0.37$ atm

Total pressure = 3.05 atm.

Exit: Partial pressure of water at $120^\circ\text{F} = 0.1152$ atmWater vapor pressure, $p_v = 0.115$ atmInert pressure, $p_g = 2.935$ atm

Total pressure = 3.05 atm

Pound mols steam inlet = 250

Pound mols steam exit = $35 \times \frac{0.115}{2.935} = 1.37$ Inert gas = 30.63Pound mols steam condensed = $250 - 1.37 = 248.63$ Heat load: Assume points at 267, 262, 255, 225, 150, 120°F , and compute the heat load q for each interval.For the interval from 267 to 262°F :From Table 7, p_v at $262^\circ\text{F} = 2.49$ atm $p_g = 3.05 - 2.49 = 0.56$ atmMol steam remaining = $35 \times \frac{2.49}{0.56} = 156$ Mol steam condensed = $250 - 156 = 94$ Heat of condensation = $94 \times 18 \times 937.3 + 0.46(267 - 262) \times 94 \times 18 = 1,590,000$ BtuHeat from uncondensed steam = $156 \times 18 \times 0.46(267 - 262) = 6,450$ Heat from noncondensable = $1544 \times 0.22 \times 5.0 = 1,700$

Total for interval = 1,598,150

Heat balance:

Interval, $^\circ\text{F}$	q
267-262	1,598,000
262-255	1,104,000
255-225	1,172,000
225-150	751,000
150-120	177,000
Total	4,802,000

Total water = $\frac{4,802,000}{115 - 80} = 137,000$ lb/hr

Water coefficient, h_i :

$$a_i = N_i \frac{a'_i}{144n} = 246 \times \frac{0.302}{144 \times 4} = 0.129 \text{ ft}^2 \quad (7.48)$$

$$G_i = \frac{w}{a_i} = \frac{137,000}{0.129} = 1,060,000 \text{ lb/(hr)(ft}^2\text{)}$$

$$V = \frac{G}{3600\rho} = \frac{1,060,000}{3600 \times 62.5} = 4.72 \text{ fps}$$

$$h_i = 1120 \quad (\text{Fig. 25})$$

$$h_{io} = h_i \times \frac{\text{ID}}{\text{OD}} = 1120 \times \frac{0.62}{0.75} = 926 \quad (6.5)$$

Now proceed to determine $U \Delta t$ from point to point in the unit by assuming temperatures for the condensate film so that Eqs. (13.32a) and (13.32b) are equal. Shell-side coefficient for entrance gas mixture.

Mean properties for Point 1:

$$\text{Mean } c = \frac{1544 \times 0.22 + 4500 \times 0.46}{6044} = 0.407 \text{ Btu/(lb)(}^\circ\text{F)}$$

$$\text{Mean } k = \frac{1544 \times 0.0128 + 4500 \times 0.015}{6044} = 0.0146 \text{ Btu/(hr)(ft}^2\text{)(}^\circ\text{F/ft)}$$

$$\text{Mean } \mu = \frac{1544 \times 0.019 + 4500 \times 0.0136}{6044} = 0.015 \times 2.42 = 0.0363 \text{ lb/(ft)(hr)}$$

$$a_s = \text{ID} \times C' \frac{B}{144P_T} = 21.25 \times 0.25 \times \frac{12}{144 \times 1.0} = 0.442 \text{ ft}^2 \quad (7.1)$$

$$G_s = \frac{W}{a_s} = \frac{6044}{0.442} = 13,650 \text{ lb/(hr)(ft}^2\text{)} \quad (7.2)$$

$$Re_s = \frac{D_s G_s}{\mu} \quad D_s = \frac{0.95}{12} = 0.0792 \text{ ft} \quad (\text{Fig. 28})$$

$$= 0.0792 \times \frac{13,650}{0.0363} = 29,800 \quad (7.3)$$

$$j_H = 102 \quad (\text{Fig. 28})$$

$$\left(\frac{c\mu}{k}\right)^{1/3} = \left(\frac{0.407 \times 0.0363}{0.0146}\right)^{1/3} = 1.0$$

$$h_o = j_H \frac{k}{D_s} \left(\frac{c\mu}{k}\right)^{1/3} = 102 \times 0.0146 \times \frac{1.0}{0.0792} = 18.9 \quad (6.15b)$$

$$(\mu/\rho k_d)^{2/3} = 0.62 \quad \left(\frac{c\mu}{k}\right)^{2/3} = \left(\frac{0.407 \times 0.0363}{0.0146}\right)^{2/3} = 1.01$$

$$K_G = \frac{h_o(c\mu/k)^{2/3}}{cp_{sf}M_m(\mu/\rho k_d)^{2/3}} = \frac{18.9 \times 1.01}{0.407 \times p_{sf} \times 21.2 \times 0.62} = \frac{3.56}{p_{sf}}$$

Point 1:

$$T_g = 267^\circ\text{F (entrance)} \quad p_v = 2.68 \text{ atm} \quad p_g = 3.05 - 2.68 = 0.37 \text{ atm}$$

$$t_w = 115^\circ\text{F} \quad \Delta t = T_g - t_w = 267 - 115 = 152^\circ\text{F}$$

Try

$$T_c = 244^\circ\text{F} \quad p_c = 1.83 \text{ atm} \quad p'_g = 3.05 - 1.83 = 1.22 \text{ atm}$$

$$p_{sf} = \frac{p'_g - p_g}{2.3 \log p'_g/p_g} = \frac{1.22 - 0.37}{2.3 \log 1.22/0.37} = 0.715 \text{ atm}$$

$$h_o(T_g - T_c) + K_G M_v \lambda (p_v - p_c) = h_{io}(T_c - t_w) \quad (13.32)$$

$$18.9(267 - 244) + \frac{3.56}{0.715} \times 18 \times 933.8(2.68 - 1.83) = 926(244 - 115)$$

$$71,400 \neq 129,000 \quad \text{No check}$$

Try

$$T_c = 220^\circ\text{F} \quad p_c = 1.17 \text{ atm} \quad p'_c = 1.88 \text{ atm} \quad p_{cf} = 0.93 \text{ atm}$$

$$18.9(267 - 220) + \frac{3.56}{0.93} \times 18 \times 933.8(2.68 - 1.17) = 926(220 - 115)$$

$$98,400 = 97,500 \quad \text{Check}$$

$$U \Delta t = \frac{98,400 + 97,500}{2} = 97,950$$

$$U = \frac{97,950}{267 - 115} = 644$$

Having determined the conditions at the inlet, proceed from point to point in the exchanger. Since the partial pressure of water changes rapidly at high temperatures, it follows that most of the condensation will occur near the inlet. (Refer to the saturation curve for water.) To obtain a reasonable distribution of heat loads over the condenser, a second point very near the inlet has been chosen.

Point 2:

$$T_c = 262^\circ\text{F} \text{ and saturated} \quad p_c = 2.49 \text{ atm} \quad p_g = 3.05 - 2.49 = 0.56$$

$$\text{Mol steam remaining} = 35 \times \frac{2.49}{0.56} = 156$$

$$\text{New gas rate} = 1544 + 156 \times 18 = 4352 \text{ lb/hr}$$

$$\text{Mean properties: } M_m = 22.8 \quad c = 0.382 \quad k = 0.0143$$

$$\mu = 0.0154 \times 2.42 = 0.0373$$

$$G_s = \frac{W}{a_s} = \frac{4352}{0.442} = 9850 \text{ lb/(hr)(ft}^2\text{)}$$

$$Re = \frac{D_c G_s}{\mu} = 0.0792 \times \frac{9850}{0.0373} = 20,900$$

$$j_H = 83.5 \quad \left(\frac{c\mu}{k}\right)^{1/3} = 1.0 \quad \left(\frac{c\mu}{k}\right)^{2/3} = 1.0$$

$$h_o = 15.0$$

$$K_G = \frac{15.0 \times 1.0}{0.382 \times p_{cf} \times 22.8 \times 0.62} = \frac{2.80}{p_{cf}}$$

$$\text{Water rise} = 1,598,000/137,000 = 11.7^\circ\text{F}$$

$$t_w = 115 - 11.7 = 103.3^\circ\text{F}$$

Try

$$T_c = 182^\circ\text{F} \quad p_c = 0.534 \text{ atm} \quad p'_c = 2.51 \text{ atm} \quad p_{cf} = 1.305 \text{ atm}$$

$$15.0(262 - 182) + \frac{2.80}{1.305} \times 18 \times 937.3(2.49 - 0.534) = 926(182 - 103.3)$$

$$71,900 = 72,800 \quad \text{Check}$$

$$U \Delta t = \frac{71,900 + 72,800}{2} = 72,350$$

$$U = \frac{72,350}{262 - 103.3} = 456$$

Points 3, 4, and 5 are calculated in the same manner.

Point 6:

$$T_c = 120^\circ\text{F} \text{ and saturated} \quad p_c = 0.115 \text{ atm} \quad p_g = 3.05 - 0.115 = 2.935 \text{ atm}$$

$$\text{Mol steam remaining} = 35 \times \frac{0.115}{2.035} = 1.37$$

$$\text{New gas rate} = 1544 + 1.37 \times 18 = 1568.7$$

Mean properties:

$$M_m = 43.1, c = 0.214, k = 0.0102,$$

$$\mu = 0.016 \times 2.42 = 0.0387$$

$$G_s = \frac{w}{a_t} = \frac{1568.7}{0.442} = 3570 \text{ lb}/(\text{hr})(\text{ft}^2)$$

$$Re = 0.0792 \times \frac{3570}{0.0387} = 7300$$

$$j_H = 47.5 \quad \left(\frac{c\mu}{k}\right)^{1/3} = 0.935$$

$$\left(\frac{c\mu}{k}\right)^{2/3} = 0.872$$

$$h_o = 5.7$$

$$K_G = \frac{0.87}{p_{of}}$$

$$\text{Water rise} = \frac{177,000}{137,000} = 1.3^\circ\text{F}$$

$$t_w = 81.3 - 1.3 = 80^\circ\text{F}$$

Try

$$T_c = 80.7^\circ\text{F} \quad p_c = 0.0352 \text{ atm} \quad p'_g = 3.05 - 0.0352 = 3.015 \quad p_{of} = 2.97$$

$$5.7(120 - 80.7) + \frac{0.87}{2.97} \times 18 \times 1025.8(0.115 - 0.0352) = 926(80.7 - 80)$$

$$654 = 648$$

$$U \Delta t = \frac{654 + 648}{2} = 651$$

$$U = \frac{651}{(120 - 80)} = 16.2$$

Having determined both q and $U \Delta t$, it need only be recalled that $A = \Sigma q / \Sigma U \Delta t$ and proceed to evaluate A . This may be done in any of several ways. More accurately, Σq should be plotted against $1/U \Delta t$, the area of the graph corresponding to A . A simpler and generally acceptable method involves tabulation of the results obtained so far and a numerical summation. This last will be demonstrated.

Point	T_g	$T_c, ^\circ\text{F}$	$U \Delta t$	$(U \Delta t)_{av}$	q	$A = \frac{q}{(U \Delta t)_{av}}$	Δt	Δt_{av}	$\frac{q}{\Delta t_{av}}$
1	267	220	97,950				152		
2	262	182	72,350	85,150	1,598,000	18.8	158.7	155.5	10,300
3	255	145	45,900	59,075	1,104,000	18.7	159.7	159.2	6,930
4	225	101	12,900	26,000*	1,172,000	45.2	138.2	149.0	7,870
5	150	84	1,710	5,560*	751,000	135.1	68.7	99.3*	7,570
6	120	80.7	651	1,098*	177,000	161.0	40.0	53.3*	3,320
					4,802,000	$A_c = 378.8 \text{ ft}^2$		$\sum \frac{q}{\Delta t_{av}} = 35,990$	

* LMTD.

The LMTD based on a shell range from 267 to 120°F and tube range from 80 to 115°F is 84.0°F, but when weighted by the summation of $\Sigma UA = Q/\Delta t$,

$$\text{Weighted } \Delta t = \frac{Q}{\Sigma q/\Delta t_{av}} = \frac{4,802,000}{35,990} = 133^\circ\text{F}$$

and

$$U_{\text{Clean}} = \frac{Q}{A \Delta t} = \frac{4,802,000}{378.8 \times 133} = 95.3$$

External surface/ft = 0.1963

(Table 10)

Total surface available = $246 \times 12'0'' \times 0.1963 = 580 \text{ ft}^2$

$$U_D = \frac{Q}{A \Delta t} = \frac{4,802,000}{580 \times 133} = 62.3$$

$$R_d = \frac{U_C - U_D}{U_C U_D} = \frac{95.3 - 62.3}{95.3 \times 62.3} = 0.0055$$

Remarks. The pressure drop for the shell side can be computed from the average based on the inlet and outlet total gas conditions. The 0.0055 dirt factor is greater

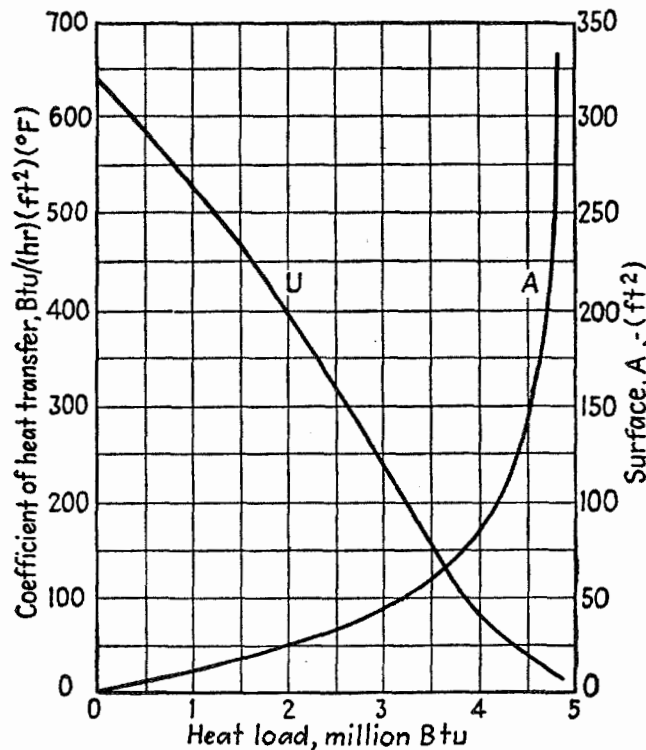


FIG. 13.7. Condensation of CO₂-steam mixture.

than is generally necessary, but large dirt factors are recommended because of the sensitivity of the calculation in the later stages. A plot of U and A vs. Q is shown in Fig. 13.7. Note that U_C varies from 644 to 16.2 from inlet to outlet.

It is not usually necessary to account for heat for subcooling all the condensate to the outlet temperature when the vapor is steam because of its high thermal conductivity when condensing on the tube wall. When the condensing vapor is organic, it will be necessary to allow extra surface as discussed in the preceding section.

5. CONDENSATION FROM A NONCONDENSABLE OF A MIXTURE IMMISCIBLE WITH WATER

The Influence of Components on the Condensing Curve. This section corresponds to Case 9 in Table 13.1. The methods used here refer particularly to steam distillation in the petroleum industry, although variations of the method can be applied to other problems as well.

When an organic vapor mixture immiscible with water enters a condenser and contains a noncondensable gas as well as steam, the calculation can be carried out using Eq. (13.20) with modifications. The total pressure is the sum of three distinct pressures: the pressure of the miscibles, pressure of the gas, and the pressure of the immiscible. The equilibrium among the miscibles is unaffected by the presence of the other components except that the equilibrium exists only at the sum of the partial pressures of the miscibles and not at the operating pressure of the condenser. Since the steam-gas and miscibles-gas mixtures attain their separate equilibrium relationships, the system may have two dew points, one for the steam and the other for the miscibles, although both need not be passed in the condenser. If there are a large number of miscible compounds, the calculation of the condensing curve will be considerably lengthier than the direct calculation of a multicomponent mixture.

A vapor mixture of miscibles is affected in definite ways by the presence of a noncondensable and steam. The noncondensable reduces the curvature of the condensation curve for the miscibles. This is due to the fact that the mols of noncondensable remain constant while the miscibles leave the vapor. The noncondensables, however, reduce the condensing coefficient for both the vapor of the miscibles and the steam. The dew point of the miscibles can be obtained from the sum of the pressures of the miscibles and the noncondensable by the summation of the K_1L_1 values with the mols of noncondensable included as a constant. Since the steam is immiscible with the miscibles, its dew point is a function only of the partial pressure and temperature at which the noncondensable becomes saturated with steam as found in tables of steam properties such as Table 7. Which of the two dew points occurs at the higher temperature depends upon the composition of the initial vapor. It is entirely possible to have either the steam or the miscibles condense first, or both may have dew points close together. Since the latent heat of the steam is about six to eight times as great as that of an oil, it is extremely important to determine just where the steam starts to condense, and the condensing curve may exhibit a greater divergence from a straight line on a heat capacity vs. T_v plot than is usually encountered with multicomponent mixtures of miscibles alone. The calculation of this type of problem

can be facilitated by empirical methods. A problem of this type will be solved presently using an empirical method developed for petroleum fractions.

Development of an Empirical Solution. In the petroleum industry it is customary to take liquid samples of complex vapors and actually distill them in the laboratory at atmospheric pressure by either of two methods. The first is the ASTM distillation, which is carried out in a standardized manner with prescribed equipment as outlined in the ASTM Standards, 1930, Part II. A 100-cm³ sample is distilled batchwise in a distillation flask with a thermometer submerged in the neck. The temperature is recorded at the *initial boiling point* (IBP) and for successive 10-cm³ portions distilled over. The percentages distilled over are plotted against the temperature. This is called the ASTM distillation curve. Another method which is more accurate but much more elaborate is the *true boiling point* (TBP) distillation. This consists of introducing the sample of material into a round-bottom flask which is then connected to a Podbielniak distilling column containing a large number of theoretical distilling plates. The material is distilled over with a reflux ratio of 10:1 to 30:1. The TBP curve is plotted in a similar manner to the ASTM distillation except that a nearly complete separation takes place and the percentages distilled can be identified by their boiling points as pure compounds. The first material to come overhead in a TBP distillation has a lower initial boiling point than the ASTM, since the most volatile fraction will have been separated almost individually from the next less volatile fraction. The presence in an ASTM distillation of the unseparated fractions shortens the range of temperatures between the initial boiling point and the end point of the distillation by comparison with the TBP.

A number of correlations have been made by Piroomov and Beiswenger¹ and Packie,² of which the latter is used here. These studies undertake the problem of correlating a batch distillation such as the ASTM or TBP with the performance to be expected in a distilling column where the vaporization occurs by a continuous process. Thus suppose a still were continuously operated at its boiling point by the introduction of liquid feed which separates continuously into a vapor rising upward and a liquid which is withdrawn at the bottom. Consider such an equilibrium system. It is assumed that boiling produces a liquid of uniform composition and that the vapor is in equilibrium with the still liquid. A similar condition exists in distilling columns, reboilers, and condensers in which liquid continuously in equilibrium with the vapor above it is formed from

¹ Piroomov and Beiswenger, *Am. Petroleum Inst. Bull.* 10, p. 52, Jan. 3, 1920.

² Packie, J. W., *Trans. AICHE*, 37, 51-78 (1941).

point to point. A curve which indicates the temperature of the liquid as a function of composition and for a given pressure when it is maintained in equilibrium with its vapor over its entire boiling or condensing range is called an *equilibrium flash* curve (EF). An EF curve is necessary for the calculation of the condensing curve, and it naturally differs from an ASTM curve in which the material previously vaporized is removed from the system. The method given below permits the conversion of an ASTM or TBP curve to an EF curve empirically as the result of a correlation of many flash distillations which have been compared with their respective ASTM and TBP distillations. The procedure for obtaining the condensing curve and the true temperature difference from the ASTM or TBP curve is as follows:

The true temperature difference:

1. Draw the ASTM distillation curve with per cent vaporized as the abscissa and temperature as the ordinate.
2. Draw the *distillation reference line* which is a straight line through the 10 and 70 per cent points of the ASTM or TBP curves.
3. Determine the slope of the distillation reference line by the following formula, where percentages represent points on the respective curves.

$$\frac{\text{Temp } ^\circ\text{F at 70\%} - \text{temp } ^\circ\text{F at 10\%}}{60} = ^\circ\text{F}/\%$$

4. Compute an average 50 per cent point for the distillation curve by averaging the 20, 50, and 80 per cent points, or if the 80 per cent point is lacking, use the 50 per cent point of the distillation reference line.
5. Enter the appropriate curves in Fig. 13.8 or 13.9, which is the key to the correlation. It relates the 50 per cent point of the ASTM distillation reference line to the 50 per cent point on the equilibrium flash reference line.
6. Using the slope calculated in 3 enter Fig. 13.10, which relates the slope of the ASTM reference line to the slope of the equilibrium flash reference line.
7. The 50 per cent point and slope determine the flash reference line, which is assumed to be straight through the 10 and 70 per cent points of the flash vaporization curve.
8. At the various percentages read temperatures from the ASTM or TBP curve and its reference line, and subtract the temperature for the reference line from that for the distillation curve noting the sign of the difference.

9. For each percentage distilled read on the appropriate curve in Fig 13.11 or 13.12 the corresponding value of the ratio

$$\frac{(\text{Temp on flash vaporization curve}) - (\text{temp on flash reference line})}{(\text{Temp on distillation curve}) - (\text{temp on distillation reference line})}$$

10. Multiply the ratio obtained in 9 by the difference obtained in 8 to obtain the difference between the temperatures on the flash vaporization curve and the flash reference line for a given percentage distilled. This enables a translation for the curvature of the ASTM to the EF curve.

11. For the same percentage distilled and used to obtain the product in 10, read a value of the temperature on the flash reference line and add

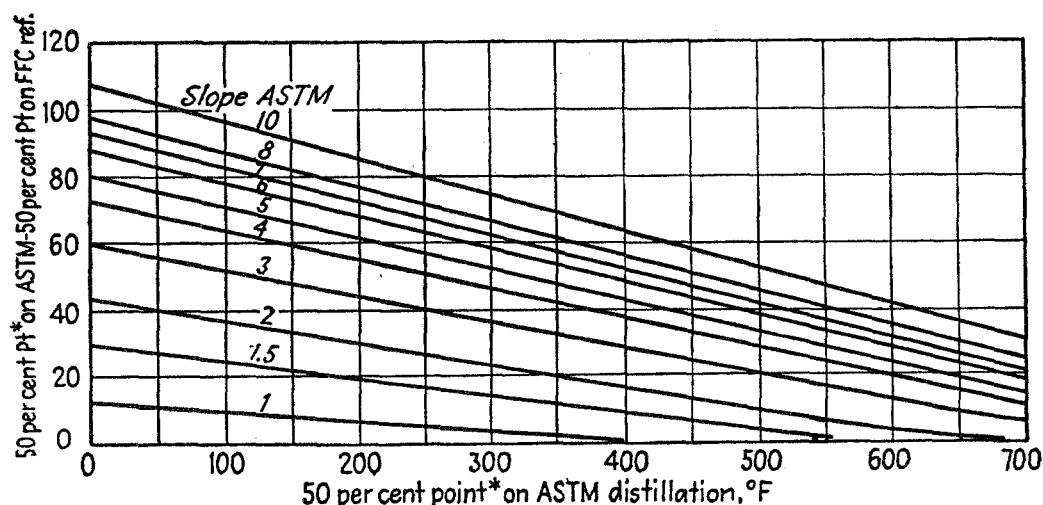


FIG. 13.8. *An average of the 20, 50, and 80 per cent values. Relation between 50 per cent points of ASTM and EF curves. (Packie, *Trans. AIChE.*)

to or subtract from it, according to the sign, the product obtained in 10. Proceed point by point to determine the entire flash vaporization curve.

12. The curve obtained thus is the EF or *flash vaporization curve* for 1 atm pressure, since it refers to the ASTM curve at atmospheric pressure. To obtain a curve at other pressures when the distilling column is to be operated at sub- or superatmospheric pressure, refer to Fig. 8 in the Appendix.¹

¹ Figure 8 is a plot of the vapor pressure of hydrocarbons. It is not plotted by individual compounds but rather by the boiling points pure compounds have or may have at atmospheric pressure. Thus, isohexane boils at 140°F at atmospheric pressure. The oblique curve at 140°F may then be considered the vapor pressure curve for isohexane. Normal hexane, however, boils at 156°F, and its vapor pressure curve will be a curve in Fig. 8 proportionately between the 140 and 160° curves. Traveling upward or downward along the 140° curve one obtains a value for the pressure of the compound as the ordinate and the saturation temperature corresponding to that pressure. Thus at 14.7 psia isohexane has a boiling point of 140° read as the abscissa. If the pressure at 100°F is desired, move downward on the 140° curve to a temperature

Observe the temperature at the intersection of the flash reference and the distillation reference line. On the vapor pressure chart (Fig. 8) estimate the number of degrees of displacement in boiling point that the change in pressure would cause for a pure hydrocarbon with an atmospheric boiling point the same as the temperature at the intersection of the two reference curves. Displace each point of the flash vaporization curve by the same number of degrees by which a pure compound with atmospheric boiling point at the temperature of the intersection would be

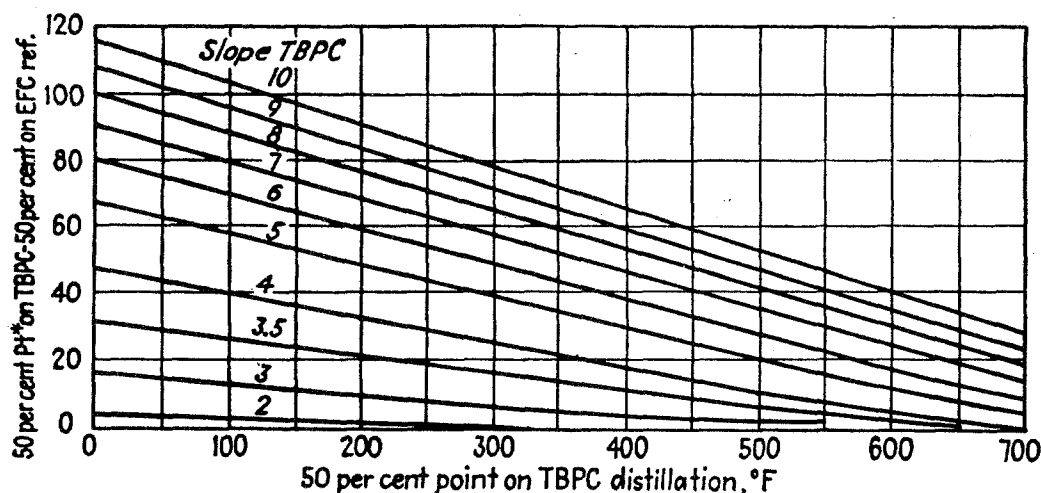


FIG. 13.9. Relation between 50 per cent points of TBP and EF curves. (Packie, *Transactions of American Institute of Chemical Engineers*.)

displaced by the change from atmospheric pressure. For operating pressures greater than 25 psi greater accuracy may be expected by employing the method of Katz and Brown.¹

The method of calculating the heat-transfer coefficients for such systems is deferred following an illustrative example on the calculation of the weighted true temperature differences, so that the problems originating in the calculation of the condensing curve can be clarified first.

Example 13.6a. Calculation of a Hydrocarbon-Noncondensable-steam Mixture. The overhead of a distilling column operating at 5 psig contains 13,330 lb/hr of oil vapor, 90 lb/hr of noncondensable gas of 50 mol. wt., and 370 lb/hr of steam. On condensation the oil vapor has an average gravity of 50°API. The overhead is at a

of 100° read on the abscissa, giving a value at the left of 6.5 psia. In a mixture of organic compounds, if its boiling temperature is known at 14.7, Fig. 8 permits with equal facility the determination of the pressure and corresponding temperature for any new condition. This type of graph has certain inherent advantages over the more conventional Cox chart, particularly when it is preferable to divide arbitrarily a mixture of hydrocarbons into a number of fictitious pure compounds so as to approximate the total composition of the mixture.

¹ Katz, D. L., and G. G. Brown, *Ind. Eng. Chem.*, **25**, 1373-1384 (1933).

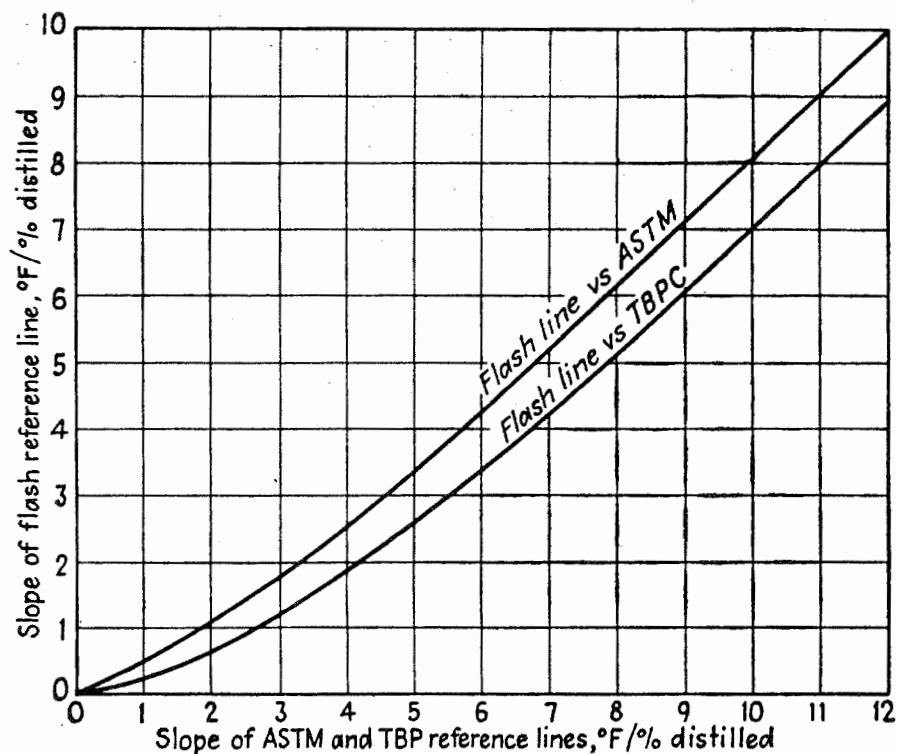


FIG. 13.10. Slope of ASTM and TBP reference lines to EFC reference lines. (Packie, *Transactions of American Institute of Chemical Engineers.*)

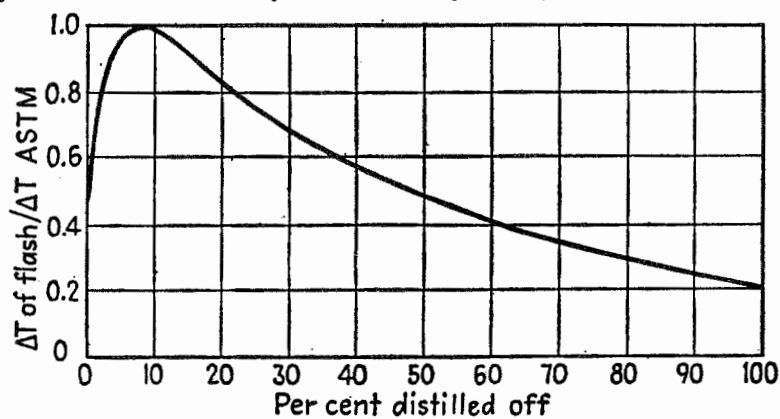


FIG. 13.11. Departure ΔT of the ASTM curve from its reference line. (Packie, *Transactions of American Institute of Chemical Engineers.*)

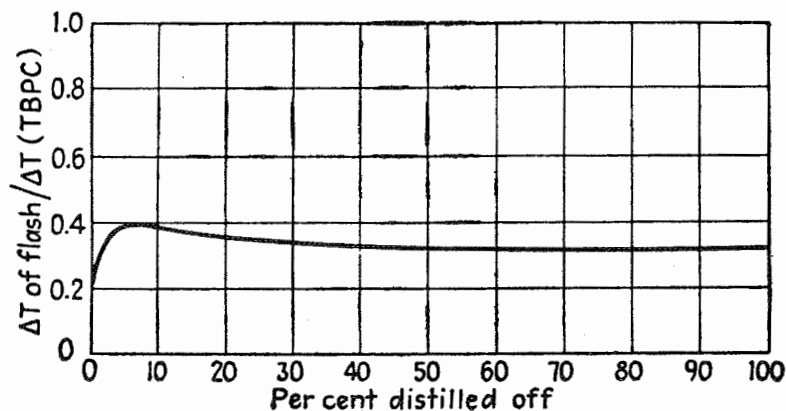


FIG. 13.12. Departure ΔT of the TBP curve from its reference line. (Packie, *Transactions of American Institute of Chemical Engineers.*)

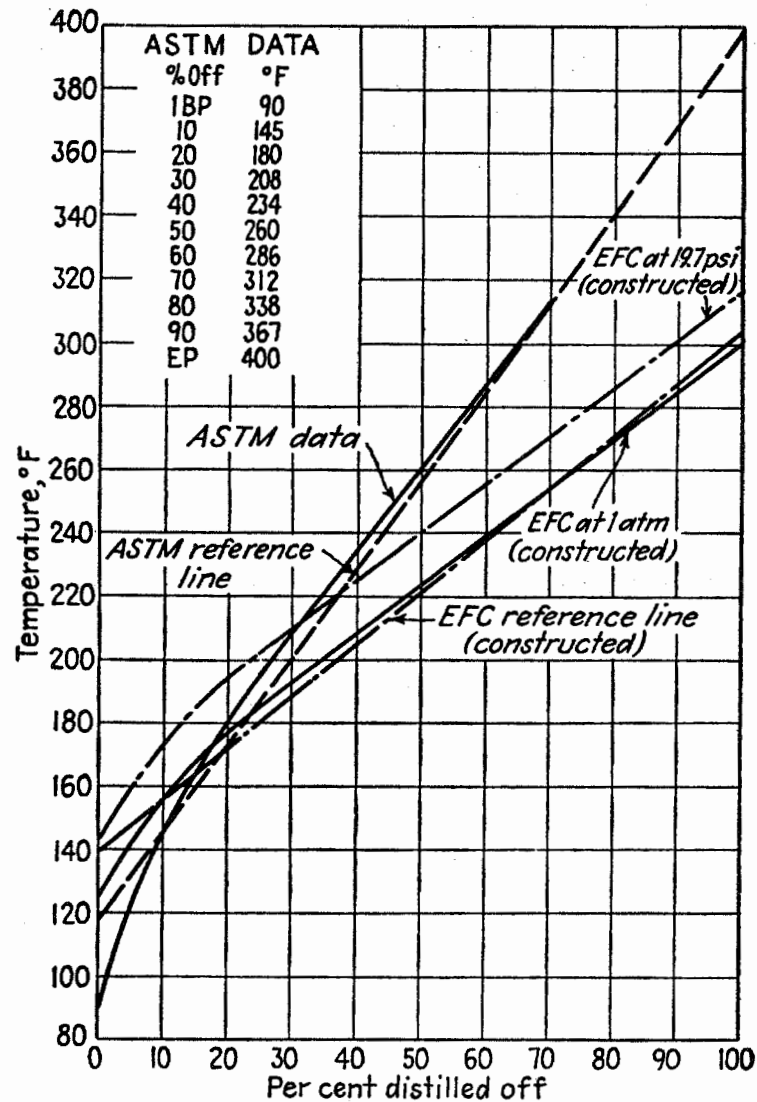


FIG. 13.13. Plot of ASTM curve and construction of the EFC.

temperature of 305°F and is to be completely condensed (as far as practicable) with water from 85 to 120°F.

Determine the weighted true temperature difference.

The ASTM distillation on the oil alone in the overhead is as follows:

% distilled	Temp, °F
IBP	90
10	145
20	180
30	208
40	234
50	260
60	286
70	312
80	338
90	367
End point	400

Solution: Basis 1 hour.

- (1) and (2) Plot the ASTM curve and reference line as shown in Fig. 13.13.
Applying the method outlined, determine the dew point of (a) oil vapor and (b) steam, (c) the condensing curve, (d) the heat load, (e) the weighted true temperature difference, and lastly (f) the condenser.
- (3) Slope of the ASTM = $\frac{70\% - 10\%}{60\%} = \frac{312 - 145}{60} = 2.79^\circ\text{F}/\%$
- (4) Average 50% point = $\frac{20 + 50 + 80\%}{3} = \frac{180 + 260 + 338}{3} = 259.3^\circ\text{F}$
- (5) For the 50 per cent point on the equilibrium flash curve (EFC) from Fig. 13.8 where the slope on the ASTM = $2.79^\circ\text{F}/\%$ and 259.3°F
50% point ASTM = 50% point flash curve = 38°F
50% on EFC = $259 - 38 = 221^\circ\text{F}$ fixing first point on EFC
- (6) From Fig. 13.10, upper curve
Slope of flash reference line = $1.65^\circ\text{F}/\%$
10% on EFC = 50% - 40% = $221 - 40 \times 1.65 = 155^\circ\text{F}$
70% on EFC = 50% + 20% = $221 + 20 \times 1.65 = 254^\circ\text{F}$
- (7) Draw this line as a reference through the 50 per cent point. Calculate the flash curve for different percentages off.
- (8) $\Delta T \text{ ASTM} = ^\circ\text{F ASTM} - ^\circ\text{F ASTM reference}$
- (9) $\Delta T \text{ ASTM} \times (\text{factor Fig. 13.11}) = \Delta T \text{ EFC}$
- (10) $^\circ\text{F EFC reference} + \Delta T \text{ EFC} = ^\circ\text{F EFC}$
- (11) 0% off:
 $\Delta T \text{ ASTM} = 90 - 117 = -27^\circ\text{F}.$
 $\Delta T \text{ EFC} = -27 \times 0.50 = 13.5^\circ\text{F}.$
 $^\circ\text{F EFC} = 139 - 13.5 = 125.5.$
- 10% off:
 $\Delta T \text{ ASTM} = 145 - 145 = 0.$
- 20% off:
 $\Delta T \text{ ASTM} = 180 - 173 = +7.0^\circ.$
 $\Delta T \text{ EFC} = +7.0 \times 0.82 = 5.7^\circ.$
 $^\circ\text{F EFC} = 172 + 5.7 = 177.7.$
- 30% off:
 $\Delta T \text{ ASTM} = 208 - 201 = +7.0^\circ.$
 $\Delta T \text{ EFC} = +7.0 \times 0.67 = 4.7^\circ.$
 $^\circ\text{F EFC} = 188 + 4.7 = 192.7.$
- 40% off:
 $\Delta T \text{ ASTM} = 234 - 229 = +5.0^\circ.$
 $\Delta T \text{ EFC} = 5.0 \times 0.57 = 2.9^\circ.$
 $^\circ\text{F EFC} = 204 + 2.9 = 206.9.$
- 70% off:
 $\Delta T \text{ ASTM} = 312 - 312 = 0.$
- (12) Correct for pressure: Intersection ASTM reference and EFC reference at 170°F .
From Fig. 8 in the Appendix follow oblique at 170° to intersection with 19.7 psi at 187°F . Add $187^\circ - 170^\circ = 17^\circ$ to each point.

Calculation of the 80 per cent point:

The following vapor enters, 13,330 lb/hr oil 50°API, 370 lb/hr steam, 90 lb/hr gas 50 mol. wt.

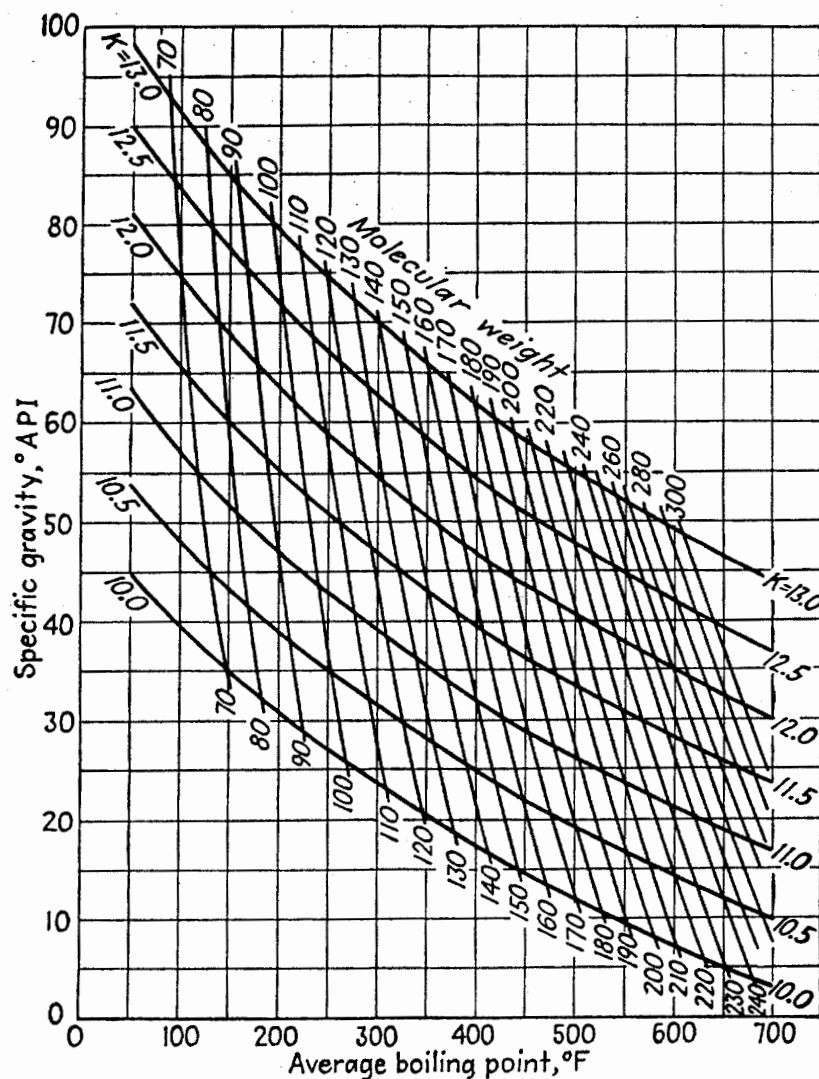


FIG. 13.14. Molecular weights of petroleum fractions. (Watson, Nelson, and Murphy, *Industrial and Engineering Chemistry*.)

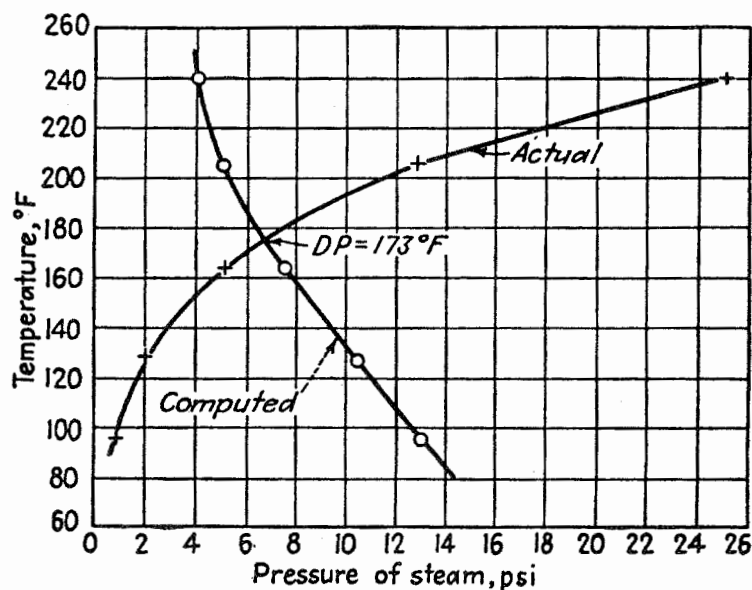


FIG. 13.15. Computed vs. actual vapor pressure of steam.

For 80 per cent, $13,330 \times 0.80 = 10,664$ lb/hr oil uncondensed.

Average boiling point from the EFC at 1 atm is 269°F from Fig. 13.13.

Average boiling point from the EFC at 19.7 psia add $17^{\circ}\text{F} = 269 + 17 = 286^{\circ}\text{F}$.

The molecular weight of the vapors from point to point is determined from Fig. 13.14.

For 50°API oil and 269°F , mol. wt. = 113.

The mols of oil still to be condensed $= 10,664/113 = 94.3$.

The following are also present: Mols gas $= 9\frac{0}{50} = 1.80$

Mols steam $= 37\frac{0}{18} = 20.6$

Mols total = 116.7

The total pressure is 19.7 psia, and the partial pressure of each component is proportional to its mol fraction.

Partial pressure of oil $= (94.3/116.7)19.7 = 15.9$ psia.

Partial pressure of NC gas $= (1.8/116.7)19.7 = 0.304$ psia.

The temperature at which the oil will condense in the presence of noncondensable corresponds to its partial pressure.

Refer to Fig. 8 in the Appendix, which relates the temperature at which a fraction will boil at a higher or lower pressure if the temperature at a given pressure is known.

At 19.7 psia the boiling point is 286°F . At the intersection of this ordinate and abscissa follow the oblique to an ordinate of 15.9 psia and read the new abscissa, 277°F . These values are the actual condensing temperatures of the oil vapor in the presence of noncondensable gas and steam.

To determine the heat load it is also necessary to determine the temperature at which the steam starts to condense, since the ultimate condensate, water, is immiscible with the condensed oil. The steam may condense before the dew point of the oil is reached, during condensation of the oil, or after nearly all the oil has condensed. The dew point of the steam is a function only of its partial pressure in the vapor and the saturation temperature of pure steam at the partial pressure.

At 19.7 psia, if only pure steam were present, it would condense at 227°F . The dew point in this case must be considerably lower.

CALCULATION OF DEW POINT OF THE STEAM

$T, ^{\circ}\text{F}$	$[p_t - (p_{\text{oil}} + p_{\text{NC}})] = p_{\text{steam}}$			p_{sat} (steam tables)
95	19.7	6.73	12.97	0.815
127	19.7	9.40	10.30	2.050
163	19.7	12.25	7.45	5.09
205	19.7	14.64	5.06	12.77
240	19.7	15.65	4.05	24.97

These are plotted in Fig. 13.15, the point of intersection, 173°F , being the dew point of the steam.

To establish the respective heat loads in the oil and steam zones the condensation of oil from inlet to the dew point of the steam is determined from Table 13.4. Where steam and oil condense together, it will be necessary to assume a temperature below the dew point of steam and the amount of oil condensed at that point. If at the assumed temperature the sum of the partial pressures adds up to the total pressure, 19.7 psia, the amounts of oil, steam, and noncondensable will be fixed.

At 173°F, the dew point of steam, $p_{\text{sat}} = 6.417$ psia

$$p_{\text{oil}} + p_{\text{NC}} = 19.7 - 6.417 = 13.29 \text{ psia}$$

$$\begin{array}{rcl} & \text{Mol/hr} & \\ \text{Oil} & = & X \\ \text{NC gas} & = 9\frac{0}{50} & = 1.8 \\ \text{Steam} & = 37\frac{0}{18} & = 20.55 \\ & \hline & 22.35 + X & \end{array}$$

But the 20.55 mol steam $\approx$ 6.417 psia.

$$\begin{array}{l} 19.7 \times 20.55 / (22.35 + X) = 6.417 \\ X = 40.75 \text{ mols oil} \end{array}$$

The partial pressures of the components, being in the mol ratios, are

$$\begin{array}{rcl} & \text{Psi} & \\ \text{Oil} & \dots\dots\dots & 12.74 \\ \text{NC gas} & \dots\dots\dots & 0.56 \\ \text{Steam} & \dots\dots\dots & 6.42 \\ & \hline & 19.72 \end{array}$$

173°F and 12.74 psia are equivalent to 180°F at 14.7. From Fig. 13.14 the molecular weight of the vapors is 85.

$$\text{Lb/hr vapor} = 40.75 \times 85 = 3470 \text{ lb}$$

$$\% \text{ Condensed} = \frac{(13,330 - 3470)100}{13,330} = 74\%$$

It is now necessary to determine the amounts of steam and oil condensed at temperatures below the steam dew point. In constructing Table 13.4 the partial pressure of

TABLE 13.4. OIL CONDENSING CURVE

%	Con- dens- ables, lb/hr	Av BP on EFC		50° API mol. wt.	Mol oil	Mol NC gas	Mol steam	Mol total	Total pres- sure, psia	Par- tial pres- sure oil, psia	Par- tial pres- sure NC gas, psia	Cond temp, °F
		14.7 psia, °F	19.7 psia, °F									
100	13,330	300	317	124	107.5	1.8	20.6	129.9	19.7	16.3	0.273	305
80	10,664	269	286	113	94.3	1.8	20.6	116.7	19.7	15.9	0.304	277
60	7,998	239	256	103	77.7	1.8	20.6	100.1	19.7	15.3	0.354	240
40	5,332	207	224	93	57.4	1.8	20.6	79.8	19.7	14.2	0.444	205
20	2,666	178	195	84	31.8	1.8	20.6	54.2	19.7	11.6	0.654	163
10	1,333	155	172	78	17.1	1.8	20.6	39.5	19.7	8.5	0.897	127
5	667	141	158	75	8.9	1.8	20.6	31.3	19.7	5.6	1.13	95

steam and the mols in the gas phase were considered constant. Actually, however, when the steam is condensed, it is the result of a continuously decreasing partial pressure. This correction is made below

The reduction in the partial pressure of steam in the mixture increases the partial pressure of oil, thereby increasing the percentage condensation for a given temperature. Thus for 80 per cent condensation if both the steam and noncondensable gas were removed, the temperature would correspond to 195°F on the 19.7 psia EFC instead of 163°F when the steam and noncondensable gas are present.

Trial 1:

At 163°F, assume 90 per cent of oil vapor is condensed.

	Mol/hr	<i>mf</i>
Oil vapor $13,300 \times 0.10/78..$	17.1	$17.1/(18.9 + X)$
NC gas.....	1.8	$1.8/(18.9 + X)$
Steam.....	<i>X</i>	$X/(18.9 + X)$
Total.....	$18.9 + X$	

Then $19.7X/(18.9 + X)$ is the partial pressure of the steam. Since water is insoluble in the hydrocarbons, the partial pressure is also fixed by the saturation pressure given in the steam tables.

For 163°F $p_{\text{steam}} = 5.09$ psia

$$19.7X/(18.9 + X) = 5.09$$

$$X = 6.58 \text{ mols steam}$$

	Mol/hr	<i>mf</i>	$mf \times p_t = p_{\text{partial}}$
Oil vapor.....	17.1	0.672	13.23
NC gas.....	1.8	0.070	1.38
Steam.....	6.58	0.258	5.09
Total.....	25.48	1.000	19.70

A temperature of 163°F at 13.23 psia is equivalent to a temperature of 188°F at 19.7 psia. From the EFC at 19.7 psia in Fig. 13.13 it will be seen that 16.5 per cent will be vaporized at 188°F or 83.5 per cent condensed. It was assumed that 90 per cent was condensed; hence the amount of oil assumed to be condensed was in error, requiring a new trial.

Trial 2:

At 163°F assume 85 per cent of vapor condensed.

	Mol/hr	<i>mf</i>	$mf \times p_t$
Oil vapor, $13,330 \times .15/81.....$	24.7	0.692	13.62
NC gas, $9\%_{50}.....$	1.8	0.050	0.99
Steam, <i>X</i>	(<i>X</i> = 9.23)	0.258	5.09
		1.00	19.70

163°F at 13.62 psia $\approx$ 185°F on the 19.7 psia EFC. This point corresponds to 15 per cent liquid or 85 per cent condensed. The assumption checks the curve.

Initial steam = $37\frac{1}{8}$ = 20.55 mols

Uncondensed steam at 163°F = 9.23

Mols condensed, 173 - 163°F = 11.32

Similarly

$T, ^\circ\text{F}$	Oil cond, %	Oil cond, lb	Steam cond, lb
173	74	9,863	0
163	85	11,350	204
127	97.5	13,000	357
95	100	13,330	370

Condensing Curve. The following enthalpy data have been taken from Fig. 11 in the Appendix and the steam tables. The limitations of this type of petroleum enthalpy data may be found in the literature. The inconsistency of the enthalpy datum temperatures for steam and the oil vapor does not affect this solution, since only differences are involved.

Oil			Steam	
$T_c, ^\circ\text{F}$	H_v , vapor	H_l , liquid	H_g or H_v , gas or vapor	H_l , liquid
305	368	242	1197.0	Superheated
277	359	225	1184.1	Superheated
240	337	204	1167.0	Superheated
205	322	185	1150.6	Superheated
173	310	168	1135.4	140.9
163	306	163	1131.4	130.9
127	293	144	1116.6	94.9
95	283	128	1103.1	63.0

Heat load

305°F:

	H	q
Oil vapor = $13,330 \times 368$	= 4,920,000	
Steam = 370×1197.0	= 443,000	
NC gas = $90 \times 0.46(273)$	= 11,300	
	5,374,300	0

277°F:

Oil vapor = $10,664 \times 354$	= 3,770,000	
Oil liquid = $2,666 \times 225$	= 595,000	
Steam = 370×1184.1	= 438,000	
NC gas = $90 \times 0.46(245)$	= 9,300	
	4,812,300	562,000

240°F:	<i>H</i>	<i>q</i>
Oil vapor = $7,998 \times 337$	= 2,695,000	
Oil liquid = $5,332 \times 204$	= 1,088,000	
Steam = 370×1167	= 432,000	
NC gas = $90 \times 0.46(208)$	= 8,600	
	<u>4,223,600</u>	588,700
205°F:		
Oil vapor = $5,332 \times 322$	= 1,715,000	
Oil liquid = $7,998 \times 185$	= 1,480,000	
Steam = 370×1150.6	= 426,000	
NC gas = $90 \times 0.46(173)$	= 7,200	
	<u>3,628,200</u>	595,400
173°F (dew point of steam):		
Oil vapor = $2,666 \times 310$	= 827,000	
Oil liquid = $10,664 \times 168$	= 1,790,000	
Steam = 370×1135.4	= 420,000	
NC gas = $90 \times 0.46(141)$	= 5,800	
	<u>3,042,800</u>	585,400
163°F:		
Oil vapor = $1,980 \times 306$	= 606,000	
Oil liquid = $11,350 \times 163$	= 1,850,000	
Steam = 166×1131.4	= 188,500	
Water = 204×130.9	= 26,700	
NC gas = $90 \times 0.46(131)$	= 5,400	
	<u>2,676,600</u>	366,200
127°F:		
Oil vapor = 330×293	= 96,600	
Oil liquid = $13,000 \times 144$	= 1,870,000	
Steam = 13×1116.6	= 14,400	
Water = 357×94.9	= 33,900	
NC gas = $90 \times 0.46(95)$	= 3,900	
	<u>2,018,800</u>	657,800
95°F:		
Oil liquid = $13,300 \times 128$	= 1,710,000	
Water = 370×63	= 23,300	
NC gas = $90 \times 0.46(63)$	= 2,600	
	<u>1,735,900</u>	282,900

These calculations are plotted in Fig. 13.16

Summary:

$$\begin{aligned}
 &\text{Inlet to steam dew point:} \\
 &\quad 5,374,300 - 3,042,800 = 2,331,500 \text{ Btu/hr} \\
 &\text{Steam dew point to outlet:} \\
 &\quad 3,042,800 - 1,735,900 = 1,306,900 \\
 &\text{Total} \quad \quad \quad 3,638,400 \text{ Btu/hr} \\
 &\text{Total water} = \frac{3,638,400}{120 - 85} = 104,000 \text{ lb/hr}
 \end{aligned}$$

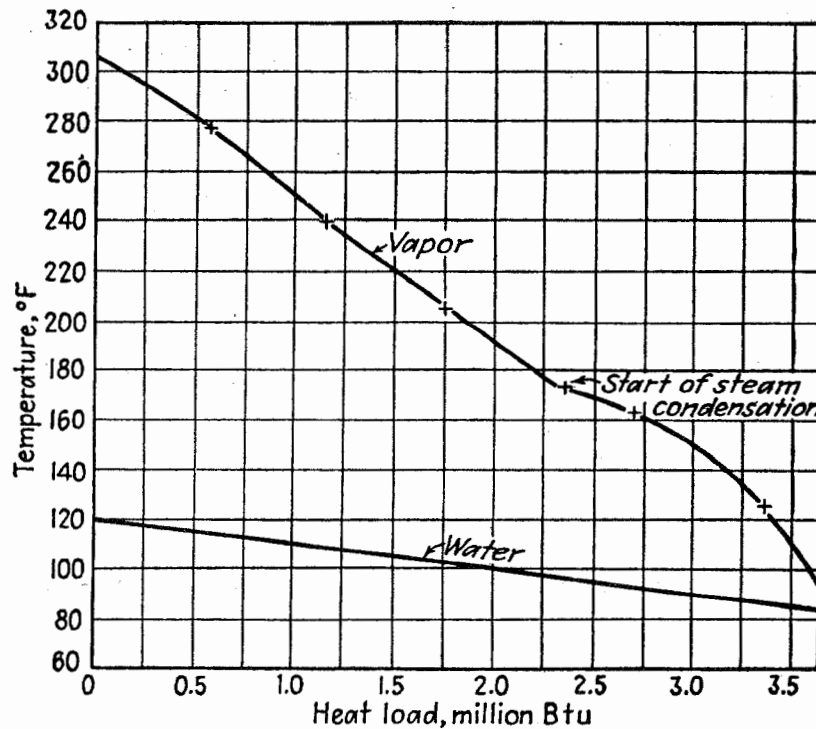


FIG. 13.16. Condensation of mixed hydrocarbons with gas and steam.

In this particular problem the condensing curve from the inlet to the dew point of the steam is nearly straight. When both the shell and tube steam temperature changes are proportional to the heat load, the LMTD may be applied.

Water temperature at dew point of steam,

$$t_w = 85 + \frac{1,306,900}{3,638,400} \times 35 = 97.5^\circ\text{F}.$$

Weighted true temperature difference, Δt :

Inlet to dew point of steam:

$$\begin{array}{ccccccc} T_1 & 305 & 120.0 & t_2 \\ T_x & 173 & 97.5 & t_y & \text{LMTD} = 122.2^\circ\text{F} \end{array}$$

$$UA = \frac{\Delta q}{\Delta t} = \frac{2,331,500}{122.2} = 19,050$$

Dew point of steam to outlet (from Fig. 13.16):

q	Δq	T_c	t_w	Δt_{av}	$\frac{\Delta q}{\Delta t_{av}} = UA$
2,331,500		173	97.5		
2,500,000	169,000	169	96	74	2,285
2,750,000	250,000	161	93	70.5	3,550
3,000,000	250,000	149	91	63	3,970
3,250,000	250,000	134	89	51.5	4,850
3,500,000	250,000	112	86	35.5	7,040
3,638,000	138,000	95	85	18.5	7,460
$UA = \sum \frac{\Delta q}{\Delta t}$					29,155

$$\text{Weighted } \Delta t = \frac{Q}{\Sigma \Delta q / \Delta t} = \frac{1,306,900}{29,155} = 44.8^\circ\text{F}$$

Overall weighted temperature difference:

$$\text{Weighted } \Delta t = \frac{3,638,500}{29,155 + 19,050} = 75.5^\circ\text{F}$$

The uncorrected LMTD is 60.1°F .

The Overall Coefficient of Heat Transfer. The condensing curve of Example 13.6a is typical of the condensation which may occur for the phase system indicated. It is repeated here that the steam dew point may be higher than the oil dew point and that the steam may start to condense first. When the latter occurs, the surface from the steam dew point to the oil dew point can be treated as the ordinary condensation of a vapor from a noncondensable with the superheated oil vapor combining with the gas to form the total noncondensables. If the vapor enters at a temperature above either dew point, it is cooled to the first dew point as a dry gas, although this does not happen after a distilling column, since the top plate vapor is always at a dew point.

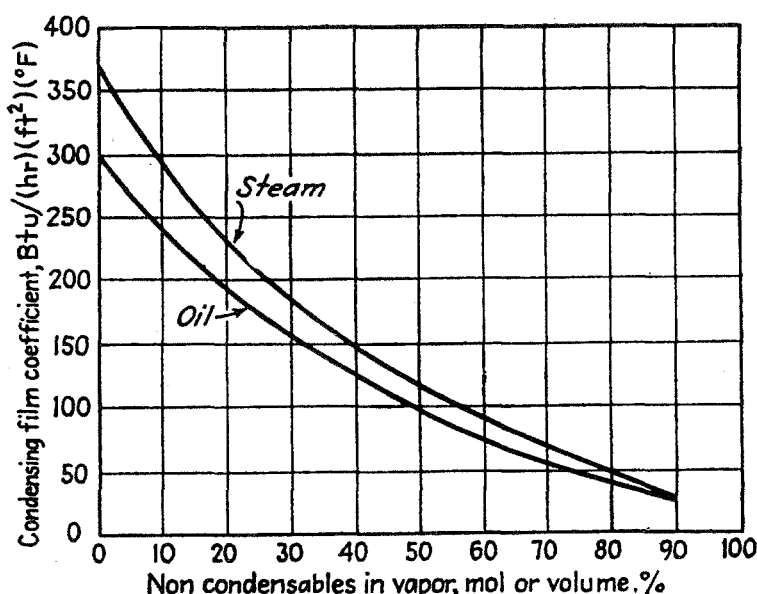


FIG. 13.17. Condensing coefficients for vapors and noncondensables.

Where steam is above its dew point, it is regarded as a noncondensable gas as a matter of safety even though the tube-wall temperature may be below the dew point of the steam and therefore wet. When oil vapor condenses in the presence of noncondensables, the problem is again one of diffusion except that the equations employed heretofore for diffusion are not very reliable when calculated from the properties of vapors of high molecular weight. A curve for the condensing and diffusion coefficient for oil vapors in the presence of noncondensable gases is plotted in Fig. 13.17 as a function of the percentage of noncondensables. Its origin is discussed presently.

When steam diffuses through a noncondensable gas and the vapor of miscibles, the rate of heat transfer is quite different from that of pure diffusion. An unpublished correlation has been made of the overall clean coefficients U_c obtained by calculating the pure diffusion of steam, using the method under Sec. 4 of this chapter, through gases with molecular weights between 16 and 50. The calculated data define a common line when U_c is plotted against the percentage of noncondensables even though the operating pressures on the systems varied. The curve obtained from this data is taken as the base curve, and to it is added a film resistance corresponding to an oil layer of 0.002 in., which gives an average resistance of 0.0020. The steam curve shown in Fig. 13.17 was arrived at in this manner. The oil diffusion curve was established by taking the pure condensing coefficient for a vapor and reducing it in the same percentage as the steam curve is reduced by the increase in the percentage of the noncondensables. The base value for the oil curve in Fig. 13.17 was 300, but any other condensing coefficient can be used and reduced accordingly. This will usually not be a necessary refinement. The use of the curves at low vacuums or very high pressures is questionable, but a large number of condensers have been designed at rates corresponding to those obtained by this method and have operated successfully.

While a method based on using fictitious coefficients as suggested above may not be so desirable as a method employing individual film calculations, it has the advantage of being rapid. The cooling of the condensate to the outlet temperature in horizontal condensers may be treated similarly to that in Sec. 2, by using the free convection rate U of about 50 and applied to the average liquid-cooling sensible heat load. The water is considered to be in the coldest passes, and the temperature rise of the water is figured accordingly.

Example 13.6b. Calculation of a Condenser for the Condensation from Noncondensable of a Mixture Immiscible with Water. The vapor of Example 13.6a is to be condensed with a fouling factor of at least 0.0030 and pressure drops not exceeding 2.0 psi for the vapor and 10.0 psi for the water.

Available for the purpose is a 27 in. ID exchanger containing 286 1 in. OD, 14 BWG tubes 12'0" long laid out on 1¼-in. triangular pitch. The bundle is arranged for eight tube passes and baffles are spaced 16 in. apart.

Solution:

Exchanger:

<i>Shell side</i>	<i>Tube side</i>
ID = 27 in.	Number and length = 286, 12'0"
Baffle space = 16 in.	OD, BWG, pitch = 1 in., 14 BWG, 1¼ in. tri.
Passes = 1	Passes = 8

Clean surface requirements:

Heat load inlet to dew point of steam = 2,331,500 Btu/hr

$\Delta t = 122.2^\circ$. h_{io} for water is 700 Btu/(hr)(ft²)(°F).

From Table 13.4 at inlet

$$\begin{aligned}\text{NC gas} + \text{steam} &= 1.8 + 20.6 = 22.4 \text{ mol/hr} \\ \text{Total} &= 129.9 \text{ mol/hr}\end{aligned}$$

$$\% \text{ NC gas} = \frac{22.4}{129.9} = 0.1735$$

From Fig. 13.17, $h_o = 205$ Btu/(hr)(ft²)(°F)

At dew point of steam

	Mol/hr
NC gas + steam.....	22.4
Oil.....	40.75
Total.....	63.2

$$\% \text{ NC} = \frac{22.4}{63.2} = 0.354$$

From Fig. 13.17, $h_o = 140$

Log mean overall coefficient = 136.5 Btu/(hr)(ft²)(°F)

$\Delta t = 122.2^\circ\text{F}$

$$A_{c1} = \frac{Q}{U \Delta t} = \frac{2,331,500}{136.5 \times 122.2} = 139.5 \text{ ft}^2$$

At the dew point of the steam the water vapor is no longer considered a noncondensable gas. And since the remainder of the oil vapor is condensing, it is not to be considered a noncondensable in the presence of water vapor although the presence of each influences the rate of diffusion of the other.

At dew point of steam to outlet:

	Mol/hr
NC gas.....	1.8
Steam.....	20.6
Total.....	22.4

$$\% \text{ NC gas} = \frac{1.8}{22.4} = 0.080$$

From Fig. 13.17, $U_c = 212$ (weighted for oil and steam)

At outlet

$$\begin{aligned}\text{NC gas} &= 1.8 \text{ mols} \\ \text{Steam} &= \text{neg}\end{aligned}$$

From Fig. 13.17, $U_c = 15$

Log mean overall coefficient = 74.5 Btu/(hr)(ft²)(°F)

$\Delta t = 44.8^\circ\text{F}$

$$A_{c2} = \frac{Q}{U \Delta t} = \frac{1,306,900}{74.5 \times 44.8} = 391 \text{ ft}^2$$

Subcooling rate: Consider all the condensate cooled over half the entire condensing range.

Heat of liquid (50°API) = $13,330 \times 0.55(305 - 95)/2 = 770,000$ Btu/hr

Assume the coldest water passes are in contact with the condensate.

$$\text{Water rise} = \frac{770,000}{3,638,400} \times 35 = 7.4^\circ\text{F}$$

Condensate, 305 to 95°F

Water, 85 to 92.4°F

LMTD = 66.3°F

Use a value of $U = 50$ for free convection.

$$A_s = \frac{Q}{U \Delta t} = \frac{770,000}{50 \times 66.3} = 232.5 \text{ ft}^2$$

Total clean surface, $A_c = 139.5 + 391 + 232.5 = 763 \text{ ft}^2$

Clean overall coefficient U_c :

$$U_c = \frac{Q}{A_c \Delta t} = \frac{3,638,400}{763 \times 75.5} = 63.2 \text{ Btu/(hr)(ft}^2\text{)(°F)}$$

Design overall coefficient U_D :

External surface/ft of length = 0.2618 ft

(Table 10)

$$A = 286 \times 12'0'' \times 0.2618 = 897 \text{ ft}^2$$

$$U_D = \frac{3,638,400}{897 \times 75.5} = 53.7$$

Dirt factor:

$$R_d = \frac{U_c - U_D}{U_c U_D} = \frac{63.2 - 53.7}{63.2 \times 53.7} = 0.0028 \text{ (hr)(ft}^2\text{)(°F)/Btu}$$

The pressure drop is computed in the usual manner.

Submerge $(232.5/763)897 = 274 \text{ ft}^2$ of surface

Summary

*	h outside	700
U_c	63.2	
U_D	53.7	
R_d Calculated 0.0028		
R_d Required 0.003		
0.8	Calculated ΔP	5.6
2.0	Allowable ΔP	10.0

* Composite. Low dirt factor.

PROBLEMS

13.1. The following vapor enters a condenser at 40 psia:

	Mols/hr
C_4	6.4
C_5	219.7
C_6	2.3
	228.4

Assume a pressure drop of 5 psi in the condenser. Available for the service is a 25 in. ID 1-2 horizontal condenser containing 222 1 in. OD, 14 BWG, 12'0'' long tubes arranged for eight passes on 1¼-in. square pitch. Baffles are spaced 18 in. apart. Cooling is by water from 85 to 120°F.

- (a) What is the true temperature difference?
 (b) What are the dirt factor and pressure drops?

13.2. The following overhead vapor enters a condenser at 20 psia.

	<i>Mols/hr</i>
C_4	10.5
C_5	150.0
C_6	39.6
C_7	63.7
C_8	191.5

Cooling is effected by water from 85 to 120°F.

Available for the service is a 23¼ in. ID 1-2 horizontal condenser containing 308 ¾ in. OD, 14 BWG, 12'0" long tubes arranged for four passes on 1-in. square pitch. Baffles are spaced 24 in. apart.

- (a) What is the true temperature difference?
 (b) What are the dirt factor and the pressure drop?

13.3. The following overhead vapor enters a condenser at 50 psia.

	<i>Mols/hr</i>
C_4	31
C_5	245
C_6	124
H ₂ O	40
	440

Cooling is effected by water from 85 to 120°F.

- (a) What is the true temperature difference?
 (b) Calculate the size of 1-2 horizontal condenser required for the service providing a minimum dirt factor of 0.004 and a pressure drop of 2.0 for the vapor and 10.0 for water. Use 1-in. tubes on 1¼-in. square pitch and a minimum water velocity of 3.0 fps.

13.4. 2800 lb/hr of air is saturated with water vapor at 203°F at 5 psi, and the mixture enters a condenser where it is cooled to 100°F by water from 70 to 100°F.

Available for the service is a 27 in. ID 1-2 condenser containing 462 ¾ in. OD, 16 BWG, 12'0" tubes arranged for four passes on 1-in. triangular pitch. Baffles are spaced 24 in. apart.

- (a) What is the true temperature difference?
 (b) What are the dirt factor and the pressure drops?

13.5. An oil has the following ASTM distillation curve:

<i>% distilled</i>	<i>°F</i>
IBP	310
10	328
20	338
30	346
40	354
50	360
60	367
70	372
80	377
90	382
100	387

28,000 lb/hr of oil (mol. wt. = 154) and 4400 lb/hr of steam enter a condenser at 20 psia. Cooling will be effected by water from 85 to 100°F. Available for the service is a 31 in. ID 1-2 vertical condenser 16'0" tall containing 728 $\frac{3}{4}$ in. OD, 16 BWG, 16'0" long tubes arranged for two passes on 1-in. triangular pitch.

(a) What is the weighted Δt ?

(b) What are the dirt factor and pressure drops?

13.6. A 48°API oil has the following ASTM distillation curve:

% distilled	°F
IBP	100
10	153
20	190
30	224
40	257
50	284
60	311
70	329
80	361
90	397
End point	423

15,300 lb/hr of oil vapor, 3620 lb/hr of steam, and 26,200 lb/hr of noncondensable gas (mol. wt. = 46) at 2.0 psi are to be condensed using cooling water with an inlet temperature of 85°F.

Available for the service are two 33 in. ID 1-2 horizontal condensers connected in series and containing 680 $\frac{3}{4}$ in. OD, 14 BWG tubes 16'0" long laid out for four passes on 1-in. square pitch. Baffles are at full pitch.

(a) Considering the condenser to be at atmospheric pressure, what is the weighted Δt ?

(b) What are the dirt factors and the pressure drops?

NOMENCLATURE FOR CHAPTER 13

A	Heat-transfer surface, ft ²
A_c	Clean heat-transfer surface, ft ²
A_c, A_s	Heat-transfer surface for condensation and subcooling respectively, ft ²
A_q	Heat-transfer surface corresponding to the heat load interval q , ft ²
a	Flow area, ft ²
B	Baffle spacing, in.
C	Specific heat of hot fluid in derivations, Btu/(lb)(°F); number of components in the phase rule, dimensionless
C'	Clearance between tubes, in.
c	Specific heat of cold fluid, Btu/(lb)(°F)
D	Inside diameter of tubes, ft
D_o	Outside diameter of tubes, ft
d	Inside diameter of tubes, in.
D_e	Equivalent diameter for heat transfer and pressure drop, ft
d_e	Equivalent diameter for heat transfer and pressure drop, in.
F	Degrees of freedom in the phase rule
f	Friction factor
f_p, f_t	Fugacity of the pure compound, fugacity at the total pressure, atm

G	Mass velocity, lb/(hr)(ft ²)
G_m	Mol mass velocity, mol/(hr)(ft ²)
G'	Condenser loading for vertical condensers, lb/(hr)(ft)
G''	Condenser loading for horizontal condensers, lb/(hr)(ft)
g'	Acceleration of gravity, ft/sec ²
H_v, H_l	Enthalpy of the vapor, enthalpy of the liquid, Btu/lb
h, h_i, h_o	Heat-transfer coefficient in general, for inside fluid, and for outside fluid, respectively, Btu/(hr)(ft ²)(°F)
h_{io}	Value of h_i when referred to the tube OD, Btu/(hr)(ft ²)(°F)
$\bar{h}$	The average condensing film coefficient, Btu/(hr)(ft ²)(°F)
j_H	Factor for heat transfer, $(hD/k)(c\mu/k)^{-1/4}$, dimensionless
j_h	Factor for heat transfer, $(h/cG)(c\mu/k)^{2/3}$, dimensionless
j_d	Factor for diffusion, dimensionless
K	Equilibrium constant, dimensionless
K_1	Equilibrium constant for any single compound or the first compound, dimensionless
K_G	Diffusion coefficient, (lb-mol)/(hr)(ft ²)(atm)
k	Thermal conductivity, Btu/(hr)(ft ²)(°F/ft)
k_d	Diffusivity, ft ² /hr
L	Tube length, ft or total liquid, mol or mol/hr
L_1	Liquid of a single compound or the first compound, mol or mol/hr
L'_1	Liquid of a single compound or the first compound condensed in an interval, mol or mol/hr
L_C	Length of tube on which condensation is effective, ft
L_c	Liquid of a single compound condensed prior to an interval, mol or mol/hr
M_A, M_B	Molecular weight of diffusing and inert gases, respectively
M_m	Mean molecular weight of vapor mixture, lb/mol
M_v	Molecular weight of diffusing component, lb/mol
mf	Mol fraction, dimensionless
N	Number of baffles, dimensionless
N_d	Number of mols transferred by diffusion, dimensionless
N_t	Number of tubes effective for condensation in a partially submerged bundle, dimensionless
n	Number of tube passes; number of compounds in a mixture
P	Number of phases in the phase rule
P_T	Tube pitch, in.
ΔP	Pressure drop in general, psi
$\Delta P_T, \Delta P_i, \Delta P_r$	Total, tube, and return pressure drop, respectively, psi
p	Partial pressure in general, atm
p_1, p_2	Partial pressures of the components; partial pressures for one compound at two different points, atm
p_t, p_p	Total pressure and pressure of the pure compound, atm
p_o, p'_o	Partial pressure of the inert gas in the gas body and at the condensate film, atm
p_{of}	Log mean pressure difference of the inert gas between p_o and p'_o , atm
Δp	Pressure difference of diffusing fluid between p_v and p_c , atm
Q	Heat flow, Btu/hr
q	Heat flow for interval, Btu/hr
q_c	Heat flow for condensation, Btu/hr

R_d	Combined dirt factor, (hr)(ft ²)(°F)/Btu
Re	Reynolds number, dimensionless
s	Specific gravity, dimensionless
T	Temperature of a gas mixture, °K
T_1, T_2	Inlet and outlet temperatures of hot fluid, °F
T_c	Temperature of condensate film; caloric temperature of hot fluid, °F
T_s	Steam temperature, °F
t, t_1, t_2	Cold-fluid temperature in general, inlet, and outlet, respectively, °F
t_c	Caloric temperature of cold fluid
t_f	Temperature of film, °F
t_w, t_w	Temperature of water, tube-wall temperature, °F
Δt	True temperature difference, °F
U, U_c, U_r	Overall coefficient of heat transfer, clean coefficient, and design coefficient, respectively, Btu/(hr)(ft ²)(°F)
U_c, U_s	Clean overall coefficient for condensation and subcooling, respectively, Btu/(hr)(ft ²)(°F)
V	Velocity, fps or total vapor, mol or mol/hr
V_1	Vapor of a single compound or the first compound, mol or mol/hr
v_A, v_B	Molar volumes of diffusing and inert gas, dimensionless
W	Weight flow of hot fluid, lb/hr
w	Weight flow of cold fluid, lb/hr
x, x_1, x_2	Mol fraction in liquid
Y, Y_1, Y_2	Total vapor feed or each component, mol or mol/hr
y, y_1, y_2	Mol fraction in vapor
α	Relative volatility
λ	Latent heat of vaporization, Btu/lb
μ	Viscosity, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
μ_w	Viscosity at the tube-wall temperature, centipoises $\times 2.42 = \text{lb}/(\text{ft})(\text{hr})$
ρ	Density, lb/ft ³
ϕ	Viscosity ratio, $(\mu/\mu_w)^{0.14}$

Subscripts (except as noted above)

a	Average
c	Condensate or condensing
f	Film
g	Gas (inerts)
i	Inert
l	Liquid
s	Shell side
t	Tube side
v	Vapor

CHAPTER 14

EVAPORATION

The Mechanism of Vaporization. Much of our present knowledge of the boiling phenomenon is obtained from the work of Jakob and Fritz¹ and later researches of Jakob.² When steam flows through a tube which is submerged in a pool of liquid, minute bubbles of vapor form at random points on the surface of the tube. The heat passing through the tube surface where no bubbles form enters the surrounding liquid by convection. Some of the heat in the liquid then flows toward the bubble, causing evaporation from its inner surface into itself. When sufficient buoyancy has been developed between the bubble and the liquid, the bubble breaks loose from the forces holding it to the tube and rises to the surface of the liquid pool. Kelvin postulated that, in order for this behavior to prevail, the liquid must be hotter than the saturation temperature in the incipient bubble. This is possible, since the spherical nature of the bubble establishes liquid surface forces on it so that the saturation pressure inside the bubble is less than that of the surrounding liquid. The saturation temperature of the bubble being lower than that of the liquid surrounding it, heat flows into the bubble. The number of points at which bubbles originate is dependent upon the texture of the tube surface, the number increasing with the roughness. Jakob and Fritz have detected the presence of superheated liquid close to the heating surface and have found the difference between the temperature of the superheated liquid and the saturation temperature of the vapor to be less for rough surfaces than for smooth surfaces.

Heat transfer by vaporization without mechanical agitation is obviously a combination of ordinary liquid free convection and the additional convection produced by the rising stream of bubbles. Under very small temperature differences between tube wall and boiling liquid the formation of bubbles proceeds slowly and the rate of heat transfer is essentially that of free convection as given in Eqs. (10.13) and (10.14). The surface tension as an influence on bubble formation and growth is another variable as shown in Fig. 14.1. The surface tension of water against air

¹ Jakob, M., and W. Fritz, *Forschr. Gebiete Ingenieurw.*, **2**, 434 (1931).

² Jakob, M., *Mech. Eng.*, **58**, 643 (1936); "Heat Transfer," Vol. 1, John Wiley & Sons, Inc., New York, 1949.

is approximately 75 dynes/cm at room temperature, whereas the majority of organics have surface tensions ranging from 20 to 30 dynes/cm at room temperature. The surface tensions of most liquids at their respective boiling points, however, are probably not so far apart as at room temperature. The surface tensions of the liquids against metals may also differ from their surface tensions against air, since the rate of vaporization of water is actually much greater than that of organics under identical conditions. If the surface tension of the liquid is low it tends to wet surfaces, so that the bubble in Fig. 14.1a is readily occluded by the liquid and rises. For liquids with intermediate surface tensions, as in Fig. 14.1b, a momentary balance may exist between the bubble and the tube wall, so that it is necessary to form larger bubbles before the buoyant force can free it from the surface. The bubble in Fig. 14.1c indicates the influence of high surface tension.

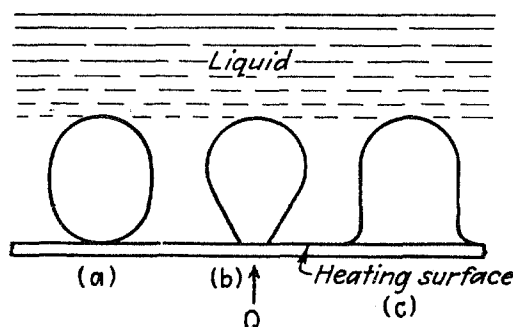


FIG. 14.1. Effect of interfacial tension on bubble formation. (After Jakob and Fritz.)

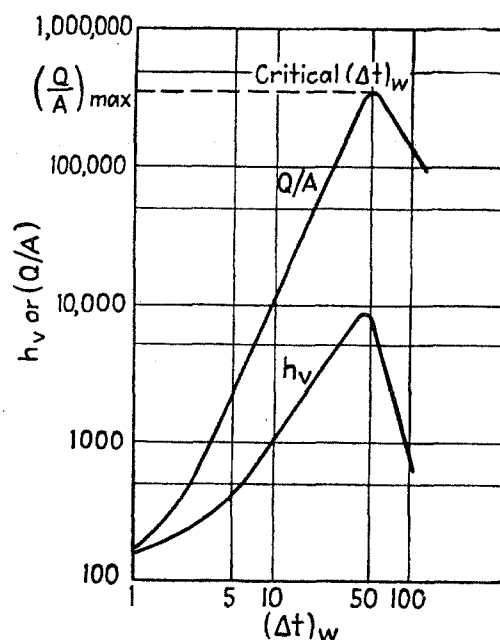


FIG. 14.2. Boiling curve of water from pools. (After McAdams.)

Consider the typical boiling coefficient curve of McAdams¹ based on the data on several investigators for water as shown in Fig. 14.2. From a $(\Delta t)_w$ above 5°F there is a relatively straight logarithmic relationship between the coefficient of vaporization and the temperature difference, where $(\Delta t)_w$ is the difference between the tube wall and the vapor temperatures. This relationship changes, however, at the *critical temperature difference*, which occurs at about 45°F for water evaporating from pools. At that temperature difference the hot surface and liquid approach the condition shown in Fig. 14.3. There is a predominance of vapor at the tube wall because of the high rate of heat throughput, and very little liquid actually contacts the hot tube wall. This condition is called

¹ McAdams, W. H., "Heat Transmission," p. 296, McGraw-Hill Book Company, Inc., New York, 1942.

blanketing or *vapor binding*, wherein the large amount of vapor formed at the tube wall actually serves as a gas resistance to the passage of heat into the liquid and reduces the film coefficient for vaporization as the temperature difference increases. Drew and Mueller¹ have reported the critical temperature differences for a number of organic compounds under varying conditions of surface and the organics exhibit a critical temperature difference of roughly 60 to 120°F. When vaporization takes place directly at the heating surface, it is called *nuclear boiling*, and when it takes place through a blanketing film of gas, it is *film boiling*.

A useful criterion of performance during vaporization is that of the maximum attainable heat flux defined as $(Q/A)_{\max}$ or $(U \Delta t)_{\max}$. The heat flux vs. $(\Delta t)_w$ is plotted for boiling water as the upper curve of Fig. 14.2. It represents the number of Btu per hour being transferred per square foot of surface with the maximum attainable flux corresponding to the heat flow at the critical temperature difference. For water at atmospheric pressure this occurs, according to McAdams,² when $h_v = 8800$ and $(Q/A)_{\max} = 400,000$ and at higher pressures both values increase. More than $(Q/A)_{\max}$ Btu/(hr)(ft²) cannot be forced through the heating surface because of the appearance of a gas film. When a higher $(\Delta t)_w$ than the critical is employed, a smaller vaporization coefficient results and the flux similarly decreases. It follows then that the large temperature differences so favorable to conduction and convection may actually be a hindrance to vaporization.

The following factors are found to affect the rate of heat transfer by vaporization from pools and have defeated, to a great extent, the possibility of obtaining one or two simple correlations applicable to the majority of liquids: (1) nature of the surface and distribution of bubbles; (2) properties of the fluid such as surface tension, coefficient of expansion, and viscosity; and (3) the influence of the temperature difference upon the evolution and vigor of the bubbles.

Effect of Pressure and Properties on the Vaporizing Coefficient. Much of the present method of calculation of vaporization from pools consists of nothing more than the use of observed coefficients for individual liquids reported at atmospheric pressure. If the coefficient for vaporization from a pool has been reported for a fluid at atmospheric pressure, it can

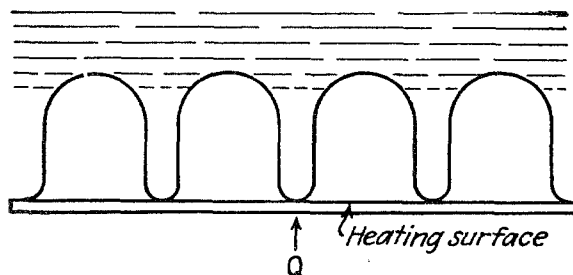


FIG. 14.3. Incipient blanketing. (After Jakob and Fritz.)

¹ Drew, T. B., and A. C. Mueller, *Trans. AIChE*, **33**, 449-471 (1939).

² McAdams, *op. cit.*, p. 297.

be converted to subatmospheric pressures by the equation of Jakob:¹

$$h_v = h_{v,14.7 \text{ psia}} \left(\frac{p}{p_{14.7 \text{ psia}}} \right)^{1/3} \quad (14.1)$$

For superatmospheric pressures up to 226 psi the coefficient is given by

$$h_v = h_{v,14.7 \text{ psia}} \left(\frac{p}{p_{14.7 \text{ psia}}} \right)^{1/6} \quad (14.2)$$

where h_v and p refer to the new conditions.

Since blanketing is caused by the accumulation of vapor bubbles, the pressure on the system is important in defining the size of individual bubbles. The influence of the viscosities and surface tensions of liquids on their respective atmospheric coefficients as functions of the absolute values of the properties has not been correlated. From experiments on single liquids outside single tubes with variable pressure Cryder and Finalborgo,² working at low fluxes, obtained a family of curves of nearly uniform slope when plotting h_v vs. $(\Delta t)_w$. Their mean equation is

$$\log \frac{h_{v,14.7 \text{ psia}}}{h_v} = 0.015(t_{14.7 \text{ psi}} - t) \quad (14.3)$$

where h_v and t refer to the new conditions.

Classification of Vaporizing Equipment. There are two principal types of tubular vaporizing equipment used in industry: *boilers* and *vaporizing exchangers*. Boilers are directly fired tubular apparatus which primarily convert fuel energy into latent heat of vaporization. Vaporizing exchangers are unfired and convert the latent or sensible heat of one fluid into the latent heat of vaporization of another. If a vaporizing exchanger is used for the evaporation of water or an aqueous solution, it is now fairly conventional to call it an *evaporator*. If used to supply the heat requirements at the bottom of a distilling column, whether the vapor formed be steam or not, it is a *reboiler*. When not used for the formation of steam and not a part of a distillation process, a vaporizing exchanger is simply called a *vaporizer*. When an evaporator is used in connection with a power-generating system for the production of pure water or for any of the evaporative processes associated with power generation, it is a *power-plant evaporator*. When an evaporator is used to concentrate a chemical solution by the evaporation of solvent water, it is a *chemical evaporator*. Both classes differ in design. Unlike evaporators it is the object of reboilers to supply part of the heat required for distillation and not a change in concentration, although a change generally cannot

¹ Jakob, M., *Tech. Bull. Armour Inst. Tech.*, **2**, No. 1 (1939).

² Cryder, D. S., and A. C. Finalborgo, *Trans. AIChE*, **33**, 346-361 (1937).

be avoided. Very often the term *evaporator* is also applied to a combination of several pieces of equipment each of which can also be defined as an evaporator.

Unfortunately certain classes of evaporators are still designed as part of an art rather than the rational summation of the individual resistances to heat flow as practiced heretofore. This is due to the high transfer coefficients with which certain classes of evaporators operate and the difficulty of identifying each of the small individual resistances which make up the overall resistance. As in the case of surface condensers in Chap. 12, numerous classes of evaporators are designed on the basis of *accepted* overall coefficients, and it is these classes and their processes which are treated in this chapter. Generally they involve vaporization from pools as compared with vaporization in the shell or tubes of a 1-2 exchanger. Evaporators which may be or are usually designed from individual coefficients will be treated in Chap. 15.

POWER-PLANT EVAPORATORS¹

Introduction. One of the main purposes of power-plant evaporators is to provide relatively pure water for boiler feed. The principal features

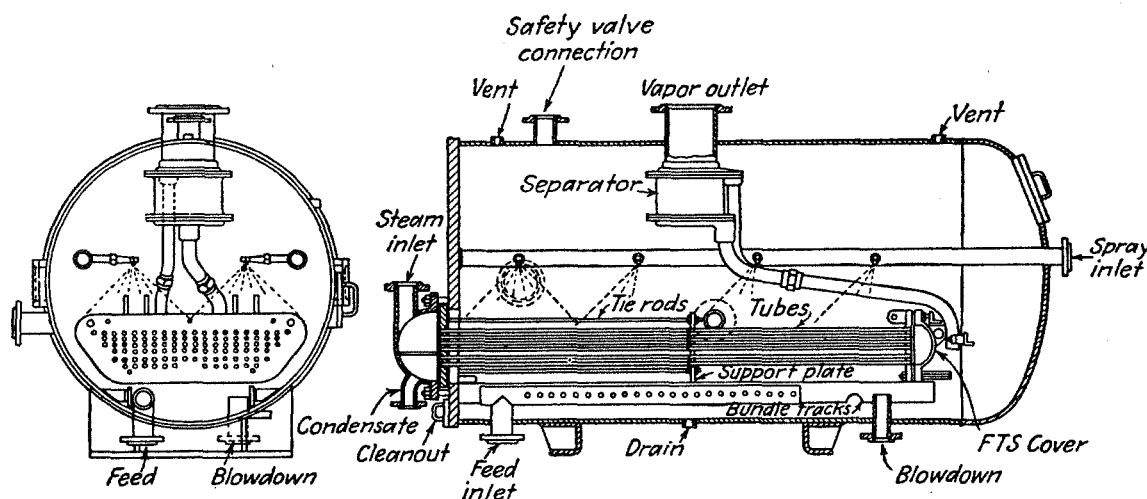


FIG. 14.4. Power-plant evaporator. (The Lummus Company.)

incorporated in power-plant evaporators are a tubular heating element, a space in which the liquid droplets which are carried up by the bursting bubbles may be *disengaged*, and a means of removing the scale from the outsides of the tubes. Three typical examples, each with merit, are shown in Figs. 14.4 to 14.6. In Fig. 14.4 the tube bundle is laid out flat and the feed is introduced at the bottom. In Fig. 14.5 the bundle is cylindrical and the feed is introduced just below the liquid level. All

¹ The author is indebted to Gerald D. Dodd of the Foster Wheeler Corporation for his generous assistance in the preparation of this section. Readers interested only in chemical evaporation may omit pages 379 to 393 without loss of context.

operate half full of water, the upper half being the disengaging space, and all are equipped with separators which return the entrained liquid below the liquid level in the evaporator.

The Treatment of Feed Water. When a pound of steam is evaporated from a pool of boiling water, most of the impurities originally in the water

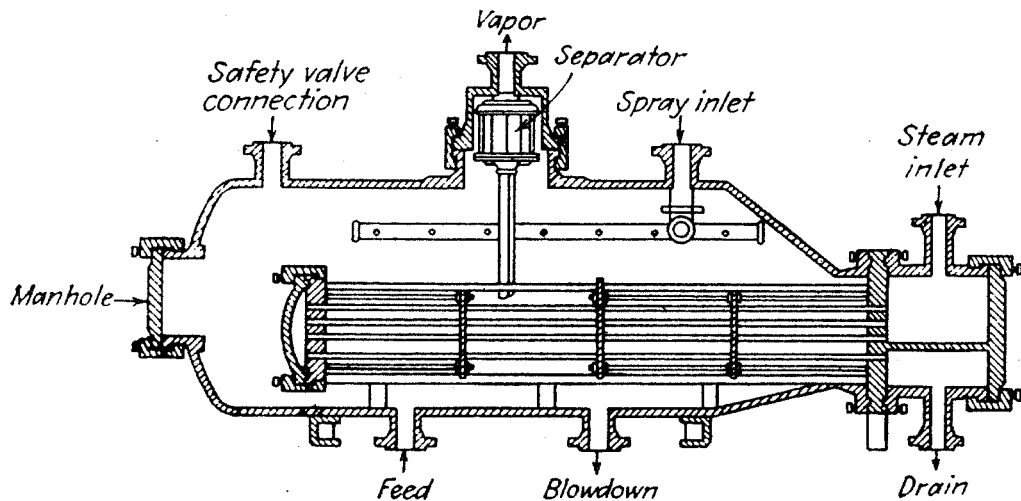


FIG. 14.5. Power-plant evaporator. (Alco Products.)

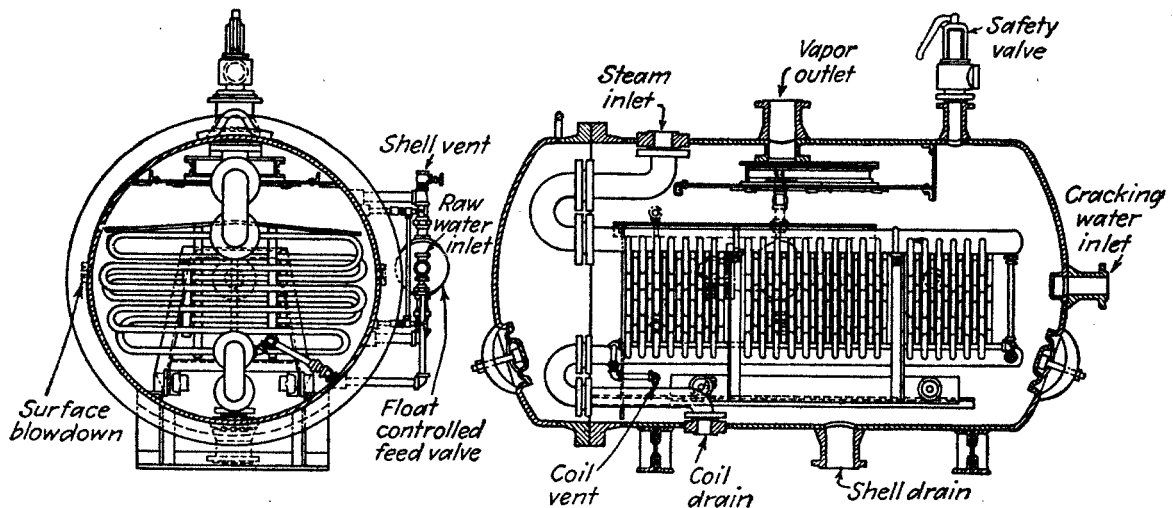


FIG. 14.6. Power-plant evaporator. (Foster Wheeler Corporation.)

remain behind and in time tend to form scale on the submerged heating surface. All natural water contains mineral salts, of which those of calcium and magnesium in particular form scale. The scale forms more rapidly on hot surfaces and is an additional resistance to the flow of heat. This is detrimental to the performance of evaporators which are designed for high rates of heat transfer. Water is classified as hard or soft principally from its behavior in household use. Soap reacts with the calcium and magnesium salts, forming insoluble compounds, but water which is considered soft in the home may still be unsatisfactory as feed to a continuous evaporator. In order to reduce the accumulation of the scale-

forming materials, it is customary to vaporize only 90 per cent of the water feed continuously, the remaining 10 per cent being drained continuously as *blowdown*.

There are three principal chemical means by which waters are softened. These are (1) the cold lime-soda process, (2) the hot lime-soda process, and (3) the zeolite process. The last is by far the commonest on new installations. In the sodium zeolite process a complex sodium silicate reacts continuously with the scale-forming compounds of the feed water, replacing the positive ions of the calcium and magnesium compounds with sodium ions and at the same time retaining the calcium and magnesium compounds as calcium and magnesium zeolites. It is therefore necessary to have two zeolite beds in operation so that the calcium and magnesium can be removed from one bed while the other accumulates calcium and magnesium as zeolites. A detailed discussion of water conditioning is beyond the scope of this chapter, but since evaporation is itself a purification process, the question arises as to the need for preliminary chemical purification. The reasons are as follows: When bubbles of vapor are disengaged from the liquid at the surface of a pool, the vapor *entrains*, or carries off, some of the liquid water with it which has not been distilled and consequently contains the impurities of the concentrated *blowdown*. When these small quantities of carry-over are continuously fed to a boiler, they cause scale deposition on the boiler tubes. In addition they increase the quantity of blowdown which must be removed from the boiler where the temperature is greater than in the evaporator and which represents an added loss in sensible heat from the power generating system.

The softening of water prior to entering the evaporator is in no sense a guarantee of the purity of the final boiler feed since a number of factors influence the amount of entrainment occurring at the surface of the liquid. Significant is the ratio of the total solids to suspended solids or the ratio of total solids to sodium alkaline solids, particularly when accompanied by a large quantity of suspended solids or a minute quantity of inorganic solids. When the softening reduces the surface tension of the water appreciably, excessive priming or foaming invariably results. The control of foaming is consequently one of the most important considerations in evaporator design. The standard required purity of a typical evaporator product in the United States contains not more than two to four parts of mineral solids per U.S. gallon. Notwithstanding these precautions the formation of scale in an evaporator is such that it is considered normal to descale a power-plant evaporator every 24 hr.

There are three principal types of scale: (1) soft scale which can be washed from the tube, (2) hard scale which can be removed only manu-

ally, and (3) hard scale which can be *shocked* from the tube. All three evaporators in Figs. 14.4 to 14.6 are equipped for the removal of the last type of scale. The principle of scale removal by shocking is as follows: In Figs. 14.4 and 14.5 the distance between the tube sheets is fixed by external bars connecting them. In order to descale, all the hot water is removed from the shell and steam is circulated through the tubes, causing expansion of the tubes alone so that they bow if originally straight or straighten if originally installed with a bend at their centers. The hot

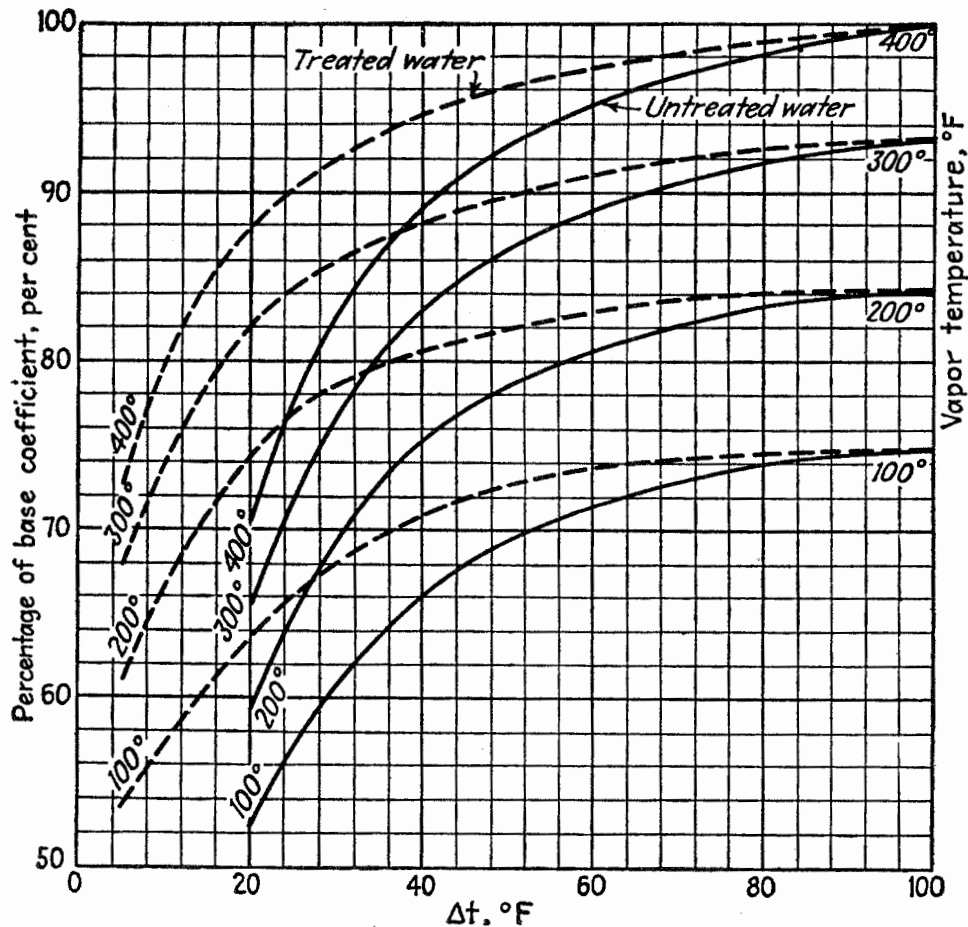


FIG. 14.7. Commercial evaporation coefficients for water.

scale is stressed owing to the expansion, and cold water is sprayed on the tubes, resulting in a sudden differential contraction which cracks the scale and causes it to be shed. If the evaporator is small, 100 to 600 ft², it is customary to flood the shell with cold water instead of spraying. For large installations the time required to fill and drain the shell would cause too long a shutdown period, and spraying is employed. Another variation of the power-plant evaporator is shown in Fig. 14.6 using serpentine tubes. The bundle is stressed between the vertical support pieces so that the scale is effectively removed from the curved portions as well as the flat portions of the tube.

Heat-transfer Coefficients in Power-plant Evaporators. The rates of heat transfer in power-plant evaporators, as mentioned before, are not treated on the basis of individual film coefficients. Because of the rapidity of scale formation and the nature of the scale resistance, the overall rates are based on the normal operating cycle of about one descaling every 24 hr. In Fig. 14.7 the percentages of a base average overall coefficient U_D are plotted against Δt which in an evaporator is the difference between the saturation temperature of the heating steam and the saturation temperature on the shell side. This temperature difference is always employed when evaporators are designed from overall coefficients. In the power-plant evaporator it is called the *temperature head*, and in chemical evaporators it is the *apparent temperature difference*, designated $(\Delta t)_a$. The values of U_D and Δt so defined permit the direct substitution of U_D in $Q = U_D A \Delta t$. The overall coefficient is greatly influenced by the pressure on the system, since the vapor volume of the bubbles is less at high pressure than at low pressures so that larger coefficients may be expected for the former. This is reflected by the curves which have been plotted in Fig. 14.7 as functions of the vapor temperature and which represent indirectly the operating pressure of the shell side. The curves level off abruptly at high-temperature heads owing to incipient vapor blanketing. The base value of the overall coefficient is varied in industry owing both to special problems suggested by a chemical analysis of the water and to changing competition among manufacturers.

A value of 700 Btu/(hr)(ft²)(°F) is a fair average for the base value of U_D , although higher base coefficients have frequently been used.

Example 14.1. Calculation of Evaporator Surface. 10,000 lb/hr of distilled water is required from untreated water. Steam is available at 300°F, and the condenser will vent to the atmosphere. How much surface is required?

Assume a pressure drop through the condenser and lines of about 5 psi. The saturation temperature in the evaporator shell will be 19.7 psia or 226°F.

Heat balance:

$$Q_{\text{evap}} = 10,000 \times 961 = 9,610,000 \text{ Btu/hr}$$

$$Q_{300^\circ\text{F}} = 10,550 \times 910 = 9,610,000 \text{ Btu/hr}$$

Temperature head:

$$\Delta t = 300 - 226 = 74^\circ\text{F}$$

Overall coefficient:

From Fig. 14.7 at a heat head of 74°F and a vapor temperature of 226°F, the coefficient is 86.5 per cent of base. Using a base of 700 Btu/(hr)(ft²)°F

$$U_D = 700 \times 0.865 = 605$$

$$A = \frac{Q}{U_D \Delta t} = \frac{9,610,000}{605 \times 74} = 2150 \text{ ft}^2$$

Temperature Differences at Less than Maximum Flux. The preceding calculation is of value only to estimate the surface requirement. The surface will fill half of the shell or less, and the method of spacing the surface varies greatly with different manufacturers. Tubes, as a rule, are spaced farther apart in evaporators than in exchangers.

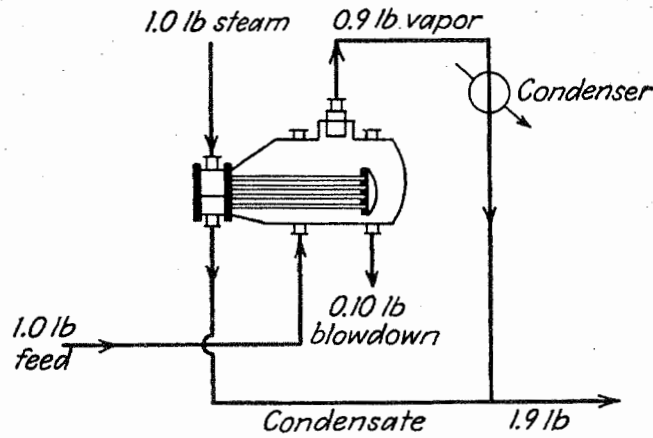
According to Fig. 14.7 it would be possible to obtain increasing overall coefficients up to a temperature head of 100°F although the vapor temperature is 212°F corresponding to atmospheric pressure. Previously it was stated that the critical temperature difference at atmospheric pressure was 45°F. A temperature head of Δt of 100°F corresponds to a critical temperature difference $(\Delta t)_w$ of about 75°F but the limitation of *the critical temperature difference holds only when operating at maximum flux*. In Fig. 14.7 the flux for 212°F vapor and $\Delta t = 100^\circ\text{F}$ is

$$700 \times 0.85 \times 100 = 60,000 \text{ Btu}/(\text{hr})(\text{ft}^2)$$

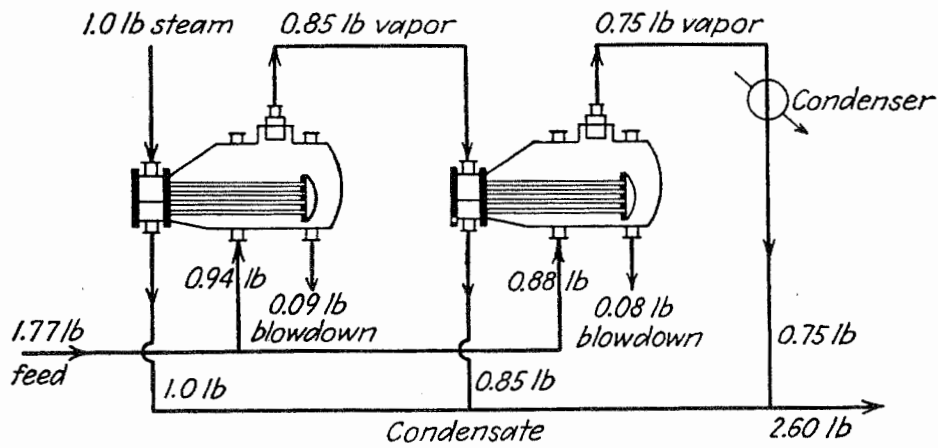
If the flux has a value less than maximum, the temperature difference $(\Delta t)_w$ may be greater than the critical temperature difference. The flux is the principal index of vapor binding, and for this reason it is always restricted to a fraction of the maximum attainable when designing commercial evaporators. Restricting the flux to a *design* maximum amounts to an increase in the total surface for the purpose of reducing the amount of vaporization per unit of heat-transfer surface.

Multiple-effect Evaporation. In the production of distilled water the vapor formed in the evaporator is useful steam as well as relatively pure water. If a pound of steam is supplied to an evaporator as shown in Fig. 14.8a, it can be used to produce about 0.9 lb of steam from a pound of water. The remaining 0.1 lb of water contains the bulk of the impurities, and it is removed from the evaporator as evaporator *blowdown*. The 0.9 lb of vapor from the evaporator can be condensed by partially preheating the evaporator feed, or in the power plant, it can be mixed directly with cold return condensate before being fed to the boiler.

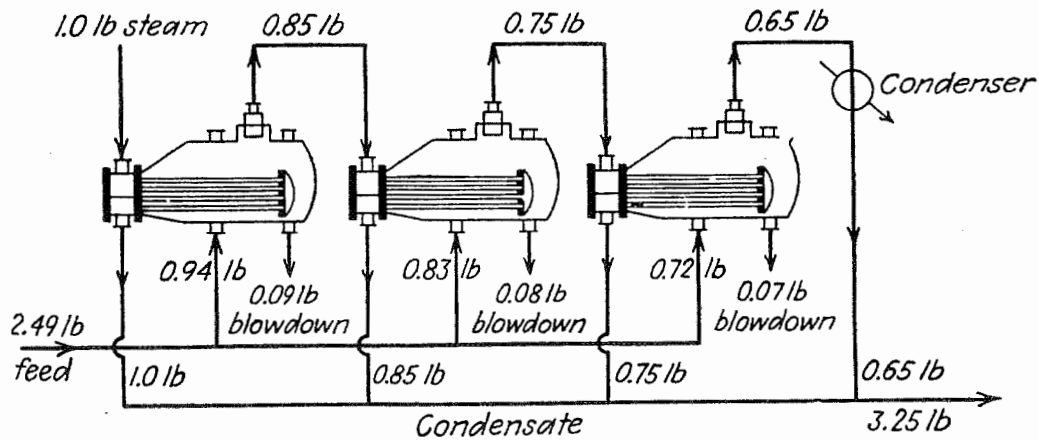
If, however, the original pound of steam was supplied to a process as shown in Fig. 14.8b and the vapor produced in the first evaporator was then used as a heat source in a second evaporator operating at a lower pressure than the first, an additional utilization could be made of most of the heat. If both evaporators in Fig. 14.8b were fed in *parallel* with raw water, about 0.85 lb of pure water would be formed in the first *effect* and about 0.75 lb would be formed in the second effect. For each pound of steam supplied about 1.6 lb of pure water would be produced. The original pound of steam can also be considered a pound of pure water. When the vapor formed in the first effect is reused as the heating medium in a second effect, it is a *double-effect* evaporator. When applied to three



(a) SINGLE EFFECT-1 LB STEAM PRODUCES 1+0.9 LB DISTILLED WATER



(b) DOUBLE EFFECT-1 LB STEAM PRODUCES 1+1.60 LB DISTILLED WATER



(c) TRIPLE EFFECT-1 LB STEAM PRODUCES 1+2.25 LB DISTILLED WATER

FIG. 14.8. Multiple-effect evaporation with parallel feed.

effects as shown in Fig. 14.8c, it is a *triple-effect* evaporator and the original pound of steam produces about 2.25 lb of pure water.

In order to maintain temperature differences for heat transfer between the vapor from one effect and the boiling liquid of the next effect, the pressure on each succeeding evaporator must be lower than its predecessor. The quantity of blowdown, which is arrived at from experience with water of different impurities, carries with it considerable sensible heat and thereby reduces the amount of vaporization which can be realized in succeeding effects. Due in part to this escape of heat from the system there is a limit to the number of effects which are justifiable. The fixed charges for additional effects ultimately dissipate the savings in energy which result from the use of a large number of effects.

Power-plant Evaporation Processes. Power-plant evaporation processes fall into four classes:

1. *Make-up* evaporators for boiler feed
2. *Process* evaporators for the production of purified water
3. *Heat-transformer* evaporators
4. *Salt-water distillers*

These processes are discussed below with typical flow sheets which contain all the necessary information computed from simple heat balances. Since only enthalpy differences are involved, it has not been felt necessary to include the individual heat balances for each case.

1. *Make-up Evaporators.* Make-up evaporators supply boiler feed water to replace leakage and losses from the system as process steam in plants or as discarded condensate. This is by far the largest class of evaporation process and is usually accomplished in a single-effect evaporator, although occasionally a double-effect evaporator may be used depending upon the characteristics of the condensate cycle in the power plant and the amount of make-up required. There is hardly a power plant built today which does not include such equipment. The evaporators themselves are small, containing about 100 to 1000 ft² of surface. Typical process examples of the use of single-effect make-up evaporators are shown in Figs. 14.9 and 14.10. In Fig. 14.9 a turbine operating on 150,000 lb/hr of steam at 400 psig and superheated to 800°F is bled in three stages to supply sufficient high-temperature superheated steam so that the make-up can be heated to the saturation temperature of the boiler corresponding to 400 psig. Naturally saturated steam at 400 psig cannot be used to heat feed up to its saturation temperature. It can be shown by an economic balance that it would be wasteful to use steam directly from the superheater at 800°F for feed-water heating in preference to the use of partially spent superheated steam from the turbine.

Bleed steam from the eighth stage of the turbine is run directly to the evaporator where it vaporizes 9000 lb/hr of make-up. The vaporized make-up and evaporator steam are both condensed in an evaporator-condenser by the bulk of boiler feed with a 5°F temperature approach. The remainder of the diagram is obtained by trial-and-error calculation for the optimum efficiency of the cycle.

A variation of this flow arrangement is shown in Fig. 14.10 in which some of the steam formed in the evaporator is combined with the make-up

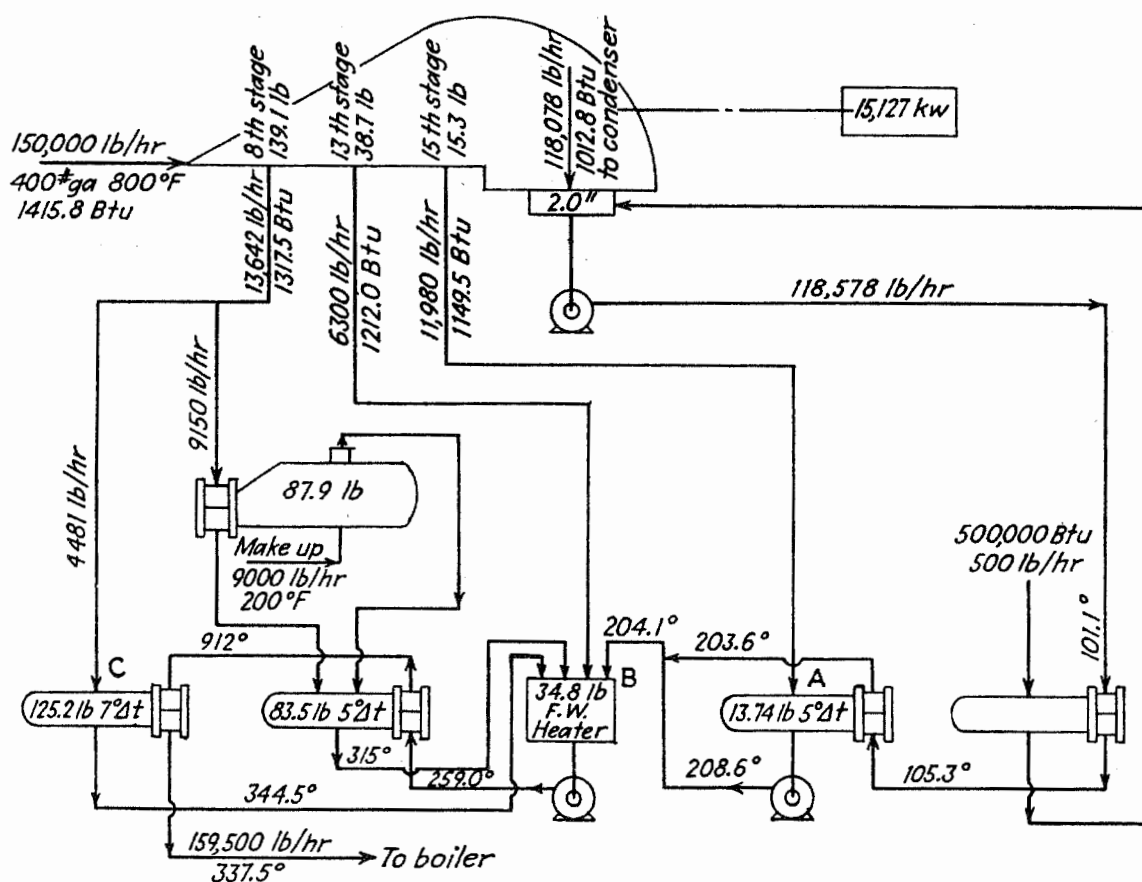


FIG. 14.9. Single-effect make-up evaporator with evaporator-condenser between two bleed heaters.

in a preheater. The remainder of the steam formed in the evaporator goes to an evaporator-condenser, where it is condensed by the boiler feed. The condensate from the evaporator tubes and the evaporator-condenser is then combined in the flash tank. In the flash tank the condensate from the evaporator comes out as superheated (339.4°F) liquid compared with the condensate from the evaporator-condenser (313.3°F), and a portion flashes back to the evaporator-condenser, thereby providing all the pressure differences necessary for the operation of the equipment. In Fig. 14.11 a triple-effect evaporator is shown for cases where an excessively large amount of make-up water is required.

between that particular stage heater and the next lower one *B*. This arrangement permits taking all the heat for heating feed water in the evaporator-condenser and for the higher of the two feed heaters *C*, from the high bleed point without displacing the bleed steam from the next lower bleed point. It can readily be seen that, if the evaporator discharges into the next lower bleed heater, the heat absorbed in the next lower heater will come from the next higher bleed point and will have by-passed several stages of the turbine in so doing, resulting in a loss of kilowatts.

2. *Process Evaporators*. There are a number of industries which continuously require large quantities of distilled water. This type of plant employs double, triple, or quadruple effects and receives heat either

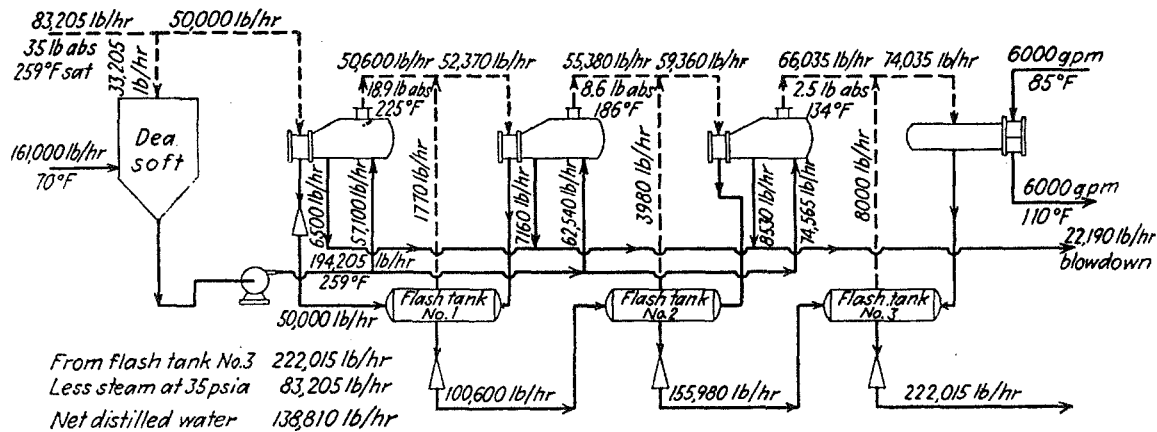


FIG. 14.12. Triple-effect process evaporator.

from a bleed point or from the boiler directly. The selection of the number of effects is closely allied to the relationship among the fixed charges and the steam operating cost. Multiple-effect evaporators with parallel feed need not have all effects simultaneously in operation and can therefore be adjusted if the demand for distilled water varies. Evaporators for this type of service are generally of medium size, about 500 to 2000 ft² per shell. A typical triple-effect process evaporator is shown in Fig. 14.12, where 83,205 lb/hr of steam at 35 psia (20 lb gage) and saturated is split to mix directly with the cold feed at 70°F and for vaporization in the evaporator. This process employs several of the elements discussed under make-up evaporators. The final product is 222,015 lb/hr of distilled treated water. In order to obtain a maximum quantity of vaporization from a given quantity of steam initially near atmospheric pressure, temperature differences are maintained from the first to the last effect by operating with the last effect under vacuum. This is accomplished by using a steam jet vacuum pump on the last effect. Thus the shell side of the first effect in Fig. 14.12 operates at 18.9 psia or 225°F, the second at 8.6 psia or 186°F, and the third at

2.5 psia or 134°F. This establishes differences in the effects of 34, 39, and 52°F. The operating pressures for each of the effects is determined by trial and error, so that all three effects will have the same surface as calculated by $A = Q/U_D \Delta t$. This procedure will be demonstrated in detail in the treatment of a chemical evaporator. If the vacuum were not applied, the maximum available temperature difference over the three effects would be from 259 to 212°F or 47 instead of 125°F as shown in the flow sheet.

The principle of vacuum evaporation is extensively used in chemical evaporation. Since the vapor from the last effect is at a low temperature, it has little value in the preheating of the feed, its temperature in this case being 134°F. The evaporator-condenser, therefore, operates with cooling-tower water instead of feed water. It will be noted that, although

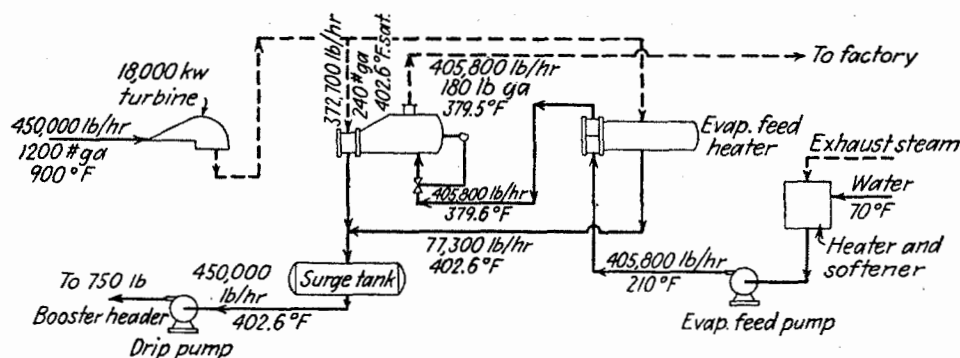


FIG. 14.13. Heat-transformer evaporator.

the heat from the last-effect vapor is rejected, the blowdown heat losses of such a system are reduced considerably by being at a low heat level. The steam required for the vacuum pump must also be taken into account in computing the efficiencies of vacuum processes.

3. Heat-transformer Evaporators. The heat-transformer evaporator is a single-effect system of one or more shells *in parallel* receiving steam from the exhaust of a high-pressure turbine or high-pressure engine. Flow sheets are shown in Figs. 14.13 and 14.14. The purpose of this type of evaporator is to condense steam from a high-pressure boiler which has passed through a high-pressure turbine and into the evaporator. The condensate is then returned directly to the high-pressure boiler by a pressure booster, thereby keeping the high-pressure circuit closed and continuously supplied with high-pressure boiler water and steam. Obviously high-pressure boiler and turbine installations are favorably affected by this circuit. By the condensation of the exhaust steam from the high-pressure turbine or engine the heat transfer in the evaporator is used to produce large quantities of process steam, all or a large part of which is never returned to the evaporator system. If the condensate is not

returned, it is because it may be difficult to collect or the vapor may be consumed in a chemical or heating process or it may be continuously contaminated.

This type of evaporator is relatively large, having been built in a single unit in sizes up to 11,000 ft² of surface and capable of producing 150,000 to 200,000 lb/hr of steam. Single evaporators of this size are 10 to 11 ft

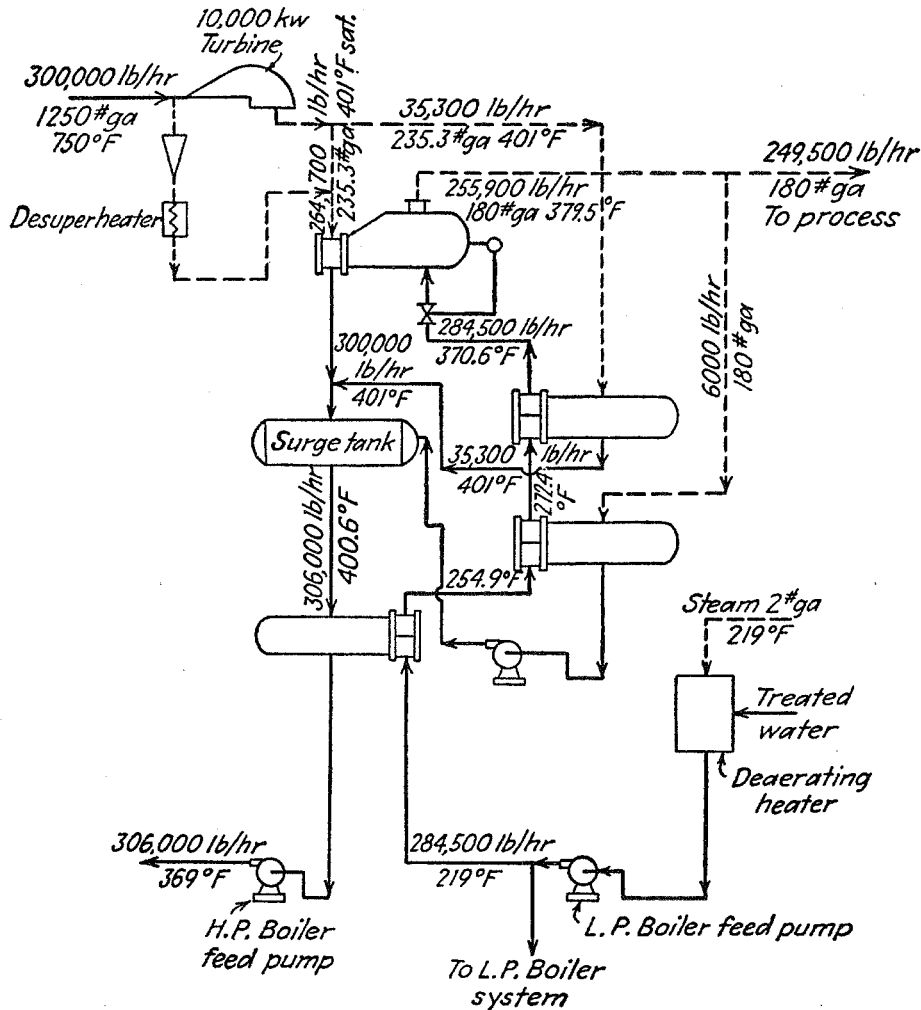


FIG. 14.14. Heat-transformer evaporator.

in diameter and 40 to 50 ft in length. There are not many of the larger installations, but where large quantities of process or heating steam are required, it is readily understood that a high-pressure boiler operating at 1400 psi throttle pressure and exhausting at 200 psi produces large quantities of "by-product" power. This power is charged only with the heat extracted on the passage of the steam through the turbine, the remaining charges being debited to the cost of producing high-pressure steam, the primary object of the steam cycle. When the energy from a high-pressure line is continuously used to produce low-pressure steam for purposes outside the original installation, it is called a *heat transformer*

from its similarity to a stepdown electrical transformer and also a *reducing-valve* transformer. It is actually the only way in which high-pressure saturated steam can be converted to low-pressure saturated steam without superheat. The reason for the reducing-valve evaporator, however, is primarily to preserve the pressure on the high-pressure side.

Sometimes the heat going to process must be transported a considerable distance. To prevent condensation the vapor should be somewhat superheated before leaving the generating system, although the heat-transformer evaporator produces only saturated steam. For this purpose

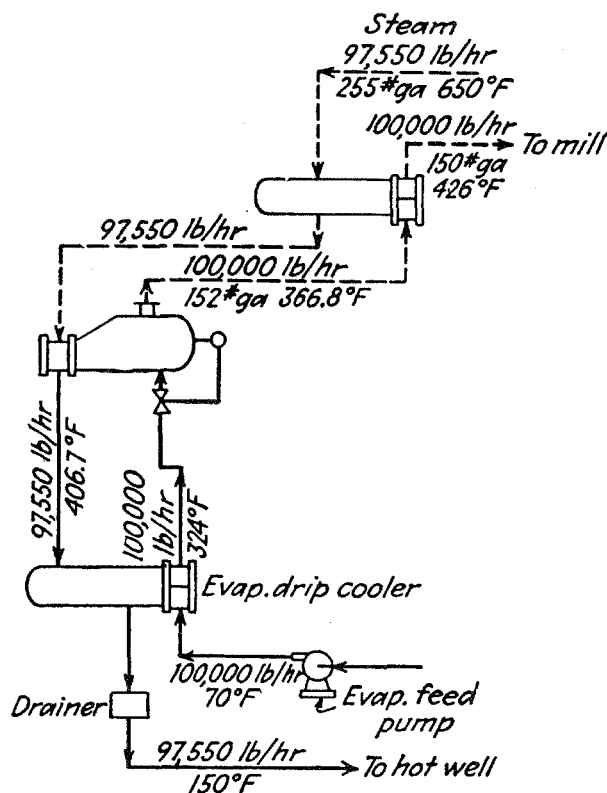


Fig. 14.15. Heat-transformer evaporator supplying superheat.

the arrangement shown in Fig. 14.15 is generally used to provide a heat exchanger or *reheater* which passes the steam to the heat-transformer evaporator through the tubes and the vapor from the evaporator through the shell, thereby providing some superheat.

4. *Salt-water Distillers.* A pound of fuel can normally yield about 10 lb of steam, and in a double-effect evaporator for use with salt water 10 lb of steam will yield a total of 18.5 lb of pure water. It is little wonder, then, that ships at sea commonly produce their own water requirements from sea water. Sea water contains about 3 per cent solids by weight corresponding to about 34,000 ppm compared with 340 ppm in fresh water. Instead of about 90 per cent vaporization it is customary to vaporize only about one-third of the feed. The remainder, containing

about 5 per cent solids or 51,000 ppm, is dumped overboard. Because of the largeness of the blowdown the use of a vacuum system with low evaporation temperatures is desirable, since low temperatures are also favorable to low rates of scaling. Unlike evaporators in stationary power

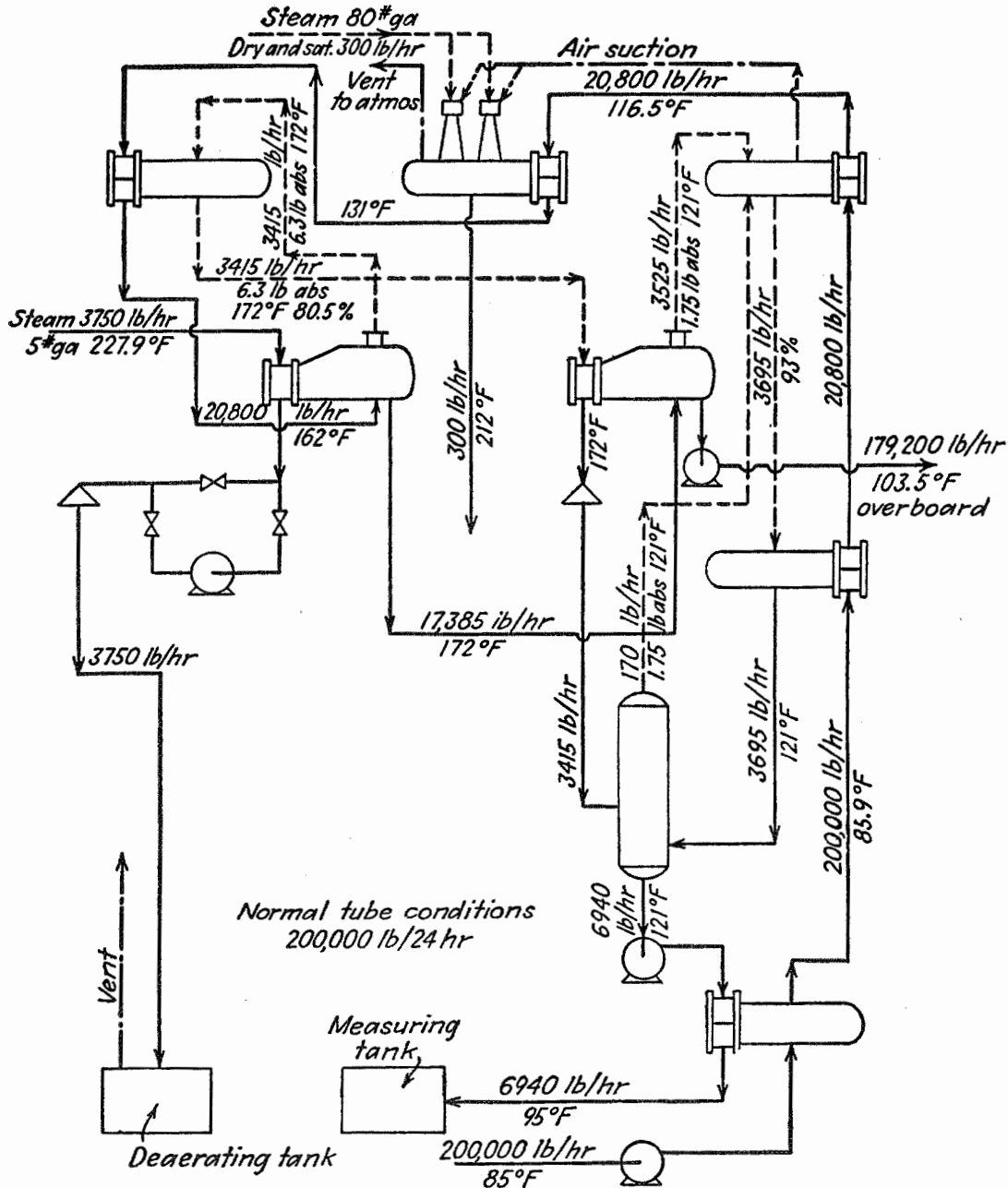


FIG. 14.16. Salt-water distiller.

plants salt-water evaporators or distillers operate for 600 to 700 hr without descaling. The flow sheet of a typical salt-water distiller is shown in Fig. 14.16.

Vacuum Operation of a Process. Evaporators frequently operate with the last effect under vacuum, and one of the important considerations is

the method of continuously maintaining the vacuum. The use of a mechanical compressor on the last stage is usually prohibitive because of the energy required for compression. The specific volume of water vapor at 2 in. Hg abs is 399.2 ft³/lb. Furthermore, there is little reason to

operate a mechanical compressor when the required reduction in volume can be partially achieved by condensation. The latter is one of the principles embodied in the

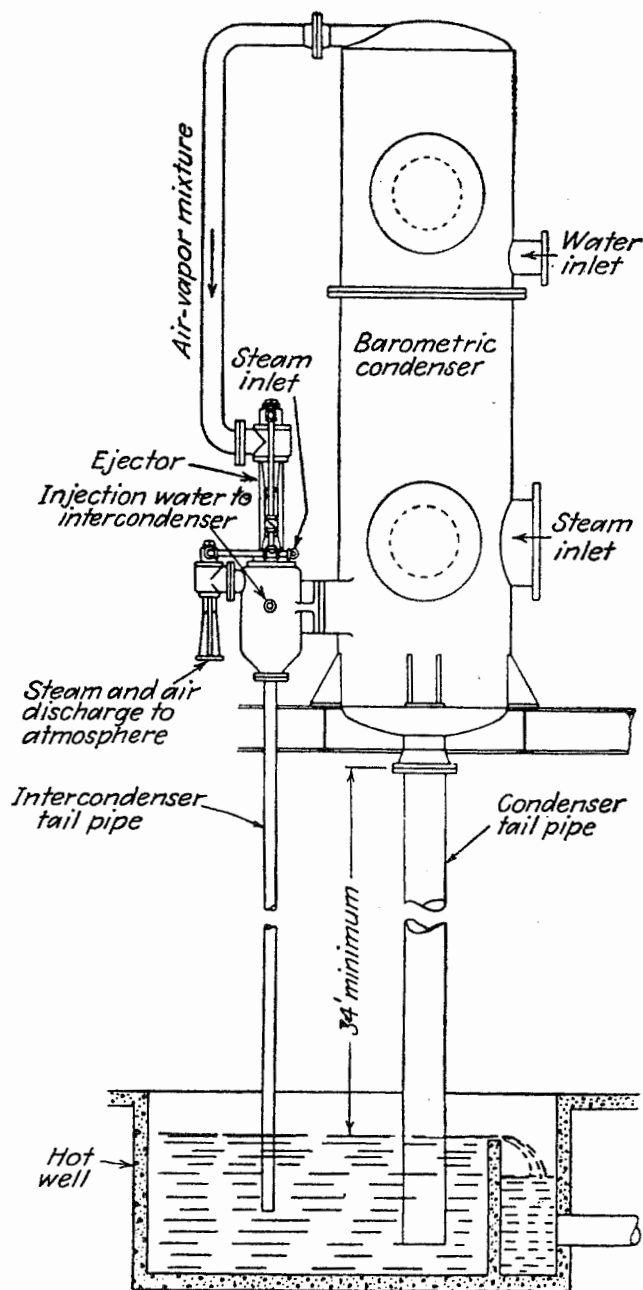


FIG. 14.17. Two-stage ejector with jet intercondenser, serving a barometric condenser. (Foster Wheeler Corporation.)

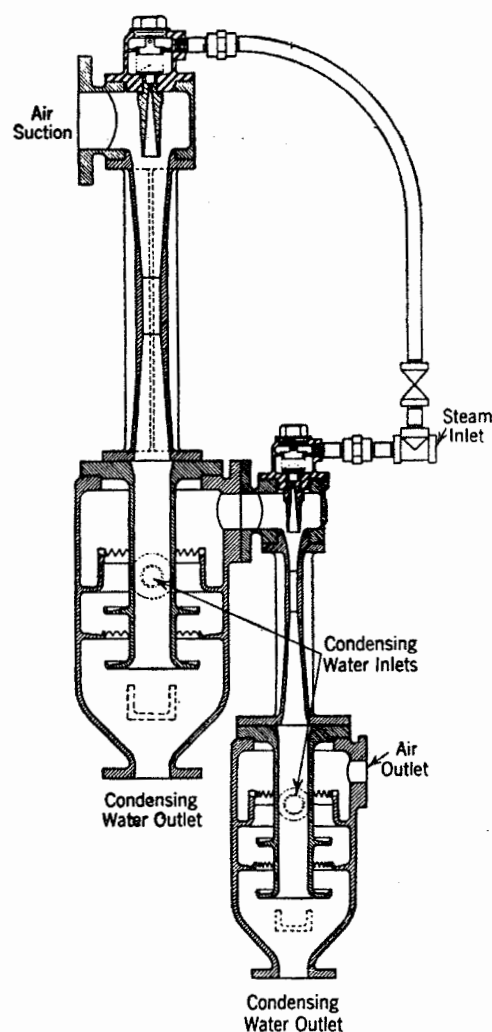


FIG. 14.18a. Two-stage ejector with jet inter- and aftercondensers. (Foster Wheeler Corporation.)

barometric condenser with air ejectors, an example of which is shown in Fig. 14.17. It is operated by the two steam jet ejectors.

A single ejector connected to a condenser is capable of maintaining a vacuum of about 26.5 in. Hg abs and can be made with several jets

replacing the single-motive steam nozzle. This provides a more uniform distribution of steam in the mixing zone. When a vacuum of 26.5 to 29.3 in. is desired, it can be accomplished by means of a two-stage ejector as shown in Fig. 14.18a and b. For higher vacuums the use of a three-stage ejector is required. The two-stage detail in Fig. 14.18a is the same as that employed in Fig. 14.17. The condenser in Fig. 14.17 is a *barometric condenser* equipped with a water inlet and distribution plates, so that the water entering the inlet cascades or sprays over the incoming

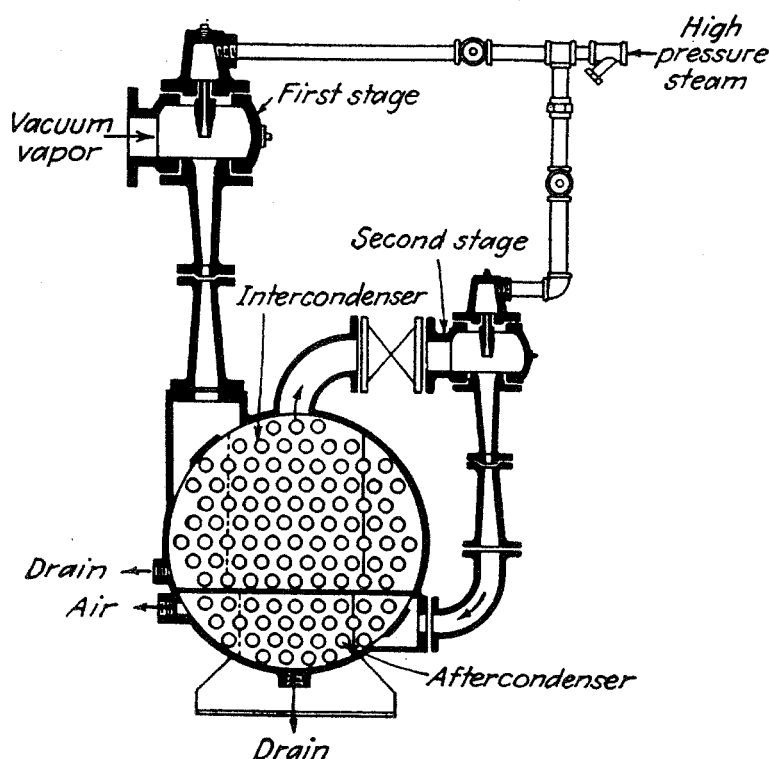


FIG. 14.18b. Two-stage ejector with surface inter- and aftercondensers. (The Lummus Company.)

stream from the evaporator or precondenser and removes a large part of the steam from the process as condensate. The remaining air with less steam flows overhead into the first stage ejector. After compression in the first stage the partial pressure of the steam will have been increased and most of the remaining vapor can be condensed by another direct contact with cooling water.

Again referring to Fig. 14.17, in order to remove the water and condensate from the assembly without losing vacuum it is necessary that a leg of liquid be maintained with a hydrostatic head zp equal to the difference between the vacuum and atmospheric pressure, where z is the height and ρ the density. In this manner the upper surface of the liquid in the tail pipe is at a pressure corresponding to the vacuum and the liquid at the bottom of the tail pipe is at atmospheric pressure due to the weight of the hydrostatic head. Thus liquid under vacuum continu-

ously enters the tail pipe, and liquid at atmospheric pressure continually leaves by way of the hot well at the bottom of the tail pipe. Atmospheric pressure corresponds to a hydrostatic head of 34 ft of water, and complete vacuum corresponds to zero hydrostatic head. To maintain a process at substantially complete vacuum requires that a leg of 34 ft of water be maintained between the barometric condenser and the hot well. If a vacuum of less than 29.92 in. Hg is to be maintained by the ejectors but a leg of 34 ft equivalent to 29.92 in. Hg has been provided, it means merely that the height of liquid in the tail pipe will automatically drop

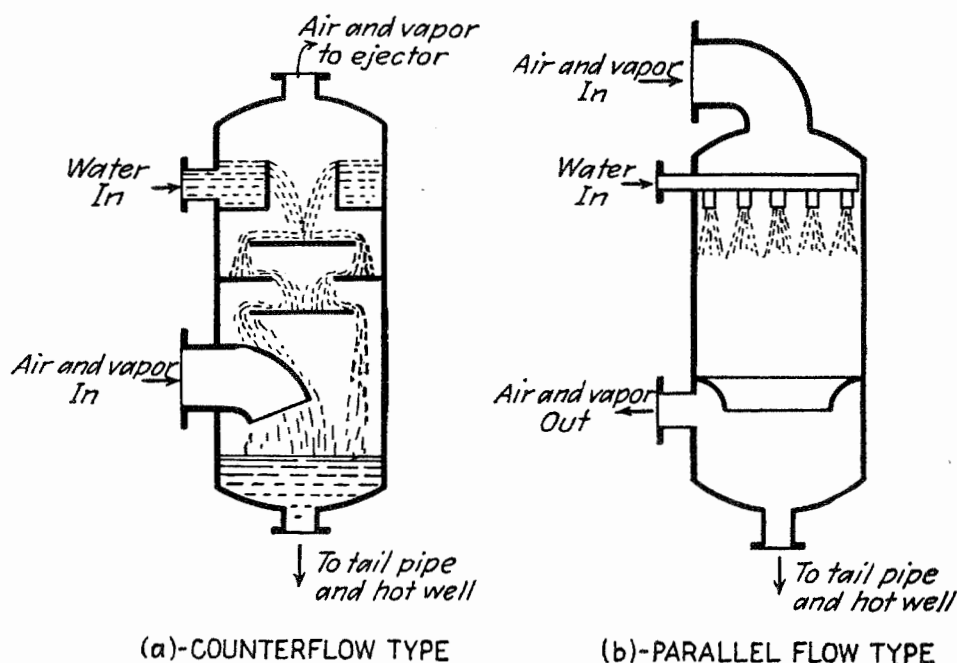


FIG. 14.19. Barometric condensers.

to provide only the necessary hydrostatic difference between the operating vacuum of the ejector and the atmospheric pressure.

Barometric condensers are of two types, counterflow and parallel flow as shown in Fig. 14.19a and b. In counterflow types the temperature of the water at the liquid level may approach the vapor temperature more closely than in parallel-flow types. Counterflow types are preferable where water is at a premium or it is difficult to have the vapor enter from the top. If a pump is used to remove the tail liquid instead of a total barometric height, whatever head is supplied by the pump can be deducted from the total barometric height and the assembly is known as a *low-level* condenser.

The quantity of water required in the barometric condenser can be computed from

$$\text{Gpm} = \frac{Q}{500(T_s - t_w - t_a)} \quad (14.4)$$

where T_s = saturation temperature of the vapor, °F

t_w = temperature of the water, °F

t_a = degrees of approach, to T_s , °F

In counterflow barometric condensers t_a is taken as 5°F.

The two-stage ejector assembly in Fig. 14.18a will produce the same results as that of Fig. 14.18b. It differs only in that the condensation after each stage is accomplished by means of tubular surface instead of the direct contact of the cooling water with the steam or vapor mixture. The tubular intercondenser and aftercondenser are combined in a single shell, since the total surface required is usually quite small. Surface condensation is mandatory where the vacuum exhaust cannot be mixed with the cooling water for reasons of corrosion or chemical reaction.

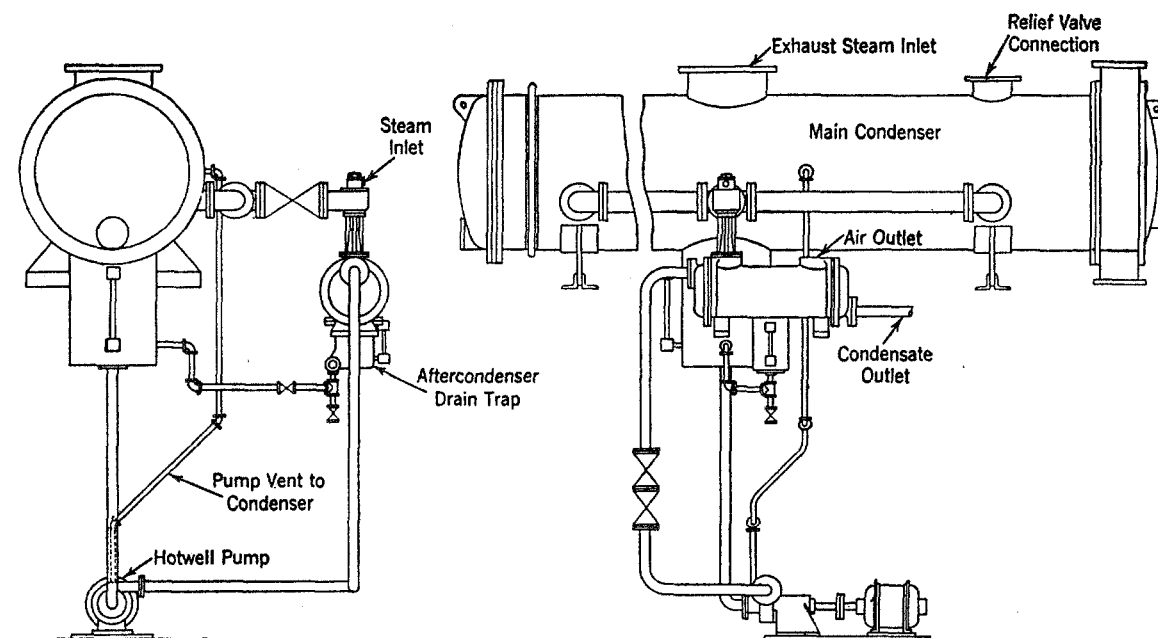


FIG. 14.20. Single-stage ejector with surface aftercondenser serving a surface condenser. (Foster Wheeler Corporation.)

In the operation of a steam turbine or engine the surface condenser discussed in Chap. 12 actually serves as the precondenser for the maintenance of vacuum across the turbine. The surface condenser yields not only condensate but also a mixture of air saturated with water vapor, which must be continuously removed. Failure to remove the air causes an increase in the pressure and temperature of the condenser, as well as a noncondensable air blanket which reduces the overall heat-transfer coefficient. The air removal can be accomplished as shown in Fig. 14.20 with a surface intercondenser and a removal pump. The hot-well pump is provided for low-level installation. A counterpart using a low-level barometric condenser can also be used. For the design and selection of ejectors Jackson¹ has presented a comprehensive discussion.

¹ Jackson, D. H., *Chem. Eng. Progress*, **44**, 347-352 (1948).

CHEMICAL EVAPORATION

Comparison between Power-plant and Chemical Evaporation. The purpose of the majority of power-plant evaporators is the separation of pure water from raw or treated water. The impurities are continuously withdrawn from the system as blowdown. In chemical industry the manufacture of heavy chemicals such as caustic soda, table salt, and sugar starts with dilute aqueous solutions from which large quantities of water must be removed before final crystallization can take place in suitable equipment. In the power-plant evaporator the unevaporated

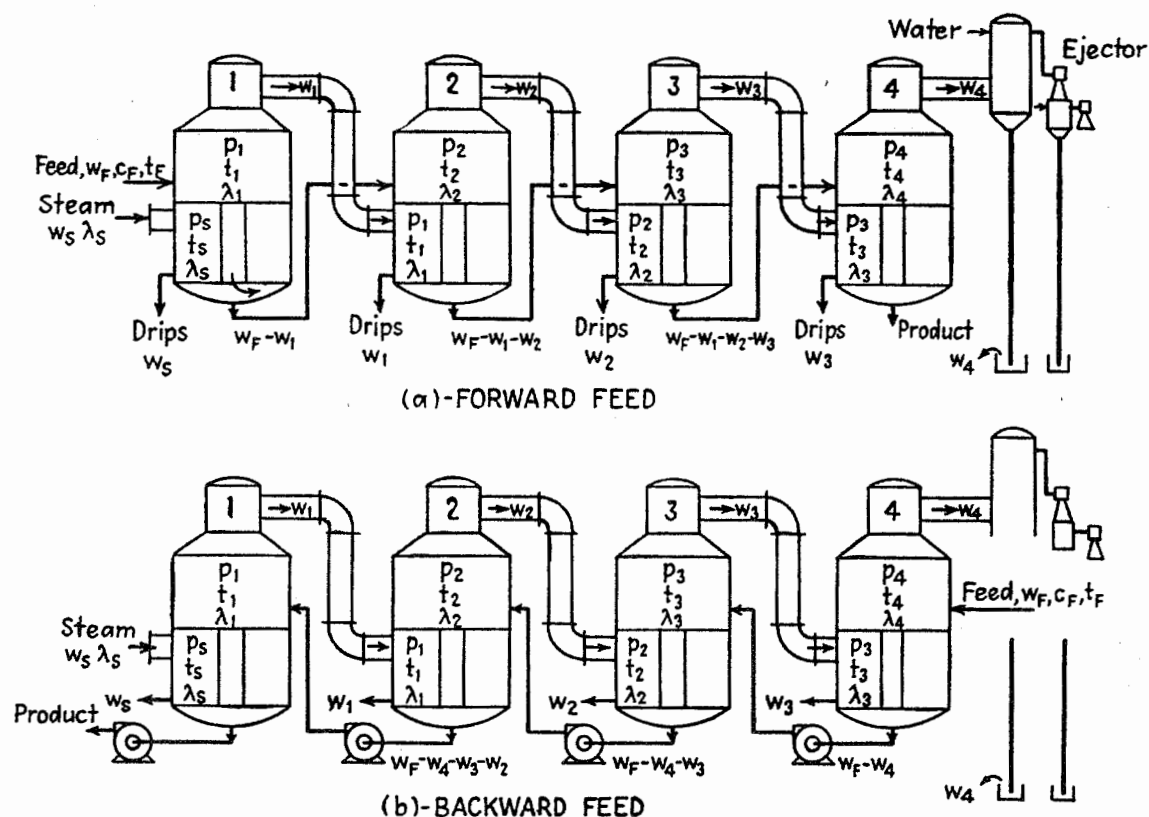


FIG. 14.21. Quadruple-effect chemical evaporator arranged for forward and backward feed.

portion of the feed is the residue, whereas in the chemical evaporator it is the product. This leads to the first of several differences between power-plant and chemical evaporation. These are as follows:

Absence of Blowdown. Chemical evaporators do not operate with blowdown, and instead of liquid being fed in parallel to each body it is usually fed to multiple-effect systems in series. Common methods of feeding are shown in Fig. 14.21a and b. The feed to the first effect is partially evaporated in it and partially in each of the succeeding effects. When the liquid feed flows in the same direction as the vapor, it is *forward feed*, and

when fed in the reverse direction, it is *backward feed*. From the standpoint of effectively using the temperature potentials forward feed is preferable. If the liquid is very viscous, there is an advantage to the use of backward feed, since the temperature of the first effect is always the greatest and the corresponding viscosity will be less. The advantages and disadvantages of both will be discussed later. The absence of blow-down enables greater heat recovery in a chemical evaporator.

Boiling Point Rise (BPR). Although chemical evaporators are capable of high heat efficiencies, they are incapable under certain conditions of a high utilization of temperature potentials and consequently require greater surfaces. This is due to the fact that a concentrated aqueous solution undergoes a *boiling point rise* above the saturation temperature corresponding to pure water at the same pressure. Suppose steam enters the tubes or chest of a chemical evaporator at 45 psia and is to evaporate water from a caustic soda solution. The steam temperature is 274°F. If pure water is evaporated at 18 in. Hg, the temperature of the vapor formed would be 169°F. But because of the dissolved salt the liquor boils at 246°F at 18 in. Hg instead of at 169°F. The temperature difference across the heat-transfer surface is only $274 - 246 = 28^\circ\text{F}$, and the difference of $246 - 169 = 77^\circ\text{F}$ represents lost potential which cannot be attained owing to the presence of the dissolved material. The difference between the temperature of heating vapor and the saturation temperature corresponding to the pressure of the evaporating vapor is the *apparent temperature drop* $(\Delta t)_a$, or $274 - 169 = 105^\circ\text{F}$ in the example above. Heat-transfer coefficients, which are reported on a basis of

$$U_D = \frac{Q}{A(\Delta t)_a},$$

are *apparent overall coefficients*. If coefficients are based on the temperature difference across the heating surface between the heating vapor and the evaporating liquid, as in most cases, $U_D = Q/A \Delta t$, where $\Delta t = 28^\circ\text{F}$ in the example above. If solutions have boiling point rises above about 5°F , the latent heat of vaporization of steam from the solution differs from that obtained from the steam tables (Table 7) at the saturation pressure of the vapor. The latent heat of vaporization for steam from a solution can be computed either from Duhring's relationship or from the equation of Othmer.¹ According to Duhring's rule,

$$\frac{\lambda_w}{\lambda_s} = \frac{\Delta t'_w}{\Delta t'_s} \left(\frac{t'_w}{t'_s} \right)^2 \quad (14.5)$$

¹ Othmer, D. F., *Ind. Eng. Chem.*, **32**, 841-856 (1940).